# Introduction to topology Lecture Notes

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清華大學 求真书院
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# Contents

**1 Topological spaces and continuous maps** **3**
1.1 What does topology study? . . . 3
1.2 Definition of a general topological space . . . 9
1.3 Basis of a topology . . . 10

**2 Further properties of topological spaces** **14**
2.1 Interior, closure and boundary of a subset . . . 14
2.2 Continuous maps . . . 18
2.3 Separation axioms . . . 22
2.4 Metric spaces are normal. . . . 24

**3 Product, disjoint union and completion** **26**
3.1 Cartesian product and disjoint union of sets . . . 26
3.2 Product topology and disjoint union topology . . . 28
3.3 Completion of metric spaces . . . 33

**4 Compactness** **37**
4.1 Key properties of compact spaces . . . 37
4.2 Sequentially compactness . . . 43
4.3 Locally compactness . . . 45

**5 Connectedness and path-connectedness** **49**
5.1 Decomposition of a set . . . 49
5.2 Connected spaces . . . 49
5.3 Path-connectedness . . . 53

**6 Quotient spaces** **59**
6.1 Definition of quotient spaces . . . 59
6.2 Quotient maps . . . 65
6.2.1 Attaching spaces . . . 67

**7 Topological groups and their actions on topological spaces** **69**
7.1 Example of topological groups . . . 69
7.2 Group action on topological spaces . . . 71

**8 Urysohn's lemma and its applications** **78**

**9 Homotopy of maps, the fundamental group** **82**
9.1 Homotopy equivalences . . . 82
9.2 The fundamental group . . . 87



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**10 Covering spaces, the fundamental group of a circle, and applications** **89**
10.1 Covering spaces . . . 89
10.2 Basic properties of fundamental group . . . 92
10.3 Some applications of $\pi_1(S^1)$ . . . 95

**11 Seifert-Van Kampen theorem** **98**
11.1 Fundamental group of a product . . . 98
11.2 Free product of groups . . . 99
11.3 Proof of the Seifert-Van Kampen theorem . . . 102
11.4 Applications of the Seifert-Van Kampen theorem . . . 104

**12 Basic properties of CW complexes** **107**
12.1 CW complexes: definition and examples . . . 107
12.2 Fundamental group of CW complexes . . . 111

**13 Classification of closed surfaces** **116**
13.1 Constructing closed surfaces by gluing edges of a polygon . . . 116
13.2 Simplicial complexes . . . 121
13.3 Orientation on manifolds . . . 123

**14 Classification of covering spaces** **126**
14.1 Lifting criteria . . . 126
14.2 Classification theorem . . . 128

**15 Introduction to the simplical homology** **133**
15.1 Definition of the simplical homology group . . . 133
15.2 The simplical approximation theorem . . . 136
15.3 Applications of the simplical homology . . . 138



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# Chapter 1

# Topological spaces and continuous maps

## 1.1 What does topology study?

Intuitively, topology is the study of geometric shapes, with regard only to those properties that are unchanged by stretching and bending. In topology, a circle is the same as a square because one can be continuously deformed to the other. However, they are not the same with a straight line. The reason is simple: if you remove a single point from a circle, the complement is still connected. But if you remove a single point from the line, the complement is not connected.

In this subsection, we will explain this idea by first studying subsets of $\mathbb{R}^n$.

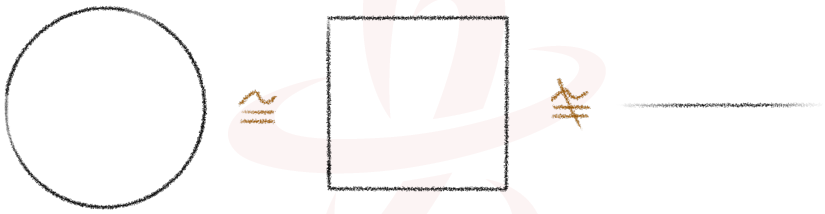

Figure 1.1

**Definition 1.1.** Given $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$, a map $f : U \to V$ is called **continuous** if the following condition holds:

- For any $x \in U$ and any $\epsilon > 0$, there exists $\delta > 0$ such that for any $y \in U$ with $|y - x| \leq \delta$, we have $|f(y) - f(x)| \leq \epsilon$.

**Theorem 1.2** (Intermediate value theorem)**.** *For any continuous map $f : [-1, 1] \to \mathbb{R}$ with $f(-1) < 0$ and $f(1) > 0$, there exists some $x \in [-1, 1]$ such that $f(x) = 0$.*

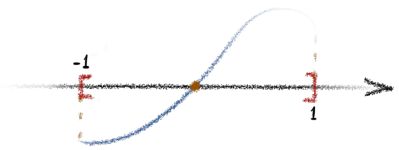


Figure 1.2



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We have seen this theorem in analysis. It is natural to try to generalize this theorem to higher dimensions. Let's make some notations:

$$D^n := \{x \in \mathbb{R}^n \mid |x| \leq 1\} \quad \text{the } n\text{-dimensional disk}$$

$$S^{n-1} := \partial D^n = \{x \in \mathbb{R}^n \mid |x| = 1\} \quad \text{the } n-1\text{-dimensional sphere.}$$

**Question 1.3.** *Given a continuous map $f : D^n \to \mathbb{R}^m$, is there a suitable condition on $f|_{S^{n-1}}$ that can guarantee that $f^{-1}(0) \neq \varnothing$? (To rule out trivial argument, let's assume that $f(x) \neq 0$ for any $x \in S^{n-1}$.)*

It turns out the answer is "no" when $n < m$ (namely, when $n < m$ any continuous from $S^{n-1}$ to $\mathbb{R}^m - \{0\}$ can be extended to a continuous map from $D^n$ to $\mathbb{R}^m - \{0\}$. When $n > m$, the answer is related to one of the deepest problem in topology (homotopy group of spheres) and still not completely solved. Let's focus on the case $m = n$ (which have been solved) and state the following special case.

**Theorem 1.4.** *Given a continuous map $f : D^n \to \mathbb{R}^n$, suppose for any $x \in S^{n-1}$, there exists some $t > 0$ such that $f(x) = tx$. Then there exists some $x \in D^n$ such that $f(x) = 0$.*

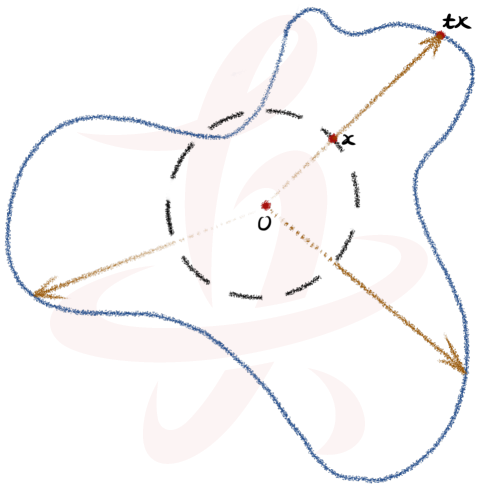


Figure 1.3

We can't prove theorem 1.4 until the end of this semester. However, let's assume it and see some of its beautiful applications.

**Corollary 1.5** (Brouwer fixed point theorem)**.** *Any continuous map from $D^n$ to itself has a fixed point. (I.e., $\forall$ continuous $f : D^n \to D^n$, there exists $x$ s.t. $f(x) = x$.)*

*Proof.* Suppose some $f : D^n \to D^n$ has no fixed points. Then we can construct a function $g : D^n \to D^n$ as follows: For any $x \in D^n$, draw a ray $l_x$ from $f(x)$ to $x$, define $g(x)$ to be the unique point in $l_x \cap S^{n-1}$.



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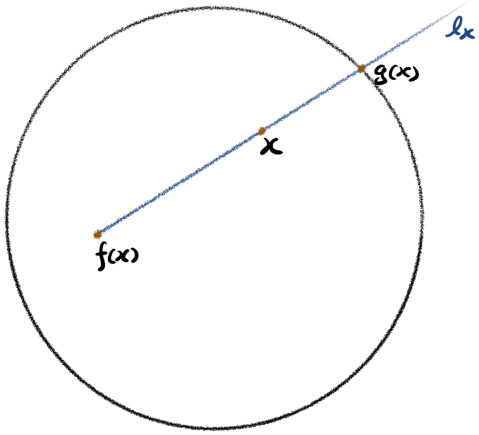


Figure 1.4

It's not hard to check:

- $g$ is well-defined and continuous.
- $g(x) = x$ for any $x \in S^{n-1}$.
- The image of $g$ is $S^{n-1}$. So $g^{-1}(0) = \varnothing$

This contradicts theorem 1.4 and finishes the proof. □

The following exercise illustrate the power of this theorem.

**Exercise A.** *Consider the map $f : \mathbb{R}^2 \to \mathbb{R}^2$ defined by*

$$f(x, y) = (x + \cos(y^2), y - \sin(xy)).$$

*Show that $f$ is surjective. (Hint: Use the Brouwer fixed theorem in some way.)*

**Question 1.6.** *Does Brouwer fixed-point theorem hold if we replace $D^n$ with the following shapes?*

- $X_1 := [-1, 1]^n = [-1, 1] \times \cdots [-1, 1]$ *(the n-dimensional cube)*
- $X_2 := \{x \in \mathbb{R}^2 \mid 0.5 \leq |x| \leq 1\}$ *(a small disk removed from a large disk)*
- $X_3 := \{x \in \mathbb{R}^2 \mid |x| \leq 1, |x - (-\frac{1}{2}, 0, \cdots, 0)| \geq \frac{1}{4}, |x - (\frac{1}{2}, 0, \cdots, 0)| \geq \frac{1}{4}\}$ *(two small disks removed from a large disk).*

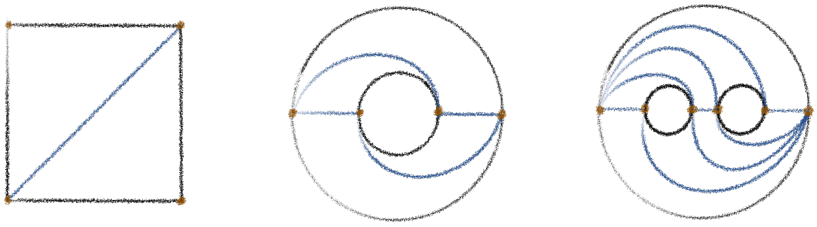

Figure 1.5

*Answer: No for $X_2$ and $X_3$ (try to construct a counter example). But yes for $X_1$.*

To prove the fixed-point theorem for $X_1$, we introduce the following definition.



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**Definition 1.7.** Given subsets $X, Y$ of Euclidean spaces, a continuous map $f : X \to Y$ is called a **homeomorphism** if $f$ is bijective and $f^{-1}$ is also continuous. (Equivalently, a continuous map $f$ is a homeomorphism if and only if there exists a continuous $g : Y \to X$ such that $f \circ g = \mathrm{id}_Y$ and $g \circ f = \mathrm{id}_X$. Here id means the identity map.) We say $X$ and $Y$ are **homeomorphic** if there exists a homeomorphism from $X$ to $Y$. (We will use the notation $X \cong Y$ to denote this.) It is easy to see that being homeomorphic is an equivalence relation.

**Example 1.8.** *$D^n$ is homeomorphic to $X_1$. To see this, we construct a map $g : D^n \to X_1$ as follows: Take any $x \in S^{n-1}$, draw a ray $l_x$ from $0$ to $x$. Then $l_x$ intersects the boundary of $X_1$ at a single point $p_x$. We define*

$$g(tx) := t \cdot p_x, \quad \forall t \in [0, 1], x \in S^{n-1}.$$

*It is not hard to check that $g$ is well-defined and it is a homeomorphism.*

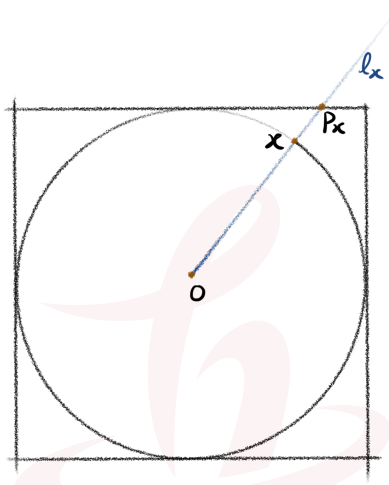


Figure 1.6

**Lemma 1.9.** *If $X$ is homeomorphic to $D^n$, then any continuous map $f : X \to X$ has a fixed-point.*

*Proof.* Let $g : D^n \to X$ be a homeomorphism. Then by corollary 1.5, $g^{-1} \circ f \circ g : D^n \to D^n$ has a fixed point $x$. So $g(x)$ is a fixed point of $f$. □

We say a property is a **topological property** if it is preserved under any homeomorphisms. For example, the following properties are topological properties:

- Every continuous map $f : X \to X$ has a fixed-point.
- Every continuous map $f : X \to \mathbb{Z}$ is a constant map. (We will study this property, called "connectedness")
- Every continuous map $f : X \to \mathbb{R}$ has a max. (This property is closely related to a property called "compactness".)

Length, angle, area are not preserved under homeomorphisms, so they are not topological properties.

Two main topics of topology are:

1. Study topological properties of "geometric shapes" (or more precisely, topological spaces, which we will define soon.)



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2. Develop invariants to distinguish different topological spaces up to homeomorphisms (or some weaker/stronger notions of equivalence, e.g., homotopy equivalence).

Let's talk a little about (2). How do we show that $X_1, X_2$ and $X_3$ in Question question 1.6 are not homeomorphic to each other? To do this, we need an invariant called "Euler number". Roughly speaking, a **triangulation** of $X$ (a closed, bounded region in $\mathbb{R}^2$) is a division of $X$ into pieces $F_1, \cdots, F_m$ by adding vertices $v_1, \cdots, v_n$ and edges $e_1, \cdots, e_k$ such that there is a homeomorphism from $F_i$ to a standard triangle which sends vertices to vertices and sends edges to edges.

The Euler number of $X$ is defined as

$$\chi(X) := \text{number of vertices} - \text{number of edges} + \text{number of faces}.$$

*Fact* 1.10. Euler number is well-defined (i.e. it does not depend on triangulation) and is a topological invariant (i.e. a quantity that is preserved under homeomorphisms.)

**Example 1.11.** $\chi(X_1) = 1$, $\chi(X_2) = 0$, $\chi(X_3) = -1$. *So they are not homeomorphic to each other.*

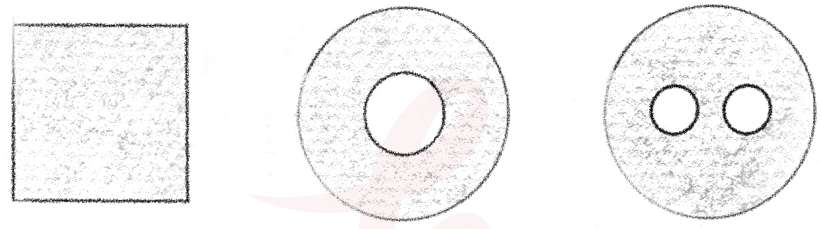

Figure 1.7

Let's end this subsection by mentioning another corollary of the Brouwer fixed point theorem. Recall that the Piano's curve is a sujective, continuous map from $[0,1]$ to $[0,1]^2$. So one may ask the "dual question": does there exists an injective, continuous map from $[0,1]^2$ to $[0,1]$? The answer is no.

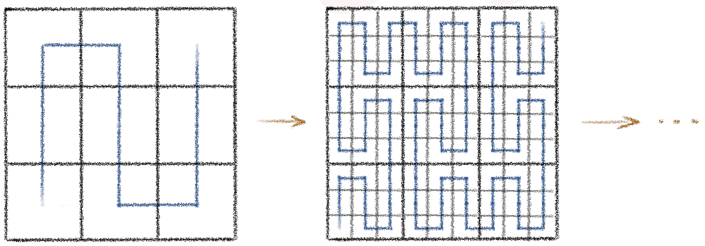

Figure 1.8

**Theorem 1.12** (Invariance of domain)**.** *Let $U$ be an open subset of $\mathbb{R}^n$. Suppose $f : U \to \mathbb{R}^n$ is continuous and injective. Then $f(U)$ must be open.*

*Remark* 1.13. Interested readers can see Tao's blog [**Tao**] for an deep introduction to this theorem and its connection with invariance of domain.

**Exercise B.** *Use invariance of domain to prove that there exists no continuous, injective map from $[0,1]^m$ to $[0,1]^n$ if $m > n$.*

*Proof.* Suppose there exists a continuous injective map $f : I^m \to I^n (m > n)$. Consider the map $g : I^n \to \mathbb{R}^m, x \mapsto (f(x), 0, \cdots, 0)$, which is a continuous injection. Hence the map $g \circ f$ is also a continuous injection, whose image is not open in $\mathbb{R}^m$. Contradiction. □



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**Exercise C.** *Without using invariance of domain, prove that there exists no injective, continuous map from* $[0,1]^2$ *to* $[0,1]$. *You may use the 1-dimensional intermediate value theorem (theorem 1.2) because we have proved it in analysis. (Hint: prove the following lemma: Given any three different points* $x, y, z \in [0,1]^2$, *there exists a continuous map* $f : [0,1] \to [0,1]^2 \setminus \{z\}$ *such that* $f(0) = x$ *and* $f(1) = y$.*)*

Let's end this subsection by considering an interesting question: the government decides to choose a location for a new school on an island $X$ (as a subset of earth $\mathbb{S}^2$) by voting. We suppose each person is allowed to vote one location on the island for school. The government wants to find an algorithm satisfies the following conditions:

- Continuity: If someone changes its vote slightly, the solution will be changed only slightly.
- Symmetry: Exchanging the votes of two people doesn't change the solution.
- If everyone votes for some location, then the out put is exactly that location.

Described in mathematical language, the question is for islands $X$ of different shapes, whether there exists such a function $f : \overbrace{X \times \cdots \times X}^{n \text{ copies}} \to X$ such that

- $f$ is continuous.
- $f(\cdots, x, \cdots, y, \cdots) = f(\cdots, y, \cdots, x, \cdots)$.
- $f(x, x, \cdots, x) = x$.

**Definition 1.14.** We say that $X$ is contractible if there exists a continuous map $f : [0,1] \times X \to X$ such that

- $f(0, x) = x$ for any $x$.
- $\exists x_0$, s.t $f(1, x) = x_0, \forall x$.

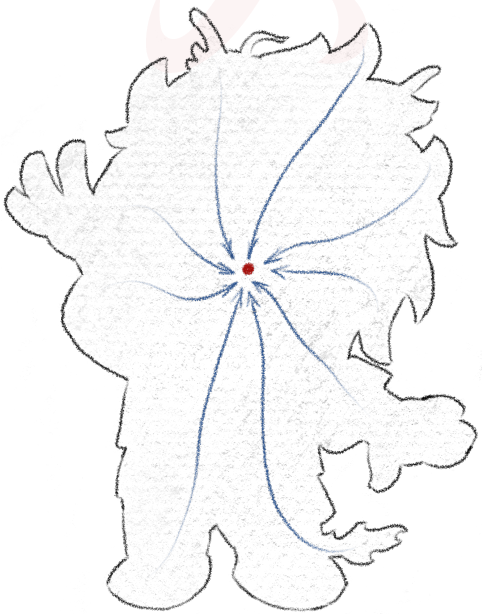

Figure 1.9

**Theorem 1.15** (Social choice function, Weinberger, 2003[**Weinberger**])**.** *$f$ exists $\iff$ $X$ is contractible.*



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## 1.2 Definition of a general topological space

So far, we have only been focused on subsets of $\mathbb{R}^n$. But there are many interesting sets that are not of this type.

**Example 1.16.** *(1)* $C^0[0,1] = \{$*continuous maps* $[0,1] \to \mathbb{R}\}$.
*(2)* $S^\infty = \{(a_1, a_2, \cdots) \mid \sum_{i=1}^\infty a_i^2 = 1\}$.
*(3)* $\mathrm{Top}(n) = \{$*homeomorphisms* $\mathbb{R}^n \to \mathbb{R}^n\}$.
*(4) Given a commutative ring* $R$ *(e.g.* $R = \mathbb{Z}$*), consider*

$$\mathrm{Spec}(R) = \{\textit{all prime ideals of } R\}.$$

It turns out these sets also carry some interesting topology! And we also want to study topological properties of them. As we can see, the first step is to generalize the notation of continuous maps, so that it make sense to say whether a map between between these sets is continuous. The original definition makes use of the distance between two points in $\mathbb{R}^n$ (i.e. a metric). In general, such distance do not exists, so we need an alternative definition of continuous map which is more suitable for generalization.

**Definition 1.17.** (1) A subset $U$ of $\mathbb{R}^n$ is "open" if for any $x \in U$, there exists $\epsilon > 0$ such that $\{y \in \mathbb{R}^n \mid |y - x| < \epsilon\} \subset U$.
(2) Given subsets $U \subset A \subset \mathbb{R}^n$, we say $U$ is an open subset of $A$ (or $U$ is open in $A$) if for any $x \in U$, there exists $\epsilon > 0$ such that $\{y \in \mathbb{R}^n \mid |y - x| < \epsilon\} \cap A \subset U$. (3) We say U is closed in $A$ if $A \setminus U$ is open in $A$.

*Remark* 1.18. Note that an open subset of $A$ is not necessarily an open subset of $\mathbb{R}^n$. For example, $[0,1)$ is an open subset of $[0,\infty)$, but not an open subset of $\mathbb{R}$.

**Definition 1.19.** Given $X \subset \mathbb{R}^m$, $Y \subset \mathbb{R}^n$, we say a map $f : X \to Y$ is continuous if for any open subset $U$ of $Y$, $f^{-1}(U)$ is an open subset of $X$. (In other words, a map is continuous if the preimage of an open subset of the domain is an open subset of the target.)

**Exercise D.** *Show that definition 1.19 is equivalent to definition 1.1.*

So given general sets $X$ and $Y$, if we can specify which subsets of them are open, then we can talk about whether a map $f : X \to Y$ is continuous. This motives the following definition. Recall that for a set $X$, one can define its power set

$$P(X) := \{\text{all subsets of } X\}.$$

**Definition 1.20.** Let $X$ be a set. A **topology** on $X$ is a subset $\mathcal{T}$ of $P(X)$ that satisfies the following 3 axioms:

1. $\varnothing \in \mathcal{T}$ and $X \in \mathcal{T}$.
2. For any $U, V \in \mathcal{T}$, we have $U \cap V \in \mathcal{T}$.
3. For any $S \subset \mathcal{T}$, we have $\bigcup_{U \in S} U \in \mathcal{T}$.

**Definition 1.21.** A **topological space** is a set $X$ together with a topology $\mathcal{T}$ on $X$. We use $(X, \mathcal{T})$ to denote a topological space. (Most of the time, $\mathcal{T}$ is clear from the context and we will ignore it from our notations.) A subset $U$ is called **open** (with respect to the topology $\mathcal{T}$) if $U \in \mathcal{T}$. We say $U$ is **closed** if $X \setminus U$ is open.

So definition 1.21 is essentially saying that a topology on a set $X$ is a collection of subsets of $X$ (called open subsets) such that (1) empty set and the whole set are both open. (2) A finite intersection of open sets is open. (3) Arbitrary union of open sets are open.

The definition of continuous maps and homeomorphisms can be easily adapted.



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**Definition 1.22.** Given a map $f : X \to Y$ between two topological spaces, we say $f$ is continuous if for any open subset $U$ of $Y$, its preimage $f^{-1}(U)$ is open.

**Definition 1.23.** A map $f : X \to Y$ between two topological spaces is called a **homeomorphism** if (i) $f$ is bijective and (ii) Both $f$ and $f^{-1}$ are continuous. We say $X$ and $Y$ are **homeomorphic** if there is a homeomorphism between them.

Let's see some examples.

**Example 1.24.** *Given any set $X$, we can always define the* ***trivial topology***

$$\mathcal{T}_{trivial} := \{\varnothing, X\}$$

*and the* ***discrete topology***

$$\mathcal{T}_{discrete} := P(X).$$

*We can also talk about the finite complement topology*

$$\mathcal{T}_{finite\ complement} := \{U \subset X \mid X \setminus U \text{ is either } X \text{ or a finite set}\}.$$

**Example 1.25.** *$X = \{1, 2, 3\}$. Then $\mathcal{T} = \{\varnothing, \{1\}, \{1, 2\}, \{1, 3\}, \{1, 2, 3\}\}$ is a topology. But $\mathcal{T}' = \{\varnothing, \{1\}, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}\}$ is not a topology.*

**Definition 1.26.** Let $\mathcal{T}, \mathcal{T}'$ be two topologies on $X$. If $\mathcal{T}' \subset \mathcal{T}$, then we say $\mathcal{T}$ is finer/stronger than $\mathcal{T}'$, and $\mathcal{T}'$ is coarser/weaker than $\mathcal{T}$.

**Definition 1.27.** Given a topological space $(X, \mathcal{T})$ and a subset $Y \subset X$, the **induced topology** (or **subspace topology**) on $Y$ is defined as

$$\mathcal{T}_Y := \{U \cap Y \mid U \in \mathcal{T}\}.$$

We say $(Y, \mathcal{T}_Y)$ is a **subspace** of $(X, \mathcal{T})$.

**Example 1.28.** *The* ***standard topology*** *on $\mathbb{R}^n$ is the topology defined in definition 1.17 (1). For $X \subset \mathbb{R}^n$, the induced topology by the standard topology is exactly the topology defined in definition 1.17 (2). (Exercise: check this claim.)*

**Example 1.29.** *Given a field $\mathbb{F}$, the* ***Zariski topology*** *on $X = \mathbb{F}^n$ is defined as follows: A subset $U \subset X$ is* ***closed*** *if there exists a set of* ***polynomial functions*** *$S = \{f : \mathbb{F}^n \to \mathbb{F}\}$ such that*

$$U = \bigcap_{f \in S} f^{-1}(0).$$

*In other words, a set is open in Zariski topology if its complement is the solution of a set of polynomial equations.*

**Exercise E.** *(1) Show that the Zariski topology is actually a topology. (2) Show that when $n = 1$, the Zariski topology on $\mathbb{F}^n$ is exactly the finite complement topology. (3) Show that the Zariski topology and the finite complement topology on $\mathbb{C}^2$ are different. (4) The closed disk $D^2$ is not closed in $\mathbb{R}^2$ with Zariski topology.*

## 1.3 Basis of a topology

So far, we define a topological space by specifying **all** open subsets (or all closed subsets). However, sometimes this is very tedious and inconvenient. We need a more efficient way that allow us to define a topology by specifying **some** open subsets.



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**Definition 1.30.** Let $X$ be a set. A collection $\mathcal{B} \subset P(X)$ of subsets is called a **basis of a topology** if the following two axioms are satisfied:

1. For any $x \in X$, there exists some $U \in \mathcal{B}$ such that $x \in U$.
2. For any $U, V \in \mathcal{B}$ and any $x \in U \cap V$, there exists some $W \in \mathcal{B}$ such that $x \in W$ and $W \subset U \cap V$.

Elements in $\mathcal{B}$ are called basis elements.

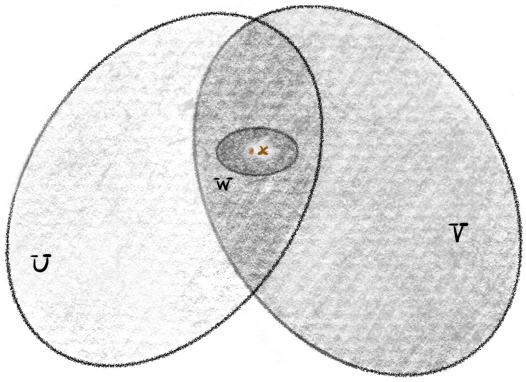


Figure 1.10

**Definition 1.31.** Given a set $X$ and a basis $\mathcal{B}$, we can define a topology $\mathcal{T}_{\mathcal{B}}$ on $X$ as follows:

$$U \in \mathcal{T}_{\mathcal{B}} \iff U = \bigcup_{V \in \mathcal{S}} V \quad \text{for some } \mathcal{S} \subset \mathcal{B}.$$

(In other words, a set is open if and only if it is the union of some basis elements.) We say $\mathcal{T}_{\mathcal{B}}$ is the the topology generated by $\mathcal{B}$ and we say $\mathcal{B}$ is a basis of $\mathcal{T}_{\mathcal{B}}$.

**Exercise F.** *Show that $T_{\mathcal{B}}$ is indeed a topology on $X$.*

**Exercise G.** *Let $\mathcal{S} \subset P(X)$ be a collection of subsets of a set $X$. Let $\mathcal{B}$ be the collection all finite intersections of elements in $\mathcal{S}$. Suppose $\bigcup_{U \in \mathcal{S}} U = X$. Show that $\mathcal{B}$ is a basis. We call $\mathcal{S}$ a subbasis for the topology generated by $\mathcal{B}$.*

**Exercise H.** *Given a topology $\mathcal{T}$ on $X$ and a subset $\mathcal{B} \subset \mathcal{T}$, show that $\mathcal{T}$ is generated by $\mathcal{B}$ if and only if the following condition holds: For any $x \in X$ and any $U \in \mathcal{T}$ that contains $x$, there exists some $V \in \mathcal{B}$ such that $x \in V \subset U$.*

**Exercise I.** *Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be two topolgies on the same set $X$. If $\mathcal{T}_1 \subset \mathcal{T}_2$, then we say $\mathcal{T}_2$ is **finer/stronger** than $\mathcal{T}_1$, and we say $\mathcal{T}_1$ is **coarser/weaker** than $\mathcal{T}_2$. Now suppose $\mathcal{T}_i$ is generated by $\mathcal{B}_i$ for $i = 1, 2$. Show that $\mathcal{T}_1$ is coarser than $\mathcal{T}_2$ if and only if the following condition holds:*

- *For any $U \in \mathcal{B}_1$ and any $x \in U$, there exists $V \in \mathcal{B}_2$ such that $x \in V \subset U$.*

Here is the reason why we call say topology is "stronger" when it has more open sets: one way to describe a topology on $X$ is to specify when a sequence $(x_n)$ converges to a limit under this topology. When a topology contains more open sets, a sequence is harder to be convergent. So being convergent is a "stronger condition".



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**Example 1.32.** *Given $x \in \mathbb{R}^n$ and $r > 0$, we consider the (open) ball*

$$B_r(x) := \{y \in \mathbb{R}^n \mid |y - x| < \epsilon\}.$$

*We define a basis of a topology on $\mathbb{R}^n$ as follows*

$$\mathcal{B} := \{B_r(x) \mid r > 0, x \in \mathbb{R}^n\}.$$

*It's not hard to Check that this is a basis and it generates the standard topology on $\mathbb{R}^n$.*

The above example can be generalized to arbitrary set with a metric. More precisely, we have the following definition

**Definition 1.33.** A **metric** on a set $X$ is a map (also called the distance function) $d : X \times X \to [0, \infty)$ such that

- $d(x, y) = 0$ if and only if $x = y$.
- $d(x, y) = d(y, x)$.
- $d(x, y) + d(y, z) \geq d(x, z)$.

The pair $(X, d)$ is called a metric space.

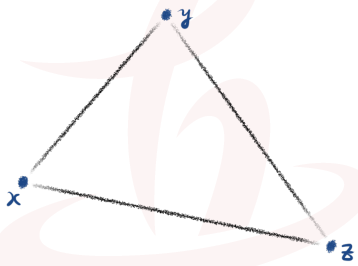


Figure 1.11

**Definition 1.34.** Let $X$ be a set with metric $d$. Let $B_r(x) := \{y \in X \mid d(x, y) < r\}$ Let

$$\mathcal{B} := \{B_r(x) \mid r > 0, x \in X\}. \tag{1.1}$$

Then $\mathcal{B}$ is a basis of a topology. The topology $\mathcal{T}$ generated by $\mathcal{B}$ is called **the metric topology**.

**Exercise J.** *Use the triangle inequality to show that the collection (eq. (1.1)) is indeed a basis.*

**Example 1.35.** *A normed vector space is a vector space $V$ over $\mathbb{F} = \mathbb{R}, \mathbb{C}$ or $\mathbb{Q}$, equipped with a norm function $\| \cdot \| : V \to \mathbb{R}$ such that*

- $\|v\| \geq 0$ *for any* $v \in V$ *and* $\|v\| = 0$ *iff* $v = 0$.
- $\|a \cdot v\| = |a|\|v\|$ *for any* $a \in \mathbb{F}$ *and any* $v \in V$.
- $\|v + w\| \leq \|v\| + \|w\|$

*A normed vector space is canonically a metric space because we can define a metric on $V$ by $d(v, w) := \|v - w\|$*

**Example 1.36.** *Let $X = \mathbb{R}^n$, then we can define the standard norm $\|x\| = |x|$. The metric topology is just the standard topology.*



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**Example 1.37.** *Let $X = \{$continuous maps $[0,1] \to \mathbb{R}\}$. We can define two different norms on $X$:*

$$\text{the } C^0\text{-norm: } \|f\|_{C^0} := \max_{x\in[0,1]} |f(x)|$$

$$\text{the } L^2\text{-norm: } \|f\|_{L^2} := \sqrt{\int_0^1 f^2(x)dx}$$

*In analysis, you will see more metrics that also include the (higher) derivatives.*

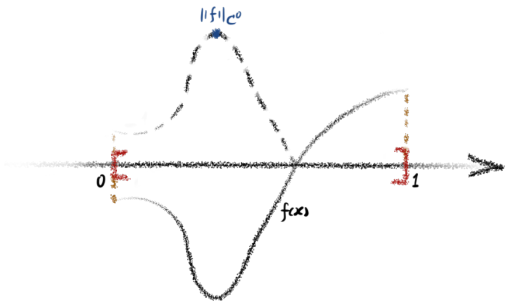


Figure 1.12

**Example 1.38.** *Let $X = \mathbb{Q}$. Then the standard metric on $\mathbb{R}$ restricts to a metric on $\mathbb{Q}$. However, there is another metric which is useful in number theory: **the $p$-adic metric** $d_p$. Here $p$ is a fixed prime. Given a reduced fractional $x \in \mathbb{Q}$, we write $x = p^k \cdot \frac{a}{b}$, where neither $a$ or $b$ is divisible by $p$. Then we define the $p$-adic norm $\|x\|_p := \frac{1}{p^k}$. (We set $\|0\|_p := 0$.) Given $x, y \in \mathbb{Q}$, we define the $p$-adic metric*

$$d_p(x,y) := \|x - y\|_p.$$

*Intuitively, in this metric, two rationals are closed to each other if their difference can be divisible by a large power of $p$.*

**Exercise K.** *Check that the $p$-adic metric is indeed a metric. The induced topology is called the **$p$-adic topology** on $\mathbb{Q}$.*



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# Chapter 2

# Further properties of topological spaces

## 2.1 Interior, closure and boundary of a subset

Let $X$ be a topological space. From now on, we will usually ignore the topology $\mathcal{T}$ from the notations unless we want to emphasize it. When we talk about subsets of $\mathbb{R}^n$, we will use the subspace topology of the standard topology unless otherwise noted.

Recall that a subset $U \subset X$ is open if it belongs to the topology $\mathcal{T}$. And $U$ is closed if its complement $X \setminus U$ is open. The following three properties correspond to the three axioms in definition 1.20:

1. $\varnothing$ and $X$ are both closed.
2. Given **finitely many** closed sets $U_1, \cdots, U_n$, the union $\bigcup_{i=1}^n U_i$ is closed.
3. Given **a arbitrary collection** of closed sets $\{U_i\}_{i\in S}$, the intersection $\bigcap_{i\in S} U_i$ is closed.

**Exercise A.** *(1) Give an example to show that an infinite collection of closed subsets of $\mathbb{R}$ is not necessarily closed. (2) Show that the Cantor set is closed.*

Of course, there are many subsets which are neither open nor closed (e.g. $\mathbb{Q}$ is neither open nor closed in $\mathbb{R}$). And there are subsets which are both open and closed (e.g. $\varnothing$ and $X$. And for disconnected spaces, there are more.) So the following question is natural:

**Question 2.1.** *Let $A$ be a subset of $X$. Not necessarily open or closed.*
*(1) Among the open sets that is contained in $A$. Is there a largest one? (That means an open subset $U$ that satisfies the following two conditions: (1) $U \subset A$, (2) For any open $U'$ such that $U' \subset A$, we must have $U' \subset U$.*

*(2) Among the closed subsets that contains $A$, is there a smallest one? (That means a closed subset $V$ that satisfies the following two conditions: (1) $A \subset V$, (2) For any open $V'$ such that $A \subset V'$, we must have $V \subset V'$.)*



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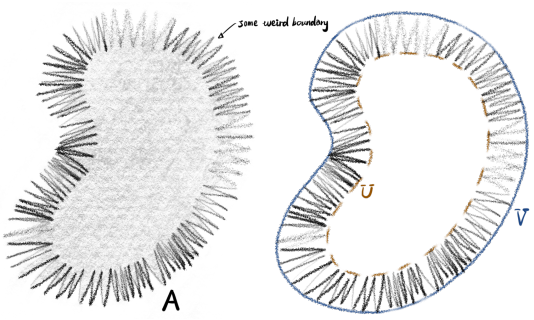


Figure 2.1

The answers to both questions are yes. (If we switch "open" and "closed" in (1) and (2), then the answers will be no. Think about why.)

**Definition 2.2.** Let $A$ be a subset of topological space $X$.

1. We say a subset $U$ of $X$ is a **neighborhood** of $A$ if it is an open set that contains $A$.
2. We say a point $x$ is an interior point of $A$ if $x$ has a neighborhood $U$ that is contained in $A$. We use $\mathring{A}$ to denote the set of all interior points. This set is called the **interior** of $A$.
3. We say a point $x \in X$ is in the **closure** of $A$ if any neighborhood of $x$ intersects $A$. We use $\overline{A}$ to denote the set of all such points and call it the **closure** of $A$.
4. It is straightforward to see that $\mathring{A} \subset A \subset \overline{A}$. We use $\partial A$ to denote $\overline{A} \setminus \mathring{A}$ and call it the **boundary** of $A$. A point in $\partial A$ is called a boundary point. Note that a boundary point may or may not be in $A$.
5. A point $x \in X$ is called a **limit point** of $A$ if $x \in \overline{A \setminus \{x\}}$. In other words, every neighborhood of $x$ intersects $A$ at some point **other then** $x$.
6. We say $x$ is an **isolated point** of $A$ if $x \in A$ but $x$ is not a limit point of $A$.
7. We say $A$ is **dense** in $X$ if $\overline{A} = X$.

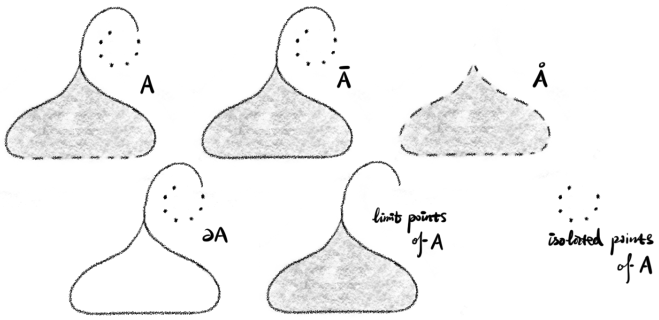


Figure 2.2

*Remark* 2.3. In some books, a neighborhood is not required to be open. Instead, they say $U$ is a neighborhood of $A$ if there is some open set $V$ such that $A \subset V \subset U$. And they use the terminology "open neighborhood" when they want $U$ to be itself open. Most of the time (99%), this change doesn't affect anything (i.e., one can interchange "neighborhood" and "open neighborhood" without affecting the definition/conclusion.) For this reason, we decide to follow the convention of Munkres and require that any neighborhood should be open itself. We won't use the terminology "open neighborhood".



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**Example 2.4.** *Let $X = \mathbb{R}$ with standard topology and let $A = \{\frac{1}{n}\}_{n\in\mathbb{N}^+}$. Then $\mathring{A} = \varnothing$. $\overline{A} = \partial A = \{0\} \cup A$. $0$ is the unique limit point. Every point in $A$ is an isolated point.*

**Exercise B.** *(1) Show that $X \setminus \mathring{A} = \overline{X \setminus A}$. (2) Show that $\partial A = \overline{A} \cap \overline{X \setminus A}$. (In some references, $\partial A$ is also called the "frontier of $A$".)*

**Example 2.5.** *Let $X = \mathbb{R}$ and let $A = [0, 1)$. Let $A'$ be the set of limit points in $A$. Then*

- *Suppose we use the standard topology. Then $\mathring{A} = (0, 1)$, $A' = \overline{A} = [0, 1]$ and $\partial A = \{0, 1\}$. There is no isolated point.*
- *Suppose we use the finite complement topology. Then $\mathring{A} = \varnothing$, $A' = \overline{A} = \partial A = \mathbb{R}$. There is no isolated point.*
- *Suppose we use the discrete topology. Then $\overline{A} = \mathring{A} = [0, 1)$. $\partial A = A' = \varnothing$. Every point in $A$ is an isolated point.*

**Exercise C.** *Check the claims in example 2.5.*

**Exercise D.** *Let $(X, d)$ be a metric space and let $A$ be a subset of $X$. Show that*

$$\overline{A} = \{x \in X \mid \inf_{y\in A} d(x, y) = 0\}$$

**Exercise E.** *Let $X = \mathbb{Q}$ be equipped with the 2-adic topology (see example 1.38). Let $A = \mathbb{Z}$. Find $\mathring{A}$ and $\overline{A}$. What are the isolated points in $A$?*

**Exercise F.** *Let $X = \mathbb{R}^2$ (with standard topology). Let $A = \{(t, \sin\frac{1}{t}) \mid t \in (0, 1]\}$. Find $\overline{A}$. ($\overline{A}$ is called the "topologists' sine curve". It is an example of set which is connected but not path-connected. We will return to it when we talk about these definitions.)*

The following lemma summarizes some basic properties of these sets.

**Lemma 2.6.** *(1) $\mathring{A} = \bigcup_{U \text{ is open and } U\subset A} U$.*
*(2) $\overline{A} = \bigcap_{V \text{ is closed and } A\subset V} V$.*
*(3) $\mathring{A}$ is the largest open set that is contained in $A$. In particular, $A$ is open if and only if $A = \mathring{A}$.*
*(4) $\overline{A}$ is the smallest closed set that contains $A$. In particular, $A$ is closed if and only if $A = \overline{A}$.*
*(5) Let $A'$ be the set of all limit points of $A$. Then $\overline{A} = A \cup A'$. In particular, $A$ is closed if and only if it contain all its limit points.*

*Proof.* (1) Let $V = \bigcup_{U \text{ is open and } U\subset A} U$. Suppose $p \in \mathring{A}$. Then by definition, there is an open set $U$ s.t. $p \in U \subset X$. Since $V$ is the union of all such $U$, we see that $p \in U \subset V$. This shows that $\mathring{A} \subset V$.

Suppose $p \in V$. Note that $V$ is a union of open sets so itself is open. Because $p \in V \subset A$, we have $p \in \mathring{A}$ by definition. This shows that $V \subset \mathring{A}$.

The proof of (2) is similar to (1). (3) and (4) follows directly from (1) and (2).

Now we prove (5): Suppose $x \in \overline{A}$ but $x \notin A$. Then every neighborhood of $x$ intersects $A = A \setminus \{x\}$. So $x \in A'$ by definition. This shows that $\overline{A} \subset A \cup A'$.

Also by definition, we have $A' \subset \overline{A}$ and $A \subset \overline{A}$. Therefore, $A' \cup A \subset \overline{A}$. This proves the first assertion in (5).

The second assertion in (5) follows directly from the first assertion and the second assertion in (4). □



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Next, we generalize the definition of the limit of a sequence.

**Definition 2.7.** Let $x_1, x_2, \cdots$ be a sequence of points in $X$. (We use $(x_i)$ to denote this sequence.) We say a point $x \in X$ is a limit of this sequence if for any neighborhood $U$ of $x$, we have $x_i \in U$ for all but finitely many $i$.

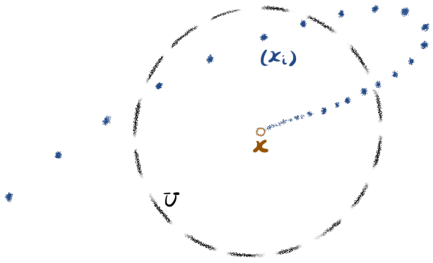


Figure 2.3

*Remark* 2.8. Given a sequence $(x_i)$ in $X$, we can define a subset of $X$ that consists of all points in the sequence and talk about the limit points of this set. This should be **not** be confused the limit of the sequence. (Think about the constant sequence.)

Recall two standard facts in analysis:

1. A point $x \in \mathbb{R}^n$ belongs to the closure of a subset $A$ if and only if $x$ is the limit of a sequence in $A$;
2. Any sequence in $x \in \mathbb{R}^n$ has at most one limit.

If we replace $\mathbb{R}^n$ by a general space $X$, then both conclusions fail.

**Example 2.9.** *(1) Equip $\mathbb{R}$ with the finite complement topology. Then every point in $\mathbb{R}$ is a limit of the sequence $1, 2, \cdots$.*
*(2) Equip $\mathbb{R}$ with the so-called "countable complement topology", namely $U$ is open in this topology if $U = \varnothing$ or $\mathbb{R} \setminus U$ is countable. Let $A = [0, 1]$. Then $-1 \in \overline{A}$ but $-1$ is not a limit of any sequence in $A$.*

To rule out these pathological situations, we need suitable conditions on $X$.

**Definition 2.10.** Given $x \in X$, a collection $\mathcal{U} = \{U_\alpha\}_{\alpha \in S}$ of neighborhoods of $x$ is called a "neighborhood basis" is for any neighborhood $U$ of $x$, there exists some $U_\alpha \in \mathcal{U}$ such that $x \in U_\alpha \subset U$.

We say a space $(X, \mathcal{T})$ is **first-countable** if any point has a countable neighborhood basis. We say $X$ is **second-countable** if the topology $\mathcal{T}$ is generated by some countable basis.

**Lemma 2.11.** *(1) Any second-countable space is also first-countable.*
*(2) Any metric space is first countable.*

*Proof.* (1) Let $\mathcal{B}$ be a basis for $\mathcal{T}$. For any $x \in X$, we let $\mathcal{B}_x = \{U \in \mathcal{B} \mid x \in U\}$. Then $\mathcal{B}_x$ is a neighborhood basis of $x$.
(2) For any $x \in X$, the collection

$$\mathcal{U} := \{B_{\frac{1}{n}}(x)\}_{n \in \mathbb{N}^+}$$

is a countable neighborhood basis for $X$. □



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**Example 2.12.** *Let $X$ be a uncountable set (for example $\mathbb{R}$). We consider the "discrete metric"*

$$d(x,y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}.$$

*Then the metric topology is the discrete topology. It is first-countable but not second-countable.*

**Lemma 2.13.** *Let $A$ be a subset of $X$. Suppose $x$ is a limit of some sequence in $A$. Then $x \in \overline{A}$. The converse is true if we assume $X$ is first-countable.*

*Proof.* Suppose $x$ is a limit of a sequence $x_1, x_2, \cdots$ in $\overline{A}$. Then for any neighborhood $U$ of $x$, $x_i \in U \cap A$ for $i$ larger enough. So $x \in \overline{A}$.

Suppose $x \in \overline{A}$ and $X$ is first-countable. Take a countable collection $\{U_1, U_2, \cdots\}$ of neighborhood basis for $x$. By replacing $U_n$ with $U_1 \cap U_2 \cap \cdots \cap U_n$, we may assume that $U_i \subset U_j$ whenever $i \geq j$. Because $x \in \overline{A}$, we have $U_i \cap A \neq \varnothing$. Take any point $x_i \in U_i \cap A$. Then $x$ is a limit of the sequence $(x_i)$ because any neighborhood $U$ of $x$ will contain a $U_n$ for some $n \geq 1$. Then for $i \geq n$, have $x_i \in U_i \subset U_n \subset U$. □

**Definition 2.14.** We say a topological space $X$ is **Hausdorff** if for any two different points $x$ and $y$. There exists a neighborhood $U$ of $x$ and a neighborhood $V$ of $y$ such that $U \cap V = \varnothing$.

**Exercise G.** *(1) Show that the standard topology on $\mathbb{R}^n$ is Hausdorff and second-countable.*
*(2) Show that the Zariski topology on $\mathbb{C}^n$ (see example 1.29) is not Hausdorff.*
*(3) Show that the Zariski topology on $\mathbb{C}^n$ is not first-countable.*

**Proposition 2.15.** *Suppose $X$ is Hausdorff. Then any sequence in $X$ has at most one limit.*

*Proof.* Suppose a sequence $x_1, \cdots$ have two limits $y$ and $y'$. Since $X$ is Hausdorff, we can find neighborhood $U$ of $y$ and $U'$ of $y'$ such that $U \cap U' = \varnothing$. By definition, there exists $N \gg 0$, such that for any $n > N$, we have $x_n \in U$ and $x_n \in U'$. This is impossible. □

We also mention the following basic result:

**Lemma 2.16.** *A subspace of a Hausdorff/first-countable/second countable space is Hausdorff/first-countable/second-countable. (See definition 1.27)*

*Proof.* We focus on the Hausdorff case. Let $Y$ be a subspace of a Hausdorff space $X$. Take two different points $x, y \in Y$. Because $X$ is Hausdorff, there exists a neighborhood $U$ of $x$ (in $X$) and a neighborhood $V$ of $y$ (in $X$) such that $U \cap V = \varnothing$. Then $U \cap Y$ and $V \cap Y$ are neighborhoods of $x$ and $y$ in $Y$ which do not intersect each other. So $Y$ is Hausdorff. □

## 2.2 Continuous maps

Let $f : X \to Y$ be a map between two topological spaces. Recall the following definitions:

1. We say $f$ is continuous if the preimage of any open subset of $Y$ is an open subset of $X$.
2. We say $f$ is a homeomorphism if $f$ is bijective and both $f, f^{-1}$ are continuous.
3. We say $f$ is an **embedding** if it is a homeomorphism to its image. In other words, the induced map $\bar{f} : X \to f(X)$ defined by $\bar{f}(x) = f(x)$ is a homeomorphism. We are using the subspace topology on $f(X)$.

*Remark* 2.17. As we will see, the image of an open/closed set under a continuous map is not necessarily open/closed. We say a continuous map $f : X \to Y$ is an **open map** if $f(U)$ is open for any open $U$, and we say it's a **closed map** if $f(V)$ is closed for any closed $V$.



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**Example 2.18.** *(1) The constant map $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 0$ for all $x$ is a closed map but not an open map.*
*(2) The map $f : \mathbb{R} \to \mathbb{R}$ defined by $g(x) = \arctan x$ is an open map but not a closed map (because $g(\mathbb{R}) = (-\frac{\pi}{2}, \frac{\pi}{2})$). It is actually an embedding.*
*(3) Consider the map $h : [0, 1) \to S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ defined by $h(x) = e^{2\pi i x}$. Then $h$ is a bijective continuous map but not a homeomorphism (because $h^{-1}$ is not continuous). It also gives an example of an injective continuous map which is not an embedding.*

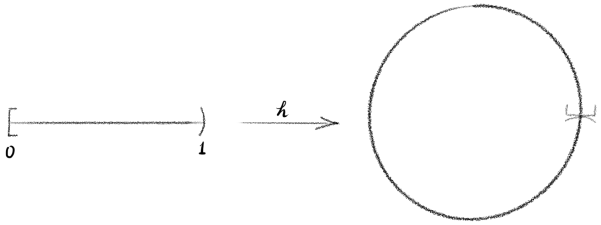


Figure 2.4

**Example 2.19.** *We have seen in example 1.8 that the $n$-dimensional cube is homeomorphic to the $n$-dimensional disk $D^n$.*
*(2) Consider the interior of the disk $\mathring{D}^n = \{x \in \mathbb{R}^n \mid |x| < 1\}$. It is homeomorphic to $\mathbb{R}^n$. A homeomorphism $f : \mathring{D}^n \to \mathbb{R}^n$ is given by*

$$f(x) = \frac{1}{1 - |x|^2} \cdot x.$$

**Example 2.20** (Stereographic projection)**.** *For any point $q \in S^n$, we will prove that $S^n \setminus \{q\}$ is homeomorphic to $\mathbb{R}^n$ via the so-called "**stereographic projection**".*

*Note that given any two points $q, q' \in S^n$, there is a homeomorphism from $S^n$ to itself that sends $q$ to $q'$. (Think about the rotation.) Therefore, without loss of generality, we may assume that $q$ is the north pole, i.e., $q = (1, 0, \cdots, 0)$ .*

*We define $p : S^n \setminus \{(1, 0, \cdots, 0)\} \to \mathbb{R}^n$ as follows: Take any point $x \in S^n \setminus \{(1, 0, \cdots, 0)\}$, there is a unique straight line $l_x$ in $\mathbb{R}^{n+1}$ that passes $x$ and $(1, 0, \cdots, 0)$. $l_x$ intersects the plane*

$$\{0\} \times \mathbb{R}^n := \{(0, x) \mid x \in \mathbb{R}^n\}$$

*at a unique point $(0, a_x)$. We define $p(x) := a_x$. We can find a concrete formula*

$$p(x_0, x_1, \cdots, x_n) = (\frac{x_1}{1 - x_0}, \cdots, \frac{x_n}{1 - x_0}), \quad \text{for } (x_0, \cdots, x_n) \in S^n.$$

*The inverse of $p$ can also be solved explicitly:*

$$p^{-1}(x_1, \cdots, x_n) = (\frac{x_1^2 + \cdots + x_n^2 - 1}{x_1^2 + \cdots + x_n^2 + 1}, \frac{2x_1}{x_1^2 + \cdots + x_n^2 + 1}, \cdots, \frac{2x_n}{x_1^2 + \cdots + x_n^2 + 1}).$$

*Since both $p$ and $p^{-1}$ are continuous, $p$ is a homeomorphism.*



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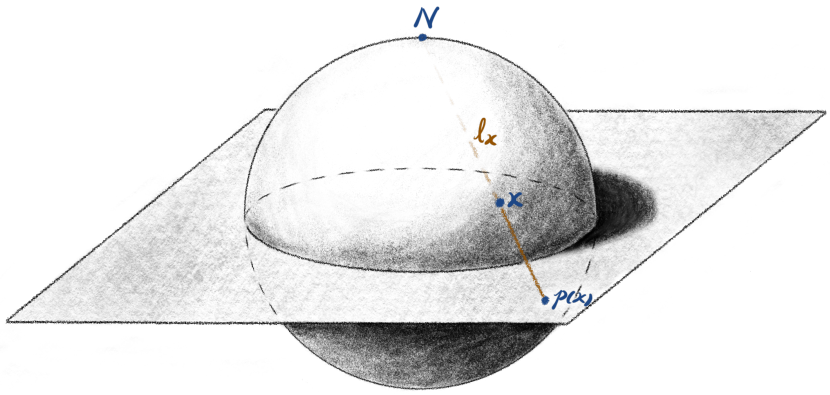


Figure 2.5

**Lemma 2.21.** *The following basic properties of continuous maps hold.*

1. *A constant map is continuous.*
2. *The identity map* $\mathrm{id} : X \to X$ *is continuous.*
3. *Let* $Y$ *be a subspace of* $X$*. Then the inclusion map* $i : Y \to X$ *is continuous.*
4. *The composition of two continuous maps is continuous.*
5. *Let* $f : X \to Y$ *be a continuous map. Take a subspace* $X'$ *of* $X$ *and a subspace* $Y'$ *of* $Y$*. Suppose* $f(X') \subset Y'$*. Then the map* $f' : X' \to Y'$ *defined by* $x \mapsto f(x)$ *is continuous. In particular, the maps*
$$f|_{X'} : X' \to Y \text{ and } f' : X \to f(X)$$
*are both continuous.*

**Exercise H.** *Prove lemma 2.21.*

**Lemma 2.22.** *Let* $f : X \to Y$ *be a map between topological spaces. Let* $\mathcal{B}$ *be a basis for the topology on* $Y$*. Then the following are equivalent:*

1. $f$ *is continuous.*
2. *For any base element* $U \in \mathcal{B}$*, the set* $f^{-1}(U)$ *is open.*
3. *For any subset* $A$ *of* $X$*, we have* $f(\overline{A}) \subset \overline{f(A)}$*.*
4. *For any closed subset* $V$ *of* $Y$*, its preimage* $f^{-1}(V)$ *is closed in* $X$*.*
5. *For any* $x \in X$ *and any neighborhood of* $V$ *of* $f(x)$*, there exists a neighborhood* $U$ *of* $x$ *such that* $f(U) \subset V$*. (We say* $f$ *is continuous at* $x$ *if this condition holds for a fixed* $x \in X$*.)*

*Proof.* (1) $\iff$ (2): Any basis element is open, and any open set is a union of some basis elements.

(1) $\implies$ (3): Take $x \in \overline{A}$, we want to show $f(x) \in \overline{f(A)}$. Take any neighborhood $U$ of $f(x)$. Then $f^{-1}(U)$ is a neighborhood of $x$ so it intersects $A$. So $U$ intersects $f(A)$. By definition, $f(x) \in \overline{f(A)}$.

(3) $\implies$ (4): Let $A = f^{-1}(V)$. By assumption, we have
$$f(\overline{A}) \subset \overline{f(A)} \subset \overline{V} = V.$$
So $\overline{f^{-1}(V)} \subset f^{-1}(V)$. So $f^{-1}(V)$ is closed.



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(4) $\Longrightarrow$ (1): Take any open $U \subset Y$. Then $V = Y \setminus U$ is closed. So $f^{-1}(V) = X \setminus f^{-1}(U)$ is closed. So $f^{-1}(U)$ is open,

(1) $\Longrightarrow$ (5): We set $U = f^{-1}(V)$. Then $U$ is a neighborhood of $x$ and $f(U) \subset V$.

(5) $\Longrightarrow$ (1): Let $V \subset Y$ be open. Take $x \in f^{-1}(V)$. Then by assumption, there is a neighborhood $U_x$ of $x$ such that $U_x \subset f^{-1}(V)$. Hence the set

$$f^{-1}(V) = \bigcup_{x \in f^{-1}(V)} U_x$$

is open. □

The following lemma states that a function being continuous is a "local property".

**Lemma 2.23.** *Let $f : X \to Y$ be a map. Suppose $X$ is the union of a collection $\{U_\alpha\}_{\alpha \in S}$ of open sets. Then $f$ is continuous if and only if the map $f|_{U_\alpha} : U_\alpha \to Y$ is continuous for any $\alpha \in S$.*

*Proof.* Suppose $f$ is continuous. Then by lemma 2.21 (5), $f|_{U_\alpha}$ is continuous.

Suppose $f|_{U_\alpha}$ is continuous for any $\alpha$. Take an open subset $V$ of $Y$. Then $(f|_{U_\alpha})^{-1}(V)$ is an open subset of the subspace $U_\alpha$. By definition of the subspace topology, this means that $(f|_{U_\alpha})^{-1}(V) = V_\alpha \cap U_\alpha$ for some open subset $V_\alpha$ of $X$. Since $U_\alpha$ is also open in $X$, we see that $(f|_{U_\alpha})^{-1}(V)$ is an open subset of $X$. Note that

$$f^{-1}(V) = \bigcup_{\alpha \in S} (f|_{U_\alpha})^{-1}(V).$$

So $f^{-1}(V)$ is an open subset of $X$. □

**Exercise I.** *(1) Show that lemma 2.23 still holds if we instead assume $X$ is a **finite** union of **closed** subsets $\{U_\alpha\}$. (2) Give an example to show that the finiteness assumption in (1) is indeed necessary.*

We have the following useful "gluing lemma":

**Lemma 2.24.** *Let $\{A_i\}_{i \in S}$ be a collection of subsets of $X$. Suppose $A_i$ is open for any $i \in S$. Then given any collection $\{f_i : A_i \to Y\}_{i \in S}$ of continuous maps that satisfy*

$$f_i|_{A_i \cap A_j} = f_j|_{A_i \cap A_j}, \quad \forall i, j \in S,$$

*there exists a unique continuous map $f : \bigcup_{i \in S} A_i \to Y$ such that $f|_{A_i} = f_i$ for any $i$. The same result holds if we assume $A_i$ are all closed and $S$ **is finite**.*

*Proof.* We define $f(x) = f_i(x)$ for any $x \in A_i$. Then by the local property of continuous maps, $f$ is continuous. □

We end this section by some enlightening examples.

**Exercise J.** *Let $\mathbb{F}$ be a field. A map $f = (f_1, f_2, \cdots, f_m) : \mathbb{F}^n \to \mathbb{F}^m$ is called a polynomial map if all of its components $f_i$ are polynomial functions. (For example, the map $f : \mathbb{C}^2 \to \mathbb{C}^2$ defined by $f(x, y) = (x^2 - y, y + 1)$ is a polynomial map.) A subset of $A \subset \mathbb{F}^n$ is called an **algebraic set** if it is closed in the Zariski topology. Let $X \subset \mathbb{F}^n, Y \subset \mathbb{F}^m$ be algebraic sets, equipped with the subspace topology of the Zariski topology. A map $g : X \to Y$ is a called an "**algebraic map**" if it is the restriction of a polynomial map $f : \mathbb{F}^n \to \mathbb{F}^m$.*

*For example, let $X = \{(x, y) \in \mathbb{F}^2 \mid x^2 = y^3\}$ and let $Y = \{(x, y) \in \mathbb{F}^2 \mid x^5 = y^3\}$. Then the map $f : X \to Y$ defined by $f(x, y) = (x, xy)$ is an algebraic map.*

***Question:** Show that any algebraic map is continuous.*



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**Exercise K.** *Consider the following algebraic sets. We use the subspace topology of the Zariski topology on all of them.*

- $X = \{(x, y) \in \mathbb{C}^2 \mid xy = 1\}$,
- $Y = \{(x, y) \in \mathbb{C}^2 \mid xy = 0\}$,
- $Z = \{(x, y) \in \mathbb{C}^2 \mid xy = x + y\}$,
- $W = \mathbb{C}$.

*(1) Show that $X$ is homeomorphic to $W$ but **not** homeomorphic to $Y$. (Hint: Show that the topology on $X$ is the finite complement topology. But the topology on $Y$ is not.)*
*(2) Show that there exists a homeomorphism $f : X \to Z$ such that $f$ and $f^{-1}$ are both algebraic maps. (So in algebraic geometry, these two algebraic sets are "isomorphic".)*
*(3) Show that there does **not** exists a homeomorphism $g : W \to X$ such that both $g$ and $g^{-1}$ are algebraic maps. (So even if $X$ and $W$ are are homeomorphic, they are not isomorphic in algebraic geometry.)*

**Exercise L.** *Equip $\mathbb{Q}$ with the p-adic topology (See example 1.38). (1) Suppose two maps $f, g : \mathbb{Q} \to \mathbb{Q}$ are both continuous. Show that $f + g$ and $f \cdot g$ are both continuous. (Hint 1: When the domain and target are both metric spaces, the $\epsilon - \delta$ definition of continuous maps holds. You can use this fact without proof, because you have essentially proved it in exercise D. Hint 2: Show that the p-adic norm satisfies a stronger version of the triangle inequality $|x+y|_p \leq \max(|x|_p, |y|_p)$.)*
*(2) Show that the any polynomial map $f : \mathbb{Q} \to \mathbb{Q}$ is continuous.*

**Exercise M.** *Let's introduce two more topolgies on $\mathbb{R}$:*

- *Consider the bases $\mathcal{B}_{lsc} = \{(a, +\infty) \mid a \in \mathbb{R}\}$. It generates the "lower semicontinuous topology", denoted by $\mathcal{T}_{lsc}$.*
- *Consider the basis $\mathcal{B}_{ll} = \{(b, a] \mid a, b \in \mathbb{R}, b < a\}$. It generates the "lower limit topology", denoted by $\mathcal{T}_{ll}$*

*Let's denote the standard topology by $\mathcal{T}_{st}$.*
*(1) Show that $\mathcal{B}_{lsc}$ and $\mathcal{B}_{ll}$ are indeed basis of a topology.*
*(2) A map $f : \mathbb{R} \to \mathbb{R}$ is called "left-continuous" if it is continuous when we use $T_{ll}$ on the domain and use $T_{st}$ on the target. Find the $\epsilon - \delta$ definition of a map being left-continuous.*
*(3) A map $f : \mathbb{R}^k \to \mathbb{R}$ is called "lower semi-continuous" if it is continuous when we use $T_{st}$ on the domain and use $T_{lsc}$ on the target. Find the $\epsilon - \delta$ definition of a map being lower semi-continuous.*
*(4) Here is a natural example: Let $M(m, n)$ be the set of all $m \times n$ matrices. By taking entries, we can identify $M(m, n)$ with $\mathbb{R}^{mn}$. Show that the map*

$$f : M(m, n) \to \mathbb{R} \quad \text{defined by } f(A) := \operatorname{rank}(A)$$

*is lower semi-continuous (but not continuous if we use the standard topology on both domain and target).*

## 2.3 Separation axioms

We have seen the definition of Hausdorff spaces: a space is Hausdorff is any two distinct points can be separated by disjoint neighborhoods. Sometimes we want to separate closed sets not just points. To do this, we introduce the following definition:



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**Definition 2.25.** Let $X$ be a topological space.

- We say $X$ satisfies the "$T_1$ axiom" if for any two different points $x \neq y$, there exists a neighborhood $U$ of $x$ such that $y \notin U$.
- We say $X$ satisfies the "$T_2$ axiom" (i.e., $X$ is **Hausdorff**) if for any two different points $x \neq y$, there exists a neighborhood $U$ of $x$ and a neighborhood $V$ of $y$ such that $U \cap V = \varnothing$.
- We say $X$ satisfies the "$T_3$ axiom" (i.e. $X$ is **regular**) if for any point $x$ and any closed set $V$ such that $x \notin V$, there exists a neighborhood $U$ of $x$ and a neighborhood $U'$ of $V$ such that $U \cap U' = \varnothing$.
- We say $X$ satisfies the "$T_4$ axiom" (i.e. $X$ is **normal**) if for any closed sets $V, V$ such that $V \cap V' = \varnothing$, there exists a neighborhood $U$ of $V$ and a neighborhood $U'$ of $V'$ such that $U \cap U' = \varnothing$.

These axioms are called "separation axioms". They describe how well can we separate disjoint points (or closed sets) using disjoint neighborhoods.

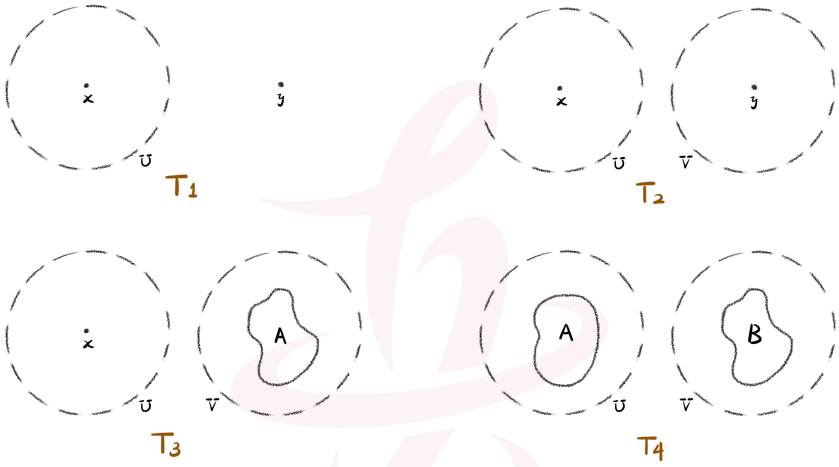


Figure 2.6

To understand the logical relation between these axioms, we need the following lemma.

**Lemma 2.26.** *The following conditions are equivalent:*

1. *$X$ satisfies the $T_1$-axiom.*
2. *Any points in $X$ is closed. (I.e. $\forall x \in X$, the set $\{x\}$ is closed.)*
3. *Any finite subset of $X$ is closed.*

*Proof.* (1) $\Longrightarrow$ (2): Fix a point $x \in X$. Consider a different point $y \in X$. Since $X_1$ is $T_1$, $y$ has a neighborhood $U_y$ s.t. $x \notin U_y$. Note that we have

$$X \setminus \{x\} = \bigcup_{y \in X} U_y.$$

So $X \setminus \{x\}$ is open and $\{x\}$ is closed.

(2) $\Longrightarrow$ (3): A finite union of closed set is closed.

(3) $\Longrightarrow$ (1): Take $x \neq y \in X$. Then $X \setminus \{x\}$ is a neighborhood of $y$ that doesn't contain $x$. So $X$ is $T_1$. □



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小红书号:3189205503 qq

**Corollary 2.27.** *(1)* $T_2 \implies T_1$.
*(2)* $T_1 + T_4 \implies T_3$, *and* $T_1 + T_3 \implies T_2$.

**Theorem 2.28** (Lindelöf)**.** *Second countable* $+T_3 \implies T_4$.

*Proof.* Let $X$ be a space that satisfies the $T_3$ axiom and has a countable basis $\mathcal{B}$. Let $A_0, A_1$ be disjoint closed subsets. Let

$$\mathcal{B}_0 = \{U \in \mathcal{B} \mid \overline{U} \cap A_1 = \varnothing\} = \{U_0, U_1, \cdots\}$$

and let

$$\mathcal{B}_1 = \{U \in \mathcal{B} \mid \overline{U} \cap A_0 = \varnothing\} = \{V_0, V_1, \cdots\}.$$

We first show that $A_0 \subset \bigcup_{i\in\mathbb{N}} U_i$: Take any $x \in A_0$. Because $X$ is $T_3$, there exists a neighborhoods $U'$ of $x$ that is disjoint from some neighborhood of $A_1$. That implies $\overline{U'} \cap A_1 = \varnothing$. Because $\mathcal{B}$ is a basis, we can find $U \in \mathcal{B}$ such that $x \in U \subset U'$. Then $U = U_i$ for some $i$. So $x \in \bigcup_{i\in\mathbb{N}} U_i$. Similarly, we can show $A_1 \subset \bigcup_{i\in\mathbb{N}} V_i$.

Now we let $U_n' = U_n \setminus \bigcup_{i=0}^n \overline{V_i}$ and $V_n' = V_n \setminus \bigcup_{i=0}^n \overline{U_i}$. They satisfy the following properties:

- For any $m, n$, $U_m'$ and $V_n'$ are open and disjoint.
- $A_0 \subset \bigcup_{i\in\mathbb{N}} U_i'$: this is because $A_0 \subset \bigcup_{i\in\mathbb{N}} U_i$ and $A_0 \cap \overline{V_i} = \varnothing$.
- $A_1 \subset \bigcup_{i\in\mathbb{N}} V_i'$: Similar as above.

Let $U = \bigcup_{i\in\mathbb{N}} U_i'$ and let $V = \bigcup_{i\in\mathbb{N}} V_i'$. Then $A_0 \subset U$ and $A_1 \subset V$ and $U \cap V = \varnothing$. □

**Exercise N** (From Munkres)**.** *Show that if $X$ is normal, then every pair of disjoint closed sets have neighborhoods whose closure are disjoint.*

**Exercise O** (From Munkres)**.** *Let $f : X \to Y$ be a continuous, surjective, closed map. (Recall that a map is closed if the image of any closed set is closed.) Suppose $X$ is normal. Show that $Y$ is also normal.*

## 2.4 Metric spaces are normal.

We have seen the definition of a metric space and the induced metric topology, A topological space $(X, \mathcal{T})$ is **metrizable** if there exists a metric $d$ on $X$ that is compatible with the topology $\mathcal{T}$ (that means the metric topology is exactly $\mathcal{T}$). It turns out that metrizable spaces are all Hausdorff (you have essentially proved this in exercise G (1)). Actually, more is true.

**Theorem 2.29.** *Let $X$ be a metrizable space. Then $X$ is first-countable and $X$ satisfies the $T_i$ axiom for $i = 1, 2, 3, 4$.*

To prove theorem 2.29, we start with the following lemma.

**Lemma 2.30.** *Let $(X, d)$ be a metric space. Then we have the following conclusions:*

1. *A sequence $x_1, x_2, \cdots$ in $X$ has limit $x_\infty$ if and only if $\lim_{i\to\infty} d(x_i, x_\infty) = 0$.*
2. *The function $d : X \times X \to \mathbb{R}$ is continuous (Here we use the product topology of the metric topology on the domain (to be defined next week) and the standard topology on the target. See next section for the )*



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3. *For any subset $A \subset X$, the function $d_A : X \to \mathbb{R}$ defined by*

$$d_A(x) = \inf_{y \in A} d(x, y)$$

*is continuous.*

4. *For any $A$, we $(d_A)^{-1}(0) = \overline{A}$.*

5. *Suppose $x \notin \overline{A}$. Then there exists some $\epsilon > 0$ such that $d(x, y) > \epsilon$ for any $y \in A$.*

*Proof.* (1) This is straightforward.
(2) If suffices to check that for any $a < b \in \mathbb{R}$, the set $f^{-1}(a, b)$ is open in $X \times X$. Take any point $(x_1, x_2) \in f^{-1}(a, b)$. Then $a < d(x, y) < b$. Take a small $\epsilon > 0$ such that

$$(d(x, y) - 2\epsilon, d(x, y) + 2\epsilon) \subset (a, b).$$

Let $B_\epsilon(x) = \{y \in X \mid d(x, y) < \epsilon\}$. Then by the triangle inequality, we have

$$B_\epsilon(x_1) \times B_\epsilon(x_2) \subset d^{-1}(a, b)$$

Since $B_\epsilon(x_1) \times B_\epsilon(x_2)$ is open in the product topology, we see that $d^{-1}(a, b)$ is open.

(3) Let $x \in d_A^{-1}(a, b)$. Take $\epsilon > 0$ small enough such that $(d_A(x) - \epsilon, d_A(x) + \epsilon) \subset (a, b)$. By the triangle inequality, for any $y \in B_\epsilon(x)$, we have

$$|d_A(x) - d_A(y)| \leq d(x, y) < \epsilon.$$

So $y \in d_A^{-1}(a, b)$. So $d_A^{-1}(a, b)$ is open. So $d_A$ is continuous.

(4) Suppose $x \in \overline{A}$. Then its neighborhood $B_{\frac{1}{n}}(x)$ intersects $A$. So $d_A(x) = 0$.

Suppose $d_A(x) = 0$. Take any neighborhood $U$ of $x$. Then $B_\epsilon(x) \subset U$ for some $\epsilon > 0$. By our assumption, $B_\epsilon(x)$ intersects $A$. So $U$ intersects $A$. So $x \in \overline{A}$.

(5) This follows directly from (4). □

*Proof of theorem 2.29.* $X$ is first-countable: For any $x \in X$, the collection $\{B_{\frac{1}{n}}(x)\}_{n \in \mathbb{N}}$ is a countable neighborhood basis.

$X$ satisfies the $T_1$-axiom: For any $x \in X$, let $A = \{x\}$. Then $\{x\} = d_A^{-1}(0)$. So $\{x\}$ is closed.

$X$ satisfies the $T_4$-axiom: Let $A$ and $B$ be disjoint closed sets. For any $x \in X$, we consider $d_A(x), d_B(x) \in [0, +\infty)$. Since $\overline{A} \cap \overline{B} = A \cap B = \varnothing$, by lemma 2.30 (4), we have $d_A(x) + d_B(x) > 0$. Consider the function $f : X \to \mathbb{R}$ defined by

$$f(x) := \frac{d_A(x)}{d_A(x) + d_B(x)}.$$

Then lemma 2.30 (3), $f$ is a continuous function and it satisfies $f(A) = 0$ and $f(B) = 1$. Consider the open sets $U = f^{-1}(-\infty, \frac{1}{3})$ and $V = f^{-1}(\frac{2}{3}, +\infty)$. Then $A \subset U$ and $B \subset V$ and $U \cap V = \varnothing$. □

*Remark* 2.31. In the above proof, we establish the $T_4$-axiom by separating two disjoint closed sets using a continuous function. This idea turns out to be quite useful. As we will see in week 7. In a space $X$ that satisfies the $T_4$-axiom, one can always "separate" two disjoint closed sets $A, B$ by finding a continuous function $f : X \to [0, 1]$ that satisfies $f(A) = 0$ and $f(B) = 1$. This is called the **Urysohn lemma**. We will discuss it in week 7. One beautiful application of this Lemma is the following theorem, whose proof will also be discussed in week 7.

**Theorem 2.32** (Urysohn metrization theorem)**.** *Suppose $X$ is second-countable and satisfies the $T_i$-axiom for $i = 1, 4$. Then $X$ is metrizable.*



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# Chapter 3

# Product, disjoint union and completion

## 3.1 Cartesian product and disjoint union of sets

Before talking about topological spaces, let's first do some set theory. Consider a finite collection of sets $\{A_\alpha\}_{\alpha\in S}$, indexed by $S=\{1,\cdots,n\}$. Let $A=\bigcup_{\alpha\in S}A_\alpha$ be their union. Then we can two other sets: the Cartesion product

$$\begin{aligned}\prod_{\alpha\in S}A_\alpha=A_1\times\cdots\times A_n&:=\{\text{all maps} f:S\to A\mid f(\alpha)\in A_\alpha,\forall\alpha\in S\}\\&=\{(x_1,\cdots,x_n)\mid x_\alpha\in A_\alpha,\forall\alpha\in S\}\end{aligned}\tag{3.1}$$

and the disjoint union

$$\bigsqcup_{\alpha\in S}A_\alpha=A_1\sqcup\cdots\sqcup A_n:=\{(\alpha,x)\in S\times A\mid x\in A_\alpha,\forall\alpha\in S\}.\tag{3.2}$$

*Remark* 3.1. Formula (eq. (3.2)) may look a little awkward: why do we add an extra component $\alpha$? The reason for doing this to "separate" $A_{\alpha_0}$ and $A_{\alpha_1}$ so the subsets

$$\{(\alpha_0,x)\mid x\in A_{\alpha_0}\}\quad\text{and}\quad\{(\alpha_1,x)\mid x\in A_{\alpha_1}\}$$

are disjoint in $\bigsqcup_{\alpha\in S}A_\alpha$ even if $A_{\alpha_0}\bigcap A_{\alpha_1}\neq\varnothing$. For this reason, the disjoint union $\bigsqcup_{\alpha\in S}A_\alpha$ and the union $\bigcup_{\alpha\in S}A_\alpha$ are different in general. However, if we assume

$$A_\alpha\bigcap A_\beta=\varnothing\quad\text{for any }\alpha\neq\beta\in S,$$

then there is a natural bijection between $\bigsqcup_{\alpha\in S}A_\alpha$ and $\bigcup_{\alpha\in S}A_\alpha$. In this case, we do not distinguish them.



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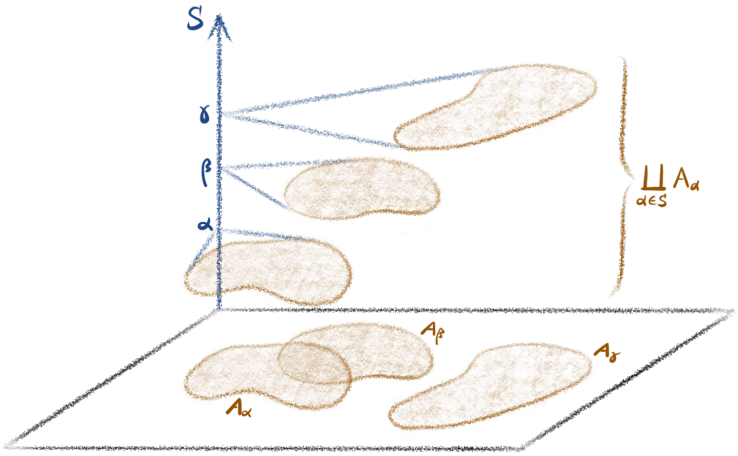


Figure 3.1

**Example 3.2.** *Let* $A_1 = \{a, b\}$ *and* $A_2 = \{b, c\}$. *Then we have*

- $A_1 \cup A_2 = \{a, b, c\}$;
- $A_1 \sqcup A_2 = \{(1, a), (1, b), (2, b), (2, c)\}$
- $A_1 \times A_2 = \{(a, b), (a, c), (b, b), (b, c)\}$.

Now we generalize these constructions to an infinite collection of sets.

**Definition 3.3.** Let $\{A_\alpha\}_{\alpha\in S}$ be a collection of sets, indexed by a set $S$ (not necessarily finite). Let $A = \bigcup_{\alpha\in S} A_\alpha$. Then we define the disjoint union

$$\bigsqcup_{\alpha\in S} A_\alpha := \{(\alpha, x) \in S \times A \mid x \in A_\alpha, \forall \alpha \in S\}.$$

and the Cartesian product

$$\begin{aligned}\prod_{\alpha\in S} A_\alpha &:= \{\text{all maps } f : S \to A \mid f(\alpha) \in A_\alpha, \forall \alpha \in S\} \\ &= \{(x_\alpha) \mid x_\alpha \in A_\alpha, \forall \alpha \in S\}.\end{aligned}$$

**Definition 3.4.** Given sets $A$ and $S$, we use $A^S$ to denote the Cartesian product $\prod_{\alpha\in S} A_\alpha$ where $A_\alpha = A$ for any $\alpha \in S$. By definition, we have $A^S = \{\text{all maps } S \to A\}$ so we call it the function set.

*Remark* 3.5. Suppose we take a subset $C_\alpha \subset A_\alpha$ for each $\alpha \in S$. Then

$$\bigsqcup_{\alpha\in S} C_\alpha = \{(\alpha, x) \in \bigsqcup_{\alpha\in S} A_\alpha \mid x \in C_\alpha\}$$

is a subset of $\bigsqcup_{\alpha\in S} A_\alpha$. And

$$\prod_{\alpha\in S} C_\alpha := \{f \in \prod_{\alpha\in S} A_\alpha \mid f(\alpha) \in C_\alpha\}.$$

is a subset of $\prod_{\alpha\in S} A_\alpha$.



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Note that for any $\beta \in S$, we have an **inclusion map**

$$i_\beta : A_\beta \to \bigsqcup_{\alpha \in S} A_\alpha, \quad x \mapsto (\beta, x)$$

and a **projection map**

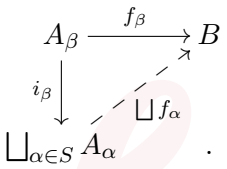
$$p_\beta : \prod_{\alpha \in S} A_\alpha \to A_\beta, \quad f \mapsto f(\beta).$$

For a modern perspective, the main reason that we care about the Cartesian product and the disjoint union is the following two lemmas.

**Lemma 3.6** (Universal property of disjoint union)**.** *Given any set $B$ and any collection of maps $\{g_\alpha : A_\alpha \to B\}_{\alpha \in S}$, there exists a unique map*

$$\bigsqcup_{\alpha \in S} f_\alpha : \bigsqcup_{\alpha \in S} A_\alpha \to B$$

*such that the following diagram commutes for any $\beta \in S$:*



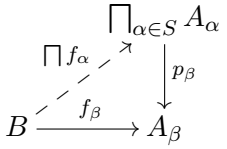

*We call $\bigsqcup f_\alpha$ the disjoint union of the maps $\{f_\alpha\}$.*

*Proof.* We define $(\bigsqcup_{\alpha \in S} f_\alpha)(\beta, x) := f_\beta(x)$. □

**Lemma 3.7** (Universal property of Cartesian product)**.** *Given any set $B$ and any collection of maps $\{f_\alpha : B \to A_\alpha\}_{\alpha \in S}$, there exists a unique map*

$$\prod_{\alpha \in S} f_\alpha : B \to \prod_{\alpha \in S} A_\alpha$$

*such that the following diagram commutes for any $\beta \in S$*




*We call $\prod f_\alpha$ the (Cartesian) product of the maps $\{f_\alpha\}$.*

*Proof.* We $(\prod f_\alpha)(b)$ to be the map $S \to A$ that sends $\alpha \in S$ to $f_\alpha(b)$. □

## 3.2 Product topology and disjoint union topology

Suppose we are given a collection of **topological spaces** $\{X_\alpha\}_{\alpha \in S}$. We want to repeat the story in topology.

**Question 3.8.** *Can we put suitable topologies on $\prod_{\alpha \in S} X_\alpha$ and $\bigsqcup_{\alpha \in S} X_\alpha$ such that the universal properties (lemma 3.7 and lemma 3.6) hold for **continuous maps**?*

The answer is yes.

**Definition 3.9.** The **disjoint union topology** on the set $\bigsqcup_{\alpha \in S} X_\alpha$ is defined as follows:



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- $\bigsqcup_{\alpha\in S} U_\alpha$ is open in $\bigsqcup_{\alpha\in S} X_\alpha \iff$ for any $\alpha \in S$, $U_\alpha$ is open in $X_\alpha$.

Unless otherwise noted, we will always put the disjoint topology on $\bigsqcup_{\alpha\in S} X_\alpha$ and call it the disjoint union of the topological spaces $\{X_\alpha\}$.

We have the following universal property.

**Lemma 3.10.** *(1) The map $i_\beta : X_\beta \to \bigsqcup X_\alpha$ is continuous for any $\beta \in S$.*
*(2) Given any topological space $Y$ and any collection of maps $\{f_\alpha : X_\alpha \to Y\}$, let*

$$\bigsqcup f_\alpha : \bigsqcup_{\alpha\in S} X_\alpha \to Y$$

*be the map provided by lemma 3.6. Then $\bigsqcup f_\alpha$ is continuous if and only if $f_\alpha$ is continuous for any $\alpha \in S$.*

*Proof.* (1) Take any open subset $\bigsqcup_{\alpha\in S} U_\alpha$ of $\bigsqcup_{\alpha\in S} X_\alpha$. Then $i_\beta^{-1}(\bigsqcup_{\alpha\in S} U_\alpha) = U_\beta$ is open. So $i_\beta$ is continuous.

(2) This follows directly from the fact that $(\bigsqcup f_\alpha)^{-1}(U) = \bigsqcup f_\alpha^{-1}(U)$ for any subset $U \subset Y$. □

**Corollary 3.11.** *The universal property of disjoint union holds for continuous maps.*

Now we discuss the topology on the product $\prod_{\alpha\in S} X_\alpha$. As a naive try, one may mimic the disjoint union topology and declare the topology to be

$$\mathcal{B}' := \{\prod_{\alpha\in S} U_\alpha \mid U_\alpha \text{ is open in } X_\alpha \text{ for all } \alpha \in S\}. \tag{3.3}$$

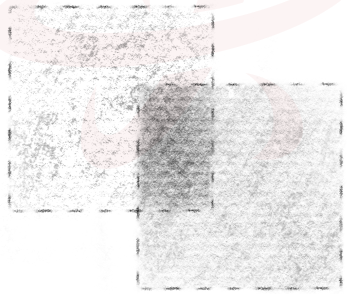

Figure 3.2

Unfortunately, the union of two elements in $\mathcal{B}'$ is not in $\mathcal{B}'$. So the best we can hope is to use $\mathcal{B}'$ as a basis. This indeed works, and the topology generated by $\mathcal{B}'$ is called the "**box topology**". When $S$ is finite, the box topology satisfies the universal property. However, we will see that when $S$ is infinite, the box topology is usually pathological for the following two reasons:

1. It doesn't satisfy the universal property.
2. It contains way too many open sets (i.e. it is too strong) and it is very hard for a sequence to be convergent under this topology.

The following example shows that the box topology does **not** satisfies the universal property.



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**Example 3.12.** *For $\alpha \in \mathbb{N}$, consider the map $f_\alpha : \mathbb{R} \to \mathbb{R}$ defined by $f_\alpha(x) = \alpha x$. Then the map*

$$\prod f_\alpha : \mathbb{R} \to \mathbb{R}^{\mathbb{N}}$$

*is **not** continuous if we use the standard topology on the domain and use the box topology (of the standard topology) on the target. This is because the preimage*

$$(\prod f_\alpha)^{-1}(-1,1)^{\mathbb{N}} = \{0\}$$

*is not open. As we can see, the essential reason is that an infinite intersection of open sets is not necessarily open.*

So we have to modify $\mathcal{B}'$ again to get the right answer.

**Definition 3.13.** The **product topology** on the set $\prod X_\alpha$ is defined as follows: Take a basis $\mathcal{B}_\alpha$ for each $X_\alpha$. Consider the following collection of subsets

$$\begin{aligned}\mathcal{B} := \{\prod U_\alpha \mid \quad & U_\alpha \in \mathcal{B}_\alpha \text{ for finitely many } \alpha \in S, \\ & \text{and } U_\alpha = X_\alpha \text{ for all remaining } \alpha \in S\}\end{aligned} \tag{3.4}$$

Then $\mathcal{B}$ is a basis it generates the product topology. The set $\prod X_\alpha$, equipped with the product topology, is called the product space of $\{X_\alpha\}$. ( Unless otherwise noted, we will always use the product topology on $\prod X_\alpha$.)

*Remark* 3.14. The product topology does not depend on the choice of $\mathcal{B}_\alpha$, as long as it generates the given topology $\mathcal{T}_\alpha$ on $X_\alpha$. In particular, we can simply set $\mathcal{B}_\alpha = \mathcal{T}_\alpha$. Then (eq. (3.4)) turns into the following basis (compare this with (eq. (3.3)))

$$\begin{aligned}\mathcal{B} := \{\prod U_\alpha \mid & U_\alpha \text{ is open in } X_\alpha \text{ for all } \alpha \in S, \\ & \text{and } U_\alpha = X_\alpha \text{ for all but finitely many } \alpha \in S\}.\end{aligned}$$

**Example 3.15.** *Let's consider the product topology on $\mathbb{R}^n$, (which is also the box topology because we have a finite product here). By definition, it is generated by the "open boxes"*

$$\mathcal{B} = \{(a_1, b_1) \times \cdots \times (a_n, b_n) \mid a_\alpha < b_\alpha \text{ for all } \alpha\}.$$

*Using exercise I, it is not hard to see that this is just the standard topology on $\mathbb{R}^n$.*

**Example 3.16.** *The product space $S^1 \times \cdots \times S^1$ is also called the $n$-dimensional torus and denoted by $T^n$. Later, we will give an alternative description of $T^n$ in terms of the quotient space. For $n = 2$, we can visualize $T^2$ as the following subspace of $\mathbb{R}^3$:*

$$\{((2 + u_1)u_2, (2 + u_1)v_2, v_1) \mid u_1^2 + v_1^2 = 1, u_2^2 + v_2^2 = 1\}.$$

*To get this formula, we start with the circle in the $xz$-plane given by*

$$\{(2 + u_1, 0, v_1) \mid u_1^2 + v_1^2 = 1\}.$$

*Then we rotate it around the $z$-axis.*



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小红书号:3189205503 qq

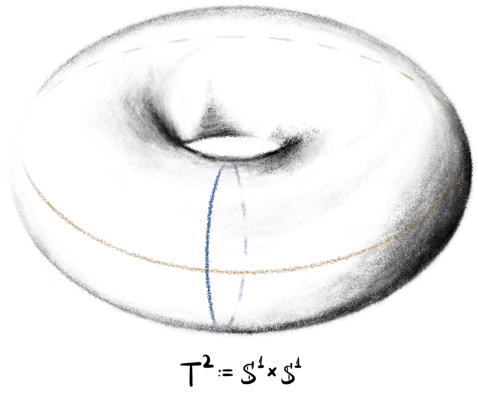


Figure 3.3

**Exercise A.** *(1) Show* $[0,1)$ *is* ***not*** *homeomorphic to* $(0,1)$.
*(2) Show that* $[0,1)\times[0,1)$ *is homeomorphic to* $[0,1)\times(0,1)$. *(Remark: From this problem, we see that* $X\times Y\cong X'\times Y$ *usually doesn't imply* $X\cong X'$. *Actually, there exists a space* $X$ *such that* $X\times\mathbb{R}\cong\mathbb{R}^4$ *but* $X$ *is* ***not*** *homeomorphic to* $\mathbb{R}^3$.*)*

**Exercise B.** *Let* $X$ *be a topological space. Show that* $X$ *is Hausdorff if and only if the diagonal*

$$\Delta=\{(x,x)\in X\times X\mid x\in X\}$$

*is a closed subset of the product space* $X\times X$.

**Exercise C.** *Consider the projection map* $p_1: X_1\times X_2\to X_1$. *Is it always an open map? Is it always a closed map? If yes, prove it. If no, give a counter example. (Recall that a map is called an open/closed map if it sends open/closed sets to open/closed sets.)*

Now we state the universal property of the product space.

**Lemma 3.17.** *Let* $\{X_\alpha\}_{\alpha\in S}$ *be a collection of topological spaces and let* $\prod X_\alpha$ *be the product space. The we have the following conclusions.*
*(1) For any* $\beta\in S$, *the projection map* $p_\beta:\prod X_\alpha\to X_\beta$ *is continuous.*
*(2) Let* $Y$ *be a topological spaces and let* $\{f_\alpha: Y\to X_\alpha\}_{\alpha\in S}$ *be a collection of maps. Let*

$$\prod f_\alpha: Y\to\prod_{\alpha\in S}X_\alpha$$

*be the map provided by lemma 3.7. Then* $\prod f_\alpha$ *is continuous if and only if* $f_\alpha$ *is continuous for any* $\alpha\in S$.

*Proof.* (1) Take any open subset $U$ of $X_\beta$. Then $p_\beta^{-1}(U)=\bigcap U_\alpha$, where $U_\alpha=X_\alpha$ when $\alpha\neq\beta$ and $U_\alpha=U$ when $\alpha=\beta$. This is a basis element of the product topology so it is open.

(2) Suppose $\prod f_\alpha$ is continuous. Then $f_\beta=p_\beta\circ(\prod f_\alpha)$ is also continuous.

Suppose $\{f_\alpha\}_{\alpha\in S}$ are all continuous. To show $\prod f_\alpha$ is continuous, it suffices to show that $(\prod f_\alpha)^{-1}(\prod U_\alpha)$ is open for all basis element $(\prod_{\alpha\in S}U_\alpha)\in\mathcal{B}$. Note that we have

$$(\prod_{\alpha\in S}f_\alpha)^{-1}(\prod_{\alpha\in S}U_\alpha)=\bigcap_{\alpha\in S}f_\alpha^{-1}(U_\alpha).$$

Since $U_\alpha\neq X_\alpha$ for only finitely many $\alpha\in S$, we see that $f_\alpha^{-1}(U_\alpha)\neq Y$ for only finitely many $\alpha$. Thus $\bigcap f_\alpha^{-1}(U_\alpha)$ is a finite intersection of open sets so itself is open. □

**Corollary 3.18.** *The universal property for Cartesian products (lemma 3.7) holds for continuous maps if we use the product topology .*



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**Exercise D.** *In this problem, we will see why box topology on an infinite product is pathological and not very useful in analysis.*

*(1) Let $S$ be any set. Suppose we use the* ***product topology*** *on $\mathbb{R}^S$. Show that a sequence $(f_n : S \to \mathbb{R})_{n\in\mathbb{N}}$ in $\mathbb{R}^S$ converges to a point $f \in \mathbb{R}^S$ if and only if $f_n$ converges to $f$* ***pointwisely****, namely*

$$\lim_{n\to\infty} f_n(\alpha) = f(\alpha), \quad \forall \alpha \in S.$$

*Because of this, we also call the product topology to be the* ***pointwise convergent topology****.*

*(2) Suppose we use the box topology on $\mathbb{R}^S$. Show that a sequence $(f_n : S \to \mathbb{R})_{n\in\mathbb{N}}$ in $\mathbb{R}^S$ converges to a point $f \in \mathbb{R}^S$ if and only if there exists a* ***finite*** *subset $S' \subset S$ and a constant $N$ such that both of the following conditions are true:*

*(i) For any $\alpha \in S'$, we have $\lim_{n\to\infty} f_n(\alpha) = f(\alpha)$.*

*(ii) For any $\alpha \notin S'$ and any $n \geq N$, we have $f_n(\alpha) = f(\alpha)$.*

*Clearly, this is an extremely strong condition and almost never holds.*

Now we discuss the following question:

**Question 3.19.** *Is the product of metrizable spaces still metrizable?*

It's not hard to prove that a **finite product** of metrizable spaces is metrizable: Let $(X_\alpha, d_\alpha)$ with $\alpha \in S = \{1, 2, \cdots n\}$ metric spaces. Then it's not hard to verify that the product topology on $\prod_{\alpha\in S} X_\alpha$ is exactly the metric topology of the metric

$$d((x_\alpha), (y_\alpha)) := \sum_{\alpha\in S} d_\alpha(x_\alpha, y_\alpha).$$

In the following exercise, we will show that the product of **countably** many metrizable spaces is again metrizable, while the product of **uncountably** many metrizable spaces is usually not metrizable.

**Exercise E.** *(1) We say a metric $d : X \times X \to [0, \infty)$ is* ***bounded*** *if the image of $d$ is bounded. We say two metrics on the same set are "****topologically equivalent****" if they induce the same metric topology. Given an arbitrary metric $d$ on $X$, show that the map $d' : X \times X \to [0, \infty)$ defined by*

$$d'(x, y) = \frac{d(x, y)}{1 + d(x, y)}$$

*is indeed a metric and it is topologically equivalent to $d$. Therefore, any metrizable space can be given a bounded metric.*

*(2) Let $\{(X_n, d_n)\}_{n\in\mathbb{N}}$ be a countably infinite collection of metric spaces. Show the following expression*

$$d((x_n), (y_n)) := \sum_{n\in\mathbb{N}} \frac{1}{2^n} \cdot \frac{d(x_n, y_n)}{1 + d(x_n, y_n)}$$

*is indeed a metric on $\prod_{n\in\mathbb{N}} X_n$ and show that the metric topology is exactly the product topology on $\prod_{n\in\mathbb{N}} X_n$. In particular, this shows that a* ***countably infinite*** *product of metrizable spaces is metrizable.*

*(3) Show that $\mathbb{R}^{\mathbb{R}}$ with the product topology is* ***not*** *metrizable. This shows that a* ***uncountably infinite*** *product of metrizable spaces is usually* ***not*** *metrizable.*



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## 3.3 Completion of metric spaces

Recall that a sequence $(x_\alpha)_{\alpha\in\mathbb{N}}$ in a metric space $(X,d)$ is called a **Cauchy sequence** if for any $\epsilon>0$, there exists $N$ such that $d(x_\alpha,x_\beta)<\epsilon$ for any $\alpha,\beta>N$.

**Definition 3.20.** A metric space $(X,d)$ is **complete** if any Cauchy sequence has a limit.

**Example 3.21.** *(1) We consider two metrics on $\mathbb{R}$. The standard metric*

$$d_1(x,y)=|x-y|$$

*and the bounded metric*

$$d_2(x,y)=\left|\frac{e^x}{1+e^x}-\frac{e^y}{1+e^y}\right|.$$

*Although the induced topology $\mathcal{T}_{d_1}$ and $\mathcal{T}_{d_2}$ are the same, $(\mathbb{R},d_1)$ is complete but $(\mathbb{R},d_2)$ is not (because the sequence $1,2,\cdots$, which is Cauchy with respect to $d_2$, has no limit). So we see that being complete depends on the metric (not just the induced topology) and is really **not** a topological property. (Note: To construct $d_2$, we consider the homeomorphism $f:\mathbb{R}\to(0,1)$ defined by $f(x)=\frac{e^x}{1+e^x}$ and then "pull back" the standard metric on $(0,1)$ to $\mathbb{R}$ using $f$.)*

**Example 3.22.** *Let $(X,d)$ be a complete metric space and let $A$ be a subset of $X$. It is not hard to prove that $(A,d|_{A\times A})$ is a complete metric space if and only if $A$ is closed in $X$. Therefore, any closed subset of $\mathbb{R}^n$, with the standard metric, is complete.*

**Example 3.23.** *Consider the normed vector space $(\mathcal{H},\|\|_{l^2})$ (see example 1.35). Here*

$$\mathcal{H}:=\{(x_n)_{n\in\mathbb{N}}\in\mathbb{R}^{\mathbb{N}}\mid\sum_{n=0}^{\infty}x_n^2<\infty\}$$

*and*

$$\|(x_n)\|_{l^2}:=\sqrt{\sum_{n=0}^{\infty}|x_n-y_n|^2}.$$

*Then $(\mathcal{H},\|\cdot\|_{l^2})$ is complete. As you will see in functional analysis, this the most important (in the unique up to isomorphism) example of a " separable Hilbert space". Separable means that it contains a countable and dense subset.*

**Exercise F.** *Show that the Hilbert space $(\mathcal{H},\|\cdot\|_{l^2})$ is complete.*

**Example 3.24.** *As we have learnt in analysis, the standard metric $d(x,y)=|x-y|$ on $\mathbb{Q}$ is not complete. Actually, more is true: the standard topology on $\mathbb{Q}$ is not compatible with any complete metric. In other words, there does not exists a complete metric $d'$ such that $\mathcal{T}_d=\mathcal{T}_{d'}$. We will see how to prove this in exercise H.*

**Example 3.25.** *We have also talked about the p-adic metric $d_p(-)$ on $\mathbb{Q}$ (which induces the p-adic topology), it is also not complete. For example, when $p\neq 2,3$, the sequence $(2^{p^n})_{n\in\mathbb{N}}$ is Cauchy but not convergent (the proof uses Fermat's little theorem and is a little tricky, see [**P-adic**]. Alternatively, you can just use Baire's theorem. (see exercise H). It's amusing the same result can be proved in two very different ways and one may wonder whether there is a deeper connection between them. (I don't know the answer.)*

**Exercise G.** *We introduce the definition of a **Baire space** in this exercise. A topological space $X$ is called a Baire space if the following condition is satisfied:*

- *For any countable collection $\{U_n\}_{n\in\mathbb{N}}$ of **open and dense** subsets, the intersection $\bigcap_{n\in\mathbb{N}}U_n$ is also dense.*



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*(1) Show that $X$ is a Baire space if and only if the following condition is satisfied:*

- *For any countable collection $\{A_n\}_{n\in\mathbb{N}}$ of closed sets such that $\mathring{A}_n = \varnothing$ for any $n$, we must have*

$$(\bigcup_{n\in\mathbb{N}} \mathring{A_n}) = \varnothing.$$

*(Hint: Use the following fact: $X \setminus \mathring{A} = \overline{X \setminus A}$ for any set $A$.)*
*(2) Show $\mathbb{Q}$, equipped with the standard topology, is not a Baire space.*
*(3) Show that $\mathbb{Q}$, equipped with the p-adic topology, is not a Baire space.*

The following deep theorem is a fundamental result in functional analysis.

**Theorem 3.26** (Baire's theorem)**.** *Any **complete** metric space is a Baire space.*

**Exercise H.** *In this exercise, we will prove Baire's theorem in several steps.*
*(1) Let $\{U_n\}_{n\in\mathbb{N}}$ be a countable collection of open and dense subsets of a metric space $(X, d)$. Let $U$ be any nonempty open set. Show that there exists a decreasing sequence of nonempty closed sets $V_0 \supset V_1 \supset \cdots$ such that $V_0 \subset U$ and for any $n \in \mathbb{N}$, we have $V_n \subset U_n$ and $\operatorname{diam}(V_n) < \frac{1}{2^n}$. Here the diameter is defined as*

$$\operatorname{diam}(V_n) := \sup_{x,y\in V_n} d(x, y).$$

*(2) Suppose $(X, d)$ is complete. Show that $\bigcap_{n\in\mathbb{N}} V_n \neq \varnothing$.*
*(3) Show that $U \bigcap(\bigcap_{n\in\mathbb{N}} U_n) \neq \varnothing$.*
*(4) Show that $\bigcap_{n\in\mathbb{N}} U_n$ is dense. This finishes the proof of theorem 3.26.*
*(5) Show that the the standard topology and the p-adic topology on $\mathbb{Q}$ are both not compatible with any complete metric on $\mathbb{Q}$.*

You can image an incomplete metric space as "a balloon with many tiny holes". To fill in these holes, we introduce an operation called "completetion".

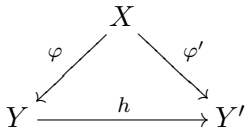
**Definition 3.27.** (1) A continuous map $f : (X, d_X) \to (Y, d_Y)$ is called an **isometric embedding** if

$$d_X(x_1, x_2) = d_Y(f(x_1), f(x_2)) \quad \forall x_1, x_2 \in X.$$

A surjective isometric embedding is called a **isometry**. (In other words, an isometric embedding is an embedding that preserves the metric, and an isometry is a homeomorphism that preserve the metric.)

(2) An isometric embedding $\varphi : (X, d_X) \to (Y, d_Y)$ is called a **completion** if $(Y, d_Y)$ is complete and $\varphi(X)$ is dense in $Y$.

**Theorem 3.28.** *(1) Any metric space $(X, d)$ has a completion.*
*(2) Completion is unique up to isometry. More precisely, let $\varphi : (X, d_X) \to (Y, d_Y)$ and $\varphi' : (X, d_X) \to (Y', d_{Y'})$ be two completions of the same metric space. Then there exists a unique isometry $h : (Y, d_Y) \to (Y', d_{Y'})$ that fits into the following commutative diagram*




*Proof.* (1) **Step 1:** Let $\widehat{X} \subset X^{\mathbb{N}}$ be the set of all Cauchy sequences in $X$. Define a map

$$\hat{d} : \widehat{X} \times \widehat{X} \to [0, +\infty)$$



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by

$$\hat{d}((x_n),(y_n)) := \lim_{n\to\infty} d(x_n, y_n)$$

(This limit exists because $\mathbb{R}$ is complete.) Then define an equivalence relation $\sim$ on $\widehat{X}$ by

$$x \sim y \iff \hat{d}(x,y) = 0.$$

Let $\overline{X} = \widehat{X}/\sim$ be the set of equivalent classes. Then $\hat{d}$ induces a map

$$\bar{d} : \overline{X} \times \overline{X} \to [0, +\infty)$$

which can be checked to be a metric.

**Step 2:** Consider the map $\varphi : X \to \overline{X}$ which sends $x \in X$ to $[(x)]$ (the equivalent class of the constant sequence $(x)$). It's straightforward to check that $\varphi$ is an isometric embedding. Furthermore, since any point $[(x_n)_{n\in\mathbb{N}}]$ is the limit of the sequence $(\varphi(x_n))_{n\in\mathbb{N}}$, we see that $\varphi(X)$ is dense in $\overline{X}$.

**Step 3:** We prove that any Cauchy sequence $(\varphi(x_n))_{n\in\mathbb{N}}$ in $\varphi(X)$ has a limit in $\overline{X}$: Because $\varphi$ is an isometric embedding, the sequence $(\varphi(x_n))_{n\in\mathbb{N}}$ being Cauchy implies that the sequence $(x_n)_{n\in\mathbb{N}}$ is Cauchy in $X$. So $[(x_n)_{n\in\mathbb{N}}]$ is a point in $\overline{X}$ and it's straightforward to check that $(\varphi(x_n))_{n\in\mathbb{N}}$ converges to $[(x_n)_{n\in\mathbb{N}}]$.

**Step 4:** Given any Cauchy sequence $(s_n)_{n\in\mathbb{N}}$ in $\overline{X}$, because $\varphi(X)$ is dense in $X$, we can find a sequence $(s'_n)_{n\in\mathbb{N}}$ in $\varphi(X)$ such that $|s'_n - s_n| \le \frac{1}{2^n}$. Then one can check $(s'_n)_{n\in\mathbb{N}}$ is Cauchy. By previous step, $(s'_n)_{n\in\mathbb{N}}$ has a limit $s_\infty \in \overline{X}$. Then it's straightforward to see that $x_\infty$ is also the limit of $(s_n)_{n\in\mathbb{N}}$. So $\overline{X}$ is complete.

(2) This is straightforward: Since $\varphi(X)$ is dense in $Y$, any point $y \in Y$ is the limit of a Cauchy sequence $(\varphi(x_n))_{n\in\mathbb{N}}$. Since $\varphi$ and $\varphi'$ are both isometric embedding, we see that $(\varphi'(x_n))_{n\in\mathbb{N}}$ is also a Cauchy sequence. Then we define $h(y)$ to be the limit of $(\varphi'(x_n))_{n\in\mathbb{N}}$. □

theorem 3.28 tells us that a completion always exists and is essentially unique. This allows us to stop saying "a completion" but instead say "**the completion**"

$$\varphi : (X,d) \to (\overline{X}, \bar{d})$$

from now on. Furthermore, by identifying $X$ with its image $\varphi(X)$, we will treat $X$ as a dense subspace of $\overline{X}$. This also justify our usage of $\overline{X}$ because it is indeed the closure of $X$.

**Example 3.29.** *The completion of $\mathbb{Q}$ with respect to the standard metric $d(x,y) = |x-y|$ gives us $\mathbb{R}$.*

**Example 3.30.** *If we do completion on $\mathbb{Q}$ with respect to the p-adic metric $d_p$. Then the result, denoted by $\mathbb{Q}_p$, is called field of p-adic numbers (it is actually a field). We can formally express a p-adic number as a power series*

$$\sum_{k=m}^{+\infty} p^k \cdot \frac{a_k}{b_k}$$

*where $m \in \mathbb{Z}$, and $a_k, b_k$ are integers not divisible by $p$. $\mathbb{Q}_p$ is an important tool in number theory. You can even do analysis on it (just like the usual analysis $\mathbb{R}$). This is an active field of research in number theory, called "p-adic analysis".*

**Example 3.31.** *Let $C^\infty([0,1])$ be the space of smooth functions from $[0,1]$ to $\mathbb{R}$. We consider the uniform metric*

$$d_{C^0}(f,g) := \max_{x\in[0,1]} |f(x) - g(x)|.$$



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*Recall that every continuous function can be uniformly approximated by a sequence of smooth functions. So the completion of $(C^\infty([0,1]), d_{C^0})$ is exactly $C^0([0,1])$, the space of continuous functions. However, in analysis, we can also complete $C^\infty([0,1])$ via some other metrics. For example, suppose we want to define the distance between $f$ and $g$ to be some "avaergae" of the distance $|f(x)-g(x)|$ as $x$ varies in $[0,1]$. For any $p \in [1,+\infty)$, we can define the $L^p$-norm*

$$\|f\|_{L^p} := (\int_{[0,1]} |f(x)|^p)^{\frac{1}{p}}$$

*and the induced metric*

$$d_{L^p}(f,g) := \|f-g\|_{L^p}.$$

*The completion of $(C^\infty([0,1]), d_{L^p})$ is denoted by $L^p([0,1])$, an important space in real analysis. Suppose you further want to take the first $k$-th derivative into account. Then you can consider the $W^{k,p}$-Sobolev norm*

$$\|f\|_{W^{k,p}} := (\int_{[0,1]} (|f(x)|^p + |f'(x)|^p + \cdots + |f^{(k)}(x)|^p))^{\frac{1}{p}}$$

*and the induced metric*

$$d_{W^{k,p}}(f,g) := \|f-g\|_{W^{k,p}}.$$

*The completion of $(C^\infty([0,1]), d_{W^{k,p}})$ is denoted by $W^{k,p}([0,1])$ and called the $W^{k,p}$-Sobolev space (some books uses $L^p_k$ instead of $W^{k,p}$). This is an important space when we study partial differential equations.*



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# Chapter 4

# Compactness

## 4.1 Key properties of compact spaces

It is well known that any map $f : S \to \mathbb{R}$ has a max if the domain $S$ is a finite set. And we know similar result holds for any **continuous** map on $[0, 1]$. Since we rarely deal with finite sets in topology, the following question is natural.

**Question 4.1.** *Is there a property on a topological space $X$ which guarantee that any **continuous** map $f : X \to \mathbb{R}$ always has a max (and a min)? **In topology, such property would replace the finiteness property in set theory.***

**Definition 4.2.** An **open cover** of a space $X$ is a collection $\mathcal{U} = \{U_\alpha\}_{\alpha \in S}$ of open subsets such that $X = \bigcup_{\alpha \in S} U_\alpha$. A subcover of $\mathcal{U}$ is a subcollection $\mathcal{U}' \subset \mathcal{U}$ **that still covers** $X$.

**Definition 4.3.** (1) A space $X$ is **compact** if any open cover of $X$ has a finite subcover.

(2) A subset $A$ of a topological space $X$ is compact if the subspace topology on $A$ is compact. (Equivalently, this means that for any covering of $A$ **by open subsets of** $X$, we can find a finite subcover.)

Compactness is extremely useful in mathematics. We summarize the three key properties of a compact space in the following three theorems.

**Theorem 4.4.** *Suppose $X$ is compact. Then any continuous map $f : X \to \mathbb{R}$ has a max and a min.*

**Theorem 4.5.** *Suppose $X$ is compact and $Y$ is Hausdorff. Then any bijective, continuous map $f : X \to Y$ must be a homeomorphism.*

**Theorem 4.6.** *Suppose $X$ is compact and first countable. Then any sequence $(x_n)_{n \in \mathbb{N}}$ has a convergent subsequence $(x_{n_i})_{i \in \mathbb{N}}$.*

*Proof of theorem 4.4.* We first prove that $f : X \to \mathbb{R}$ has an upper bound: Because $f$ is continuous, for any $x \in X$, there is a neighborhood $U_x$ of $x$ such that $f(y) \leq f(x) + 1$ for any $y \in U_x$. Then $\mathcal{U} = \{U_x\}_{x \in X}$ is an open cover of $X$. Since $X$ is compact, we can find finite subcover $\mathcal{U}' = \{U_{x_i}\}_{1 \leq i \leq n}$. Then for any $y \in X$, we have $y \in U_{x_i}$ for some $i$. Therefore, $f(y) \leq \max_{1 \leq i \leq n}\{f(x_i) + 1\}$.

Because $f$ has an upper bound, it must have a least upper bound $C \in \mathbb{R}$. We claim that $C = f(x)$ for some $x \in X$. Otherwise, we can define a new continuous map $g : X \to \mathbb{R}$ by $g(x) := \frac{1}{C - f(x)}$. Then $g$ is continuous and has no upper bound. This contradicts with our previous conclusion. Therefore, $C = f(x)$ is a max of $f$.

By a similar argument, we can prove that $f$ has a min. □



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*Remark* 4.7. In exercise G, we will see that any noncompact metrizable space has a continuous, unbounded function to $\mathbb{R}$. So the inverse of theorem 4.4 holds for metrizable space. (It does not hold in general.)

**Exercise A.** *Let $A, B$ be subsets of a metric space $(X.d)$. We define the distance between the two sets by*

$$d(A,B) = \inf_{x\in A, y\in B} d(x,y).$$

*In this whole problem, we assume that $A, B$ are closed and disjoint. We further assume that $A$ is compact.*

*(1) Show that we have* $\operatorname{diam}(A) = d(x,y)$ *for some $x, y \in A$. Here* diam *denotes the diameter (see definition 4.27).*

*(2) Show that $d(A,B) > 0$.*

*(3) Suppose $B$ is also compact. Show that $d(A,B) = d(x,y)$ for some $x \in A, y \in B$.*

*(4) Suppose $X = \mathbb{R}^n$ and $d$ is the standard metric. Show that $d(A,B) = d(x,y)$ for some $x \in A, y \in B$. (We don't need to assume $B$ is compact.)*

We will prove theorem 4.5 and theorem 4.6 later.

**Theorem 4.8.** *A subset $A$ of $\mathbb{R}^n$ is compact if and only if $A$ is closed and bounded (i.e. $\exists R > 0$ such that $|x| < R$ for any $x \in A$).*

To prove this theorem, we need some preparation. Let's look at some examples and nonexamples first.

**Example 4.9.** *(0) Any topology on a finite set is compact.*

*(1) The sphere $S^n$, the closed disk $D^n$, the cube $I^n$ and the torus $T^n$ are all compact. Because they are closed and bounded subset of some Euclidean space.*

*(2) The Cantor set is compact (although it doesn't look very "finite") because it is a closed and bounded subset of $\mathbb{R}$.*

*(3) Consider the disjoint union of a collection of nonempty space $\{X_\alpha\}_{\alpha\in S}$. It's not hard to show that $\bigsqcup_{\alpha\in S} X_\alpha$ is compact if and only if $S$ is finite and $X_\alpha$ is compact for any $\alpha \in S$. However, by a result called the Tychonoff theorem (see Tychonoff theorem 4.16) states that the product space $\prod_{\alpha\in S} X_\alpha$ (with product topology) is compact if and only if $X_\alpha$ is compact for any $\alpha \in S$. We don't even need $S$ to be finite or countable! This is somewhat suprising and we will explain it when we talk about the Tychonoff theorem.*

*(4) Consider the unit disk*

$$D^\infty = \{(x_i)_{i\in\mathbb{N}} \mid \sum_{i\in\mathbb{N}} x_i^2 \le 1\}$$

*in the Hilbert space $\mathcal{H}$ (see example 3.23). Then it is not compact. This is because the open cover $\{V\} \cup \{U_n\}_{n\in\mathbb{N}}$, where $V = \mathring{D^\infty}$ and $U_n = \{(x_i)_{i\in\mathbb{N}} \in D^\infty \mid x_n \neq 0\}$, has no finite subcover. So theorem 4.8 doesn't hold for an infinite dimensional normed space. We will discuss an interesting compact subset of $\mathcal{H}$ in exercise F.*

*(5) While an "infinite dimensional" object* ***tends to be*** *noncompact, this is not always the case. For example, by the Tychonoff theorem, the product space $[0,1]^{\mathbb{N}}$ is compact (with product topology). But if we use the box topology, then it is noncompact. We will have an exercise that proves this claim.*

*(6) For any $\mathbb{F}$ and $n$, the Zariski topology on $\mathbb{F}^n$ is always compact. Actually, it satisfies an even stronger condition called "Noetherian": That means any decreasing sequence of closed subsets*

$$V_1 \supset V_2 \supset V_3 \supset \cdots$$



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*must eventually stabilize (i.e. $\exists N$ such that $V_n = V_N$ for any $n \geq N$). In other words, you can't cut an algebraic set smaller and smaller for infinitely many times. We won't prove this claim here because it requires a fundamental result in commutative algebra called* ***the Hilbert's basis theorem****, which implies that any increasing sequence of ideals of $\mathbb{F}[x_1, \cdots, x_n]$ must stabilize. (Exercise: (i) Why Noetherian implies compactness? (ii) Prove that the Zariski topology on $\mathbb{F}$ is compact. (iii) We have learnt that any polynomial map $f : \mathbb{R}^n \to \mathbb{R}$ is continuous with respect to the Zariski topology. And the Zariski topology on $\mathbb{R}^n$ is compact. By theorem 4.4, this seems to imply that any polynomial function has a max.* ***This is obviously wrong.*** *What's going wrong?)*

**Exercise B.** *(1) Consider the following subspace of $\mathbb{R}^2$*

$$X := \bigcup_{n \in \mathbb{N}_{>0}} \{(x, y) \in \mathbb{R}^2 \mid x^2 + (y - \frac{1}{n})^2 = \frac{1}{n^2}\}.$$

*We call $X$ the* ***Hawaiian earring****. It is an infinite collection of circles (with smaller and smaller radius) that are tangent to other at the same point. Prove that $X$ is compact (although it looks complicated).*

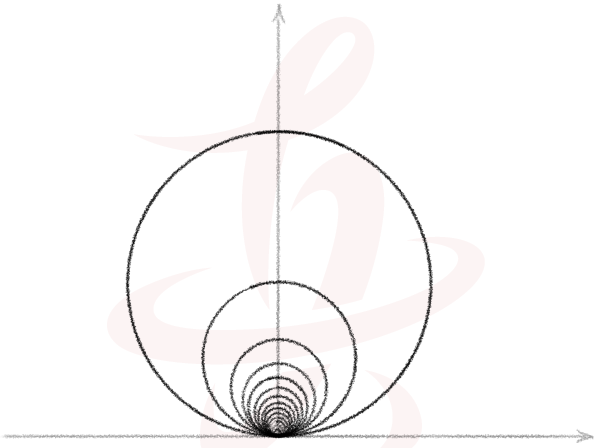

Figure 4.1

*(2) For $n \in \mathbb{N}$, consider the circle*

$$C_n = \{(x, y) \in \mathbb{R}^2 \mid (x - a_n)^2 + (y - b_n)^2 = \frac{1}{2^{2n}}\}$$

*with center $(a_n, b_n)$ and radius $\frac{1}{2^n}$. Let $X = \bigcup_{n \in \mathbb{N}} C_n$. Suppose we have $(a_n, b_n) \in C_{n-1}$ for any $n \geq 1$. Show that either $X$ is compact, or $X \cup \{p\}$ is compact for some $p \in \mathbb{R}^2$.*

Now we prove the following baby case of theorem 4.5 (which also turns out to be the most essential case).

**Theorem 4.10** (Heine-Borel)**.** *A closed interval $[a, b]$ is compact.*

*Proof.* Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in S}$ be an open cover of $[a, b]$. We consider the set

$$A = \{x \mid [a, x] \text{ is contained in a finite collection of } \mathcal{U}\}$$

$\sup A$ exists because $A$ is bounded above by $b$. Let $c = \sup A$. Since $a \in U$ for some $U \in \mathcal{U}$. We have $[a, a + \epsilon) \subset U$ when $\epsilon$ is small enough. So $a + \epsilon \in A$. So $c > a$.



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We first prove that $c \in A$. $c$ is contained in some $U_{\alpha_0}$. Since $U_{\alpha_0}$ is open in $[a, b]$ and $c > a$, there exists $\epsilon > 0$ s.t. $(c - \epsilon, c] \subset U_{\alpha_0}$. Since $c = \sup A$, there exists some $x_0 \in A \cap (c - \epsilon, c]$. Then $[0, x_0]$ is contained in a finite subcollection of $\mathcal{U}$. Since $[0, c] \subset [0, x_0] \cup U_\alpha$, $[0, c]$ is also contained in a finite subcollection of $\mathcal{U}$. So $c \in A$.

Suppose $c < b$. Then $U_{\alpha_0}$ contains $[c, c + \epsilon')$ for some $0 < \epsilon' < b - c$. By the same argument as above, we have $c + \epsilon' \in A$. This is impossible because $c = \sup A$. So $c = b$. □

**Proposition 4.11.** *Let $f : X \to Y$ be a continuous map. Let $A$ be a compact subspace of $X$. Then $f(A)$ is compact.*

*Proof.* Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in S}$ be a collection of open sets in $Y$ that covers $f(A)$. Then $\{f^{-1}(U_\alpha)\}_{\alpha \in S}$ is a collection of open subsets that covers $A$. Since $A$ is compact, we can find a finite subcollection $\{f^{-1}(U_\alpha)\}_{\alpha \in S'}$ such that that still covers $A$. Then $f(A) \subset \bigcup_{\alpha \in S'} U_\alpha$. □

**Corollary 4.12.** *Let $f : X \to Y$ be a continuous, surjective map. Suppose $X$ is compact. Then $Y$ is compact.*

*Remark* 4.13. In general, the preimage of a compact set under a continuous map is not necessarily compact. (For example, consider the constant map $\mathbb{R} \to \{0\}$.) We say a continuous map $f : X \to Y$ is **proper** if the preimage $f^{-1}(A)$ of for any compact subset $A$ of is also compact.

**Exercise C.** *(1) Show that a continuous map $f : \mathbb{R}^m \to \mathbb{R}^n$ is proper if and only if $|f(x)| \to +\infty$ when $|x| \to +\infty$. (That means for any $R > 0$, there exists $R' > 0$ such that for any $|x| > R'$ we have $|f(x)| > R$.)*

*(2) Show that any **nonconstant** polynomial map $f : \mathbb{C} \to \mathbb{C}$ is proper.*

*(3) Suppose $n \geq 2$, show that any polynomial map $f : \mathbb{C}^n \to \mathbb{C}$ is not proper. (Note that similar result is **not** true for $\mathbb{R}$. For example, the map $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x, y) = x^2 + y^2$ is proper.)*

**Theorem 4.14.** *The product space $X_1 \times X_2$ is compact if and only both $X_1$ and $X_2$ are compact.*

*Proof.* For $i = 1, 2$, the projection map $p_i : X_1 \times X_2 \to X_i$ is a surjective, continuous map. Suppose $X_1 \times X_2$ is compact. By corollary 4.12, both $X_1$ and $X_2$ are compact.

Suppose $X_1$ and $X_2$ are both compact. Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in S}$ be an open cover of $X_1 \times X_2$.

We have the following claim: For any $x \in X_1$, there exists a neighborhood $V_x$ of $x$ such that $V_x \times X_2$ is covered by a finite subcollection of $\mathcal{U}$. To see this, we take any point $y \in X_2$. Then $(x, y) \in U_{\alpha(y)}$ for some $\alpha(y) \in S$. By the definition of product topology, we have $(x, y) \in U_{1,y} \times U_{2,y} \subset U_\alpha$, where $U_{i,y}$ is open in $X_i$. Note that $\{U_{2,y}\}_{y \in X_2}$ is an open cover of $X_2$. Since $X_2$ is compact, we can find a finite subcover $\{U_{2,y_1}, U_{2,y_2}, \cdots, U_{2,y_n}\}$. Set

$$V_x = U_{1,y_1} \cap \cdots U_{1,y_n}.$$

Then $V_x \times X_2 \subset \bigcup_{1 \leq i \leq n} U_{\alpha(y_i)}$. This proves the claim.

Note that $\{V_x\}_{x \in X_1}$ is an open cover of $X_1$. Because $X_1$ is compact, we can find a finite subcover $\{V_{x_1}, \cdots V_{x_m}\}$. Then $X_1 \times X_2$ is covered by $\{V_{x_i} \times X_2\}_{1 \leq i \leq m}$. Since each $V_{x_i} \times X_2$ is covered by a finite subcollection of $\mathcal{U}$, $X_1 \times X_2$ is also covered by a subcollection of $U$.



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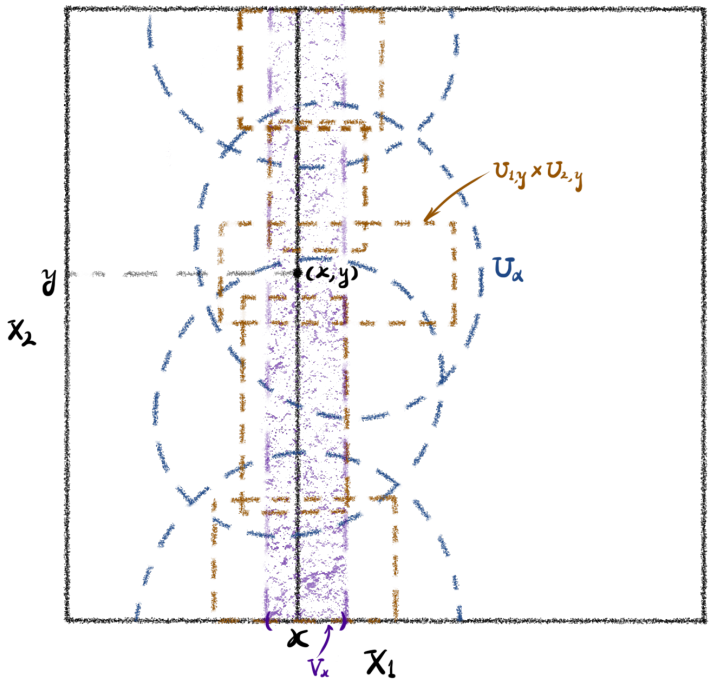


Figure 4.2

□

**Corollary 4.15.** *For any $n \in \mathbb{N}^{>0}$, the space $[a,b]^n$ is compact.*

theorem 4.14 implies that a finite product of compact space is compact. It turns out that the same result holds even for infinite products.

**Theorem 4.16** (Tychonoff)**.** *The product of **any collection** of compact spaces is compact with respect to the **product topology.***

*Remark* 4.17. theorem 4.16 is some what surprising (and counter intuitive), because there are infinitely many "directions" in an infinite product $\prod_{\alpha \in S} X_\alpha$ and its hard to image how to use a finite subcollection to cover all of them. However, note that we are using product topology here. So any open set $U$ contains some $\prod_\alpha U_\alpha$ where $U_\alpha \neq X_\alpha$ for only finitely many $\alpha \in S$. So $U$ already "fill" all but finitely many directions. Although we won't do a full proof of Tychonoff theorem here because it goes too far in set theory. (When $S$ is uncountable, the theorem is essentially equivalent to the axiom of choice. See Munkres.) We refer to Munkres for a complete proof.

**Proposition 4.18.** *Any closed subset of a compact space is compact.*

*Proof.* Let $A$ be a closed subset of a compact space $X$ and let $\mathcal{U}$ be a collection of open subsets of $X$ that covers $A$. Then $\mathcal{U}' = \mathcal{U} \cup \{X - A\}$ is an open cover of $X$. Since $X$ is compact, we can find finite subcollection $\mathcal{U}''$ of $\mathcal{U}$ that covers $X$. Then $\mathcal{U}'' \setminus \{X - A\}$ is a finite subcollection of $\mathcal{U}$ that covers $A$. □

**Proposition 4.19.** *Let $A$ be a compact subset of a Hausdorff space $X$. Then $A$ must be closed.*

To prove proposition 4.19, we need the following lemma.

**Lemma 4.20.** *Let $A$ be a compact subset of a Hausdorff space $X$ and let $x$ be a point in $X \setminus A$. Then there exists a neighborhood $U$ of $A$ and a neighborhood $U'$ of $x$ such that $U \cap U' = \varnothing$.*



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*Proof.* Because $X$ is Hausdorff, for each $y \in A$, there exists a neighborhood $U_y$ of $y$ and a neighborhood $V_y$ of $x$ such that $U_y \cap V_y = \varnothing$. Then $\{U_y\}_{y' \in A}$ covers $A$. Because $A$ is compact, we can find $\{U_{y_1}, \cdots U_{y_n}\}$ that still covers $A$. Then we set $U = U_{y_1} \cup \cdots \cup U_{y_n}$ and $U' = V_{y_1} \cap \cdots \cap V_{y_n}$.

□

*Proof of proposition 4.19.* Let $x$ be a point in $X \backslash A$. By lemma 4.20, there exists a neighborhood $V$ such that $V \subset X \setminus A$. So $X \setminus A$ is open. □

*Proof of theorem 4.8.* Let $A$ be a subset of $\mathbb{R}^n$.

Suppose $A$ is compact. Then by theorem 4.4, the continuous map $A \to \mathbb{R}$ defined by $x \mapsto |x|$ has a max. So $A$ is bounded. By proposition 4.19, $A$ is closed.

Suppose $A$ is bounded and closed. Then $A$ is a closed subset of $[-R, R]^n$ for some $R \gg 0$. By corollary 4.15, $[-R, R]^n$ is compact. So by proposition 4.18, $A$ is compact. □

proposition 4.19 has another useful corollary.

**Corollary 4.21.** *Let $f : X \to Y$ be a continuous map from a compact space $X$ to a Hausdorff space $Y$. Then $f$ is a closed map.*

*Proof.* Let $A$ be a closed subset of $X$. Then by proposition 4.18, $A$ is compact. By proposition 4.11, $f(A)$ is compact. By proposition 4.19, $f(A)$ is closed. □

corollary 4.21 directly implies theorem 4.5.

*Proof of theorem 4.5.* By corollary 4.21, $f$ is a closed map. So $f^{-1} : Y \to X$ is continuous. □

theorem 4.5 is very powerful: To prove two spaces $X, Y$ are homeomorphic, it is usually easy to construct a continuous bijection in one direction. But this is not enough because we still have to prove that its inverse is also continuous. (Think about the example $[0, 1) \to S^1, x \mapsto e^{2\pi i x}$.) In a lot of situations, we can use theorem 4.5 to avoid this additional (tricky) check.

**Corollary 4.22.** *Let $f : X \to Y$ be an injective, continuous map from a compact space $X$ to a Hausdorff space $Y$. Then $f$ is an embedding.*

**Exercise D.** *(1) Let $f : S^1 \to (0, \infty) \times \mathbb{R}$ be a continuous, injective map given by*

$$f(x, y) = (f_1(x, y), f_2(x, y)).$$

*Here we set $S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$. Prove that the following subspace*

$$\{(f_1(x, y)z, f_1(x, y)w, f_2(x, y)) \mid x^2 + y^2 = 1, z^2 + w^2 = 1\} \subset \mathbb{R}^3$$

*is homeomorphic to the torus $T^2$. (This generalizes example 3.16, which correspond to the case $f(x, y) = (x + 2, y)$.)*

*(2) We let $S^3 = \{(u, v) \in \mathbb{C}^2 \mid |u|^2 + |v|^2 = 1\}$ be the unit sphere in $\mathbb{C}^2$. Show that the following subspace of $S^3$*

$$\{(u, v) \in \mathbb{C}^2 \mid |u|^2 + |v|^2 = 1, u^p + v^q = 0\}$$

*is homeomorphic to $S^1$. Here $p, q$ is a pair of positive, co-prime integers. (Note that an embedding of $S^1$ into $S^3$ is called a **knot**. And the example given here is called the $(p, q)$-torus knot. Denoted by $K_{p,q}$ in the picture below.)*



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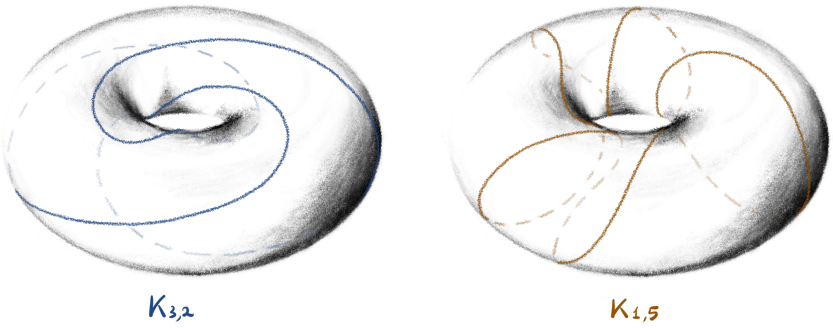


Figure 4.3

## 4.2 Sequentially compactness

Now we introduce a variation of the compactness, which may sounds more familiar in analysis.

**Definition 4.23.** A topological $X$ space is **sequentially compact** is any sequence in $X$ has a convergent subsequence (i.e. a subsequence that has a limit).

Without any assumptions, sequential compactness and compactness doesn't imply each other. For example, $[0,1]^{\mathbb{R}}$ is compact by Tychonoff theorem but it is not sequentially compact. However, by the following theorems, the difference between these two definitions is quite mild.

**Theorem 4.24.** *For first-countable spaces, compactness implies sequentially compactness.*

**Theorem 4.25.** *For metric spaces, compactness is equivalent to sequentially compactness.*

*Proof of theorem 4.24.* Let $(x_i)_{i\in\mathbb{N}}$ be a sequence in a compact space $X$.

First, we prove that there exists a point $x_\infty \in X$ such that any neighborhood of $x_\infty$ contains $x_i$ for infinitely many $i \in \mathbb{N}$. Suppose this were not true. Then any $x \in X$ has a neighborhood $U_x$ that only contains $x_i$ for finitely many $i$. Then $\{U_x\}$ is an open cover of $X$. Since $X$ is compact, there is a finite subcover $\{U_{x_1}, \cdots, U_{x_n}\}$. So $X$ also contains $x_i$ for finitely many $i\mathbb{N}$. This is a contradiction because $\mathbb{N}$ is an infinite set.

Next, we prove that $x_\infty$ is a limit of a subsequence of $(x_i)_{i\in\mathbb{N}}$: We let $\{U_1, U_2, \cdots\}$ be countable neighborhood basis of $x_\infty$. By replacing $U_n$ with $\bigcap_{i=1}^n U_i$, we may assume that $U_m \subset U_n$ whenever $m > n$. For any $n$, $U_n$ contains infinitely many terms in the sequence $(x_i)$. We let $x_{i_n}$ be the $n$-th term in the sequence $(x_i)_{i\in\mathbb{N}}$ that is contained in $U_n$. Consider the sequence $(x_{i_1}, x_{i_2}, \cdots)$. Then we have $x_{i_m} \in U_n$ whenever $m \geq n$. So this subsequence converges to $x_\infty$. □

Since all metric spaces are first-countable, one direction of theorem 4.25 is directly implied by theorem 4.24. The other direction is harder and we need some preparations.

**Definition 4.26.** Let $A$ be a subset of metric space $(X, d)$. The **diameter** of $A$ is defined to be $\operatorname{diam}(A) := \sup\{d(x, y) \mid x, y \in A\}$. We set $\operatorname{diam}(A) = \infty$ if the supreme doesn't exist.

**Definition 4.27.** Let $\mathcal{U}$ be an open cover of a metric space $(X, d)$. We say $\delta > 0$ is a **Lebesgue number** of $\mathcal{U}$ if for any subset $A$ with $\operatorname{diam}(A) < \delta$, we have $A \subset U$ for some $U \in \mathcal{U}$.

**Lemma 4.28.** *Suppose $X$ is a sequentially compact metric space. Then any open cover of $X$ has a (positive) Lebesgue number.*



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*Proof.* Suppose this were not true for some open cover $\mathcal{U}$. Then for any positive integer $n$, there exists some subset $A_n$ such that $\operatorname{diam}(A_n) < \frac{1}{n}$ and $A_n$ is not contained in any $U \in \mathcal{U}$. Take any point $x_n \in A_n$. Then $A_n \subset B_{\frac{2}{n}}(x_n)$ so $B_{\frac{2}{n}}(x_n)$ is not contained in any $U \in \mathcal{U}$.

Now we consider the sequence $(x_n)_{n\in\mathbb{N}}$. Since $X$ is sequentially compact, this sequence has a subsequence $(x_{n_i})_{i\in\mathbb{N}}$ that converges to some point $x_\infty \in X$. Then $x_\infty \in U$ for some $U \in \mathcal{U}$ and there exists $\epsilon > 0$ such that $B_{2\epsilon}(x_\infty) \subset U$. For $i \gg 0$, we have $\frac{2}{n_i} < \epsilon$ and $d(x_{n_i}, x_\infty) < \epsilon$. By the triangle inequality, we have

$$B_{\frac{2}{n_i}}(x_{n_i}) \subset B_{2\epsilon}(x_\infty) \subset U.$$

This is a contradiction. □

**Definition 4.29.** For $\delta > 0$, a subset $A$ of a metric space $X$ is called an $\delta$-net if the open balls $\{B_\delta(x)\}_{x\in A}$ cover $X$.

**Lemma 4.30.** *Suppose $X$ is a sequentially compact metric space. Then $X$ has a **finite** $\epsilon$-net for any $\epsilon > 0$.*

*Proof.* Suppose $X$ has no finite $\delta$-net. We construct a sequence $(x_n)_{n\in\mathbb{N}}$ inductively as follows: Pick any point $x_0 \in X$ and pick $x_{n+1}$ in $X \setminus (\bigcup_{i=1}^n B_\delta(x_i))$ (which is nonempty because $X$ has no finite $\delta$-net). By this construction, we have

$$d(x_n, x_{n'}) \geq \delta, \quad \forall n \neq n'.$$

Since $X$ is sequentially compact, the sequence $(x_n)_{n\in\mathbb{N}}$ has a subsequence $(x_{n_i})_{i\in\mathbb{N}}$ that converges to some point $x_\infty$. Then for $i$ large enough, both $x_{n_i}$ and $x_{n_{i+1}}$ belongs to $B_{\delta/2}(x_\infty)$. So $d(x_{n_i}, x_{n_{i+1}}) < \delta$. This is a contradiction. □

*Proof of theorem 4.25.* As we mentioned, since any metric space is first-countable, by theorem 4.24, compactness implies sequentially compactness.

Suppose $X$ is a sequentially compact metric space. Let $\mathcal{U}$ be an open cover of $X$. Then by lemma 4.28, $\mathcal{U}$ has a Lebesgue number $\delta > 0$. By lemma 4.30, $X$ has a finite $\frac{\delta}{2}$-net $\{x_1, \cdots, x_n\}$. By the definition of a Lebesgue number, each $B_{\frac{\delta}{2}}(x_i)$ is contained in some $U_i \in \mathcal{U}$. Then $\{U_1, \cdots, U_n\}$ is a finite subcover of $\mathcal{U}$. □

lemma 4.28 is useful in other places. So we rephrase it as follows:

**Lemma 4.31** (The Lebesgue number lemma)**.** *Any open cover of a compact metric space has a Lebesgue number.*

**Exercise E.** *Let $\{X_\alpha\}_{\alpha\in S}$ be a collection of sequentially compact spaces. Let $X = \prod_{\alpha\in S} X_\alpha$, equipped with product topology.*

*(1) Suppose $S$ is countable. Show that $X$ is sequentially compact. (Hint: Convergence in product topology means pointwise convergence. See Section 3 exercise D)*

*(2) Show that the box topology on $[0,1]^{\mathbb{N}}$ is not compact.*

**Exercise F.** *Consider the following subset of the Hilbert space $\mathcal{H}$ (see example 3.23).*

$$A = \{(x_n)_{n\in\mathbb{N}} \in \mathcal{H} \mid \sum_{n=0}^{\infty} (n^2+1)x_n^2 \leq 1\}$$

*Show that $A$ is compact.*



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**Exercise G.** *Let $(X, d)$ be a noncompact metric space. Then $X$ contains a sequence $\{x_n\}_{n\in\mathbb{N}}$ that doesn't have any convergent sequence. We consider the following function $f : X \to \mathbb{R}$*

$$f(x) := \inf_{n\in\mathbb{N}}(d(x, x_n) + \frac{1}{n}).$$

*(1) Show that $f$ is continuous.*

*(2) Show that $f(x) > 0$ for any $x \in X$.*

*(3) Consider the function $g : X \to \mathbb{R}$ defined by $g(x) = \frac{1}{f(x)}$. Show that $g$ is a continuous function without an upper bound. Therefore, the converse of theorem 4.4 is true for metrizable spaces.*

**Exercise H.** *Let $f : X \to Y$ be a closed map. Suppose $f^{-1}(y)$ is compact for any point $y \in Y$. Show that $f$ is proper. Give an example to show that this is not true if you don't require $f$ to be closed.*

## 4.3 Locally compactness

Although compactness is really a useful property, it is not satisfied for many spaces we are interested in (most remarkably, $\mathbb{R}^n$ is not compact). On the other hand, we have the following observation:

- Any bounded sequence in $\mathbb{R}^m$ has a convergent subsequence.
- There exists a bounded sequence in the Hilbert space $\mathcal{H}$ (see example 3.23) that do not have any convergent subsequence. (For example, any two points in the sequence

  $$(1, 0, 0, \cdots),\ (0, 1, 0\cdots),\ (0, 0, 1, \cdots),\ \cdots$$

  have distance 1, so it doesn't have any convergent subsequence.)

So we want to define a weaker notation of compactness so that is satisfied for $\mathbb{R}^m$ but not for $\mathcal{H}$. This motivates the following definition.

**Definition 4.32.** A space $X$ is **locally compact** if for any point $x \in X$, there is a compact subset $K$ of $X$ that such that $x \in \mathring{K}$.

**Example 4.33.** *$\mathbb{R}^n$ is locally compact because for any point $x \in \mathbb{R}^n$, the closed disk $\{y \in \mathbb{R}^n \mid |y - x| \le 1\}$ is a compact set that contains a neighborhood of $x$.*

**Definition 4.34.** A Hausdorff space $X$ is called a (topological) manifold if for any point $x \in X$, there is an neighborhood $U$ of $x$ that is homeomorphic to $\mathbb{R}^n$ for some $n$.

Manifold is a generalization of points, curves and surfaces to higher dimensions. For example, $T^n$ and $S^n$ are both manifolds, and any open subspace of $\mathbb{R}^n$ is a manifold, but the subspace $\{(x, y) \in \mathbb{R}^2 \mid xy = 0\}$ is not a manifold.

Manifold is an extremely important object of study in topology. One of the central topics in topology is to classify manifolds. We will return to it in later weeks.

**Example 4.35.** *Any manifold is locally compact because $\mathbb{R}^n$ is locally compact.*

The following lemma is useful when we try to find a "arbitrary small" compact set in a locally compact space.

**Lemma 4.36.** *Let $X$ be a locally compact, Hausdorff space. Then for any point $x \in X$ and any neighborhood $U$ of $x$, there exists a compact set $K \subset U$ such that $x \in \mathring{K}$.*



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小红书号:3189205503 qq

*Proof.* Since $X$ is locally compact. There exists a compact set $K'$ such that $x \in \mathring{K'}$. We consider $K'' = K' \setminus U$. This is a compact set that doesn't contain $x$. By lemma 4.20, we can find a neighborhood $U_2$ of $x$ and a neighborhood $U_3$ of $K''$ such that $U_2 \cap U_3 = \varnothing$. Then we set $K = K' \setminus U_3$. Since $K' \setminus U \subset U_3$, we have

$$K = K' \setminus U_3 \subset K' \setminus (K' \setminus U) = U$$

and $K$ contains $U_2 \cap \mathring{K'}$, which is a neighborhood of $x$. So $x \in \mathring{K}$. □

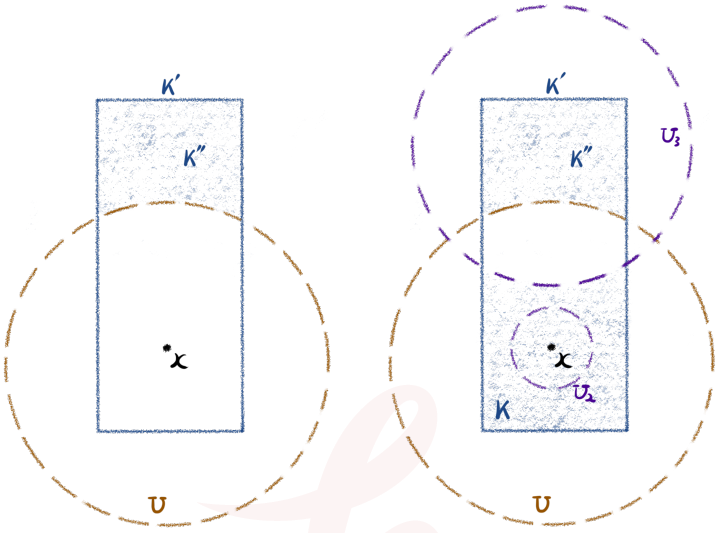


Figure 4.4

One can use lemma 4.36 to deduce the following useful proposition, which is a variation of corollary 4.21 in the noncompact setting.

**Proposition 4.37.** *Let $f : X \to Y$ be a continuous map. Suppose $f$ is proper and $Y$ is Hausdorff and locally compact. Then $f$ is closed.*

*Proof.* Let $V$ be a closed subset of $X$. Take any point $y \in Y \setminus f(V)$. By lemma 4.36, we can find a compact set $K \subset Y$ that contains a neighborhood of $y$. Since $f$ is proper, $f^{-1}(K)$ is compact. So $f^{-1}(K) \cap V$ is compact. Since continuous map must send compact set to compact set, $f(f^{-1}(K) \cap V)$ is compact. Because $Y$ is Hausdorff, $f(f^{-1}(K) \cap V)$ is closed. We consider the open set

$$U := \mathring{K} \setminus f(f^{-1}(K) \cap V).$$

This is a neighborhood of $y$ that doesn't intersect $f(V)$. So $Y \setminus f(V)$ is open. Hence $f(V)$ is closed. □

The following corollary is a noncompact version of theorem 4.5.

**Corollary 4.38.** *Let $f : X \to Y$ be a continuous, injective, proper map. Suppose $Y$ is Hausdorff and locally compact. Then $f$ is an embedding.*

*Proof.* By proposition 4.37, $f$ is a closed map. So the inverse of the continuous bijection $f : X \to f(X)$ is also continuous. □

**Exercise I.** *(1) Show that $\mathbb{Q}$ (with the subspace topology of $\mathbb{R}$) is not locally compact.*

*(2) Show that the Hilbert space $\mathcal{H}$ in example 3.23 is not locally compact.*

*(3) Let $f : \mathbb{R}^m \to \mathbb{R}^n$ be an injective, continuous map. Suppose $|f(x)| \to \infty$ when $|x| \to \infty$. Show that $f$ is an embedding.*



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It turns out that for a locally compact, Hausdorff space $X$, either $X$ itself is compact, or we can turn it into a compact, Hausdorff space by adding a single point.

**Definition 4.39.** Let $X$ be a space which is Hausdorff, locally compact but not compact. The **one-point compactification** of $X$ is defined as

$$X^+ := X \cup \{\infty\}.$$

The topology on $X^+$ is defined as follows: Let $U$ be a subset of $X^+$. Suppose $\infty \in U$. Then $U$ is open if and only if $X^+ \setminus U$ is compact. Suppose $\infty \notin U$. Then $U$ is open if and only if $U$ is open in $X$.

*Remark* 4.40. When $X$ is Hausdorff but not locally compact, one can still define $X^+$ as in definition 4.39. In this case, $X^+$ is still compact but no longer Hausdorff. When $X$ is not Hausdorff, the "topology" given in definition 4.39 is even not necessarily a topology.

**Lemma 4.41.** *(1) The topology described in definition 4.39 is indeed a topology.*
*(2) $X^+$ is compact and Hausdorff.*

*Proof.* (1) We will verify the three axioms for closed sets. It is clear that $\varnothing$ and $X^+$ are both closed. To check the other two axioms, we note that a subset $A$ of $X^+$ is closed if and only if it is one of the following two types:

(i) $A$ is a compact subset of $X$. Note that $A$ must be closed in $X$ because $X$ is Hausdorff.

(ii) $A = A' \cup \{\infty\}$, where $A'$ is a closed subset of $X$.

Using this description, one can verify the other two axioms via a case by case argument. (Note that a finite union of compact sets is still compact.)

(2)$X^+$ is Hausdorff: Let $x, y$ be distinct points in $X^+$. Suppose $x, y \neq \infty$. Then they have some disjoint neighborhoods because $X$ is Hausdorff. Suppose $x = \infty$. Because $X$ is locally compact, there is a compact subset $K$ of $X$ that contains some neighborhood $V$ of $y$. We let $U = X^+ \setminus K$. Then $U$ is a neighborhood of $x$ and $U \cap V = \varnothing$.

$X^+$ is compact: Let $\mathcal{U}$ be an open cover of $X^+$. Then $\infty$ is contained in some $U_0 \subset \mathcal{U}$. By definition, $X^+ \setminus U_0$ is compact. So $X^+ \setminus U_0$ is covered by a finite sub-collection $\{U_1, \cdots U_n\} \subset \mathcal{U}$. Then $X^+$ is covered by $\{U_0, U_1, \cdots U_n\}$. □

**Exercise J.** *Let $Y$ be a compact and Hausdorff space and let $y$ be a point in $Y$.*
*(1) Show that $Y \setminus \{y\}$ is locally compact.*
*(2) Suppose $Y \setminus \{y\}$ is not compact. Show that $Y \cong (Y \setminus \{y\})^+$.*
*(3) Suppose $Y \setminus \{y\}$ is compact. Show that $Y \cong (Y \setminus y) \bigsqcup \{y\}$. Here $\{y\}$ denotes the single-point space with trivial topology.*
*(4) Show that $S^n$ is homeomorphic to $(\mathbb{R}^n)^+$.*
*(5) Given an example to show that the one-point compatification of a manifold is not necessarily a manifold.*

**Exercise K.** *Given topological spaces $X, Y$, we let $\mathcal{C}(X, Y)$ be the set of **continuous** maps from $X$ to $Y$. We want a suitable topology on $\mathcal{C}(X, Y)$. Since $\mathcal{C}(X, Y)$ is a subset of $Y^X$ (the set of all maps), we can use the subspace topology of the product topology on $Y^X$. This is called the **pointwise convergence topology** (See exercise D for an explanation for this name.). However, because the product topology doesn't take the topology of the domain $X$ into account, this pointwise convergence topology is not suitable to for algebraic topology (as we will discuss*



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*soon). So we will define another topology on $\mathcal{C}(X,Y)$ as follows: Let $K$ a compact subset of $X$ and let $U$ an open subsets of $Y$, we define the following subset*

$$S(K,U) := \{f \in \mathcal{C}(X,Y) \mid f(K) \subset U\}.$$

*Consider the following collection of subsets of $\mathcal{C}(X,Y)$*

$$\mathcal{B} := \{S(K_1,U_1) \cap \cdots \cap S(K_n,U_n) \mid K_i \text{ is compact and } U_i \text{ is open}\}.$$

*Clearly, $\mathcal{B}$ a basis for some topology. We call the topology generated by $\mathcal{B}$ the* ***compact-open topology*** *on $\mathcal{C}(X,Y)$.*
*(1) Show that when $X$ is space with discrete topology, the compact-open topology on $\mathcal{C}(X,Y)$ is the same as the pointwise convergence topology.*
*(2) We have an evaluation map*

$$\mathrm{ev} : X \times \mathcal{C}(X,Y) \to Y$$

*defined by*

$$\mathrm{ev}(x,f) := f(x).$$

*Suppose we use the compact-open topology on $\mathcal{C}(X,Y)$. And suppose $X$ is locally compact and Hausdorff. Show that* ev *is continuous.*

*(3) Given any topological spaces $X,Y,Z$ and any map $f : X \times Y \to Z$, we have an induced map $\widetilde{f} : X \to Z^Y$ defined by $\widetilde{f}(x) = f(x,-) \in Z^Y$.*

*Suppose $Y$ is locally compact and Hausdorff. Show that $f$ is continuous if and only if* ***both*** *of the following conditions hold:*

- *The map $f(x,-)$ is continuous for any $x$. So $\widetilde{f}$ is actually a map from $X$ to $\mathcal{C}(Y,Z)$.*
- *The map $\widetilde{f} : X \to \mathcal{C}(Y,Z)$ is continuous. Here we use the compact-open topology on $\mathcal{C}(Y,Z)$.*

*So we have a bijection $\varphi : \mathcal{C}(X \times Y, Z) \to \mathcal{C}(X, \mathcal{C}(Y,Z))$ defined by $\varphi(f) := \widetilde{f}$.*

*(4) Suppose both $X,Y$ are locally compact Hausdorff. Show that $\varphi$ is a homeomorphism. Here we are using the compact-open topology on all three function spaces.*



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# Chapter 5

# Connectedness and path-connectedness

## 5.1 Decomposition of a set

To better discuss connectedness, we do a little more set theory.

**Definition 5.1.** A **partition** of a set $A$ is a collection $\{A_\alpha\}_{\alpha\in S}$ of **non-empty and disjoint** subsets whose union equals $A$. The expression $A=\bigcup_{\alpha\in S}A_\alpha$ is called a **decomposition** of $A$.

*Remark* 5.2. Given any partition $\mathcal{P}$ of $A$, we can define an equivalence relation setting $\sim_{\mathcal{P}}$ on $A$ by setting $x\sim_{\mathcal{P}} y$ if $x,y$ belongs to the same element in $\mathcal{P}$. Conversely, given any equivalence relation $\sim$ on $A$, we can define a partition $\mathcal{P}_\sim$ on $A$ by taking equivalent classes of $\sim$. So we have a 1-1 correspondence between partitions of a set and equivalence relations on a set.

**Lemma 5.3.** *(1) Let $X$ be a topological space and let $X=\bigcup_{\alpha\in S}X_\alpha$ be a decomposition. Suppose for any $\alpha\in S$, $X_\alpha$ is open in $X$. Then $X\cong\bigsqcup_{\alpha\in S}X_\alpha$.*

*(2) If $S$ is finite, then the same result holds a decomposition into closed subsets.*

*Proof.* (1) Let $X'=\bigsqcup_{\alpha\in S}X_\alpha$. We define a map $f:X\to X'$ by setting $f(x):=(x,\alpha)$ for any $x\in X_\alpha$. Clearly $f$ is bijective. Given any open subset $\bigsqcup_{\alpha\in S}U_\alpha$ of $X'$, its preimage $f^{-1}(\bigsqcup_{\alpha\in S}U_\alpha)=\bigcup_{\alpha\in S}U_\alpha$ is open in $X$. So $f$ is continuous. Given any open subset $U$ of $X$, we have $f(U)=\bigsqcup_{\alpha\in S}(U\cap X_\alpha)$. Since $U$ is open in $X$, $U\cap X_\alpha$ is open in $X_\alpha$. So $f$ is an open map. So $f$ is a homeomorphism.

(2) Suppose $S$ is finite. Then $\bigsqcup_{\alpha\in S\setminus\{\beta\}}X_\alpha$ is closed for any $\beta$. So $X_\beta$ is open. □

## 5.2 Connected spaces

Given two spaces $X$ and $Y$, we can form a larger space $Z=X\bigsqcup Y$. One can ask the inverse question, given a space $Z$, is it homeomorphic to the disjoint union of two nonempty spaces?

**Definition 5.4.** A space is **connected** if it is **not** homeomorphic to the disjoint union of two **nonempty** spaces. Clearly, this is a topological property.

**Example 5.5.** *(1) As we will see soon, a subspace of $\mathbb{R}$ is connected if and only if it is an interval, so $\mathbb{R},(a,b),[a,b],(a,b]$ are all connected.*

*(2) For any $n\geq 0$, $D^n,T^n,I^n$ are all connected.*

*(3) $S^n$ is connected when $n\geq 1$ but $S^0=\{-1,1\}$ is not connected.*

*(3) $\mathbb{R}^n\setminus\{0\}$ is connected when $n\geq 2$, but $\mathbb{R}\setminus\{0\}$ is not connected because it is homeomorphic to $(-\infty,0)\bigsqcup(0,\infty)$.*

*(4) $\mathbb{Q}$ is not connected because it is homeomorphic to $(\mathbb{Q}\cap(-\infty,\sqrt{2}))\bigsqcup(\mathbb{Q}\cap(\sqrt{2},+\infty))$.*



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There are several equivalent definitions of connected.

**Lemma 5.6.** *Let $X$ be a topological space. The following conditions are equivalent so any of them can be used as a definition of connectedness.*

1. *$X$ is connected.*
2. *$X$ does not have a decomposition $A \cup B$ into open sets. (Note: by our definition of a decomposition, $A, B$ must be disjoint and nonempty.)*
3. *$X$ does not have a decomposition $A \cup B$ into closed sets.*
4. *$X$ does not have a decomposition $A \cup B$ such that $X = A \cup B$ and $A \cap \overline{B} = B \cap \overline{A} = \varnothing$.*
5. *$\varnothing$ and $X$ are the only subsets of $X$ that are both open and closed.*
6. *All continuous maps $X \to \{0, 1\}$ are constant maps. (Here we are using the discrete topology on $\{0, 1\}$.)*

*Proof.* The proof is routine. Here is a sketch: (1) $\implies$ (2): This directly follows from lemma 5.3.

(2) $\implies$ (3): Suppose $X$ has a decomposition $A \cup B$ with $A, B$ closed. Then $A = X \setminus B$ is open and $B = X \setminus A$ is open.

(3) $\implies$ (4): Suppose $X$ has a decomposition $A \cup B$ with $A \cap \overline{B} = B \cap \overline{A} = \varnothing$. Then $\overline{B} \subset X \setminus A = B \subset \overline{B}$ so $B = \overline{B}$ and $B$ is closed. Similarly, $A$ is closed.

(4) $\implies$ (5): Suppose $A \neq \varnothing, X$ is both open and closed. Let $B = X \setminus A$. Then $X = A \cup B$ and $A \cap \overline{B} = B \cap \overline{A} = \varnothing$.

(5) $\implies$ (6): Suppose there is a nonconstant map $f : X \to \{0, 1\}$. Then $f^{-1}(0)$ is both open and closed and $f^{-1}(0) \neq \varnothing, X$ because $f$ is nonconstant.

(6) $\implies$ (1): Let $A, B$ be two nonempty spaces. Then there always a nonconstant, continuous map $f : A \bigsqcup B \to \{0, 1\}$ that sends $A$ to 0 and sends $B$ to 1. Suppose $X \cong A \bigsqcup B$. Then we can compose $f$ with the homeomorphism to get a nonconstant, continuous map from $X$ to $\{0, 1\}$. □

*Remark* 5.7. lemma 5.6 (6) is an extremely useful criterion of connectedness. (We will use it to prove a many results in this section.) And it reflects an important philosophy: To understand the "inner structure" of a topological space $X$ (e.g. whether $X$ can be decomposed into two disjoint open subsets), we can take a simple "testing object" $Y$ (e.g. the discrete space $\{0, 1\}$) and study the set of maps between $X$ and $Y$. This idea will be used more and more as we go deeper in topology (we will consider more and more interesting testing objects like $[0, 1]$, the triangle, the cube, the sphere, the Eilenberg-Maclane space, the classifying space...) This is also a naive (but important !) idea of the Category theory: instead of studying the "inner structure" of an object, we can just treat each object as a black box and study the maps (morphisms) between different objects.

**Proposition 5.8.** *Let $A$ be a subset of $\mathbb{R}$. Then $A$ is connected if and only if $A$ is a (open/closed/half-open-half-closed/finite/infinite) interval. (A single point and the emptyset are both intervals because they can be expressed as $[a, a]$ and $[a, a - 1]$ respectively.)*

*Proof.* It's straightforward to see that $A$ is an interval if and only if it satisfies the following property:

$$\forall a, b \in A, a < c < b \implies c \in A. \tag{5.1}$$

Suppose (eq. (5.1)) is satisfied. Let $f : A \to \{0, 1\}$ be a continuous, nonconstant map. Take any $a \in f^{-1}(0)$ and $b \in f^{-1}(1)$. Without loss of generality, we assume $a < b$. Then by (i), we have $[a, b] \subset A$. Let $c = \sup\{x \in [a, b] \mid f(x) = 0\}$. Because $f^{-1}(0)$ and $f^{-1}(1)$ are both open



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in $A$. We must have $a < c < b$. Then for any $\epsilon > 0$, we can find an element $a' \in (c, c-\epsilon)$ such that $f(a') = 0$ and an element $b' \in (c, c+\epsilon)$ such that $f(b') = 1$. By continuity of $f$, the former implies that $f(c) = 0$ and the latter implies that $f(c) = 1$. This contradiction shows that a nonconstant $f$ doesn't exist, so $A$ is connected.

Suppose (eq. (5.1)) is not satisfied. Then there exists $a, b \in A$ and $c \notin A$ such that $a < b < c$. Then $A$ is the union of $(-\infty, c) \cap A$ and $(c, +\infty) \cap A$, which are open, nonempty and disjoint. So $A$ is not connected. □

**Lemma 5.9.** *(1) Let $f : X \to Y$ be a continuous map. Suppose $A \subset X$ is connected. Then $f(A)$ is connected.*

*(2) Suppose $A$ is dense in $X$ and $A$ is connected. Then $X$ is also connected.*

*(3) Given $A, B \subset X$, suppose $A$ is connected and $A \subset B \subset \overline{A}$. Then $B$ is also connected.*

*Proof.* (1) Suppose $f(A)$ is not connected. Then there exists a nonconstant, continuous map $g : f(A) \to \{0, 1\}$. Then the composition

$$A \xrightarrow{f|_A} f(A) \xrightarrow{g} \{0, 1\}.$$

is a continuous, nonconstant map. This leads to a contradiction.

(2) Given a continuous map $f : X \to \{0, 1\}$. Then $f|_A$ must be constant. Then $f^{-1}(f(A))$ is closed so it contains $\overline{A} = X$. So $f$ is also constant.

(3) $A \subset B \subset \overline{A}$ implies that $A$ is dense in $B$. □

**Example 5.10.** *Let $X = \mathbb{R}^2$ and let $A = \{(x, \sin\frac{1}{x}) \mid x \in (0, 1]\}$. Then $A$ is the image of a continuous map $f : (0, 1] \to X$ defined by $f(x) = (x, \sin\frac{1}{2})$ so $A$ is connected by lemma 5.9 (1). As we have proved in Exercise 2.exercise F, its closure is the topologists's sine curve*

$$\overline{A} := \{(0, x) \mid x \in [-1, 1]\} \cup A$$

*So by lemma 5.9, $\overline{A}$ is connected as well. This is because any point in $\{(0, x) \mid x \in [-1, 1]\}$ is the limit of sequence in $A$. So the two subsets $\{(0, x) \mid x \in [-1, 1]\}$ and $A$ are not separated. We will return to this example soon (when we talk about path-connectedness).*

Next, we introduce a lemma that allows us to show that a space is connected by covering it with connected subspaces.

**Lemma 5.11.** *Let $\{A_\alpha\}_\alpha$ be a collection of subspace of a space $X$. Let $A = \bigcup_{\alpha \in S} A_\alpha$. Suppose the following conditions are satisfied:*

*(i) $A_\alpha$ is connected for all $\alpha \in S$;*

*(ii) For any $\alpha, \alpha'$, there exists a finite sequence $\alpha = \alpha_0, \alpha_1, \cdots, \alpha_n = \alpha'$ in $S$ such that $A_{\alpha_i} \cap A_{\alpha_{i+1}} \neq \varnothing$ for any $0 \leq i \leq n-1$.*

*Then $A$ is connected.*

*Proof.* Suppose there is a nonconstant, continuous map $f : \bigcup_{\alpha \in S} A_\alpha \to \{0, 1\}$. Take any $a \in f^{-1}(0)$ and $b \in f^{-1}(1)$. Then $a \in A_\alpha$ and $b \in A_{\alpha'}$ for some $\alpha, \alpha' \in S$. Let $\{\alpha_i\}_{1 \leq i \leq n-1}$ be the sequence provided by (ii). Then $f|_{A_{\alpha_i}}$ must be constant because $A_{\alpha_i}$ is connected. And $f(A_{\alpha_i}) = f(A_{\alpha_{i+1}})$ because $A_i \cap A_{i+1} \neq \varnothing$. This leads to a contradiction because $f(A_{\alpha_0}) = 0 \neq 1 = f(A_{\alpha_n})$. □

**Proposition 5.12.** *$\prod_{\alpha \in S} X_\alpha$ (with product topology) is connected if and only if $X_\alpha$ is connected for any $\alpha \in S$.*



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*Proof of proposition 5.12.* We focus on the case $S$ is finite. The general case can be proved similarly. ( Exercise: try it.) By induction, we can reduce to the case that $S = \{1, 2\}$. Let $X = X_1 \times X_2$.

Suppose $X$ is connected. Because the projection map $p_\alpha : X \to X_\alpha$ is surjective and continuous. $X_\alpha$ is also connected by lemma 5.9.

Suppose $X_1$ and $X_2$ are both connected. We consider the following collection of subspaces of $X$

$$\{\{a\} \times X_2\}_{a \in X_1} \bigcup \{X_1 \times \{b\}\}_{b \in X_2}$$

Then each subspace in this collection is homeomorphic to $X_1$ or $X_2$ so is connected. Note that $\{a\} \times X_2 \cap X_1 \times \{b\} \neq \varnothing$. So we apply lemma 5.11 to show that $X$ is connected. □

**Exercise A.** *(1) Let $A$ be a proper subset of $X$ (a subset is proper means that its complement is nonempty) and let $B$ be a proper subspace of $Y$. Suppose $X, Y$ are both connected. Show that $(X \times Y) \setminus (A \times B)$ is connected.*

*(2) Suppose $\mathbb{R} \cong X \times Y$. Show that either $X$ or $Y$ is a single point.*

**Exercise B.** *Without using path-connectedness, show that the following spaces are connected.*

*(1) $S^n$ for any $n \geq 1$.*

*(2) $T^n$ for any $n \geq 1$.*

*(3) $\mathbb{R}^n \setminus A$, where $n \geq 2$ and $A$ is a finite set.*

Using connectedness, we can define an equivalence relation $\sim$ on $X$ as follows: Two points $x, y \in X$ are equivalent if and only if there exists a connected subspace $A$ of $X$ that contains both $a$ and $b$. By lemma 5.11, this is indeed an equivalence relation. This equivalence relation induces a decomposition

$$X = \bigcup_{\alpha \in S} X_\alpha \tag{5.2}$$

into equivalence classes. Each $X_\alpha$ is called a **connected component** of $X$. (Recall that two members in a decomposition never intersect.)

**Example 5.13.** *Let $X = \{(x, y) \in \mathbb{R}^2 \mid x \neq 0, y \neq 0\}$. Then $X$ has four connected components (corresponds to the four quadrants): $(-\infty, 0) \times (-\infty, 0)$, $(-\infty, 0) \times (0, \infty)$, $(0, \infty) \times (-\infty, 0)$, $(0, \infty) \times (0, \infty)$.*

**Proposition 5.14.** *Let $\{X_\alpha\}_{\alpha \in S}$ be the connected components of a space $X$. Then we have the following result:*

*(1) Let $A$ be a connected subspace of $X$. Then $A \subset X_\alpha$ for some $\alpha \in S$.*

*(2) Each $X_\alpha$ is connected.*

*(3) Each $X_\alpha$ is closed in $X$.*

*(4) $X$ is connected if and only if it has only one connected component.*

*Proof.* (1) This follows from the our definition of $\sim$.

(2) Take $a \in X_\alpha$. Then for any $b \in X_\alpha$, there is a connected subspace $A_b \subset X$ that contains both $a$ and $b$. Then we have $A_b \subset X_\alpha$. So $X_\alpha = \bigcup_{b \in X_\alpha} A_b$. Then we apply lemma 5.11 to show that $X_\alpha$ is connected.

(3) Since $X_\alpha$ is connected, $\overline{X}_\alpha$ is connected too. By (1), $\overline{X}_\alpha \subset X_\beta$ for some $\beta$. So $X_\alpha \subset X_\beta$. That implies $\alpha = \beta$. So $\overline{X}_\alpha \subset X_\alpha$. That implies $X_\alpha$ is closed.

(4) This follows from (2). □

*Remark* 5.15. Suppose $X$ only has finitely many connected components. Then by lemma 5.3, the decomposition eq. (5.2) gives a homeomorphism

$$X \cong \bigsqcup_{\alpha \in S} X_\alpha$$



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from $X$ to the disjoint union of its connected components. In particular, each connected component $X_\alpha$ is both open and closed. As we can see from the example below. This is not true in general if $X$ has infinitely many connected components.

**Example 5.16.** *Let $X = \mathbb{Q}$. Then the only nonempty, connected subspace of $X$ is the singleton. So each point is a connected component by itself. From this, we see that* ***a connected component is not necessarily open****. And a space $X$ is not necessarily homeomorphic to the disjoint union of all its connected components. This is a bad situation, and we will rule in out by introducing a condition called "locally connectedness".*

## 5.3 Path-connectedness

Now let's try to use $[0, 1]$ as the "testing object". This time we put it on the domain.

**Definition 5.17.** Let $X$ be a topological space. Let $x, y$ be two points in $X$. We make the following definitions:

- A **path** in $X$ from $x$ to $y$ is a continuous map $\gamma : [0, 1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$.
- $\gamma(0) \in X$ is called the **start point** of $\gamma$.
- $\gamma(1) \in X$ is called the **final point** of $\gamma$.
- Both $\gamma(0)$ and $\gamma(1)$ are called the **end points** of $\gamma$.
- We say the path $\gamma$ is **closed** if $\gamma(0) = \gamma(1)$. (The word "closed" here has nothing to do a closed set.)
- A closed path with $\gamma(0) = \gamma(1) = x$ is also called a **loop based at $x$.**

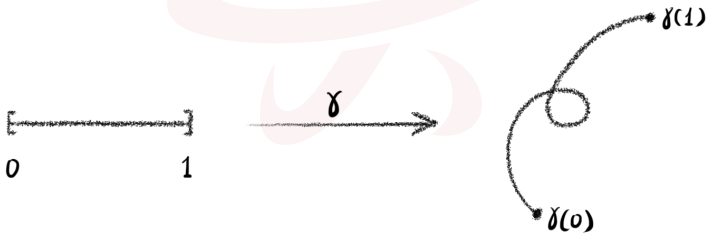


Figure 5.1

**Definition 5.18.** A space $X$ is called **path-connected** if given any two points $x, y \in X$, there exists a path $\gamma$ from $x$ to $y$.

*Remark* 5.19. A pair of points in $X$ can be equivalently treated as a continuous map from $\{0, 1\} = \partial I$ to $X$. So $X$ is path-connected if and only if any continuous map $\partial I \to X$ can be extended to a continuous map $I \to X$. This equivalent definition turns out to be an extremely useful. For example, it has a natural generalization to higher dimensions: For $n \geq 0$, we say a space $X$ is **$n$-connected** if for any $1 \leq m \leq n + 1$ and any continuous map $\gamma' : \partial I^m \to X$, there exists a continuous map $\gamma : I^m \to X$ such that $\gamma|_{\partial I^m} = \gamma'$. 0-connected just means path-connected. And 1-connected is also called "simply-connected". In algebraic topology, we will be able to prove that $S^n$ is $(n-1)$-connected but not $n$-connected. So this is a topological property that can distinguish spheres of different dimensions. (Exercise: can you guess a continuous map $\partial I^{n+1} \to S^n$ that can not be extended to $I^{n+1}$? Hint: we have seen this map in the first lecture.)

**Lemma 5.20.** *A path-connected space is always connected.*



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*Proof.* Suppose there is a nonconstant continuous function $f : X \to \{0,1\}$. Take any point $x \in f^{-1}(0)$ and $y \in f^{-1}(1)$. Since $X$ is path-connected, there is a path $\gamma : [0,1] \to X$ from $x$ to $y$. Then $f \circ \gamma : [0,1] \to \{0,1\}$ is a nonconstant, continuous function. This is impossible because $[0,1]$ is connected. □

In an exercise, we will see that for open subsets of $\mathbb{R}^n$, being connected is the same as being path-connected.

**Example 5.21.** *As we have seen in example 5.10, the topolgist's sine curve*

$$X = \{(0,y) \mid y \in [-1,1]\} \cup \{(x, \sin\frac{1}{x}) \mid x \in (0,1]\} \tag{5.3}$$

*is connected. Now we show that it is not path-connected. Suppose there were a path $\gamma : [0,1] \to X$ with $\gamma(0) = (0,0)$ and $\gamma(1) = (\frac{1}{\pi}, 0)$. We use $\gamma_1 : [0,1] \to \mathbb{R}$ and $\gamma_2 : [0,1] \to \mathbb{R}$ to denote the first and second component of $\gamma$. Let $A = \gamma_1^{-1}(0)$ and let $t_0 = \sup A$. Because $A$ is closed, we have $t_0 \in A$. So $\gamma_1(t_0) = 0$. Because $\gamma_2$ is continuous, there exists $t_1 > 0$ such that $|\gamma_2(t) - \gamma_2(t_0)| < \frac{1}{2}$ for any $t \in (t_0, t_1)$. So we have*

$$|\gamma_2(t) - \gamma_2(t')| < 1, \quad \forall t, t' \in (t_0, t_1). \tag{5.4}$$

*Note that $\gamma_1(t_0) = 0 < \gamma_1(t_1)$. We take $n \gg 0$ such that $\frac{1}{(n+\frac{1}{2})\pi}, \frac{1}{(n-\frac{1}{2})\pi} \in (\gamma_1(t_0), \gamma_1(t_1))$. Then by the mean value theorem, there exists $t, t' \in (t_0, t_1)$ such that $\gamma_1(t) = \frac{1}{(n+\frac{1}{2})\pi}$ and $\gamma_1(t') = \frac{1}{(n-\frac{1}{2})\pi}$. However, this implies that $\gamma_2(t) = 1$ and $\gamma_2(t') = -1$. This leads to a contradiction with (eq. (5.4)).*

**Exercise C.** *In this exercise, we consider the following "generalized topologist's sine curve"*

$$X = \{(0,y) \mid y \in \mathbb{R}\} \cup \{(x, f(\frac{1}{x})) \mid x \in (0,1]\}$$

*where $f : \mathbb{R} \to \mathbb{R}$ is a continuous function. We will see that for many choices of $f$, $X$ is connected but not path-connected.*

*(1) Show that $X$ is a closed subset of $\mathbb{R}^2$.*

*(2) Show that $X$ is **not** connected if and only if $\lim_{t\to+\infty} f(t) = \pm\infty$.*

*(3) Show that $X$ is path-connected if and only if $\lim_{t\to+\infty} f(t) = c$ for some finite $c$.*

We summarize some basic properties of path-connected spaces in the following lemma.

**Lemma 5.22.** *(1) Let $f : X \to Y$ be continuous. Suppose $A \subset X$ is path-connected. Then $f(A)$ is path-connected.*

*(2) The product topology on $\prod_{\alpha\in S} X_\alpha$ is path-connected if and only if $X_\alpha$ is path-connected for any $\alpha \in S$.*

*Proof.* (1) Take any two points $f(x), f(x') \in f(A)$. Since $A$ is connected, there exists a path $\gamma : [0,1] \to X$ from $x$ to $x'$. Then $f \circ \gamma : [0,1] \to Y$ is a path in $f(A)$ from $f(x)$ to $f(x')$.

(2) Let $X = \prod_{\alpha\in S} X_\alpha$. The projection map $p_\alpha : X \to X_\alpha$ is continuous and surjective. Suppose $X$ is path-connected, then by (1), $X_\alpha$ is also path-connected.

Suppose $X_\alpha$ is path-connected for any $\alpha$. Take two points $(x_\alpha)$ and $(y_\alpha)$ in $X$. Then there is a path $\gamma_\alpha : [0,1] \to X_\alpha$ from $x_\alpha$ to $y_\alpha$. Consider the product $\prod_{\alpha\in S} \gamma_\alpha : [0,1] \to X$. This is a path from $(x_\alpha)$ to $(y_\alpha)$.

□

Next, we introduce an equivalence relation $\sim_p$ on a space $X$: We set $x \sim_p y$ if there is a path $\gamma : [0,1] \to X$ from $x$ to $y$. To see $\sim_p$ is an equivalence relation, we give the following definition.



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**Definition 5.23.** We introduce two operations on paths.

- Given a path $\gamma : [0,1] \to X$, we define the **inverse path** $\bar{\gamma} : [0,1] \to X$ by $\bar{\gamma}(t) = \gamma(1-t)$.
- Given two paths $\gamma_1, \gamma_2 : [0,1] \to X$ that satisfies with $\gamma_1(0) = \gamma_2(1)$. We define a new path $\gamma_1 \cdot \gamma_2 : [0,1] \to X$ by

$$(\gamma_1 \cdot \gamma_2)(t) := \begin{cases} \gamma_1(2t) & \text{if } t \in [0, \frac{1}{2}] \\ \gamma_2(2t-1) & \text{if } t \in [\frac{1}{2}, 1] \end{cases}$$

  This new path $\gamma_1 \cdot \gamma_2$ called the **composition** of $\gamma_1$ and $\gamma_2$.

Figure 5.2

**Lemma 5.24.** *$\sim_p$ is an equivalence relation.*

*Proof.* We need to check three axioms:

- $x \sim_p x$: The constant path with value $x$ goes from $x$ to itself.
- $x \sim_p y \implies y \sim_p x$: Let $\gamma$ be a path from $x$ to $y$. Then $\bar{\gamma}$ is a path from $y$ to $x$.
- $x \sim_p y, y \sim_p z \implies x \sim_p z$: Let $\gamma$ be path from $x$ to $y$ and let $\gamma'$ be a path from $y$ to $z$. Then $\gamma \cdot \gamma'$ is a path from $x$ to $z$.

□

The equivalence relation $\sim_p$ induces a decomposition $X = \bigcup_{\alpha \in S} X_\alpha$ into equivalent classes. Each equivalent class $X_\alpha$ is called a **path-component** of $X$.

**Lemma 5.25.** *(1) Any path-component is path-connected. (So it is always contains in a connected component.)*
*(2) Any path-connected subset $A$ of $X$ is contained in some path-component of $X$.*
*(3) $X$ is path-connected if and only if it has a single path component.*

*Proof.* The proof is very straightforward and skipped. □

**Example 5.26.** *Let $X$ be the topologist's sine curve (See equation (eq. (5.3))). Then $X$ has a single connected component because it is connected. However, $X$ has two path-components $X_1 = \{(0, y) \mid y \in [-1, 1]\}$ and $X_2 = \{(x, \sin \frac{1}{x}) \mid x \in (0, 1]\}$. This also shows that a path-component can be neither open nor closed. (Recall that a connected component is always closed but not always open.)*

We end this section by stating two fundamental results in topology: Jordan's curve theorem. Let's introduce some more terminology:

- A simple closed curve in $X$ is a subspace of $X$ that is homeomorphic to $S^1$.



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- A simple closed curve in $\mathbb{R}^2$ is called a Jordan curve.

Note that a Jordan curve need not be smooth and can be extremely complicated. For example, we have the following Kock curve.

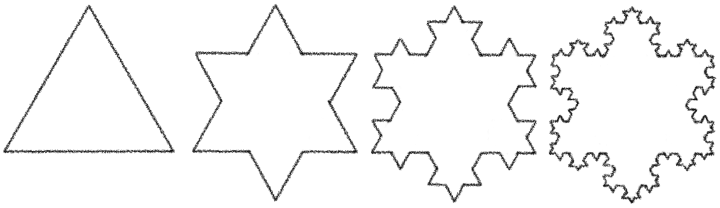

Figure 5.3: Kock curve is the limit after infinitly many iterations (the picture shows the fist four). It has an infinite length and "Hausdorff dimension" $\ln 4/\ln 3$.

**Theorem 5.27** (Jordan's curve theorem)**.** *Let $C \subset \mathbb{R}^2$ be a Jordan curve. Then $\mathbb{R}^2 \setminus C$ has two path-components $X_1$ and $X_2$. One of them is bounded and the other one is unbounded.*

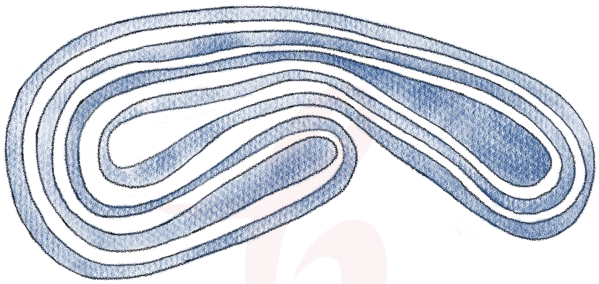

Figure 5.4: Thurston's simple closed curve, it was painted on the wall of the UC Berkeley Math Department.

**Theorem 5.28** (Jordan–Schoenflies theorem)**.** *Let $C \subset \mathbb{R}^2$ be a Jordan curve. Then there exists a homeomorphism $h : \mathbb{R}^2 \to \mathbb{R}^2$ such that $h(C)$ equals the standard circle*

$$S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}.$$

*So in particular, the bounded and the unbounded components of $X \setminus C$ are homeomorphic to $\mathring{D}^2$ and $\mathbb{R}^2 \setminus D^2$ respectively.*

At a first glance, one can easily believe the conclusion of these theorems because they fits so nicely with our intuition. However, a Jordan curve can be extremely complicated (for example, the Koch curve). So it's not so obvious why it must divide the plane into two components. It's even less clear (even unbelievable) why there exits a homeomorphism from $\mathbb{R}^2$ to itself that sends it to the standard $S^1$.

The proof of these theorems are very delicate (but elementary). So we are not going to give them. See Munkres for a proof of the Jordan's curve theorem.

*Remark* 5.29. You may wonder what happens in higher dimensions. In algebraic topology, we will learn a theorem called the "Alexander duality theorem", which implies the following generalization of theorem 5.27:

- Let $S$ be subspace of $\mathbb{R}^n$ homeomorphic to $S^{n-1}$. Then $\mathbb{R}^n \setminus S$ has two path-components, one is bounded and the other one is not.

However, the naive generalization of Jordan-Schoenflies theorem to higher dimensions is not correct. Actually, Alexander (the same Alexander as in the Alexander duality theorem, J.W.Alexander) found a famous example called the "Alexander horned sphere":



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- There exists an embedding $f : S^2 \to \mathbb{R}^3$ such that the unbounded component of $\mathbb{R}^3 \setminus f(S^2)$ is **not** homeomorphic to $\mathbb{R}^3 \setminus D^3$ (actually, it is not even simply-connected). So there exists **no** homeomorphism $h : \mathbb{R}^3 \to \mathbb{R}^3$ that takes $f(S^2)$ to the standard $S^2$.

Figure 5.5

Figure 5.6

One may ask whether we can put suitable condition on the embedding $f : S^{n-1} \to \mathbb{R}^n$ so that $f(S^{n-1})$ can be sent to the standard sphere by a homeomorphism on $\mathbb{R}^n$. This is the famous Schoenflies problem. There are a lot of exciting researches about this problem. For example, if one requires $f$ to be smooth, then the conclusion is known to hold for all $n \neq 4$.



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For $n = 4$, this is the famous 4-dimensional Schoenflies conjecture (still open). Dimension 4 is a really "exotic" dimension.

**Exercise D.** *In this exercise, we introduce the local version of connectedness and path-contentedness.*

- *A space $X$ is **locally connected** at a point $x$ if $x$ has a neighborhood basis $\{U_\alpha\}_{\alpha\in S}$ such that $U_\alpha$ is connected for any $\alpha$. This is equivalent to saying that for any neighborhood $V$ of $x$, there exists a connected neighborhood $U$ such that $x \in U \subset V$. We say $X$ is locally connected if it is locally connected at every point.*
- *A space $X$ is **locally path-connected** at a point $x$ if $x$ has a neighborhood basis $\{U_\alpha\}_{\alpha\in S}$ such that $U_\alpha$ is path-connected for any $\alpha$. This is equivalent to saying that for any neighborhood $V$ of $x$, there exists a path-connected neighborhood $U$ such that $x \in U \subset V$. We say $X$ is locally path-connected if it is locally path-connected at every point. Clearly, every locally path-connected space is locally connected. (But the converse is not true.)*
- *While compactness $\Longrightarrow$ locally compactness, we will see an example to show locally connectedness and connectedness do not imply each other, and locally path-connectedness and path-connectedness do not imply each other as well.*

*(1) Show that any open subset of $\mathbb{R}^n$ is locally path-connected.*
*(2) Let $M$ be a manifold. Show that $M$ is locally path-connected.*
*(3) Consider the following "extended topologist's sine curve":*

$$X = (\{0\} \times [-2, 2]) \cup \{(x, \sin\frac{1}{x}) \mid x \in (0, \frac{1}{\pi}]\} \cup (\{\frac{1}{\pi}\} \times [0, -2]) \cup ([0, \frac{1}{\pi}] \times \{-2\}). \tag{5.5}$$

*Draw a picture of $X$ and show that $X$ is path-connected but not locally connected.*
*(4) Consider the following "topologist's comb"*

$$X = ([0, +\infty) \times \{0\}) \cup \{(x, y) \mid x \in \mathbb{Q} \cap [0, +\infty) \text{ and } y \in [0, 1]\}. \tag{5.6}$$

*Draw a picture of $X$ and show that $X$ is path-connected but not locally connected.*

**Exercise E.** *(1) Let $X$ be a locally connected space. Show that every connected component of $X$ is both open and closed.*
*(2) Let $X$ be a locally connected space. Show that $X$ is homeomorphic to the disjoint union of its connected components.*
*(3) Let $X$ be a locally path-connected space. Show that path-components and connected components are the same.*
*(4) Let $X$ be a locally path-connected space (for example, $X$ is an open subset of $\mathbb{R}^n$). Show that $X$ is path-connected if and only if $X$ is connected.*

**Exercise F.** *(1) Let $X$ be the extended topologist's sine curve defined in (eq. (5.5)). Is $\mathbb{R}^2 \setminus X$ connected? Is $\mathbb{R}^2 \setminus X$ path-connected? Prove your answer.*
*(2) Let $A = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \notin \mathbb{Q}\}$ and let $X = \mathbb{R}^3 \setminus A$. Describe all the path-components of $X$. You need to prove that they are indeed path-components.*



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# Chapter 6

# Quotient spaces

## 6.1 Definition of quotient spaces

Given a space $X$, we can identify multiple points in $X$ as a single point and form a new space $X'$. For example, by identifying the end points $0, 1$ of $[0, 1]$ as a single point, we can form a space homeomorphic to $S^1$. This operation is called "taking quotient spaces" and it is an extremely useful in topology.

**Definition 6.1.** Let $X$ be a topological space and let $Y$ be a set. Given a surjective map $\pi : X \to Y$. We define the **quotient topology** on $Y$ (induced by $\pi$) as follows: A subset $U$ of $Y$ is open if and only if $\pi^{-1}(U)$ is open in $X$.

The most important (and essentially the unique) case of the quotient topology is when $Y$ is the set of equivalence classes in $X$. Let's first review some set-theoretic definitions.

**Definition 6.2.** (1) A **binary relation** on a set $X$ is a subset $R$ of $X \times X$. We use $x \sim y$ to denote $(x, y) \in R$.

(2) We say the binary relation is an **equivalence relation** if

- $x \sim x$ for any $x \in X$;
- $x \sim y \implies y \sim x$;
- $x \sim y, y \sim z \implies x \sim z$.

(3) Any equivalence relation $\sim$ on $X$ induces a partition $\{X_\alpha\}$ of $X$, where $x, y$ belongs to the same $X_\alpha$ if and only if $x \sim y$. Each $X_\alpha$ is called an **equivalence class**. We use $X/\sim$ to denote the set of equivalence classes. So we have the decomposition

$$X = \bigcup_{\alpha \in X/\sim} X_\alpha.$$

(4) Given a binary relation $R$, **the equivalence relation generated by** $R$ is defined as follows: Two points $x, y$ are equivalent to each other if and only if there exists a finite sequence $x_0, \cdots, x_n$ in $X$ such that:

- $x = x_0$, $y = x_n$;
- For any $1 \le i \le n$, either $(x_{i-1}, x_i)$ or $(x_i, x_{i-1})$ belongs to $R$.

**Definition 6.3.** Let $\sim$ be an equivalence relation on a space $X$. The quotient space $X/\sim$ is the set $X/\sim$ equipped with the quotient topology induced by the map

$$\pi : X \to X/\sim \text{ defined by } \pi(x) := [x]. \tag{6.1}$$



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From now on, we will always use the quotient topology on $X/\sim$ (unless otherwise noted). Unlike previous operations we have seen (e.g., taking subspaces, disjoint union, finite product), by taking the quotient space of a subspace of $\mathbb{R}^n$, we get a space that is not necessarily a subspace of $\mathbb{R}^n$. This extra flexibility allows us to construct many extremely interesting examples of topological spaces. And starting from now, the flavor of our class will be less analytical but more topological (Because we start to deal with more and more examples which are not naturally subspaces of $\mathbb{R}^n$.)

**Example 6.4.** *Let $X = [0,1]$ and let $\sim$ be generated by $0 \sim 1$. Then the quotient space $X/\sim$ is homeomorphic to $S^1$. A specific homeomorphism $f : X/\sim \to S^1$ is given by $f([x]) := e^{2\pi x i}$.*

**Example 6.5.** *We can take the product of example 6.4 with $[0,1]$: Let $X = [0,1] \times [0,1]$ and let $\sim$ be generated by*

$$(0,t) \sim (1,t), \quad \forall t \in [0,1].$$

*The resulting space $A := X/\sim$ is obtained by identifying the edges $\{0\} \times [0,1]$ and $\{1\} \times [0,1]$ and it is homeomorphic to the annulus $S^1 \times [0,1]$.*

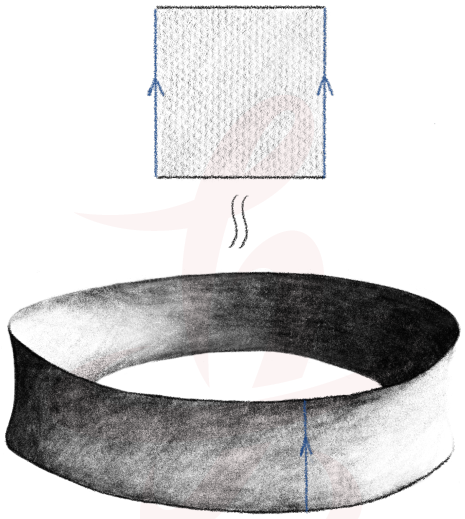

Figure 6.1

**Example 6.6.** *Let $X = [0,1] \times [0,1]$ and let $\sim$ be generated by*

$$(0,t) \sim (1,1-t), \quad \forall t \in [0,1].$$

*So we are gluing the edges $\{0\} \times [0,1]$ and $\{1\} \times [0,1]$ together after rotating one of them by $180°$. The resulting space $M := X/\sim$ is called the **Mobius band** (or the Mobius strip). It can be embedded as a subspace of $\mathbb{R}^3$.*



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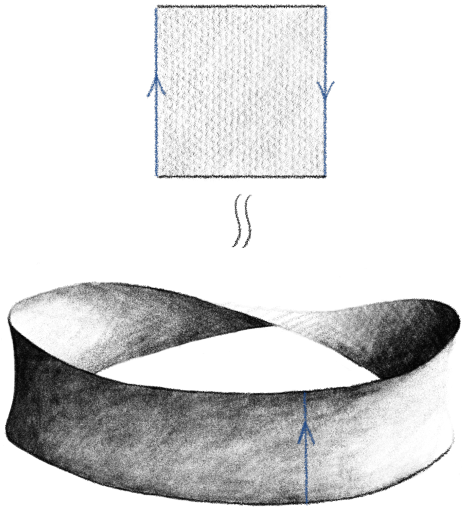

Figure 6.2

Both the Mobius band, the annulus and $D^2$ are examples of surfaces with boundary.

**Definition 6.7.** Let $X$ be a Hausdorff space. We say $X$ is a **$n$-dimensional (topological) manifold with boundary** ($n \in \mathbb{N}$) if for any $x \in X$, at least one of the following two conditions hold:

(i) $x$ has a neighborhood $U$ such that $U \cong \mathbb{R}^n$, or

(ii) $x$ has a neighborhood $U$ such that $U \cong [0, +\infty) \times \mathbb{R}^{n-1}$.

We call $x$ is **an interior point** if (i) holds for $x$, otherwise we call $x$ a **boundary point**. We use $\mathring{X}$ to denote the subspace of $X$ that consists of all interior points and we use $\partial X$ to denote the subspace that consists of all boundary points. We call $\mathring{X}$ the **interior** of the manifold $X$ and call $\partial X$ the **boundary of the manifold** $X$. It's straightforward to see that $\mathring{X}$

A compact manifold without boundary is called a **closed manifold**. A 2-dimensional manifold (with/without boundary) is a called a **surface** (with/without boundary).

*Remark* 6.8. Using invariance of domain, one can show that $\mathbb{R}^n \ncong [0, +\infty) \times \mathbb{R}^{n-1}$. So Condition (ii) in definition 6.7 is not redundant. However, an interior point can also satisfy (ii). To address this issue, we can replay (ii) with the following condition

- (ii') $x$ has a neighborhood $U$ such that there exists a homeomorphism $U \xrightarrow{\cong} [0, +\infty) \times \mathbb{R}^{n-1}$ that sends $x$ to $\vec{0}$.

Then $x \in \partial X$ if and only if $x$ satisfies (ii'). Using this new condition, it's not hard to prove that $\partial X$ is always a $(n-1)$-dimensional manifold without boundary. Note that it's straightforward to show that $\mathring{X}$ is always a $n$-dimensional manifold without boundary (but it's not compact unless $\partial X = \varnothing$.)

*Remark* 6.9. Given a subset $A$ of a topological space $Y$, we have defined the interior $\mathring{A}$ and the boundary $\partial A$. This shouldn't be confused with the interior/boundary of a manifold because they don't always agree with each other (although they have the same name and notation) even if $A$ is a manifold with boundary.

We emphasis that a Mobius band $M$ is not homeomorphic to an annulus $A = [0, 1] \times S^1$. To see this, we note that $\partial M \cong S^1$ (in Example *example* 6.6, $\partial M = \pi(\{0, 1\} \times [0, 1])$) but $\partial A = \{0, 1\} \times S^1 \cong S^1 \bigsqcup S^1$. We will see in exercise A that $M$ is also not homeomorphic to $D^2$.



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**Example 6.10.** *Let $X = [0,1] \times [0,1]$. and let $\sim$ be generated by*

$$(0,t) \sim (1,t) \text{ and } (t,0) \sim (t,1), \quad \forall t \in [0,1]. \tag{6.2}$$

*To construct $X/\sim$, we first identify the edges $\{0\} \times [0,1]$ and $\{1\} \times [0,1]$. As we have seen in example 6.5, this gives the annulus $X' = S^1 \times [0,1]$. The other two edges $[0,1] \times \{0\}$ and $[0,1] \times \{1\}$ of $X$ becomes the circles $S^1 \times \{0\}$ and $S^1 \times \{1\}$ in $X'$. To get $X/\sim$, we identify these two circles, this will give us the torus. So we conclude that $X/\sim$ is homeomorphic to $T^2$. We will give a rigorous proof of this claim soon.*

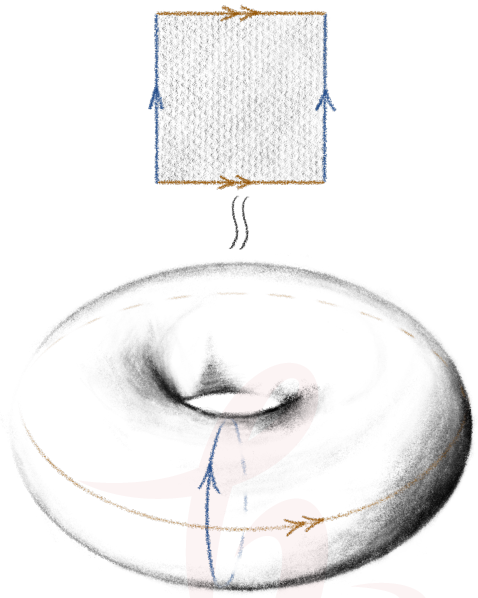

Figure 6.3

**Example 6.11.** *Let $X = [0,1] \times [0,1]$. and let $\sim$ be generated by*

$$(0,t) \sim (1,t) \text{ and } (t,0) \sim (1-t,1), \quad \forall t \in [0,1]. \tag{6.3}$$

*To construct $X/\sim$, we also first identify the edges $\{0\} \times [0,1]$ and $\{1\} \times [0,1]$ to get the annulus $X' = S^1 \times [0,1]$. The other two edges $[0,1] \times \{0\}$ and $[0,1] \times \{1\}$ of $X$ becomes the circles $S^1 \times \{0\}$ and $S^1 \times \{1\}$ in $X'$. To get $X/\sim$, we glue these two circles together after rotating one of them by $180°$. The resulting space is called the **Klein bottle**.*



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Figure 6.4

In example 6.10 and example 6.11, we have taken the quotient space in several steps. This implicitly rely on the following useful lemma:

**Lemma 6.12.** *Let $\sim$ and $\sim'$ be two equivalence relations on $X$. Suppose $x \sim' y \implies x \sim y$. Then $\sim$ induces an equivalence relation on $X/\sim'$, which we still denote by $\sim$. Then we have*

$$(X/\sim')/\sim \cong X/\sim .$$

*Proof.* The underlying set of both $(X/\sim')/\sim$ and $X/\sim$ are equivalence classes in $X$ under $\sim$. In both topologies, a set $U$ is open if and only if its preimage in $X$ is open. □

**Exercise A.** *Let $M$ be the Mobius band and let $\pi : [0,1]\times[0,1] \to M$ be the quotient map in example 6.6. Consider the subspace $N := \pi([0,1]\times\{\frac{1}{2}\})$.*

*(1) Show that $N \cong S^1$.*

*(2) Show that $M \setminus N \cong S^1 \times [0,1)$. (Geometrically, this means that if we cut a Mobius band open along a circle in the middle, then we get a half-open-half-closed annulus. )*

*(3) Let $N'$ be a subspace of $D^2 \setminus \partial D^2$. Suppose $N' \cong S^1$. Show that $D^2 \setminus N'$ is not connected. From this, we see that the Mobius band and the disk are not homeomorphic. (Hint: use the Jordan curve theorem.)*

**Example 6.13.** *Consider the equivalence relation on $X = S^n$ generated by*

$$x \sim -x, \quad \forall x \in S^n.$$

*Then the quotient space $X/\sim$ is called the $n$-dimensional **real projective space**, denoted by $\mathbb{RP}^n$. So a point in $\mathbb{RP}^n$ is a pair of antipodal points in $S^n$. Note that for any straight line in $\mathbb{R}^{n+1}$ that passes the origin, it intersects $S^n$ at a pair of antipodal points. So points in $\mathbb{RP}^n$ one-one correspond to straight lines in $\mathbb{R}^{n+1}$ that passes the origin.*



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小红书号:3189205503 qq

**Exercise B.** *(1) Show that $\mathbb{RP}^n$ is a path-connected and closed manifold. (Don't forget to check the Hausdorff condition.)*

*(2) Consider the equivalence relation $\sim$ on $D^n$ generated by*

$$x \sim -x \text{ for any } x \in \partial D^n. \tag{6.4}$$

*Show that $X/\sim \cong \mathbb{RP}^n$. In other words, we can also obtain $\mathbb{RP}^n$ from $D^n$ by identifying antipodal points on $\partial D^n$. (Hint: construct a continuous bijection between them and show that it is a homeomorphism.)*

*(3) Consider the map $f : S^2 \to \mathbb{R}^4$ defined by*

$$f(x, y, z) = (x^2 - y^2, xy, xz, yz).$$

*Show that $f(S^2) \cong \mathbb{RP}^2$. This gives an embedding of $\mathbb{RP}^2$ into $\mathbb{R}^4$. (Note that $\mathbb{RP}^2$ can **not** be embedded into $\mathbb{R}^3$. But we can't prove it for now.)*

**Example 6.14.** *Given a subspace $Y$ of $X$, we let $\sim_Y$ be generated by*

$$x \sim_Y x', \quad \forall x, x' \in Y.$$

*Then $X/\sim_Y$ is the space obtained by "collapsing" all points in $Y$ into a single point. We also denote $X/\sim_Y$ by $X/Y$. Then example 6.4 just states that $[0, 1]/\{0, 1\} \cong S^1$.*

**Exercise C.** *Consider the annulus $A = S^1 \times [0, 1]$. We have the following equivalence relations on $A$:*

- *$\sim_1$: generated by $(x, 0) \sim_1 (x', 0)$ for any $x, x' \in S^1$;*
- *$\sim_2$: generated by $(x, 1) \sim_2 (-x, 1)$ for any $x \in S^1$;*
- *$\sim_3$: generated by $(x, 0) \sim_3 (x', 0)$ for any $x, x' \in S^1$ and $(x, 1) \sim_3 (-x, 1)$ for any $x \in S^1$.*

*Prove the following conclusions:*

*(1) $A/\sim_1 \cong D^2$.*

*(2) $A/\sim_2 \cong M$ ($M$ denotes the the Mobius band).*

*(3) $A/\sim_3 \cong \mathbb{RP}^2$.*

*(4) $M/\partial M \cong \mathbb{RP}^2$ (See example 6.6.)*

*(Hint: You solution can be some pictures. lemma 6.12 should be useful here.)*

**Exercise D.** *Consider the equivalence relation $\sim_{\mathbb{Z}}$ on $\mathbb{R}$ generated by:*

$$x \sim_{\mathbb{Z}} y \text{ for any } x, y \in \mathbb{Z}.$$

*Let $X = \mathbb{R}/\sim_{\mathbb{Z}}$. We also consider the Hawaiian ear ring:*

$$X' := \bigcup_{n \geq 1} \{(x, y) \in \mathbb{R}^2 \mid (x - \frac{1}{n})^2 + y^2 = \frac{1}{n^2}\}.$$

*(we use the subspace topology on $X'$.) Draw a picture of $X$ and a picture of $X'$. Then **rigorously** prove that they are not homeomorphic.*



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## 6.2 Quotient maps

**Definition 6.15.** Let $p : X \to Y$ be a map between two topological spaces. We say $p$ is a **quotient map** if the following conditions are satisfied:

1. $p$ is surjective and continuous.
2. For a subset $V$ of $Y$, $p^{-1}(V)$ is open $\implies$ $V$ is open.

Note that we can replace (2) by the following condition:
(2') For a subset $V$ of $Y$, $p^{-1}(V)$ is closed $\implies$ $V$ is closed.

**Lemma 6.16.** *The following properties of quotient maps holds:*

1. *If a surjective, continuous map is open or is closed, then it is a quotient map.*
2. *A map is an injective quotient map if and only if it is a homeomorphism.*
3. *The composition of two quotient maps is a quotient map.*
4. *$p : X \to Y$ is a quotient map if and only if the topology on $Y$ is exactly the quotient topology induced by $f$. (In particular, the map $\pi : X \to X/\sim$ is automatically a quotient map.)*

*Proof.* The proof is straightforward. To prove (1), we note that $p(p^{-1}(V)) = V$ because $p$ is surjective. □

**Lemma 6.17.** *Let $p : X \to Y$ be a surjective, continuous map. Suppose $X$ is compact and $Y$ is Hausdorff, then $p$ is a quotient map.*

*Proof.* By corollary 4.21, $p$ is a closed map. □

The restriction of a quotient map to a subspace is usually not a quotient map. However, we have the following useful lemma:

**Lemma 6.18.** *Let $f : X \to Y$ be a quotient map and let $B$ be a subspace of $Y$ and let $A = f^{-1}(B)$. Consider the map*

$$f_A : A \to B \text{ defined by } f_A(x) = f(x).$$

*If $B$ is open or closed (equivalently, if $A$ is open or closed), then $f_A$ is also a quotient map.*

*Proof.* It's straightforward to see that $f_A$ is continuous and surjective. Suppose $B$ is open in $Y$. We take any subset $V$ of $B$ such that $f_A^{-1}(V)$ is open in $A$. Since $A$ is open in $X$, we see that $f_A^{-1}(V) = f^{-1}(V)$ is open in $X$. Since $f$ is a quotient map, we see that $V$ is open in $Y$, so $V$ is open in $B$.

The argument for closed $B$ is similar. □

**Proposition 6.19.** *Let $X, Y, Z$ be topological spaces. Let $p : X \to Y$ be a quotient map. And let $f : Y \to Z$ be a general map (not necessarily continuous). Then we have:*

1. *$f$ is continuous $\iff$ $f \circ p$ is continuous;*
2. *$f$ is a quotient map $\iff$ $f \circ p$ is a quotient map.*

*Proof.* (1) Suppose $f$ is continuous. Then $f \circ p$ is continuous because the composition of continuous maps is continuous.

Suppose $f \circ p$ is continuous. Take any open $V \subset Z$, then $p^{-1}(f^{-1}(V)) = (f \circ g)^{-1}(V)$ is open. So $f^{-1}(V)$ is open. So $f$ is continuous.

The proof of (2) is similar. □



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**Corollary 6.20.** *Let $\sim$ be an equivalence relation on $X$ and $f : X \to Z$ be a continuous map. Suppose we have $f(x) = f(x')$ for any $x \sim x'$. Then there is a unique continuous map $\widetilde{f} : X/\sim \to Z$ that fits into the following commutative diagram:*

$$\begin{array}{ccc} X & & \\ \downarrow{\scriptstyle \pi} & \searrow^{f} & \\ X/\sim & \xrightarrow{\widetilde{f}} & Z \end{array} \quad . \tag{6.5}$$

*Proof.* We define $\widetilde{f}([x]) = f(x)$. By our assumption, $\widetilde{f}$ is well-defined. Since $\widetilde{f} \circ \pi = f$ is continuous, by proposition 6.19, $\widetilde{f}$ also is continuous. □

**Corollary 6.21.** *Let $f : X \to Y$ be a quotient map. Define an equivalence relation $\sim_f$ on $X$ by $f(x) = f(x') \iff x \sim_f x'$. Then $f$ induces a homeomorphism $\widetilde{f} : X/\sim_f \to Y$.*

*Proof.* By Corollary (corollary 6.20), we have the continuous bijection $\widetilde{f} : X/\sim_f \to Y$. By proposition 6.19, $\widetilde{f}$ is a quotient map. Since $\widetilde{f}$ is injective, $\widetilde{f}$ is a homeomorphism. □

corollary 6.21 is very useful to determine the homeomorphism type of a quotient space.

**Example 6.22** (example 6.10 continued)**.** *Let $\sim$ be the equivalence relation (eq. (6.3)) on $X = [0,1] \times [0,1]$. We consider the map $f : X \to S^1 \times S^1$ defined by*

$$f(x_1, x_2) := (e^{2\pi x_1 i}, e^{2\pi x_2 i}).$$

*Then $f$ is surjective and $f(x) = f(x') \iff x \sim x'$. Since $D^n$ is compact and $T^2$ is Hausdorff, $f$ is a quotient map by lemma 6.17. By corollary 6.21, we get a homeomorphism $\widetilde{f} : X/\sim \to T^2$.*

**Example 6.23.** *Consider $D^n = \{x \in \mathbb{R}^n \mid |x| \leq 1\}$ and its boundary $\partial D^n = \{x \in \mathbb{R}^n \mid |x| = 1\}$. To determine the homeomorphism type of $D^n/\partial D^n$, we consider the map*

$$f : D^n \to S^n = \{x \in \mathbb{R}^{n+1} \mid |x| = 1\}$$

*defined by*

$$f(r \cdot x) = (\cos \pi r, \sin \pi r \cdot x) \text{ for any } r \in [0,1], |x| = 1.$$

*Then it's straightforward to check that $f(\partial D^n) = (-1, 0, \cdots, 0)$ and that $f$ induces a homeomorphism from $D^n \setminus \partial D^n$ to $S^n \setminus (-1, 0, \cdots, 0)$. By corollary 6.21 and lemma 6.17 again, we get a homeomorphism $\widetilde{f} : D^n/\partial D^n \to S^n$. Note that there is a homeomorphism from $D^n$ to $I^n$ that sends $\partial D^n$ to $\partial I^n$. So we also see that*

$$I^n/\partial I^n \cong D^n/\partial D^n \cong S^n.$$

**Exercise E** (From Munkres)**.** *Consider the following equivalence relations on $\mathbb{R}^2$:*

1. $(x, y) \sim_1 (x', y') \iff x^2 + y = x'^2 + y'$.
2. $(x, y) \sim_2 (x', y') \iff x^2 + y^2 = x'^2 + y'^2$.
3. $(x, y) \sim_3 (x', y') \iff x^2 - y^2 = x'^2 - y'^2$.

*Find the homeomorphism type of $\mathbb{R}^2/\sim_1$, $\mathbb{R}^2/\sim_2$ and $\mathbb{R}^2/\sim_3$.*

Next, we discuss the topological properties of quotient spaces. Recall that the image of a compact /connected /path-connected space under a continuous map is also compact /connected /path-connected, we see that that the quotient of a compact /connected /path-connected space is also compact /connected /path-connected. However, the quotient space of a Hausdorff space is not necessarily Hausdorff. As we can see from the example below:



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**Example 6.24.** *Let $X_1 = X_2 = \mathbb{R}$ and let $X = X_1 \bigsqcup X_2 = \{(i, x) \mid i \in \{1, 2\}, x \in \mathbb{R}\}$. Consider the equivalence relation $\sim$ generated by*

$$(1, x) \sim (2, x), \forall x \neq 0.$$

*Then $X/\sim$ is a non-Hausdorff space, called the "real axis with two origins".*

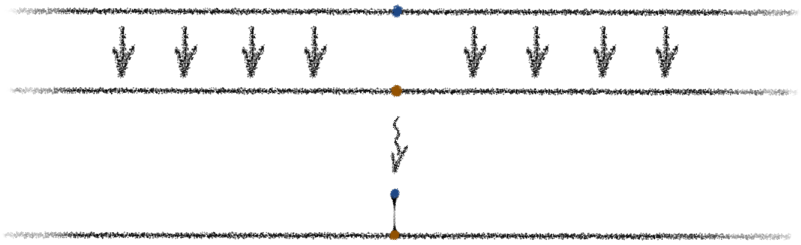

Figure 6.5

So when we take a quotient space, we have to be extra careful not to construct some pathological space. In general, it is a delicate question to determine when the quotient space is Hausdorff. We will return to this point later. For now, we just mention the following result:

**Lemma 6.25.** *Let $\sim$ be an equivalence relation on $X$. Then $X/\sim$ is Hausdorff if and only if for any $x, y \in X$ such that $x \not\sim y$, there exists a neighborhood $U_x$ and a neighborhood $U_y$ of $y$ that satisy the following conditions:*

- *$U_x \cap U_y = \varnothing$;*
- *Both $U_x$ and $U_y$ are unions of some equivalence classes. (Equivalently, this means that $x_1 \in U_x, x_2 \sim x_1 \implies x_2 \in U_x$, and $y_1 \in U_y, y_2 \sim y_1 \implies y_2 \in U_y$.)*

*Proof.* This follows from the definition of quotient topology. □

**Exercise F.** *Let $Y$ be a **closed** subspace of $X$. Then we have:*

1. *$X$ satisfies the $T_1$-axiom $\implies$ $X/Y$ satisfies the $T_1$-axiom;*
2. *$X$ satisfies the $T_4$-axiom $\implies$ $X/Y$ satisfies the $T_4$-axiom.*

*Because of these results, we usually prefer to take a quotient by a closed subspace.*

### 6.2.1 Attaching spaces

Now we use quotient space to introducing an important construction called "attaching spaces". Consider a space $A$ and its subspace $B$. We call $(A, B)$ a space pair. Given such a pair $(A, B)$ and a continuous map $f : B \to X$ to some space $X$, we consider the equivalence relation $\sim$ on the space $X \sqcup A$ generated by

$$f(x) \sim x \text{ for any } x \in B.$$

We denote the quotient space $(X \sqcup A)/\sim$ by $X \cup_f A$ (or $X \cup_B A$ if $f$ is clear from the context). We call $X \cup_f A$ the space obtained by "attaching $A$ to $X$ along its subspace $B$ via the attaching map $f$."



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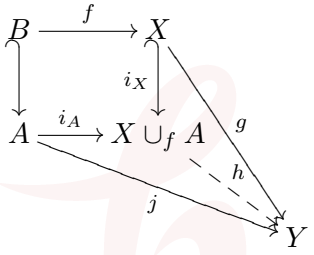
*Remark* 6.26. Although the notation looks symmetric. The role played by $X$ and $A$ in $X \bigcup_f A$ are essentially different. For example, we may consider the maps

$$i_X : X \hookrightarrow X \sqcup A \xrightarrow{\text{quotient map}} X \cup_f A.$$
$$i_A : A \hookrightarrow X \sqcup A \xrightarrow{\text{quotient map}} X \cup_f A.$$

Then $i_X$ is always an embedding so we may treat $X$ as a subspace of $X \cup_f A$. But $i_A$ is usually not an embedding, so $A$ is not a subspace of $X \cup_f A$. An exception is when the attaching map $f$ is also an embedding. In this case, we let $f^{-1} : f(B) \to B$ be the inverse of $f$. Then $X \cup_f A \cong A \cup_{f^{-1}} X$.

Attaching spaces is a special case of a more general construction called "push out". It satisfies the following universal property.

**Lemma 6.27.** *Let $(A, B)$ be a space pair and let $f : B \to X$ be an continuous map. Suppose there is a continuous map $g : X \to Y$ and a continuous map $j : A \to Y$ such that $j|_B = g \circ f$. Then there exits a unique map $h : X \cup_f A \to Y$ that satisfies $h \circ i_X = g$ and $h \circ i_A = j$. (We also write $h$ as $g \cup_f j$ to indicate that it is obtained by gluing together $g$ and $j$.)*

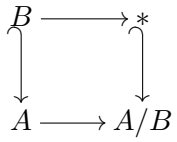



*Proof.* We just define

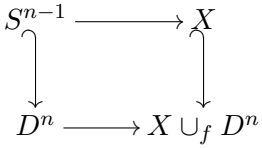
$$h(p) := \begin{cases} g(x) & \text{if } p = i_X(x) \\ j(a) & \text{if } p = i_A(a) \end{cases}$$

This is well-defined and continuous by corollary 6.20. □

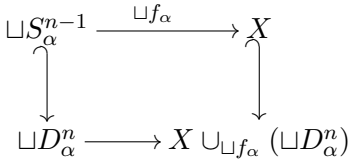
**Example 6.28.** *The space $* \cup_B A$ is exactly the quotient space $A/B$.*




**Example 6.29.** *When $(A, B) = (D^n, S^{n-1})$, we call $D^n$ a **$n$-dimensional cell** (or just an $n$-cell) and say the space $X \cup_f D^n$ is obtained from $X$ by **attaching a $n$-cell** via the attaching map $f : S^{n-1} \to X$.*




*Of course, we can generalize this a little by attaching a (finite or infinite) collection of $n$-cells $\sqcup_{\alpha \in S} D^n_\alpha$ **simultaneously** to $X$, via an attaching map $\sqcup_{\alpha \in S} f_\alpha : S^{n-1}_\alpha \to X$.*



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# Chapter 7

# Topological groups and their actions on topological spaces

## 7.1 Example of topological groups

**Definition 7.1.** A topological group is a topological space $G$ with a group structure such that the multiplication map $G \times G \to G$ defined by $(x, y) \mapsto x \cdot y$ and the inverse map $G \to G$ defined by $x \mapsto x^{-1}$ are both continuous.

Here are some examples of topological groups:

**Example 7.2.** *Given any group $G$, we can equip it with the discrete topology. Then resulting topological group is called a discrete group.*

**Example 7.3.** *$\mathbb{Z}^n$, $\mathbb{Q}^n$ and $\mathbb{R}^n$ are all topological groups under $+$.*

**Example 7.4.** *$\mathbb{C} \setminus \{0\}$ is a topological group under $\times$.*

**Example 7.5.** *$S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ is a topological group under $\times$.*

In the following examples, we let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

**Example 7.6.** *The space*

$$M_n(\mathbb{F}) := \{n \times n \text{ matrices with entries in } \mathbb{F}\}$$

*is a topological group under **matrix addition**. Here we identity $M_n(\mathbb{F})$ with $\mathbb{F}^{n^2}$ and use the standard topology on $\mathbb{F}^{n^2}$.*

In the following examples, we will consider various subspace of $M_n(\mathbb{F})$. They are all topological groups under **matrix multiplication**.

**Example 7.7.** *The the general linear group*

$$GL_n(\mathbb{F}) := \{A \in M_n(\mathbb{F}) \mid \det(A) \neq 0\},$$

*is a topological group that describes all invertible linear transformations $\mathbb{F}^n$ to $\mathbb{F}^n$. We also have the special linear group*

$$SL_n(\mathbb{F}) := \{A \in M_n(\mathbb{F}) \mid \det(A) = 1\},$$

**Example 7.8.** *The orthogonal group*

$$O(n) := \{A \in GL_n(\mathbb{R}) \mid AA^T = I\}$$



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*is a topological group that describes all linear transformations on $\mathbb{R}^n$ that preserve the standard inner product. We also have the special orthogonal group*

$$SO(n) := \{A \in GL_n(\mathbb{R}) \mid AA^T = I, \det(A) = 1\}$$

*that describes linear transformations on $\mathbb{R}^n$ that preserve both the inner product and the orientation on $\mathbb{R}^n$.*

**Example 7.9.** *As the complex version $O(n)$, we have the unitary group*

$$U(n) := \{A \in GL_n(\mathbb{C}) \mid AA^* = I\}.$$

*(Here $A^*$ means $\overline{A^T}$.) It describes the linear transformations on $\mathbb{C}^n$ that preserve the standard hermitian product. We also have the special unitary group*

$$SU(n) := \{A \in GL_n(\mathbb{C}) \mid AA^* = I, \det(A) = 1\}.$$

All previous examples are actually Lie groups (a topological group whose underlying topological space is a manifold[1]). Here is one example of topological group which is **not** a Lie group:

**Example 7.10.** *Consider the group*

$$\boldsymbol{Top}(n) := \{\textit{homeomorphisms on } \mathbb{R}^n\}$$

*equipped with the compact-open topology (see exercise K in chapter 4). This is an important object of study in manifold theory.*

**Definition 7.11.** Let $X$ be a topological space. We define $\mathrm{Homeo}(X) := \{\text{homeomorphisms on } X\}$ which is a topological group under compositions. (It's not hard to check that the group structure is compatible with the compact-open topology.)

**Definition 7.12.** $\mathrm{Homeo}_0(X)$ is a topological subgroup defined by path-component of $\mathrm{Homeo}(X)$ that contains the unit element $Id_X$.

**Theorem 7.13** (Thurston)**.** *If $X$ is a manifold, then $Homeo_0(X)$ is a simple group.*

**Theorem 7.14** (Edwards-Kirby)**.** *If $X$ is a compact manifold, then $Homeo(X)$ is locally path-connected.*

*Remark* 7.15. A path $\gamma$ in $\mathrm{Homeo}(X)$ from $f_0$ to $f_1$ is called an isotopy. This theorem tells us two homeomorphisms that are closed to each other are isotopic.

**Exercise A.** *(1) Show that $U(n)$ is compact. (Hint: what's the length of a row vector?) (You can use similar arguments to show that $SO(n), O(n), SU(n)$ are compact. But you don't need to write them).*

*(2) Show that $SL_n(\mathbb{R})$ is not compact for any $n \geq 2$. (You can use similar arguments to show that $GL_n(\mathbb{R}), GL_n(\mathbb{C})$ and $SL_n(\mathbb{C})$ are not compact. But you don't need write them.)*

[1] Most books require a compatible smooth structure on the Lie group. However, by the Hilbert fifth problem (solved by Gleason and Montgomery-Zippin), given any topological group which is a topological manifold, there exists a unique smooth structure compatible with the group structure. So we don't need a smooth structure in order to define a Lie group.



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**Exercise B.** *(1) Show that $O(n) \cong SO(n) \times \mathbb{Z}/2$. Here $\cong$ means homeomorphism as topological spaces.*

*(2)* ***When $n$ is even****, show that the homeomorphism in (1) can never be made into a homeomorphism that preserves the group structure. (So for even $n$, $O(n)$ and $SO(n) \times \mathbb{Z}/2$ are the same as topological spaces. But different as topological groups.)*

*(3) Show that $U(1) \cong SO(2) \cong S^1$.*

*(4) Show that any element $A \in SU(2)$ can be written as*

$$\begin{pmatrix} u & v \\ -\bar{v} & \bar{u} \end{pmatrix}$$

*for some $u, v \in \mathbb{C}$ with $|u|^2 + |v|^2 = 1$. Further show that $SU(2) \cong S^3$. (Hint: what's the formula for the inverse of a $2 \times 2$ matrix?)*

**Example 7.16.** *We show that $SO(3) \cong \mathbb{RP}^3$ in this example. We consider a map $f : D^3 \to SO(3)$ defined as follows: for any $x \in \partial D^3$, we draw a ray $l_x$ in $\mathbb{R}^3$ from 0 to $x$. For any $r \in [0, 1)$, we let $A_{r,x}$ be the matrix corresponding to the rotation on $\mathbb{R}^3$ with axis $l_x$ and angle $\pi \cdot x$. Then we define*

$$f(rx) := A_{r,x}, \forall x \in \partial D^3, r \in [0, 1].$$

*It's not hard to see $f$ is a quotient map and $f(x_1) = f(x_2)$ if and only if $x_1 \sim x_2$. Here $\sim$ is the equivalence relation given in (eq. (6.4)). So $SO(3) \cong D^n/\sim \cong \mathbb{RP}^3$.*

*Remark* 7.17. We have seen that $S^0 \cong O(1)$, $S^1 \cong SO(2)$ and $S^3 \cong SU(2)$. One may wonder for which $n$ is $S^n$ is a topological group. We have the following deep theorem:

*Theorem* 7.18 (Samelson, Cartan, Adams). *$S^n$ is a topological group if and only if $n = 0, 1, 3$*[2]

## 7.2 Group action on topological spaces

We first give some definitions and then discuss many interesting examples.

**Definition 7.19.** Let $G$ be a topological group and let $X$ be a topological space. A **left action** of $G$ on $X$ is a continuous map

$$\rho : G \times X \to X$$

that satisfies the following two conditions:

1. For any $g, h \in G$ and any $x \in X$, we have $\rho(g, \rho(h, x)) = \rho(gh, x)$;
2. For any $x \in X$, we have $\rho(e, x) = x$. Here $e \in G$ denotes the identity element.

Suppose we change (1) into the following condition:

(1') For any $g, h \in G$ and any $x \in X$, we have $\rho(g, \rho(h, x)) = \rho(hg, x)$.

Then we say $\rho$ is a **right action**.

*Remark* 7.20. In this section, we only focus on left actions.

**Definition 7.21.** Given two actions $\rho, \rho'$ of $G$ on $X$, we say they are conjugate if there exists a **homeomorphism** $f : X \to X$ such that for any $g \in G$, the following diagram commutes

$$\begin{array}{ccc} X & \xrightarrow{\rho(g,-)} & X \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f} \\ X & \xrightarrow{\rho'(g,-)} & X. \end{array}$$

If two actions are conjugate to each other, we essentially treat them as the same action.

[2]When $n = 7$, $S^n$ is an $H$-space (a generalization of a topological group that allows a more general multiplication map). However, since this multiplication map on $S^7$ is not associative, it is **not** a topological group.



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All the results discussed here hold for both left actions and right actions. For simplicity, we will always assume our actions are left actions. We will often denote $\rho(g,x)$ by $g\cdot x$.

**Example 7.22.** *The matrix group $GL_n(\mathbb{R})$ acts on $\mathbb{R}^n$ by*

$$A\cdot x := Ax.$$

*Similarly, $GL_C(\mathbb{R})$ acts Other matrix groups acts on $\mathbb{R}^n$ or $\mathbb{C}^n$ in a similar way. We call them "linear actions" because $\rho(g,-)$ is a linear map for any $g\in G$.*

**Example 7.23.** *Let $G=\mathrm{Iso}(\mathbb{R}^2)$ be isometry group of $\mathbb{R}^2$, namely the group of homeomorphisms on $\mathbb{R}^2$ that preserve the standard distance on $\mathbb{R}^2$ (but not necessarily the orientation). Then there is a homeomorphism*

$$f:\mathbb{R}^2\times O(2)\to G$$

*where $f(v,A)\in G$ is defined by*

$$x\mapsto Ax+v.$$

*We emphasize that $f$ does not preserve group structure. Actually, $G$ is a semidirect product of $\mathbb{R}^2$ and $O(2)$ (i.e. $G=\mathbb{R}^2\rtimes O(2)$) and it is not isomorphic $\mathbb{R}^2\times O(2)$ as a group.*

**Example 7.24.** *An action of $\mathbb{Z}/2$ on $X$ is just a homeomorphism $g:X\to X$ such that $g^2=1_X$ ($1_X$ denotes the identity map on $X$). When $X=\mathbb{R}^2$, we have the following two natural $\mathbb{Z}/2$-actions:*

$$g\cdot(x,y):=(-x,-y)$$

*or*

$$g\cdot(x,y):=(x,-y).$$

*Geometrically, the first action means the rotation by $\pi$ and the second action means reflection along $x$-axis. It can be proved that any nontrivial $\mathbb{Z}/2$-action (not necessarily linear) on $\mathbb{R}^2$ is conjugate to one of these two actions. Similarly, any nontrivial, smooth action of $\mathbb{Z}/2$ on $\mathbb{R}^3$ is conjugate to a reflection along a plane or a rotation along an axis (this is a deep theorem, called the "Smith conjecture" and proved by Waldhausen in 1969). However, staring in dimension 4, the situation is much more complicated and analogous result fails. We also mention the following Smith fixed point theorem:*

**Theorem 7.25.** *Let $p$ be a prime number. Then any $\mathbb{Z}/p^m$-action on $\mathbb{R}^n$ has a fixed point.*

**Definition 7.26.** Let $\rho$ be an action of $G$ on $X$.

1. We say the action is **transitive** if for $x,y\in X$, there exists $g\in G$ such that $\rho(g,x)=y$.
2. We say the action is **effective** (or **faithful**) if for $g\neq e\in G$, the map $\rho(g,-):X\to X$ (which is always a homeomorphism) is **not** the identity map on $X$. In other words, $\rho$ is effective if we have
$$\rho(g,x)=x \text{ for all } x\in X \implies g=e.$$
3. We say the action is **free** if for any $g\neq e\in G$, the map $\rho(g,-):X\to X$ has no fixed point. In other words, $\rho$ if free if
$$\rho(g,x)=x \text{ for some } x\in X \implies g=e.$$

**Example 7.27.** *Since the linear action of $SO(n)$ on $\mathbb{R}^n$ preserves the length of vectors, it restricts to an action of $SO(n)$ on $S^{n-1}$ (the unit sphere in $\mathbb{R}^n$). For any $n\geq 1$, this action is effective. When $n\geq 2$, this action is transitive (because you can rotate any unit vector to any other unit vector by a suitable rotation). But for any $n\geq 3$, the action is not free. (For example, any non-identity element $g\in SO(3)$ fixes exactly two points on $S^2$.)*



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**Example 7.28.** *Similarly, the linear action of $U(n)$ on $\mathbb{C}^n$ preserves the unit sphere $S^{2n-1} := \{(z_1, \cdots, z_n) \in \mathbb{C}^n \mid |z_1|^2 + \cdots + |z_n|^2 = 1\}$. Again, this action is effective, transitive for any $n \geq 1$ but not free unless $n = 1$.*

**Definition 7.29.** Given a $G$ action on $X$, we define an equivalence relation $\sim_G$ on $X$ as follows:

$$x \sim_G y \iff y = gx \text{ for some } g \in G.$$

We use $Gx$ to denote the equivalence class of $x$ and call it the **orbit** of $x$. We denote the quotient space $X/\sim_G$ by $X/G$ and call it the **orbit space**.

*Remark* 7.30. In previous sections, we have used $X/Y$ to denote the quotient space of $X$ by collapsing the whole subspace $Y$ into a single point. And here we are using $X/G$ to denote the orbit space. These two operations are different even if $G$ is a subspace of $X$. (For example, consider the translation action of $\mathbb{Z}$ on $\mathbb{R}$, the orbit space is a single circle $S^1$ but if we just collapse all points in $\mathbb{Z}$, we will get a wedge of infinitely many circles.) This is again an unfortunate abuse of (standard) notations and it can cause confusion sometimes. When there is a possible confusion, we will explicitly say whether we are taking the orbit space or we are just collapsing the subspace.

In the following examples, we will construct many interesting examples of orbit spaces.

**Example 7.31.** *Fix a basis $\{v_1, \cdots, v_n\}$ for $\mathbb{R}^n$. We consider the translation action of $\mathbb{Z}^n$ on $\mathbb{R}^n$ given by*

$$(a_1, \cdots, a_n) \cdot x := x + \sum_{i=1}^{n} a_i v_i.$$

*(So each element in $\mathbb{Z}^n$ acts as a translation.) This action is free. For any two choices of basis, the actions are conjugate to each other so have homeomorphic orbit spaces. Therefore, we may assume that $\{v_*\}$ is the standard basis $\{e_1, \cdots, e_n\}$. In this case, the action is the product of the action of $\mathbb{Z}$ on $\mathbb{R}$. Since $\mathbb{R}/\mathbb{Z} \cong S^1$, we see that $\mathbb{R}^n/\mathbb{Z}^n \cong T^n$. From now on, when we mention $\mathbb{R}^n/\mathbb{Z}^n$, we always assume we are using the translation action with standard basis.*

We use the following notations in the examples below: Let $V = \mathbb{R}^n$ or $\mathbb{C}^n$, we use $V^*$ to denote $V \setminus \{0\}$; use $S(V)$ to denote the unit sphere in $V$; and use $V^+$ to denote the one-point compatification of $V$.

**Example 7.32.** *Consider the free action of $\mathbb{Z}/2 = \{0, 1\}$ on $S^n = S(\mathbb{R}^{n+1})$ given by $1 \cdot x := -x$. This action is called the antipodal action. Then we have*

$$S^n/(\mathbb{Z}/2) \cong \mathbb{RP}^n.$$

*Alternatively, we consider the action of $\mathbb{R}^*$ on on $(\mathbb{R}^{n+1})^*$, given by $r \cdot x := rx$. Then we also have*

$$(\mathbb{R}^{n+1})^*/\mathbb{R}^* \cong \mathbb{RP}^n.$$

**Example 7.33.** *As the complex analogue of example 7.32, we consider the action of $S^1 = S(\mathbb{C})$ on $S^{2n+1} = S(\mathbb{C}^{n+1})$, and the action of $\mathbb{C}^*$ on $(\mathbb{C}^{n+1})^*$. Both actions are given by*

$$w \cdot (z_0, \cdots, z_n) := (wz_0, \cdots, wz_n).$$

*Both actions are free and the orbit spaces $S^{2n+1}/S^1$ and $(\mathbb{C}^{n+1})^*/\mathbb{C}^*$ are homeomorphic. We call this orbit space **the complex projective space**, denoted by $\mathbb{CP}^n$. There is a natural one-one correspondence*

$$\{\textit{points in } \mathbb{CP}^n\} \leftrightarrow \{\textit{1-dimensional complex subspace of } \mathbb{C}^{n+1}\}$$

*that sends a point $[(z_0, \cdots, z_n)]$ to its span $\mathbb{C} \cdot (z_0, \cdots, z_n) \subset \mathbb{C}^{n+1}$. Note that $\mathbb{CP}^n$ is a $2n$-dimensional manifold.*



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**Exercise C.** *Show that $\mathbb{CP}^n$ is a manifold and it is closed. (Hint: Consider the cover of $(\mathbb{C}^{n+1})^*$ by open sets $\{U_i\}_{0\leq i\leq n}$ where $U_i = \{(z_0, \cdots, z_n) \in \mathbb{C}^{n+1} \mid z_i \neq 0\}$. What's the image of $U_i$ under the quotient map $(\mathbb{C}^{n+1})^* \to \mathbb{CP}^n$?)*

**Example 7.34.** *We actually have a homeomorphism*

$$f : \mathbb{CP}^1 \cong \mathbb{C}^+$$

*defined by*

$$f[(z_0, z_1)] = \begin{cases} \infty & \text{if } z_0 = 0 \\ \frac{z_1}{z_0} & \text{if } z_0 \neq 0. \end{cases}.$$

*$f$ is a bijective, continuous map from a compact space to a Hausdorff space. So it is a homeomorphism. The quotient map*

$$\eta : S^3 \to S^3/S^1 = \mathbb{CP}^1 \cong S^2$$

*is called the **Hopf map** (or Hopf fibration). It is an extremely important map in modern topology (we will come back to it several times later). If we let $S^3 = S(\mathbb{R}^4)$ and $S^2 = S(\mathbb{R}^3)$, then the Hopf map can be expressed in Euclidean coordinates as*

$$\eta(x_1, x_2, x_3, x_4) = (2(x_1x_3 + x_2x_4), 2(x_1x_4 - x_2x_3), x_1^2 + x_2^2 - x_3^2 - x_4^2).$$

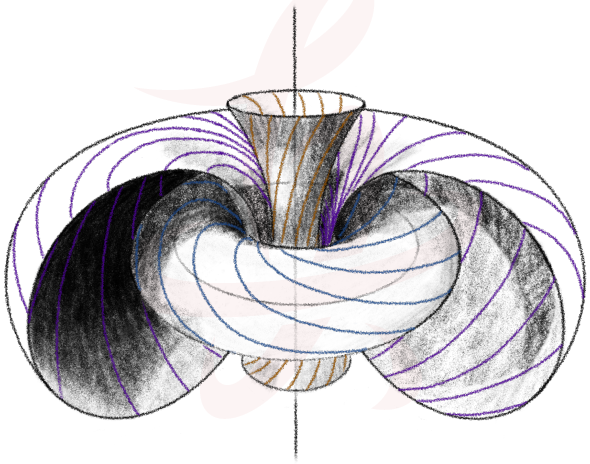

Figure 7.1

**Example 7.35.** *Given any coprime $p, q$, we can define an action of $\mathbb{Z}/p = \{0, \cdots, p-1\}$ on $S^3 = S(\mathbb{C}^2)$ by*

$$a \cdot (z_1, z_2) := (e^{\frac{2\pi ia}{p}} z_1, e^{\frac{2\pi iqa}{p}} z_2).$$

*This is a free action. The orbit space $S^3/(\mathbb{Z}/p)$ is a 3-dimensional manifold called the "lens space" and denoted by $L(p, q)$. Note that $L(2, 1)$ is just $\mathbb{RP}^3$.*

*Actually, there are many interesting actions on $S^3$ by non-cyclic finite groups. For example, the binary icosahedral group $\langle a, b \mid a^3 = b^5 = (ab)^2\rangle$ acts freely on $S^3$. And the orbit space is an important 3-manifold, called the Poincaé homology 3-sphere.*

**Exercise D.** *(1) Show that the lens space $L(p, q)$ and $L(p, -q)$ are homeomorphic.*
*(2) Show that $L(p, q_1)$ and $L(p, q_2)$ are homeomorphic if $p|(q_1 - q_2)$.*
*(3) Show that $L(p, q_1)$ and $L(p, q_2)$ are homeomorphic if $p|(q_1q_2 - 1)$.*
*(4) Show that $L(p, q_1)$ and $L(p, q_2)$ are homeomorphic if $p|(q_1 \pm q_2)$ or $p|(q_1q_2 \pm 1)$.*[3]

[3]This is actually an if and only if. So we have complete classification of Lens spaces. Proving this requires some tools from algebraic topology (Reidemeister torsion).



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Now we give an example of the orbit space of a non-free action.

**Example 7.36.** *Consider the action of $\mathbb{C}^*$ on $\mathbb{C}^2$ given by*

$$w \cdot (z_0, z_1) := (wz_0, wz_1).$$

*From example 7.33 we can deduce that the space $\mathbb{C}^2/\mathbb{C}^*$ is $[0] \cup \mathbb{CP}^1$ with the topology defined as follows: a subset $U$ is open if either $U$ is an open subset of $\mathbb{CP}^1$ or $U = [0] \cup \mathbb{CP}^1$. This is a pathological space even though the action is very natural.*

**Example 7.37.** *Pick $s \in \mathbb{R}$, we define an action of $\mathbb{R}$ on $\mathbb{R}^2$ as*

$$t \cdot (x, y) := (x + t, y + st).$$

*(So this is the translation in the direction $(1, s)$.) Since this action commutes with the translation action of $\mathbb{Z}^2$, it induces an action of $\mathbb{R}$ on $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ (See example 7.31). We call this action as the translation action on $T^2$ with slope $s$ and denote it by $\rho_s$. We use $X_s$ to denote the orbit space $T^2/\mathbb{R}$.*

*We will see in exercise E that $X_s \cong S^1$ when $s \in \mathbb{Q}$. However, when $s \notin \mathbb{Q}$ (say $s = \sqrt{2}$), $X_s$ is a space with nontrivial topology. To see this, we take any point $[x] \in X_s$, it correspond to an orbit in $T^2$, which is the image of a straight line under the quotient map $\mathbb{R}^2 \to T^2$. Since slope of this line equals $s$ and is not rational, the orbit $\mathbb{R}x$ is dense in $\mathbb{T}^2$. As a consequence, $[x]$ is contained in any nonempty open subset of $X_s$.*

**Exercise E.** *(1) Let $G = SL_2(\mathbb{Z})$ be the group of $2 \times 2$ integer matrices with determinant 1. Show that the linear action of $G$ on $\mathbb{R}^2$ induces an action of $G$ on $T^2 = \mathbb{R}^2/\mathbb{Z}^2$. Is this action free? Is the action effective? Is this action transitive?*

*(2) Let $p = (x, y)$ be a point in $T^2$. Then the orbit $Gp$ gives a point $[p] \in T^2/G$. Consider the subset $\{[p]\} \subset T^2/G$ with a single point. Show that it is closed if $x, y$ are both rational numbers and it is dense otherwise.*

*(3) Let $\rho_s$ be the translation action of $\mathbb{R}$ on $T^2$ (See example 7.37). Show that for any two rational numbers $s, s'$. There is a homeomorphism $T^2 \to T^2$ that takes each orbit of $\rho_s$ to an orbit of $\rho_{s'}$.*

*(4) Let $X_s$ be the orbit space of $\rho_s$. Suppose $s \in \mathbb{Q}$. Show that $X_s \cong S^1$.*

**Exercise F.** *(1) Let $G$ be a finite group that acts freely on a manifold $X$. Show that the orbit space $X/G$ is also a manifold. (Don't forget to check the Hausdorff condition.)*

*(2) Let $G$ be a compact topological group that acts on a Hausdorff space $X$. Show that the orbit space $X/G$ is Hausdorff. (Hint: start with the case that $G$ is finite.)*

**Example 7.38.** *Consider the free group $G = \mathbb{Z} * \mathbb{Z}$ generated by $g, h$. That means any element in $G$ can be written as*

$$g^{m_1} h^{n_1} g^{m_2} h^{n_2} \cdots g^{m_k} h^{n_k}, \quad m_i, n_i \in \mathbb{Z}.$$

*Consider the action on $\rho : G \times \mathbb{H} \to \mathbb{H}$ on the upper-half plane*

$$\mathbb{H} := \{z \in \mathbb{C} \mid \mathrm{Im}(z) > 0\}.$$

*defined by*

$$\rho(g, z) = z + 2, \quad \rho(h, z) = \frac{z}{1 - 2z}.$$

*We will prove in Exercise H that this action $\rho$ is effective. (Actually, more is true: the action is free. But proving this requires more argument so we won't discuss.) Moreover, consider the region*

$$R = \{z \in \mathbb{H} \mid |\mathrm{Re}\, z| \leq 1, |z \pm \frac{1}{2}| \geq \frac{1}{2}\}.$$

*It is a fundamental domain of $\rho$. That means the following two conditions hold:*



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- *Every orbit contains at least one point in $R$.*
- *$z, z' \in R$, $z \neq z'$, $z \sim_G z'$ implies $z, z' \in \partial R$. In other words, suppose two different points in $R$ belongs to the same orbit. Then both of them must be in the boundary.*

*In particular, we have*

$$\mathbb{H}/G \cong R/\sim_G .$$

*Furthermore, one can show that the restriction of $\sim_G$ on $R$ is generated by*

$$-1+yi \sim_G 1+yi \quad \text{for any } y>0,$$
$$\frac{1+e^{i\theta}}{2} \sim_G -\frac{1+e^{-i\theta}}{2} \quad \text{for any } \theta \in (0,\pi).$$

*Using this, one can prove that*

$$H/G \cong R/\sim_G \cong S^2 \setminus \{\text{three points}\}$$

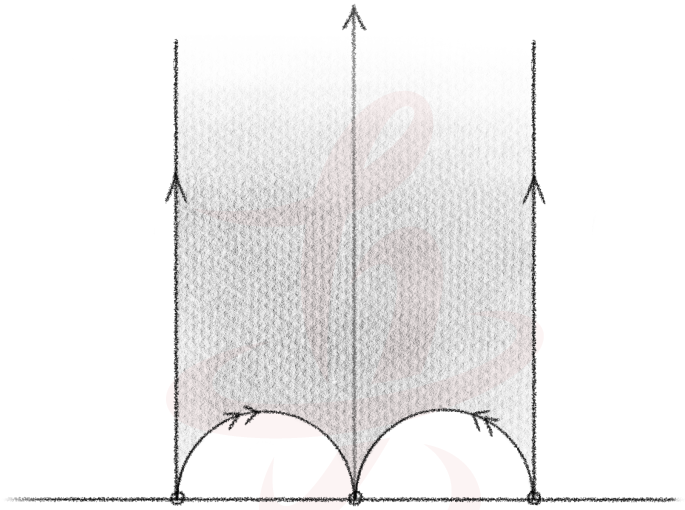

Figure 7.2

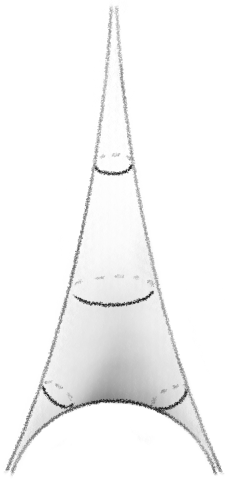

Figure 7.3



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**Exercise G.** *Consider a topological group $G$ acting on a Hausdorff space $X$. For $x \in X$, we define the* ***stabilizer*** *of $x$ as*

$$H_x := \{g \in G \mid g \cdot x = x\}.$$

*(1) Show that $H_x$ is a closed subgroup of $G$.*

*(2) Consider the set $G/H_x = \{gH_x \mid g \in G\}$ of the* ***right-cosets****, equipped with the quotient topology induced by the map $G \to G/H_x$. Suppose $G$ is compact. Show that $G/H_x$ is homeomorphic to the orbit $Gx$ (with the subspace topology).*

*(3) Give an example that $G/H_x$ may not be homeomorphic to $Gx$ when $G$ is noncompact.*

**Exercise H.** *(1) Prove the following algebraic result:*

**Lemma 7.39** (Ping-Pong lemma)**.** *Let $G$ be the free group generated by $g, h$. Consider an action $\rho : G \times X \to X$. Suppose we can find subsets $A, B$ such that the following conditions are satisfied*

*(i) $A \not\subset B$;*

*(ii) For any $n \neq 0$, $\rho(g^n, A) \subset B$;*

*(iii) For any $n \neq 0$, $\rho(h^n, B) \subset A$.*

*Then the action $\rho$ is an effective action.*

*(Hint: First show that*

$$\rho(g^{n_1}h^{m_1}g^{n_2}h^{m_2}\cdots g^{n_k}h^{m_k}g^{n_{k+1}}, -) \neq \mathrm{Id}_X$$

*if $n_i, m_i \neq 0$. Then use conjugation to reduce the general case to this case. )*

*(2) Show that the action $\rho$ in Example 7.38 is effective. You can consider the following two regions*

$$A = \{z \in \mathbb{H} \mid |z + \frac{1}{2}| < \frac{1}{2} \text{ or } |z - \frac{1}{2}| < \frac{1}{2}\}$$

$$B = \{z \in \mathbb{H} \mid |\mathrm{Re}(z)| > 1\}.$$

*(Hint: Consider the homeomorphism $f : \mathbb{H} \to \mathbb{H}$ defined by $z \to -\frac{1}{z}$, which interchanges $A$ and $B$.)*



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# Chapter 8

# Urysohn's lemma and its applications

In this section, we discuss Urysohn's lemma and mention some of its applications. The natural logical order for doing this is right after we introduced the separation axioms in Section 3. But we decide to postpone this part here so that we have learnt more examples.

Recall that a space $X$ is **normal** if it satisfies the $T_4$-axiom, namely for any two disjoint closed subsets $V_0, V_1$, there exists a neighborhood $U_i$ of $V_i$ such that $U_0 \cap U_1 = \varnothing$.

**Lemma 8.1** (Urysohn)**.** *Suppose $X$ is normal. Then for any two **disjoint, closed** subsets $A, B$, there exists a continuous function $f : X \to [0, 1]$ such that $f(A) = 0$ and $f(B) = 1$.*

The Urysohn's lemma is the first "deep theorem" in these notes that we provide a full proof. The proof is contains some really original idea and the conclusion is somewhat surprising: being able to separate disjoint closed sets using continuous function is not stronger than being able to separate them using open sets. Also note that it's easy to see that the inverse of the Urysohn's lemma is true (i.e. if one can separate disjoint closed sets by continuous functions, then the space must be normal.).

*Remark* 8.2. lemma 8.1 is easy to prove if one assumes that $X$ is metrizable because one can simply set

$$f(x) = \frac{d(x, A)}{d(x, A) + d(x, B)}.$$

We have seen this function when we prove that metrizable implies normal.

*Remark* 8.3. One may ask whether the Urysohn's lemma can be adapted to the algebraic setting or in the smooth setting. The answers are both "yes". For example, if $A, B$ are disjoint closed subsets of $\mathbb{R}^n$ (or a smooth manifold). Then there exists a smooth function $f : \mathbb{R}^n \to \mathbb{R}$ such that $f(A) = 0$ and $f(B) = 1$. (Proving this uses partition of unity.) If $A, B$ are disjoint, Zariski closed subsets of $\mathbb{C}^n$, then by Hilbert Nullstellensatz (a fundamental result in commutative algebra), we can a find a polynomial $f : \mathbb{C}^n \to \mathbb{C}$ such that $f(A) = 0$ and $f(B) = 1$. (It's important that we are working with an algebraically closed field here. Certainly not true on $\mathbb{R}^n$.)

*Proof of lemma 8.1.* **Step 0:** $X$ satisfies the following condition: For any closed $V$ and any open $U$ that contains $V$, there exists an open $U'$ such that

$$V \subset U' \subset \overline{U'} \subset U.$$

This is because $V$ and $X \setminus U$ are disjoint closed sets. So we can find a neighborhood $U'$ of $V$ and a neighborhood $U''$ of $X \setminus U$ such that $U' \cap U'' = \varnothing$.

**Step 1:** Let $\mathbb{Q}_I = \mathbb{Q} \cap [0, 1]$. We will construct a collection of open sets $\{U_r\}_{r \in \mathbb{Q}_I}$ such that the following conditions hold:



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小红书号:3189205503 qq

(i) For any $r, s \in \mathbb{Q}_I$ with $r < s$, we have $\overline{U_r} \subset U_s$.

(ii) For any $r \in \mathbb{Q}_I$, we have $A \subset U_r \subset (X \setminus B)$.

To construct this collection, we list all elements in $\mathbb{Q}_I$ as $\{r_1, r_2, \cdots\}$ and construct $U_{r_n}$ inductively. We set $r_1 = 1$ and $r_2 = 0$. Then we let $U_1 = X \setminus B$ and let $U_0$ be a neighborhood of $A$ such that $\overline{U_0} \subset U_1$ (This is possible because of Step 0.)

Now suppose we have constructed $U_{r_1}, \cdots, U_{r_n}$. We set

$$r_{i(n)} = \min\{r_i \mid 1 \le i \le n, r_i < r_{n+1}\}$$

and

$$r_{j(n)} = \max\{r_j \mid 1 \le j \le n, r_{n+1} < r_j\}.$$

Then $\overline{U_{r_{i(n)}}} \subset U_{r_{j(n)}}$ by our inductive hypothesis. By Step 0, we can find some open $U'$ such that

$$\overline{U_{r_{i(n)}}} \subset U' \subset \overline{U'} \subset U_{r_{j(n)}}.$$

Then we set $U_{r_{n+1}} = U'$. This finishes the induction.

**Step 2:** We define $f : X \to [0, 1]$ by

$$f(x) = \begin{cases} 1 & \text{if } x \notin U_1 \\ \inf\{r \in \mathbb{Q}_I \mid x \in U_r\} & \text{if } x \in U_1 \end{cases}.$$

By (ii), we have $f(A) = 0$ and $f(B) = 1$.

**Step 3:** It remains to show that $f$ is continuous. It suffices to show that for any $0 \le a < b \le 1$, the sets

$$f^{-1}([0, b)), f^{-1}((a, 1]), f^{-1}((a, b))$$

are open. Take any $x \in f^{-1}(a, b)$. Then there exists $r, r' \in \mathbb{Q}_I$ such that

$$a < r < f(x) < r' < b$$

Then $U_{r'} \setminus \overline{U_r}$ is a neighborhood of $x$ that is contained in $f^{-1}(a, b)$. Hence $f^{-1}(a, b)$ is open. The other two cases are similar.

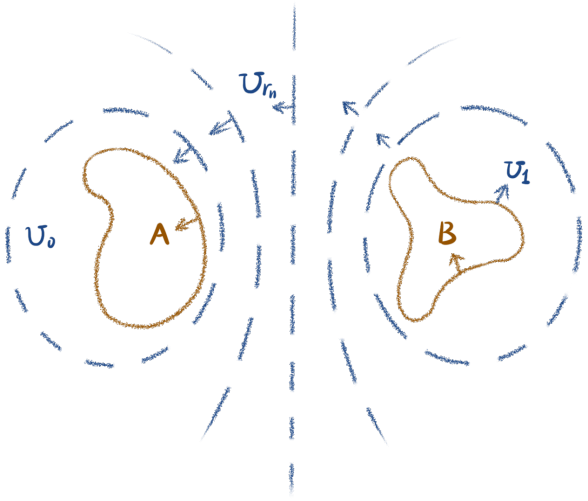


Figure 8.1

□



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**Exercise A.** *A space $X$ is called **perfectly normal** if it is normal and every closed set is the intersection of countable many open sets.*

*(1) Show that all metrizable spaces are perfectly normal.*

*(2) Show that $X$ is perfectly normal if and only if the following condition holds:*

- *For any disjoint closed set $V_0, V_1$, there exists a continuous function $f : X \to [0,1]$ such that $f^{-1}(i) = V_i$ for $i = 0, 1$.*

*(Hint: try to modify the function you obtained from the Urysohn's lemma by adding some more functions.)*

*(3) Show that $[0,1]^{\mathbb{R}}$ (with product topology) is normal.*

*(4) Show that $[0,1]^{\mathbb{R}}$ (with product topology) is not perfectly normal.*

Next, we discuss two important applications of the Urysohn's lemma.

**Theorem 8.4** (Tietze extension theorem)**.** *Let $V$ be a **closed subset** of a **normal space** $X$. Then any continuous function $f : V \to \mathbb{R}$ can be extended to a continuous function $\widetilde{f} : X \to \mathbb{R}$.*

*Proof of theorem 8.4.* **Step 1:** We first assume that $f(V) \subset [-1,1]$ and construct an extension $\widetilde{f} : X \to [-1,1]$. Let $A = f^{-1}([-1,-\frac{1}{3}])$ and $B = f^{-1}([\frac{1}{3},1])$. Then $A, B$ are closed in $V$ and $V$ is closed in $X$. Hence $A, B$ are closed in $X$. By Urysohn's lemma, there exists a function $\varphi_1 : X \to [-\frac{1}{3}, \frac{1}{3}]$ such that $\varphi_1(A) = -\frac{1}{3}$ and $\varphi_1(B) = \frac{1}{3}$. Note that for any $x \in V$, we have

$$f(x) - \varphi_1(x) \in [-\frac{2}{3}, \frac{2}{3}].$$

By replacing $f$ with $f - \varphi_1|_V$ and repeating this argument, we obtain $\varphi_2 : X \to [-\frac{2}{9}, \frac{2}{9}]$ such that

$$f(x) - \varphi_1(x) - \varphi_2(x) \in [\frac{4}{9}, \frac{4}{9}], \forall x \in V.$$

Repeating this argument, we obtain continuous function $\{\varphi_n : X \to \mathbb{R}\}$ such that

(i) $\varphi_n(X) \subset [-\frac{2^{n-1}}{3^n}, \frac{2^{n-1}}{3^n}]$;

(ii) $|f(x) - \sum_{i=1}^{n} \varphi_i(x)| \leq \frac{2^n}{3^n}$ for any $x \in V$.

Then we define $\widetilde{f} : X \to [-1,1]$ by setting

$$\widetilde{f}(x) = \sum_{i=1}^{\infty} \varphi_i(x), \quad \forall x \in X.$$

By (i), $\widetilde{f}$ is well-defined and continuous. By (ii), $\widetilde{f}$ is an extension of $f$.

**Step 2:** We assume that $f(V) \subset (-1,1)$ and construct and extension $\widetilde{f} : X \to (-1,1)$. By Step 1, there exists a continuous function $\widetilde{f'} : X \to [-1,1]$ that extends $f$. Let $A = \widetilde{f'}^{-1}\{-1,1\}$. Then $A, V$ are disjoint closed sets. By Urysohn's lemma, there exists a continuous $g : X \to [0,1]$ such that $g(A) = 0$ and $g(V) = 1$. Then we define $\widetilde{f} : X \to (-1,1)$ by setting

$$\widetilde{f}(x) = g(x) \cdot \widetilde{f'}(x).$$

**Step 3:** Given a continuous $f : V \to \mathbb{R}$, we apply the result of Step 2 to the function $g : V \to (-1,1)$ defined by

$$g(x) := \frac{2}{\pi} \arctan(f(x)).$$

This gives us a continuous $\widetilde{g} : X \to (-1,1)$ that extends $g$. Then we define $\widetilde{f} : X \to \mathbb{R}$ by setting

$$\widetilde{f}(x) := \tan(\frac{\pi}{2} \cdot \widetilde{g}(x)).$$

□



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**Exercise B.** *(1) Let $P$ be a convex polyhedron in $\mathbb{R}^n$. Show that $P$ is a **retract** of $\mathbb{R}^n$. That means there exists a continuous map $r : \mathbb{R}^n \to P$ such that $r(x) = x$ for any $x \in P$.*

*(2) Let $X$ be a normal space and let $V$ be a closed subset. Show that any continuous map $f : V \to P$ can be extended to a continuous map $\widetilde{f} : X \to P$.*

Another powerful application of the Urysohn's lemma is the following result:

**Theorem 8.5** (Urysohn metrization theorem)**.** *Suppose $X$ satisfies the $T_1, T_4$ axioms and is second-countable. Then $X$ is metrizable.*

*Remark* 8.6. By the Lindelöf theorem (theorem 2.28), second-countable and $T_3$ implies $T_4$. So we can replace the $T_4$ condition in theorem 8.5 by the $T_3$ condition.

*Proof.* **Step 1:** Let $\mathcal{B} = \{U_i\}_{i\in\mathbb{N}}$ be a countable basis for $X$. We call $(U_i, U_j)$ a "typical pair" if $\overline{U_i} \subset U_j$. Since $\mathbb{N} \times \mathbb{N}$ is countable, we can order all typical pairs as

$$(U_{i(1)}, U_{j(1)}), (U_{i(2)}, U_{j(2)}), \cdots$$

By Urysohn's lemma, for any $n \in \mathbb{N}$ there exists a continuous function $f_n : X \to [-1, 1]$ such that $f(U_{i(n)}) = 0$ and $f(X \setminus U_{j(n)}) = 1$.

**Step 2:** Let $\mathcal{H}$ be the Hilbert space (See example 3.23). We consider the map $f : X \to \mathcal{H}$ defined by

$$f(x) = (f_1(x), \frac{1}{2}f_2(x), \cdots, \frac{1}{n}f_n(x), \cdots).$$

$f$ is well-defined because $\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$.

**Step 3:** We prove that $f$ is injective. Given any $x \neq x' \in X$, because $X$ is $T_1$, $X \setminus \{x'\}$ is a neighborhood of $x$. So there exists $U_j \in \mathcal{B}$ such that $x \in U_j \subset S \setminus \{x'\}$. Because $X$ is normal, we can further find $U_i \in \mathcal{B}$ such that

$$x \in U_i \subset \overline{U_i} \subset U_j.$$

Then $(U_i, U_j)$ is a typical pair and we have $i = i(n)$ and $j = j(n)$ for some $n$. Then we see that $f_n(x) = 0$ and $f_n(x') = 1$. This shows that $f(x) \neq f(x')$.

**Step 4:** We show that $f$ is an embedding. We apply the following lemma

**Lemma 8.7.** *Let $f : Z \to W$ be a map between two first countable spaces. Then $f$ is an embedding if and only if the following condition is satisfied:*

- *For any sequence $(x_i)$ in $Z$ and any point $x \in Z$, we have $(x_i) \to x \iff (f(x_i)) \to f(x)$.*

*Proof.* The proof is straightforward. □

Let $(x_i)_{i\in\mathbb{N}}$ be a sequence in $X$. Suppose it converges to $x$. Then we have $f_n(x_i) \to f_n(x)$ for any $n \in \mathbb{N}$. For any $\epsilon > 0$, we pick $N \gg 0$ such that $\sum_{n=N}^{\infty} \frac{1}{n^2} \leq \frac{\epsilon}{2}$. Then for $i$ large enough, we have $|f_n(x) - f_n(x_i)| \leq \frac{\epsilon}{2N}$. This implies that $|f(x) - f(x_i)|^2 < \epsilon$ when $i$ is large enough. So $(f(x_i))_{i\in\mathbb{N}}$ converges to $f(x)$.

Suppose $(x_i)$ does not converge to $x$. Then there exists a neighborhood $U$ of $x$ such that $x_i \notin U$ for infinitely many $i$. Because $X$ is normal, we can find a typical pair $(U_{i(n)}, U_{j(n)})$ such that

$$x \subset U_{i(n)} \subset U_{j(n)} \subset U.$$

Then $f_n(x) = 0$ and $f_n(x_i) = 1$ for infinitely many $i$. This implies that $(f(x_i))$ does not converges to $(f(x))$.

**Step 5:** Since any subspace of a metrizable space is metrizable, $f(X)$ is metrizable. Since $f$ is an embedding, $X$ is metrizable. □



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# Chapter 9

# Homotopy of maps, the fundamental group

Starting this lecture, we will discuss basic algebraic topology. The main reference is Hatcher's book Chapter 1.

## 9.1 Homotopy equivalences

As we discussed in the first lecture, a main topic in topology is to define invariants to distinguish different topological space. An the ultimate goal, one might hope to classify topological spaces up to homeomorphisms. However, this is extremely complicated and not really doable (think how pathological can a closed subset of $\mathbb{R}^2$ be).

There are several ways to handle this problem. One way is to restrict to certain nice families of topological spaces (e.g. closed manifolds). We will actually classify closed surfaces in a few weeks. (Classification of higher dimensional manifolds is also a tractable question.) Alternatively, we can try to classify spaces up to a certain equivalence relation (called "homotopy equivalence") which is weaker than being homeomorphic.

We first talk about homotopy of maps. Roughly speaking, a homotopy is a continuous deformation between two maps.

**Definition 9.1.** (1) Let $f, g : X \to Y$ be two continuous maps. A homotopy from $f$ to $g$ is a continuous map $H : I \times X \to Y$ that satisfies the following condition

- $H(0, x) = f(x), \quad H(1, x) = g(x)$ for any $x \in X$.

If such a homotopy exists, than we say $f$ and $g$ are "homotopic", denoted as $f \simeq g$. (We write $f \overset{H}{\simeq} g$ if we want to specify a homotopy $H$.)

**Definition 9.2.** A homotopy from $f$ to a constant map is called a "null-homotopy". A map that is homotopic to a constant map is called "null-homotopic".

**Definition 9.3.** Let $A$ be a subspace of $X$ and two maps $f, g : X \to Y$ such that $f|_A = g|_A$. **A homotopy from $f$ to $g$ relative to $A$** is a continuous map $H : I \times X \to Y$ that satisfies the following conditions:

- $H(0, x) = f(x), \quad H(1, x) = g(x)$ for any $x \in X$
- $H(t, x) = f(x)$ for any $t \in I$ and $x \in A$.

If such a homotopy exists, we say $f$ and $g$ are homotopic relative to $A$, denoted by $f \simeq g$ rel$A$.



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*Remark* 9.4. Given a homotopy $H$ from $f$ to $g$, we obtain a family of maps $\{f_t : X \to Y\}_{t\in I}$ defined by $f_t(x) := H(t, x)$. Since $f_0 = f$ and $f_1 = g$, we can think a homotopy $H$ as a "continuous deformation" $\{f_t\}$ from $f$ to $g$. If any point in $A$ is not moved in this process, then we say the homotopy $H$ is a homotopy relative to $A$.

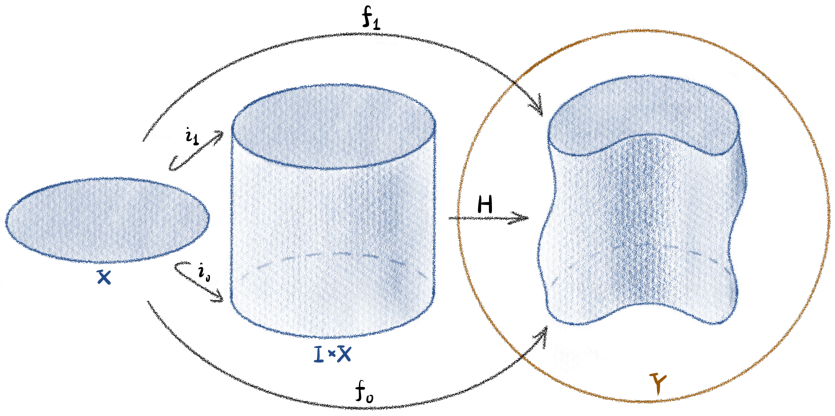


Figure 9.1

Let's see some examples.

**Example 9.5.** *(1) Let $X = \{a\}$ be a single point. Then a map $f : X \to Y$ is uniquely determined by $f(a)$. And a homotopy between from $f$ to $g$ is just a path $I \to Y$ from $f(a)$ to $g(a)$.*

*(2) Consider two maps $f, g : X \to Y$. Suppose $Y$ is a convex subset of $\mathbb{R}^n$. Then we always have "straight-line homotopy" from $f$ to $g$, defined by*

$$H : I \times X \to Y, \quad H(t, x) := tf(x) + (1 - t)g(x).$$

*(3) Given two maps $f, g : X \to S^n$. Suppose $f(x) \neq -g(x)$ for any $x \in X$. Then we have a homotopy from $f$ to $g$ defined by*

$$H : I \times X \to S^n, \quad H(t, x) := \frac{tf(x) + (1 - t)g(x)}{|tf(x) + (1 - t)g(x)|}.$$

*This is the homotopy through the big circle. (The denominator is never zero by our assumption.)*

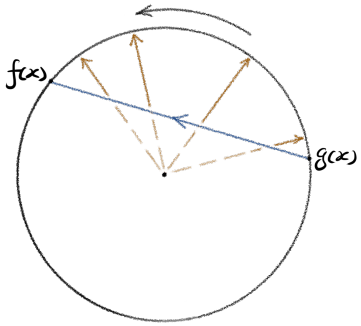


Figure 9.2

*(4)*

- *Define $f_n : S^1 \to S^1, z \mapsto z^n$. Then $f_n$ is null-homotopic if and only if $n = 0$. Moreover, for all $m \neq n$, we have $f_m \not\simeq f_n$. (We will prove this using fundamental group.)*



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- *Any maps $f : S^k \to S^1$ for $k > 1$ are null-homotopic.*
- *Any maps $f : S^m \to S^n$ for $m < n$ are null-homotopic. So from homotopy theoretic point of view, there is no "interesting" map from a low dimensional sphere to a high dimensional sphere.*
- *The Hopf map in Example 7.34 is **not** null-homotopic. You will learn how to prove this in algebraic topology class. This is one of the first nontrivial examples in homotopy theory: there can be interesting maps from a sphere of high dimension to a sphere of low dimension.*

We summarize some basic properties of homotopy in the following lemma:

**Lemma 9.6.** *(1) Being homotopic is an equivalence relation. In other words, we have (i) $f \simeq f$; (ii) $f \simeq g \iff g \simeq f$; (iii) $f \simeq g$, $g \simeq l \implies f \simeq l$. The equivalent class of $f$ is called the **homotopy class of** $f$, denoted by $[f]$.*

*(2) Given $f, f' : X \to Y$ and $g, g' : Y \to Z$, we have*

$$f \simeq f', \quad g \simeq g' \implies (g \circ f) \simeq (g' \circ f').$$

*(3) Parallel results hold for relative homotopy.*

*Proof.* (1) (i) is straightforward. To see (ii), we note that any homotopy $H : I \times X \to Y$ from $f$ to $g$ induces the inverse homotopy $\overline{H} : I \times X \to Y$ from $g$ to $f$, which is defined by

$$\overline{H}(t, x) := H(1 - t, x).$$

To prove (iii), we suppose $H_1 : I \times X \to Y$ is a homotopy from $f$ to $g$ and $H_2 : I \times X \to Y$ is a homotopy from $g$ to $l$. Then we consider the following "composition of homotopies"

$$H_1 \cdot H_2 : I \times X \to Y$$

defined by

$$(H_1 \cdot H_2)(t, x) = \begin{cases} H_1(2t, x) & \text{if } t \in [0, \frac{1}{2}] \\ H_2(2t - 1, x) & \text{if } t \in [\frac{1}{2}, 1]. \end{cases}$$

This is a homotopy from $f$ to $l$. Note that $H_1 \cdot H_2$ is continuous because of the gluing lemma (lemma 2.24. Also note that $H_1 \cdot H_2$ doesn't mean the composition of the maps $H_1, H_2$. For this reason, we use $\cdot$ instead $\circ$.

(2) Let $H : I \times X \to Y$ be a homotopy from $f$ to $f'$ and let $H' : I \times Y \to Z$ be a homotopy from $g$ to $g'$. We consider the composition of maps

$$g \circ H : I \times X \to Z.$$

This is a homotopy from $(g \circ f)$ to $(g \circ f')$.

Next, we show that $(g \circ f') \simeq (g' \circ f')$. To see this, we consider the map

$$F' : I \times X \to I \times Y$$

defined by $F'(t, x) = (t, f'(x))$. Then the composition

$$H' \circ F' : I \times X \to Z$$

is a homotopy from $(g \circ f')$ to $(g' \circ f')$.

By transitivity of homotopy, we see that $(g \circ f) \simeq (g' \circ f')$.

(3) The proof can be repeated for relative homotopy without any changes. □



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*Remark* 9.7. There is an alternative way of thinking about homotopy, which will be useful once we go deeper in algebraic topology: Let $\mathcal{C}(X,Y)$ be the space of continuous map from $X$ to $Y$, equipped with the compact-open topology (see Problem 4.K). Then any map $f : X \to Y$ represents a point $f \in \mathcal{C}(X,Y)$. Furthermore, if we assume $X$ is locally compact, then we have the following fact (follows from Problem 4.K.3):

- $f$ and $g$ are homotopic as maps from $X$ to $Y$ $\iff$ $f$ and $g$ belongs to the same path-component of $\mathcal{C}(X,Y)$.

Therefore, we have the following 1-1 correspondences:

$$\text{path-connected components of } \mathcal{C}(X,Y) \longleftrightarrow \text{homotopy classes of maps from } X \text{ to } Y.$$

In particular, all maps from $X$ to $Y$ are null-homotopic if and only if $\mathcal{C}(X,Y)$ is path-connected. As we will see, this happens if $X$ or $Y$ is contractible, and it also happens in other situations (for example we will see that $\mathcal{C}(S^n, S^1)$ is path-connected if $n \geq 2$.)

Recall that two spaces $X, Y$ are homeomorphic if and only if there exists continuous maps $f : X \to Y$ and $g : Y \to X$ such that $g \circ f = \mathrm{Id}_X$ and $f \circ g = \mathrm{Id}_Y$. By replacing "=" with the weaker condition "$\simeq$", we get the following definition:

**Definition 9.8.** We say two spaces $X, Y$ are **homotopy equivalent**, denoted by $X \simeq Y$, if there exists continuous maps $f : X \to Y$ and $g : Y \to X$ such that

$$g \circ f \simeq \mathrm{Id}_X, \text{ and } f \circ g \simeq \mathrm{Id}_Y .$$

In this case, both $f$ and $g$ are called **homotopy equivalences** and we say $f$ and $g$ are **homotopy inverse** of each other.

*Remark* 9.9. There are some simple facts:

- The composition of homotopy equivalences is a homotopy equivalence.
- Homotopy inverse is unique up to homotopy.
- Homotopy equivalences of spaces is a equivalence relation.

**Definition 9.10.** A space $X$ is **contractible** if it is homotopy equivalent to a space with a single point.

**Exercise A.** *Let $f : X \to Y$ be a homotopy equivalence. Show that $f$ induces a bijection between the set of path-connected components of $X$ and the set of path-connected components of $Y$. In particular, $X$ is path-connected if and only if $Y$ is path-connected.*

It's straightforward to see that the composition of homotopy equivalences is a homotopy equivalence. So being homotopy equivalent is an equivalence relation. This equivalent relation is actually weaker than being homeomorphic. The following definition is useful in constructing spaces that are homotopy equivalent but not homeomorphic.

**Definition 9.11.** Let $A$ be a subspace of $X$.

- A retraction from $X$ to $A$ is a continuous map $f : X \to A$ such that $r(x) = x$ for any $x \in A$. We say $A$ is a retract of $X$ is there exists a retraction from $X$ to $A$.
- A (strong) deformation retraction from $X$ to $A$ is a homotopy $F : I \times X \to X$ relative $A$ from $\mathrm{Id}_X$ to a map whose image equals $A$. In other words, $F$ satisfies the following conditions: (i) For any $x \in X$ , we have $F(0, x) = x$ and $F(1, x) \in A$; (ii) For any $t \in I$ and $x \in A$, we have $F(t, x) = x$. We say $A$ is a **deformation retract** of $X$ if there exists a deformation retraction from $X$ to $A$.



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**Example 9.12.** *(1) Any point $p \in X$ is a retract. In order for $p$ to be a deformation retract of $X$, a necessary (but not sufficient) condition is that $X$ is contractible.*

*(2) Let $X = \mathbb{R}^{n+1} \setminus \{p\}$ and let $A = S^n$. When $p = 0$, $A$ is a deformation retract (hence also a retract) of $X$ because we have the following deformation retraction*

$$F : I \times X \to S^n, \quad F(t, x) := (1 - t)x + t\frac{x}{|x|}.$$

*When $|p| < 1$, there is a homeomorphism from $\mathbb{R}^{n+1} \setminus \{p\}$ to $\mathbb{R}^{n+1} \setminus \{0\}$ that fixes $S^n$ pointwisely. So $S^n$ is still a deformation retract of $\mathbb{R}^{n+1} \setminus \{p\}$ when $|p| < 1$.*

*When $|p| > 1$, we have $D^{n+1} \subset X$. By theorem 1.4, there does not exist a retraction from $D^{n+1}$ to $S^n$. So $S^n$ is not a retract of $X$.*

**Lemma 9.13.** *Suppose $A$ is a deformation retract of $X$. Then the inclusion map $i_A : A \to X$ is a homotopy equivalence.*

*Proof.* We let $F : I \times X \to X$ be the deformation retraction and define $f : X \to A$ by $f(x) = F(1, x)$. Then we have $f \circ i_A = \mathrm{Id}_A$ and $\mathrm{Id}_X \overset{F}{\simeq} i_A \circ f$. □

**Example 9.14.** *(1) It's straightforward to see that the following conditions are equivalent:*

- *$X$ is contractible;*
- *$\mathrm{Id}_X$ is null-homotopic;*
- *Any continuous map $f : X \to Y$ is null-homotopic.*
- *Any continuous map $g : Y \to X$ is null-homotopic.*

*In particular, any convex subspace of $\mathbb{R}^n$ (e.g. $D^n, \mathbb{R}^n$) is contractible.*

*(2) The annulus $I \times S^1$ deformation retracts to the circle $\{\frac{1}{2}\} \times S^1$. Similarly, the Mobius strip*

$$M := I \times I/(0, x) \sim (1, 1 - x)$$

*deformation retracts to the circle $(I \times \{\frac{1}{2}\})/\sim \cong S^1$. So although an annulus is not homeomorphic to a Mobius band, they are indeed homotopy equivalent.*

*(3) Let $\Gamma$ be a graph with $k$ vertices and $l$ edges. We embedded $\Gamma$ as a subspace of $\mathbb{R}^2$ (so that the edges are disjoint line segments). By Euler's formula, $\mathbb{R}^2 \setminus \Gamma$ has $2 + l - k$ path-components, and exactly one of them is unbounded. For each bounded component $R_i$ $(1 \le i \le 1 + l - k)$, we take a point. Since $\overline{R_i} \cong D^2$, we can construct a deformation retraction from $\mathbb{R}^2 \setminus \{p_1, \cdots, p_{1+l-k}\}$ to $\Gamma$. So $\Gamma$ is homotopy equivalent to $\mathbb{R}^2$ minus $l + 1 - k$ points. (Draw a picture.)*

*(4) We will soon learn that $S^1, T^2, \mathbb{RP}^2$ are **not** contractible. By the end of this semester, we will be able to prove that $S^n$ is **not** contractible for any $n$ (and they are not homotopy equivalent to each other). In algebraic topology class, you will prove that any **closed** manifold of positive dimension is **not** contractible.*

**Exercise B.** *Let $X = S^{n+m-1}$ be the unit sphere in $\mathbb{R}^n \times \mathbb{R}^m$. Let $A = S^{n-1}$ and $B = S^{m-1}$ be the unit sphere in $\mathbb{R}^n \times \{0\}$ and $\{0\} \times \mathbb{R}^m$ respectively. Show that $X \setminus A$ deformation retracts to $B$.*

**Exercise C.** *A **pointed space** is a pair $(X, x)$, where $X$ is a space and $x$ is a chosen point (called base point) in $X$. We define the **wedge sum** (or one-point union) of two pointed spaces $(X, x)$ and $(Y, y)$ as*

$$X \vee Y := (X \sqcup Y)/(x \sim y).$$

*Let $p$ be any point in $T^2$. Show that $T^2 \setminus \{p\} \simeq S^1 \vee S^1$.*



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**Exercise D.** *(1) Let $f : X \to Y$ be a continuous map. Show that $f$ is null-homotopic if and only if $f$ can be extended to a continuous map $\widetilde{f} : CX \to Y$. Here $CX = (I \times X)/(\{1\} \times X)$ denotes the cone of $X$.*

*(2) Show that a continuous map $f : S^n \to Y$ is null-homotopic if and only $f$ can be extended to a continuous map $\widetilde{f} : D^{n+1} \to Y$.*

**Exercise E.** *Consider the antipodal map $f : S^n \to S^n$ defined by $f(x) = -x$. Show that $f$ is homotopic to the identity map if $n$ is odd. (Hint: start with the case $S^1 = S(\mathbb{C})$. We will later see that $f$ is **not** homotopic to the identity map if $n$ is even.)*

**Exercise F.** *Let $S^\infty = \{(x_i) \in \mathbb{R}^N \mid \sum x_i^2 = 1\}$ be the unit sphere in the Hilbert space $\mathcal{H}$, equipped with the metric topology.*

*(1) Show that $\mathrm{Id}_{S^\infty}$ is homotopic to the following map*

$$f : S^\infty \to S^\infty, f(x_0, x_1, x_2, \cdots) \mapsto (0, x_0, x_1, x_2, \cdots).$$

*(Hint: look at Example example 9.5 (3)).*

*(2) Show that $f$ is homotopic to the constant map sending $S^\infty$ to $(1, 0, \cdots)$.*

*(3) Show that $S^\infty$ is contractible.*

## 9.2 The fundamental group

Now let's narrow down our discussion and talk about homotopy of maps from $I$ to $X$, (i.e., paths in $X$).

**Definition 9.15.** We say two paths $\alpha, \beta : I \to X$ with $\alpha|_{\partial I} = \beta|_{\partial I}$ are homotopic if $\alpha \simeq \beta \quad \mathrm{rel}\ \partial I$. (From now on, we will usually ignore $\mathrm{rel}\,\partial I$ in our notation when we talk about homotopy of paths. So we always assume our homotopy of path is relative to the end points $\partial I$ (unless otherwise noted).) As before, we use $[\alpha]$ to denote the homotopy class of $\alpha$.

Recall that we have the following two operations on paths: the inverse path $\overline{\gamma}$ of a path $\gamma$, and the composition $\gamma \cdot \gamma'$ of two paths that satisfy $\gamma(1) = \gamma'(0)$. (See definition 5.23.) Both of them behave well under homotopy.

**Lemma 9.16.** *The following conclusion holds:*

1. *$\alpha \simeq \beta \implies \overline{\alpha} \simeq \overline{\beta}$. So it make sense to take the inverse of the homotopy class $[\alpha]$, denoted by $\overline{[\alpha]}$.*
2. *$\alpha \simeq \alpha'$ and $\beta \simeq \beta' \implies \alpha \cdot \beta \simeq \alpha' \cdot \beta'$. So it makes sense to compose two homotopy classes of paths $[\alpha], [\beta]$, denoted by $[\alpha] \cdot [\beta]$.*
3. *Let $c_p$ be the constant path at $p \in X$. Then we have $c_{\gamma(0)} \cdot \gamma \simeq \gamma$ and $\gamma \cdot c_{\gamma(1)}$. In other words, composing with a constant path does not change the homotopy class.*
4. *We have $\gamma \cdot \overline{\gamma} \simeq c_{\gamma(0)}$ and $\overline{\gamma} \cdot \gamma \simeq c_{\gamma(1)}$.*
5. *We have $(\alpha \cdot \beta) \cdot \gamma \simeq \alpha \cdot (\beta \cdot \gamma)$.*

*Proof.* (1) and (2) are straightforward. (Use the gluing lemma to glue together homotopy.)

To prove (3), we note that $c_{\gamma(0)} \cdot \gamma$ is a "reparameterization of $\gamma$". I.e., it is the composition of $\gamma$ with a map $\rho : I \to I$ that preserve both end points. Since any such $\rho$ is homotopic to $\mathrm{Id}_I$ by a straight-line homotopy relative to $\partial I$, we see that reparametrization of a path doesn't change its homotopy class.



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To prove (4), we construct a homotopy $H : I \times I \to X$ from $c_{\gamma(0)}$ to $\gamma \cdot \overline{\gamma}$ as follows:

$$H(t,s) = \begin{cases} \gamma(2st) & \text{if } t \in [0, \frac{1}{2}] \\ \gamma(s(2-2t)) & \text{if } t \in [\frac{1}{2}, 1] \end{cases}.$$

(5) holds because $(\alpha \cdot \beta) \cdot \gamma$ is a reparametrazation of $\alpha \cdot (\beta \cdot \gamma)$.

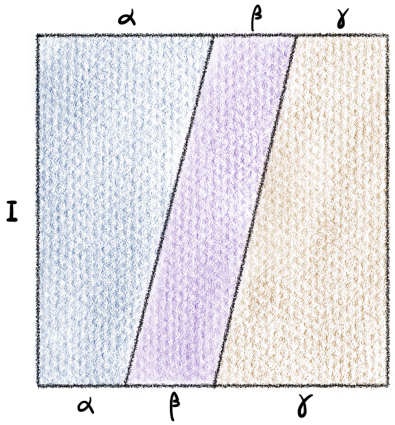


Figure 9.3

□

Now we are ready to define the fundamental group of a space $X$. Take a point $b \in X$ (called the **base point**). We consider the set

$$S := \{\gamma : I \to X \mid \gamma(0) = \gamma(1) = b\}/\text{homotopy}.$$

We have a map

$$\overline{(-)} : S \to S, \text{defined by } [\gamma] \mapsto \overline{[\gamma]}$$

and a map

$$\cdot : S \times S \to S \text{ defined by} ([\alpha], [\beta]) \mapsto [\alpha] \cdot [\beta].$$

**Lemma 9.17.** *$S$ is a group under the multiplication map $\cdot$ and the inverse map $\overline{(-)}$. The identity element equals $[c_b]$. We denote this group by $\pi_1(X, b)$, called **the fundamental group of $X$ based at** $b$.*

*Proof.* By lemma 9.16 (5), the map $\cdot$ is associative. By lemma 9.16 (3), $[c_b]$ is the unit. By lemma 9.16 (4), $\overline{(-)}$ is the inverse. □

We sometimes just write $\pi_1(X)$ when $b$ is clear from the context or when the specific choice of $b$ doesn't matter. The fundamental group is an important invariant for topological spaces.

**Exercise G.** *Let $G$ be a topological group. Let $\alpha, \beta : I \to G$ be two loops based at the identity element $e$.*

*(1) Consider the following loop*

$$\gamma : I \to G, \text{ defined by } \gamma(t) := \alpha(t)\beta(t).$$

*Show that $\gamma \simeq \alpha \cdot \beta$.*

*(2) Consider the loop $\alpha' : I \to G$ defined by $\alpha'(t) = \alpha(t)^{-1}$ for any $t$. Show that $\overline{\alpha} \simeq \alpha'$.*

*(3) We will see that the fundamental group is not abelian in general. However show that $\pi_1(G, e)$ is indeed abelian. (Hint: Given $\alpha, \beta : I \to G$. Consider the map $I \times I \to G$ defined by $(s,t) \mapsto \alpha(s)\beta(t)$.)*



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# Chapter 10

# Covering spaces, the fundamental group of a circle, and applications

The first major goal today is to prove that $\pi_1(S^1) \cong \mathbb{Z}$. Intuitively, this is easy to believe: a homotopy class of a loop is $S^1$ is determined by how many times it "winds around the circle". However, in order to prove this rigorously. We need the help of the following map

$$\pi : \mathbb{R} \to S^1 \text{ defined by } \pi(x) = e^{2\pi i x}. \tag{10.1}$$

Clearly, this map is not a homeomorphism. However, it does belong to a nice family of maps, called "covering maps".

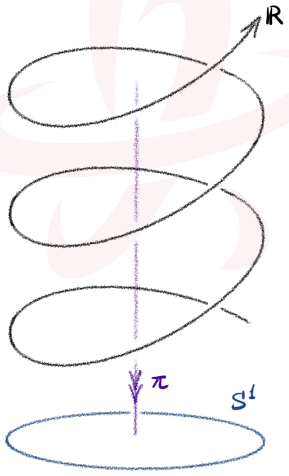


Figure 10.1

## 10.1 Covering spaces

**Definition 10.1.** Let $f : \widetilde{X} \to X$ be a continuous map. We call $f$ a **covering map** if the following two conditions are both satisfied:

- $f$ is surjective;
- Any point $x \in X$ has a neighborhood $U$ that satisfies the following condition: we have a **decomposition** $f^{-1}(U) = \bigcup U_\alpha$ into open subsets of $\widetilde{X}$ such that for any $\alpha$, the restriction $f|_{U_\alpha} : U_\alpha \to U$ is a homeomorphism. (Such $U$ are called evenly covered open sets.)



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If $f^{-1}(x)$ consists of $n$ points for any $x \in X$, then we say $f$ is an $n$-fold covering map. And we say the pair $(\widetilde{X}, f)$ is a covering space of $X$. (Sometimes we ignore $f$ and just say $\widetilde{X}$ is a covering space of $X$.)

**Example 10.2.** *Let $S$ be a **discrete** space. Then the projection map $\mathrm{pr}_1 : X \times S \to X$ is a covering map. Actually, $X$ itself is evenly covered. We call this the **trivial covering map**. Actually, when we restrict a general covering map $f : \widetilde{X} \to X$ to $f^{-1}(U)$ of some evenly covered open set $U$, we get this trivial covering map. (Some book require $A$ to be a single point when talking about trivial covering map. We won't do this here.)*

**Example 10.3.** *The map (eq. (10.1)) is a covering map. To see this, we take any $x_0 \in \mathbb{R}$ and let $U = S \setminus \{e^{2\pi i x_0}\}$. Then $\pi^{-1}(U) = \bigcup_{\alpha\in\mathbb{Z}} U_\alpha$, where $U_\alpha = (x_0 + \alpha, x_0 + \alpha + 1)$. And $\pi|_{U_\alpha} : U_\alpha \to U$ is a homeomorphism. So $U$ is evenly covered. Since we can cover $S^1$ by evenly covered open sets, $f$ is a covering map.*

**Example 10.4.** *Similarly, we can take $\pi_n : S^1 \to S^1$ defined by $z \mapsto z^n$. Then this is an $n$-fold covering map from $S^1$ to itself. This example shows that there can be many different covering maps from a fixed $\widetilde{X}$ to a fixed $X$. (And even from $X$ to itself.)*

**Example 10.5.** *Let $\rho : G \times \widetilde{X} \to \widetilde{X}$ be an action of a **discrete group** on a Hausdorff space $\widetilde{X}$. We call $\rho$ a **covering space action** if for any $x \in \widetilde{X}$, there exists a neighborhood $V$ such that*

$$V \cap gV = \varnothing, \forall g \neq e \in G.$$

*Note that covering space action $\implies$ free action, and for finite group $G$, these two conditions are equivalent.*

*As we will see in exercise A, if $\rho$ is a covering space action, then the quotient map $\widetilde{X} \to \widetilde{X}/G$ is a covering map. For example, the quotient map $S^n \to \mathbb{RP}^n$ is a 2-fold covering map. And $\mathbb{R}^n \to \mathbb{R}^n/\mathbb{Z}^n = T^n$ is a covering map. (When $n = 1$, this is another interpretation of the map (eq. (10.1)).*

**Exercise A.** *Consider an group action $\rho : G \times \widetilde{X} \to \widetilde{X}$. Let $q : \widetilde{X} \to \widetilde{X}/G$ be the quotient map to the orbit space.*

*(1) Suppose $\rho$ is a covering space action. Show that $q$ is a covering map.*

*(2) Suppose $G$ is a finite, $\rho$ is free and $\widetilde{X}$ is Hausdorff. Show that $\rho$ is a covering space action.*

Now we introduce the definition of a lift.

**Definition 10.6.** Let $f : Y \to X$ and $g : Z \to X$ be two continuous maps. We say a continuous map $\widetilde{f} : Y \to Z$ is a **lift** of $f$ (along $g$) if it satisfies the following commutative diagram:

$$\begin{array}{ccc} & & Z \\ & \overset{\widetilde{f}}{\nearrow} & \big\downarrow g \\ Y & \xrightarrow{f} & X \end{array} .$$

Now we state an important property of covering maps.

**Theorem 10.7** (Unique homotopy lifting property)**.** *Let $p : \widetilde{X} \to X$ be a covering map. Let $H : I \times X \to Y$ be a homotopy. Let $\widetilde{f} : Y \to \widetilde{X}$ be a lift of $H(0, -) : Y \to X$. Then there exists a unique $\widetilde{H} : I \times Y \to \widetilde{X}$ that is a homotopy of $\widetilde{f}$ and is a lift of $H$. In other words, there exists a unique $\widetilde{H}$ that fits into the following commutative diagram:*

$$\begin{array}{ccc} Y & \xrightarrow{\widetilde{f}} & \widetilde{X} \\ \big\downarrow \mathrm{in}_0 & \overset{\widetilde{H}}{\nearrow} & \big\downarrow p \\ I \times Y & \xrightarrow{H} & X \end{array} . \tag{10.2}$$



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*Here* $\mathrm{in}_0 : Y \to I \times Y$ *is the "time-0 inclusion map" defined by* $\mathrm{in}_0(y) = (0, y)$.

*Proof.* This is a deep theorem. We will divide the proof in to several steps:

**Step 0. The unique homotopy lifting property (UHLP) holds for trivial covering map** $p : X \times S \to X$: We let $\widetilde{f} = (f, g) : Y \to X \times S$. Then we define $\widetilde{H} : I \times Y \to X \times S$ by

$$\widetilde{H}(t, y) := (H(t, y), g(y)).$$

**Step 1. UHLP holds if** $H(I \times X) \subset U$ **for some evenly covered** $U$: We let $p^{-1}(U) = \bigcup_{\alpha \in S} U_\alpha$. We equip $S$ with the discrete topology. Then we have the following commutative diagram

$$\begin{array}{ccc} U \times S & \xrightarrow{\cong} & p^{-1}(U) \\ & {\scriptstyle \mathrm{pr}_1} \searrow \quad \swarrow {\scriptstyle p} & \\ & U & \end{array}.$$

So the covering map $p|_{p^{-1}(U)} : p^{-1}(U) \to U$ is a trivial covering map.

**Step 2. For any** $y \in Y$**, we can find a neighborhood** $V$ **of** $y$ **such that** $\widetilde{H}$ **exists on** $I \times V$: We divide $I \times \{y\}$ into $\bigcup_{1 \le m \le N} I_m \times \{y\}$, where $I_m = [\frac{m-1}{N}, \frac{m}{N}]$. We can pick $N \gg 0$ so that for any $m$, $H(I_m \times \{y\})$ is contained in some $U_m$ which is evenly covered. (This is a baby case of the lemma 4.28.) Then there exists a neighborhood $V_m$ of $y$ such that $H(I_m \times V_m) \subset U_m$. We let $V = \bigcap_{1 \le m \le N} V_m$. Then $H(I_m \times V) \subset U_m$ for any $m$. By step 1, there exists $\widetilde{H}_1 : I_1 \times Y \to Y$ that lifts $H|_{I_1 \times V}$ and satisfies

$$\widetilde{H}_1(0, -) = \widetilde{f}|_V.$$

By step 1 again, there exists $\widetilde{H}_2 : I_2 \times Y \to Y$ that lifts $H|_{I_2 \times V}$ and satisfies

$$\widetilde{H}_2(\frac{1}{m}, -) = \widetilde{H}_1(\frac{1}{m}, -).$$

We keep doing this and define $\widetilde{H}_m$ for all $m$. Then we glue them together to get $\widetilde{H} : I \times V \to Y$.

**Step 3. Suppose** $Y$ **is a single point** $*$**. Then any two maps** $\widetilde{H}_1, \widetilde{H}_2$ **that both fits into (eq. (10.2)) must be equal**: The proof is similar to Step 2. We break $* \times I$ into $* \times \bigcup I_m$ and prove the uniqueness inductively on $m$.

**Step 4. For a general** $Y$**, any two maps** $\widetilde{H}_1, \widetilde{H}_2$ **that both fits into (eq. (10.2)) must be equal**: By Step 3 for any $y \in Y$, we have $\widetilde{H}_1|_{\{y\} \times I} = \widetilde{H}_2|_{\{y\} \times I}$.

**Step 5. For a general** $Y$**,** $\widetilde{H}$ **exists**: By step 2, for any $y \in Y$, we can find a neighborhood $V_y$ such that there exists $\widetilde{H}_y : I \times V_y \to \widetilde{X}$ that fits into (eq. (10.2)). By Step 4, for any $y, y'$, we have

$$\widetilde{H}_y|_{I \times (V_y \cap V_{y'})} = \widetilde{H}_{y'}|_{I \times (V_y \cap V_{y'})}.$$

So $\{\widetilde{H}_y : I \times V_y \to \widetilde{X}\}_{y \in Y}$ can be glued together and form $\widetilde{H} : I \times Y \to \widetilde{X}$ that fits into (eq. (10.2)). □

**Corollary 10.8.** *Let* $p : \widetilde{X} \to X$ *be a covering map. Take* $a \in X$ *and* $\widetilde{a} \in p^{-1}(a)$*. Then we have the following conclusions:*

1. *For any path* $\gamma : I \to X$ *with* $\gamma(0) = a$*, there exists a unique lift* $\widetilde{\gamma} : I \to \widetilde{X}$ *with* $\widetilde{\gamma}(0) = \widetilde{a}$.
2. $\gamma \simeq \gamma' \implies \widetilde{\gamma} \simeq \widetilde{\gamma}' \implies \widetilde{\gamma}(1) = \widetilde{\gamma}'(1)$.

*Proof.* (1) This follows from theorem 10.7 by setting $Y = \{*\}$.

(2) Let $H : I \times I \to X$ be a homotopy from $\gamma$ to $\gamma'$ (i.e. $H(0, -) = \gamma$ and $H(1, -) = \gamma'$). Then $H$ can also be treated as a homotopy from $H(0, -) = c_a$ to itself. By theorem 10.7 with $Y = I$, there exists $\widetilde{H} : I \times I \to \widetilde{X}$ such that it lifts $H$ and $\widetilde{H}(-, 0) = c_{\widetilde{a}}$. Then $\widetilde{H}(0, -)$ is a lift of $\gamma$ that starts at $\widetilde{a}$, so $\widetilde{H}(0, -) = \widetilde{\gamma}$. Similarly, $\widetilde{H}(1, -) = \widetilde{\gamma}'$. Moreover, let $b = \gamma(1)$. Then $\widetilde{H}(-, 1)$ is a lift of $c_b$ so is a constant path. So we see that $\widetilde{H}$ is a homotopy from $\widetilde{\gamma}$ to $\widetilde{\gamma}'$. □



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Now we are ready to compute $\pi_1(S^1)$. Let $\gamma : I \to S^1$ be a loop and let $\widetilde{\gamma} : I \to \mathbb{R}$ be any lift of $\gamma$. We define the **winding number**

$$w(\gamma) := \gamma(1) - \gamma(0).$$

Given two lifts $\widetilde{\gamma}$ and $\widetilde{\gamma}'$. Assume $\widetilde{\gamma}(0) - \widetilde{\gamma}'(0) = k \in \mathbb{Z}$. We consider a third lift

$$\widetilde{\gamma}''(t) := \widetilde{\gamma}'(t) + k.$$

which is just a translation of $\widetilde{\gamma}'$ by $k$. Since $\widetilde{\gamma}''$ and $\widetilde{\gamma}$ are both lifts and $\widetilde{\gamma}''(0) = \widetilde{\gamma}(0)$, by uniqueness in corollary 10.8, we see that $\widetilde{\gamma}'' = \widetilde{\gamma}$. Therefore, $\widetilde{\gamma}$ and $\widetilde{\gamma}'$ are related by a translation on $\mathbb{R}$ and they give the same winding number. This shows that $w(\gamma)$ is well-defined.

**Example 10.9.** *Consider $\gamma_n : I \to S^1$ defined by $\gamma_n(x) = e^{2\pi i n x}$. Then it has a lift $\widetilde{\gamma}_n : I \to \mathbb{R}$ given by $\widetilde{\gamma}_n(x) = nx$. So $w(\gamma_n) = n - 0 = n$.*

**Theorem 10.10.** *There is a well-defined isomorphism $\rho : \pi_1(S^1, 1) \to \mathbb{Z}$ defined by $\rho([\gamma]) = w(\gamma)$.*

*Proof.* $\rho$ is well-defined: Suppose $\gamma \simeq \gamma'$. We pick the lifts $\widetilde{\gamma}$ and $\widetilde{\gamma}'$ such that $\widetilde{\gamma}(0) = \widetilde{\gamma}'(0)$. Then by corollary 10.8, we have $\widetilde{\gamma}(1) = \widetilde{\gamma}'(1)$. So $w(\gamma) = w(\gamma')$.

$\rho$ is a group homomorphism: Given $\gamma, \gamma'$, we pick $\widetilde{\gamma}$ and $\widetilde{\gamma}'$ such that $\widetilde{\gamma}(1) = \widetilde{\gamma}'(0)$. Then $\widetilde{\gamma} \cdot \widetilde{\gamma}'$ is a lift of $\gamma \cdot \gamma'$ and we have $w(\gamma \cdot \gamma') = w(\gamma) + w(\gamma')$.

$\rho$ is injective: Given $\gamma, \gamma'$ with $w(\gamma) = w(\gamma')$, we pick $\widetilde{\gamma}$ and $\widetilde{\gamma}'$ such that $\widetilde{\gamma}(0) = \widetilde{\gamma}'(0)$. Then $\widetilde{\gamma}(1) = \widetilde{\gamma}'(1)$. Because $\mathbb{R}$ is convex, we have a straight-line homotopy from $\widetilde{\gamma}$ to $\widetilde{\gamma}'$. This induces a homotopy from $\gamma$ to $\gamma'$.

$\rho$ is surjective: $\rho([\gamma_n]) = n$, where $\gamma_n$ is given in example 10.9. □

## 10.2 Basic properties of fundamental group

Now we study some basic properties of fundamental group. First of all, we noticed that in order to compose loops, we have to pick a base point $b \in X$ in the definition of the fundamental group. One may ask the following natural question:

**Question 10.11.** *What's the relation between $\pi_1(X, b)$ and $\pi_1(X, b')$? Are they isomorphic?*

To answer this question, we introduce the following construction: Let $\gamma : I \to X$ be a path from $b'$ to $b$. It induces an isomorphism $\gamma^\sharp : \pi_1(X, b) \to \pi_1(X, b')$ defined by

$$\gamma^\sharp([\alpha]) = [\gamma \cdot \alpha \cdot \overline{\gamma}].$$

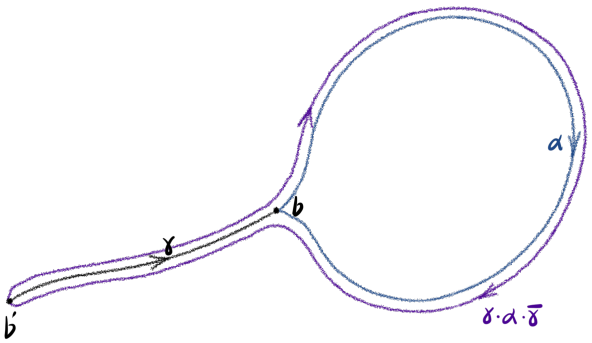


Figure 10.2

The following facts are straightforward:



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(i) $\gamma \simeq \gamma' \implies \gamma^\sharp = \gamma'^\sharp$, so it also makes sense to write $[\gamma]^\sharp$.

(ii) $c_b^\sharp = \mathrm{Id}_{\pi_1(X,b)}$.

(iii) $(\gamma \cdot \gamma')^\sharp = \gamma^\sharp \circ \gamma'^\sharp$.

Now given a path $\gamma$ from $b$ to $b'$, consider the maps

$$\gamma^\sharp : \pi_1(X,b) \to \pi_1(X,b') \text{ and } \gamma'^\sharp : \pi_1(X,b') \to \pi_1(X,b).$$

By (i) $\sim$ (iii), we see that $(\gamma^\sharp)^{-1} = (\overline{\gamma})^\sharp$. In particular, we have proved the following important result:

**Proposition 10.12.** *Any homotopy class of path $[\gamma]$ from $b'$ to $b$ induces an isomorphism $[\gamma]^\sharp : \pi_1(X,b) \to \pi_1(X,b')$. In particular, for different choice of base points in the same path-component, the resulting fundamental group are isomorphic.*

proposition 10.12 implies that up to isomorphism, $\pi_1(X,b)$ only depends on the path-component of $b$. In particular, if $X$ is path-connected and we only care about the isomorphism type of $\pi_1(X,b)$ (rather than a specific element in $\pi_1(X,b)$), we can safely ignore $b$ from our notation.

*Remark* 10.13. In general, if we choose the base points in different path-components, the resulting fundamental group will not be related to each other. Also, even if we choose $b, b'$ to be in the same path-component, there is no "canonical isomorphism" between $\pi_1(X,b)$ and $\pi_1(X,b')$. This means that we have to pick a homotopy class $[\gamma]$ of path from $b'$ to $b$ in order to define this isomorphism $[\gamma]^\sharp$. And if we choose a different homotopy class, the isomorphism will be be changed by **a conjugation**. Actually, let $\gamma, \gamma' : I \to X$ be paths from $b'$ to $b$ that are not homotopic to each other. Consider the loop $\gamma \cdot \overline{\gamma'}$ based at $b'$. Then one can check

$$\gamma^\sharp([\alpha]) = [\gamma \cdot \overline{\gamma'}] \cdot \gamma'^\sharp([\alpha]) \cdot \overline{[\gamma \cdot \overline{\gamma'}]}, \quad \forall [\alpha] \in \pi_1(X,b).$$

In particular, if $\pi_1(X,b)$ is abelian, then this ambiguity does not exist and we indeed have a natural isomorphism $\pi_1(X,b) \cong \pi_1(X,b')$.

What makes the fundamental group so useful is that it is a "functor". We won't explain the precise meaning of this terminology in category theory. In our context, it means the following:

(i) Any continuous map $f : X \to Y$ induces a group homomorphism $f_* : \pi_1(X,b) \to \pi_1(Y, f(b))$ between the fundamental groups, defined by

$$f_*([\alpha]) = [f \circ \alpha],$$

where $\alpha : I \to X$ is a loop in $X$ based at $b$;

(ii) If $f = \mathrm{Id}_X$, then $f_* = \mathrm{Id}_{\pi_1(X,b)}$;

(iii) $(f \circ g)_* = f_* \circ g_*$.

Here is another natural question:

**Question 10.14.** *Suppose $f \simeq g$. What's the relation between the maps $f_* : \pi_1(X,b) \to \pi_1(Y, f(b))$ and $g_* : \pi_1(X,b) \to \pi_1(Y, g(b))$?*

A natural guess is that "$f \simeq g \implies f_* = g_*$". This turns out to be roughly correct. But there is an obvious problem here: $f(b)$ and $g(b)$ may be different, so these two maps may have different codomain. The following proposition resolves this issue.



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**Proposition 10.15.** *Let $H : I \times X \to Y$ be a homotopy from $f$ to $g$. Consider the path $\gamma = H(-, b) : I \to Y$ from $f(b)$ to $g(b)$. Then the following diagram commutes:*

$$\begin{array}{ccc} & & \pi_1(Y, g(b)) \\ & \overset{g_*}{\nearrow} & \cong \big\downarrow \gamma^\sharp \\ \pi_1(X, b) & \xrightarrow{f_*} & \pi_1(Y, f(b)) \end{array}.$$

*Proof.* Take $[\alpha] \in \pi_1(X, b)$. Then $g \circ \alpha$ and $f \circ \alpha$ are loops in $Y$ based at $g(b)$ and $f(b)$ respectively. Then $\gamma^\sharp(g_*[\alpha])$ is represented by the loop $\gamma \cdot (g \circ \alpha) \cdot \overline{\gamma}$, while $f_*([\alpha])$ is represented by the loop $f \circ \alpha$. To see these two loops are homotopic to each other, we consider the following composition

$$H' : I \times I \to I \times X \xrightarrow{H} Y$$

where the first map sends $(t, s)$ to $(t, \alpha(s))$. We consider the path $\beta : I \to I \times I$ defined by $\beta(t) = (0, t)$ and the path $\beta' : I \to I \times I$ defined by

$$\beta'(t) = \begin{cases} (3t, 0) & \text{if } t \in [0, \frac{1}{3}] \\ (1, 3t - 1) & \text{if } t \in [\frac{1}{3}, \frac{2}{3}] \\ (3 - 3t, 1) & \text{if } t \in [\frac{2}{3}, 1] \end{cases}$$

(Geometrically, $\beta$ goes along the bottom edge and $\beta'$ goes around the other three edges. Draw a picture.) It's straight forward to check that

$$H' \circ \beta = f \circ \alpha, \quad H' \circ \beta' = \gamma \cdot (g \circ \alpha) \cdot \overline{\gamma}.$$

Since $\beta \simeq \beta'$, we have $f \circ \alpha \simeq \gamma \cdot (g \circ \alpha) \cdot \overline{\gamma}$. □

proposition 10.15 has the following extremely useful corollary.

**Corollary 10.16.** *Let $f : X \to Y$ be a map and let $b \in X$ be a point. Consider the induced map $f_* : \pi_1(X, b) \to \pi_1(Y, f(b))$. Then we have:*

1. *If $f$ is a homotopy equivalence, then $f_*$ is an isomorphism.*
2. *If $f$ is null-homotopic, then $f_* = 0$.*

*Proof.* (1) We first assume that $X = Y$ and $f \simeq \mathrm{Id}_X$. Because $\mathrm{Id}_X$ induces the identity map on $\pi_1(X, b)$. By proposition 10.15, $f_* : \pi_1(X, b) \to \pi_1(X, f(b))$ is also an isomorphism.

Now we consider the general case. Let $g : Y \to X$ be the homotopy inverse of $f$. Then we have $g \circ f \cong \mathrm{Id}_X$. Consider the induced map

$$g_* : \pi_1(Y, f(b)) \to \pi_1(X, g(f(b))).$$

Then by previous case, $g_* \circ f_*$ is an isomorphism. So $f_*$ is injective. Similarly, we can the fact that $f \circ g \cong \mathrm{Id}_Y$ to prove $f_*$ is surjective.

(2) By definition, the constant map induces the trivial map between the fundamental groups. So our conclusion follows from proposition 10.15 again. □

**Corollary 10.17.** *Let $A$ be a subspace of $X$ and let $i : A \to X$ be the inclusion map. We pick a base point $b \in A$ and consider the induced map $i_* : \pi_1(A, b) \to \pi_1(X, b)$. Then we have:*

1. *Suppose $A$ is a deformation retract of $X$. Then $i_*$ is an isomorphism.*
2. *Suppose $A$ is a retract of $X$. Then $i_*$ has a left inverse. In particular, $i_*$ is a injection.*



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*Proof.* (1) By lemma 9.13, $i$ is a homotopy equivalence. So $i_*$ is an isomorphism by corollary 10.16.

(2) Let $r : X \to A$ be a retraction from $X$ to $A$. Then $r \circ i = \mathrm{Id}_A$. That means $r_* \circ i_* = \mathrm{Id}_{\pi_1(A,b)}$. So $r_*$ is a left inverse of $i_*$. □

**Example 10.18.** *$\mathbb{R}^2 \setminus \{0\}$ and $D^2 \setminus \{0\}$ both deformation retracts to $S^1$. So*

$$\pi_1(\mathbb{R}^2 \setminus \{0\}) \cong \pi_1(D^2 \setminus \{0\}) \cong \pi_1(S^1) \cong \mathbb{Z}.$$

*Both isomorphisms are given by the winding number. (In general, a winding number of a loop $\gamma : I \to \mathbb{R}^2 \setminus \{0\}$ is defined as the winding number of the loop $t \mapsto \frac{\gamma(t)}{|\gamma(t)|}$ in $S^1$.)*

**Definition 10.19.** A space $X$ is called **simply-connected** if $X$ is path-connected and $\pi_1(X) = 0$.

**Example 10.20.** *Contractible spaces are all simply connected. So $\mathbb{R}^n, D^n$ are all simply-connected. We will see that when $n \geq 2$, $S^n$ is simply-connected but not contractible.*

**Corollary 10.21.** *Let $A$ be a retract of $X$. Then $X$ is simply-connected $\implies$ $A$ is simply-connected.*

*Proof.* We use $\pi_0(X)$ to denote the set of path-components of $X$. Then similar to $\pi_1$, the inclusion $i : A \to X$ induces a map $i_* : \pi_0(A) \to \pi_1(X)$. When $A$ is a retract of $X$, the retraction gives a left inverse of $i_*$. So $i_*$ is injective. Therefore, if $X$ is path-connected, $A$ must also be path-connected.

Similarly, the induced map $i_* : \pi_1(A) \to \pi_1(X)$ is an injective by corollary 10.17. So $\pi_1(A) = 0$. So $A$ is simply connected. □

**Exercise B.** *Given a continuous map $f : S^1 \to S^1$, it induces a homorphism*

$$f_* : \mathbb{Z} \cong \pi_1(S^1, 1) \to \pi_1(S^1, f(1)) \cong \mathbb{Z}$$

*. We Using winding number to identify the fundamental groups with $\mathbb{Z}$. The mapping degree, denoted by $\deg(f)$ is defined to be the integer $f_*(1) \in \mathbb{Z}$.*

*(1) Suppose $f$ is even. (I.e. $f(-x) = f(x)$.) Show that $\deg(f)$ is even.*

*(2) Suppose $f$ is odd. (I.e. $f(-x) = -f(x)$.) Show that $\deg(f)$ is odd. (Hint: check the proof of the Borsuk-Ulam theorem (Theorem 1.10 in Hatcher).)*

**Exercise C.** *Show that there does not exists an odd, continuous function from $f : S^2 \to S^1$. (Again, odd means that $f(-x) = -f(x)$.)*

## 10.3 Some applications of $\pi_1(S^1)$

Now we are ready to give some important applications of the fact that $\pi_1(S^1) \cong \mathbb{Z}$. As an immediate corollary of corollary 10.21, we get:

**Corollary 10.22.** *$S^1$ is **not** a retract of $D^2$ or $\mathbb{R}^2$.*

By the argument in the proof of corollary 1.5. This implies the Brouwer fixed point theorem for $D^2$.

**Theorem 10.23.** *Any continuous map $f : D^2 \to D^2$ has a fixed point.*

Next, we give a topological proof of the fundamental theorem of algebra:

**Theorem 10.24.** *Any non-constant polynomial $p : \mathbb{C} \to \mathbb{C}$ has a root.*



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*Proof.* Suppose $p$ does not have a root. Then it gives a map $p : \mathbb{C} \to \mathbb{C} \setminus \{0\}$. Without loss of generality, we may assume

$$p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0.$$

For any $R > 0$, we can define two maps

$$f_R : S^1 \to \mathbb{C} \setminus \{0\}, \quad f_R(z) = p(Rz),$$

and

$$g_R : S^1 \to \mathbb{C} \setminus \{0\}, \quad g_R(z) = (Rz)^n.$$

When $R > 1 + |a_{n-1}| + \cdots |a_0|$, the straight-line homotopy $H : I \times S^1 \to \mathbb{C} \setminus \{0\}$

$$H(t, z) := tf_R(z) + (1 - t)g_R(z)$$

is well-defined (because the image doesn't hit 0), so $f_R \simeq g_R$. Since $f_R$ can be extended to $D^2$. $f_R$ is null-homotopic. So $g_R$ is also null-homotopic. However, this is impossible, because

$$(g_R)_* : \mathbb{Z} \cong \pi_1(S^1) \to \pi_1(\mathbb{C} \setminus \{0\}) \xrightarrow{\cong} \pi_1(S^1) \cong \mathbb{Z}$$

sends 1 to $n$. □

The next application concerns the boundary points of a surface with boundary. Recall that for a surface $S$ with boundary. A point $x \in S$ is called a boundary point if it has a neighborhood homeomorphic to $[0, +\infty) \times \mathbb{R}$ but it doesn't a neighborhood homeomorphic to $\mathbb{R}^2$. To make sense of this definition, we need the following result:

**Proposition 10.25.** *$[0, +\infty) \times \mathbb{R}$ is not homeomorphic to any open subset of $\mathbb{R}^2$. (In particular, it is not homeomorphic to $\mathbb{R}^2$.)*

*Proof.* Let $U$ be a nonempty open subset of $\mathbb{R}^2$. Take any point $x \in U$. We claim that $U \setminus \{x\}$ is not simply-connected. To see this, we pick a small ball $D_\epsilon(x) \subset U$. Then the composition of inclusion maps

$$\partial D_\epsilon(x) \hookrightarrow U \setminus \{x\} \hookrightarrow \mathbb{R}^2 \setminus \{x\}$$

is a homotopy equivalence, so it induces an isomorphism

$$\mathbb{Z} = \pi_1(\partial D_\epsilon(x)) \xrightarrow{\cong} \pi_1(\mathbb{R}^2 \setminus \{x\}) = \mathbb{Z}.$$

Since this isomorphism factors through $\pi_1(U \setminus \{x\})$, we see that $U \setminus \{x\}$ is not simply-connected.

However, we consider the space $([0, +\infty) \times \mathbb{R}) \setminus \{0\}$. It deformation retracts to the half-circle $S^1 \cap ([0, +\infty) \times \mathbb{R}) \cong I$. So $([0, +\infty) \times \mathbb{R}) \setminus \{0\}$ is contractible. □

**Corollary 10.26.** *Let $S$ be a surface with boundary. Then a point $x \in S$ is a boundary point if and only if there is a neighborhood $U$ of $x$ and a homeomorphism $f : U \to [0, +\infty) \times \mathbb{R}$ with $f(x) = 0$.*

*Proof.* Suppose $x$ is a boundary point. Then by definition, there exists a neighborhood $U$ of $x$ and a homeomorphism $f : U \to [0, +\infty) \times \mathbb{R}$. Suppose $f(x) \notin \{0\} \times \mathbb{R}$. Then $f^{-1}((0, +\infty) \times \mathbb{R})$ is a neighborhood of $x$ homeomorphic to $\mathbb{R}^2$. This is not possible by our definition of boundary points. So $f(x) \in \{0\} \times \mathbb{R}$. After a composing $f$ with a suitable translation, we may arrange that $f(x) = 0$.

To prove the other direction, we assume there is a neighborhood $U$ and a homeomorphism $f : U \to [0, +\infty) \times \mathbb{R}$ such that $f(x) = 0$. It suffices to prove that $x$ has no neighborhood homeomorphic to $\mathbb{R}^2$. Suppose there were such a neighborhood $V$. We consider the set

$$A = \{x \in [0, +\infty) \times \mathbb{R} \mid |x| < \epsilon\}.$$



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Then $f^{-1}(A)$ and $A$ are both homeomorphic to $[0, +\infty) \times \mathbb{R}$. For $\epsilon$ small enough, $f^{-1}(A)$ is an open subset of $V$. So $f^{-1}(A)$ is homeomorphic to an open subset of $\mathbb{R}^2$. This contradicts with proposition 10.25. □

We end this section by noting that the Jordan curve theorem and the invariance of domain (in dimension 2) can also be proved using $\pi_1(S^1) \cong \mathbb{Z}$. The argument is a little involved so we will not discuss it here. See Armstrong Chapter 5. In a future algebraic topology class, you will learn a proof that works in any dimension.

**Exercise D.** *Consider two nonconstant, polynomial maps $f, g : \mathbb{C} \to \mathbb{C}$. Suppose the degree of $f$ does not equals the degree of $g$. Show that there exists no proper homotopy homotopy $H : I \times \mathbb{C} \to \mathbb{C}$ from $f$ to $g$. Recall that proper means the preimage of any compact set is compact.*



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# Chapter 11

# Seifert-Van Kampen theorem

We have learnt how to show $\pi_1(S^1) \cong \mathbb{Z}$. What about $\pi_1(S^n)$ when $n \geq 2$ ? Today, we will learn a powerful tool in computing the fundamental group, called the **Seifert-Van Kampen theorem**. Using this theorem, we can compute $\pi_1(X)$ by taking an open cover $X = \bigcup_{\alpha \in S} U_\alpha$ (that satisfies some mild conditions) and recover $\pi_1(X)$ from $\pi_1(U_\alpha)$.

## 11.1 Fundamental group of a product

Before talking about Seifert-Van Kampen theorem, let's start with the easier case of a product.

Since a base point is involved in the definition of a fundamental group, $\pi_1(-)$ is really an invariant of a **pointed space**. Here a pointed space $(X, x)$ means a topological space $X$ together with a chosen base point $x$. The notion of product can be easily generalized to pointed spaces

$$\prod_{\alpha \in S} (X_\alpha, x_\alpha) := (\prod_{\alpha \in S} X_\alpha, (x_\alpha))$$

In other words, we take the product space and choose the base point to be the product of the base points.

**Theorem 11.1.** $\pi_1(\prod_{\alpha \in S} X_\alpha, (x_\alpha)) = \prod_{\alpha \in S} \pi_1(X_\alpha, x_\alpha)$.

*Proof.* We will focus on the case $S = \{a, b\}$. The general case is essentially the same. Let $p_a : X_a \times X_b \to X_a$ and $p_b : X_a \times X_b \to X_b$ be the projection map.

By the universal property of product, we have a bijection

$$\{\text{loops in } X_a \times X_b \text{ based at } (x_a, x_b)\} \leftrightarrow \{\text{loops in } X_a \text{ based at } x_a\} \times \{\text{loops in } X_b \text{ based at } x_b\}$$

given by

$$\gamma \leftrightarrow (p_a \circ \gamma, p_b \circ \gamma).$$

Furthermore, by the universal property again, we have a bijection

$$\{\text{homotopies } I \times I \to X_a \times X_b\} \leftrightarrow \{\text{homotopies } I \to X_a\} \times \{\text{homotopies } I \to X_b\}.$$

So $\gamma \simeq \gamma' \iff p_a \circ \gamma \simeq p_a \circ \gamma'$ and $p_b \circ \gamma \simeq p_b \circ \gamma'$. Therefore, we see that the map

$$(p_a^*, p_b^*) : \pi_1(X_a \times X_b, (x_a, x_b)) \to \pi_1(X_a, x_a) \times \pi_1(X_b, x_b).$$

is an isomorphism. □

**Example 11.2.** $\pi_1(T^n) \cong \pi_1(S^1) \times \cdots \times \pi_1(S^1) \cong \mathbb{Z}^n$.

**Exercise A.** *Let $X = S^1$. Show that there does not exist a continuous map $f : X \times X \to X$ that satisfies both of the following conditions:*



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小红书号:3189205503 qq

*(i)* $f(x,y) = f(y,x)$ *for any* $x, y$.

*(ii)* $f(x,x) = x$ *for any* $x$.

*(This is a toy case of Theorem 1.15 on social choice function.)*

At this point, it's natural to ask about the topology of a disjoint union. Recall that

$$\pi_1(X,x) \cong \pi_1(\text{the path-component of } X \text{ that contains } x, x)$$

So we see that

$$\pi_1(X_a \bigsqcup X_b, x_a) \cong \pi_1(X_a, x_a), \quad \pi_1(X_a \bigsqcup X_b, x_b) \cong \pi_1(X_b, x_b).$$

This answer is quite boring. The problem here is that disjoint union is not a well-defined operation on pointed spaces because it will produce multiple base points (and we don't know which one to choose). To resolve this issue, we just take a quotient space of the disjoint that identifies all base points as a single one.

**Definition 11.3.** Let $\{(X_\alpha, x_\alpha)\}_{\alpha\in S}$ be a collection of pointed spaces. The one-point union is defined as

$$\bigvee_{\alpha\in S} X_\alpha := (\bigsqcup_{\alpha\in S} X_\alpha)/\sim$$

where $\sim$ is generated by $x_\alpha \sim x_\beta$ for all $\alpha, \beta \in S$. And the base point in $\bigvee_{\alpha\in S} X_\alpha$ is chosen to be $[x_\alpha]$.

Using Seifert-Van Kampen theorem, we will see that under some mild assumptions, we have

$$\pi_1(\bigvee_{\alpha\in S} X_\alpha, [x_\alpha]) \cong \ast_{\alpha\in S} \pi_1(X_\alpha, x_\alpha).$$

Here $\ast$ denotes the free product. Let's review this important notion in group theory first.

## 11.2 Free product of groups

Let $\{G_\alpha\}_{\alpha\in S}$ be a collection of groups. We make a series of definitions:

1. A **word** in $\{G_\alpha\}_{\alpha\in S}$ is a finite sequence $(g_1, \cdots, g_n)$ such that $g_i \in \bigsqcup_{\alpha\in S} G_\alpha$ for any $i$. The integer $n$ is called the **length** of the word and each $g_i$ is called a letter of the word. The empty word, denoted by (), is the unique word of length 0.

2. We treat $\{G_\alpha\}_{\alpha\in S}$ as disjoint subsets of $\bigsqcup_{\alpha\in S} G_\alpha$. Then each letter $g_i$ in a word belongs to a group $G_{\alpha_i}$ for a unique $\alpha_i \in S$. We say the word $(g_1, \cdots, g_n)$ is **reduced** if any $i$, we have $\alpha_i \neq \alpha_{i+1}$ and $g_i \neq e_{\alpha_i}$. Here $e_{\alpha_i}$ denotes the identity element in $G_{\alpha_i}$. In other words, a word is reduced if all letters are not identity elements, and two adjacent letters do not belong to the same group. (The empty word is reduced.)

3. Consider the set
$$\mathcal{W} := \{\textbf{all words in } \{G_\alpha\}_{\alpha\in S}\}.$$
We let $\sim$ be the equivalence relation on $\mathcal{W}$ generated by the following two collection of relations:

   - $(\cdots, g_i, g_{i+1}, \cdots) \sim (\cdots, g_i g_{i+1}, \cdots)$ if $\alpha_i = \alpha_{i+1}$. That means if two adjacent letters belong to the same group, then we can "merge" them into a single letter, which is their product.



小红书
小红书号:3189205503 qq

- $(\cdots, g_{i-1}, g_i, g_{i+1}, \cdots) \sim (\cdots, g_{i-1}, g_{i+1}, \cdots)$ if $g_i = e_{\alpha_i}$. That means we can remove the identity element from the word.

Note that each equivalent class contains exactly one reduced word.

4. Given two words $(g_1, \cdots, g_n)$ and $(h_1, \cdots, h_m)$, we define their product as the word $(g_1, \cdots, g_n, h_1, \cdots, h_m)$. This induces a well-defined binary operation on

$$\cdot : (\mathcal{W}/\sim) \times (\mathcal{W}/\sim) \to (\mathcal{W}/\sim).$$

**Lemma 11.4.** *$(\mathcal{W}/\sim, \cdot)$ is a group. The identity element is given by $[(\varnothing)]$ and the inverse is given by*

$$[(g_1, \cdots, g_n)]^{-1} = [(g_n^{-1}, \cdots, g_1^{-1})].$$

*We call this group the **free product** of $\{G_\alpha\}_{\alpha \in S}$ and denote it by $\underset{\alpha \in S}{*} G_\alpha$.*

From now on, we will always use $g_1 g_2 \cdots g_n$ to denote the element $[(g_1, g_2, \cdots, g_n)] \in \underset{\alpha \in S}{*} G_\alpha$.

**Example 11.5.** *Let $G_\alpha = \mathbb{Z}$ for any $\alpha \in S$. Then $\underset{\alpha \in S}{*} \mathbb{Z}$ is called the **free group** generated by $S$ and denoted by $F_S$. An element in $F_S$ can be formally written as $\alpha_1^{a_1} \alpha_2^{a_2} \cdots \alpha_n^{a_n}$, where $\alpha_i \in S$ and $a_i \in \mathbb{Z}$. (Here we use $\alpha$ to denote the the generator of $G_\alpha = \mathbb{Z}$. So $\alpha^a$ means $a \in G_\alpha$.) When $S = \{1, \cdots, n\}$, we denote $F_S$ by $F_n$ and call it the free group of rank $n$.*

Note that for any $\beta \in S$, we can treat $G_\beta$ as a subgroup of $\underset{\alpha \in S}{*} G_\alpha$ via the inclusion map

$$i_\beta : G_\beta \to \underset{\alpha \in S}{*} G_\alpha, \quad i_\beta(g) := [(g)].$$

The following lemma summarizes the universal property of the free product of groups.

**Lemma 11.6** (Universal property of free product)**.** *Let $H$ be a group and let $\{f_\alpha : G_\alpha \to H\}_{\alpha \in S}$ be a collection of group homomorphisms. Then there exists a unique group homomorphism*

$$\underset{\alpha \in S}{*} f_\alpha : \underset{\alpha \in S}{*} G_\alpha \to H$$

*such that the following diagram commutes for any $\beta \in S$*

$$\begin{array}{ccc} \underset{\alpha \in S}{*} G_\alpha & & \\ {\scriptstyle i_\beta}\uparrow & \searrow^{\underset{\alpha \in S}{*} f_\alpha} & \\ G_\beta & \xrightarrow{f_\beta} & H \end{array} . \tag{11.1}$$

*Proof.* We define $\underset{\alpha \in S}{*} f_\alpha(g_1 g_2 \cdots g_n) := f_{\alpha_1}(g_1) f_{\alpha_2}(g_2) \cdots f_{\alpha_n}(g_n)$. It's straightforward to check that this is the only map that fits the diagram. □

*Remark* 11.7. You may notice that lemma 11.6 is very similar to the universal property of the disjoint union of spaces (lemma 3.6). This is not a coincidence. In category theory, they are both special cases of a much more general construction called "colimit".

Next, we recall the notation of the normal closure:

**Definition 11.8.** Let $A$ be a subset of a group $G$. The normal closure of $A$ in $G$, denoted by $N_G(A)$, is defined as the subgroup that consists of the product of finitely many elements in the set

$$\{e\} \cup \{gag^{-1} \mid g \in G, a \in A\} \cup \{ga^{-1}g^{-1} \mid g \in G, a \in A\}.$$

Note that $N_G(A)$ equals the intersection of all normal subgroups of $G$ that contains $A$. In particular, $N_G(A)$ itself is a normal subgroup of $G$.



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**Definition 11.9.** Given a set $S$ and a subset $R$ of the free group $F_S$, we defined **the group generated over $S$ with relation $R$** as the quotient group

$$\langle S \mid R\rangle := F_S/(N_{F_S}(R)).$$

In other words, $\langle S \mid R\rangle$ is obtained from $F_S$ by imposing the relations $r = e$ for any $r \in R$.

**Example 11.10.** *(1)* $\langle a, b \mid aba^{-1}b^{-1}\rangle \cong \mathbb{Z} \oplus \mathbb{Z}$.
*(2)* $\langle a, b \mid a^2, b^2\rangle \cong \mathbb{Z}/2 * \mathbb{Z}/2$.
*(3)* $\langle a, b \mid ab, aba\rangle \cong \langle a, b \mid ab, a\rangle \cong \langle a, b \mid b, a\rangle \cong \{e\}$.
*(4) Consider the even permutation group $A_5$ and its two elements*

$$a = (1, 2)(3, 4), \quad b = (1, 2, 3, 4, 5).$$

*Then $a^2 = b^5 = (ab)^3 = e$. So we have a homomorphism $\langle a, b \mid a^2, b^5, (ab)^3\rangle \to A_5$, which can be proved to be an isomorphism.*

**Definition 11.11.** A presentation of a group $G$ is an isomorphism $G \cong \langle S \mid R\rangle$ for some $S$ and $R$. If both $S$ and $R$ are finite, we say it is a finite presentation. The rank of $G$ is the minimal size of $S$ such that $G \cong \langle S \mid R\rangle$. We say $G$ is finitely generated if its rank is finite, and we say $G$ is finitely presentable if it has a finite presentation.

**Example 11.12.** *The group $\langle a, b \mid [a^{-n}ba^n, b]$ for all $n \in \mathbb{Z}\rangle$ is finitely generated but not finitely presentable. (The proof is hard. See [***Stackexchange1***]) Here $[g, h] = ghg^{-1}h^{-1}$.*

It seems tempting to study a group by studying its presentations. However, most of the time this is difficult, due to the fact that the presentation of a group is far from unique. Actually, there is the following famous problem:

**Question 11.13** (The uniform word problem)**.** *Given a finite presentation $\langle S \mid R\rangle$, determine whether $\langle S \mid R\rangle$ is isomorphic to the trivial group $\{e\}$.*

**Theorem 11.14** (Adian-Rabin 1955-1958)**.** *The uniform word problem is undecidable.*

A problem is decidable we can construct a Turing machine which will halt in finite amount of time for every input and give answer as 'yes' or 'no'. theorem 11.14 has the following amazing application in topology.

**Theorem 11.15** (Markov 1958)**.** *For any $n \geq 4$, the problem of deciding whether two $n$-dimensional closed manifolds are homeomorphic is undecidable. In particular, a complete classification result of closed manifold of dimension $\geq 4$ does not exist.*

*Idea of proof.* Given any element $w$ of a finite presented group $G = \langle S \mid R\rangle$, there is an explicit construction of an $n$-dimensional manifold $X_\omega$ such that:

- If $w = e \in G$, then $X_w \cong S^n$;
- If $w \neq e \in G$, then $X_w$ is not-simply connected. (As we will see, this implies that $X_w \ncong S^n$.)

So the uniform word problem can be reduced to checking whether the manifolds

$$X_{s_1}, \cdots, X_{s_n},$$

are all homeomorphic to $S^4$. Here $S = \{s_1, \cdots, s_n\}$. Since the uniform word problem is undecidable, the manifold classification problem is also undecidable. □



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## 11.3 Proof of the Seifert-Van Kampen theorem

Now we are ready to state and prove the Seifert-Van Kampen theorem. Our setting is as follows: Let $X$ be a space with an open cover $\bigcup_{\alpha\in S} A_\alpha$. We assume $\bigcap_{\alpha\in S} A_\alpha \neq \varnothing$ and we pick a base point $x_0 \in \bigcap_{\alpha\in S} A_\alpha$. For any $\alpha, \beta \in S$, we let

$$f_\alpha : \pi_1(A_\alpha, x_0) \to \pi_1(X, x_0)$$

be the map induced by the inclusion $A_\alpha \hookrightarrow X$ and let

$$f_{\alpha,\beta} : \pi_1(A_\alpha \cap A_\beta, x_0) \to \pi_1(A_\alpha, x_0)$$

be the map induced by the inclusion $A_\alpha \cap A_\beta \hookrightarrow A_\alpha$. Using lemma 11.6, we obtain a map

$$\underset{\alpha\in S}{*} f_\alpha : \underset{\alpha\in S}{*} \pi_1(A_\alpha, x_0) \to \pi_1(X, x_0)$$

**Lemma 11.16.** *Consider the set*

$$B = \bigcup_{\beta,\beta'\in S} \{f_{\beta,\beta'}(g) \cdot (f_{\beta',\beta}(g))^{-1} \mid g \in \pi_1(A_\beta \cap A_{\beta'}, x_0)\}.$$

*Then* $B \subset \ker(\underset{\alpha\in S}{*} f_\alpha)$.

*Proof.* For any $g \in \pi_1(A_\beta \cap A_{\beta'}, x_0)$, we have $f_{\beta'}(f_{\beta,\beta'}(g)) = f_\beta(f_{\beta,\beta'}(g))$ because they both equal the image of $g$ under the map

$$\pi_1(A_\beta \cap A_{\beta'}, x_0) \to \pi_1(X, x_0)$$

induced by the inclusion $A_\beta \cap A_{\beta'} \hookrightarrow X$. By (eq. (11.1)), this implies that

$$(\underset{\alpha\in S}{*} f_\alpha)(f_{\beta,\beta'}(g)) = (\underset{\alpha\in S}{*} f_\alpha)(f_{\beta',\beta}(g)).$$

□

Let $N(B)$ be the normal closure of $B$ in $\underset{\alpha\in S}{*} \pi_1(A_\alpha, x_0)$. By lemma 11.16, we have a well-defined map

$$\varphi : (\underset{\alpha\in S}{*} \pi_1(A_\alpha, x_0))/N(B) \to \pi_1(X, x_0).$$

**Theorem 11.17** (Seifert-Van Kampen)**.** *Given an open cover $X = \bigcup_{\alpha\in S} A_\alpha$ with $\bigcap_\alpha A_\alpha \neq \varnothing$, pick any $x_0 \in \bigcap_\alpha A_\alpha$. Then the following conclusions hold:*

1. *Suppose $A_\alpha, A_\alpha \cap A_\beta$ is path-connected for any $\alpha, \beta \in S$. Then $\varphi$ is surjective.*
2. *Suppose $A_\alpha, A_\alpha \cap A_\beta, A_\alpha \cap A_\beta \cap A_\delta$ is path-connected for any $\alpha, \beta, \delta \in S$. Then $\varphi$ is an isomorphism.*

*Remark* 11.18. The idea underlying theorem 11.17 is simple and intuitive: to show $\varphi$ is surjective, we break a loop $\gamma : I \to X$ into small paths such that each of them is contained in some $A_\alpha$. To show $\varphi$ is injective, we break a homotopy $H : I \times I \to X$ into small pieces such that each of them is contained in some $A_\alpha$. However, the actual proof is tedious, largely due to the fact we are working with loops instead of general paths. As we have seen, this is necessary in order to get a group structure. There is a ways to break this limitation by working with the so-called "fundamental groupoid", which takes all paths into account and doesn't involve a choice of base point. (Groupoid is a definition in category theory that generalizes the notation of a group.) In [**May**], there is a categorical proof of theorem 11.17 for fundamental groupoid.



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小红书号:3189205503 qq

*Proof.* We fix a base point $x_0$ and ignore it from our notations. We make a few definitions:

- Let $\gamma : I \to X$ be a loop based at $x_0$. A decomposition of $\gamma$ is a sequence $0 = t_0 < t_1 < \cdots < t_n = 1$ such that for any $1 \leq i \leq n$, we have $\gamma([t_{i-1}, t_i]) \subset A_{\alpha_i}$ for some $A_{\alpha_i}$.
- A factorization of an element $g \in \pi_1(X)$ is a sequence $(g_1, \cdots, g_n)$ such that $g_i \in \pi_1(A_{\alpha_i})$ and such that $g = f_{\alpha_1}(g_1) \cdots f_{\alpha_n}(g_n)$.

Now we prove (1). Let $g = [\gamma] \in \pi_1(X)$. Then $g$ is in the image of $\varphi$ if and only if $g$ has a factorization. By using the Lebesgue lemma (lemma 4.28) on the open cover $\{\gamma^{-1}(A_\alpha)\}_{\alpha \in S}$ of $I$, we see that $\gamma$ has a decomposition $(t_0, \cdots, t_n)$ such that $\gamma|_{[t_{i-1}, t_i]} \subset A_{\alpha_i}$. For $1 \leq i \leq n-1$, we have $\gamma(t_i) \subset A_{\alpha_i} \cap A_{\alpha_{i+1}}$. By our assumption that $A_{\alpha_i} \cap A_{\alpha_{i+1}}$ is path-connected, we can find a path $\xi_i$ in $A_{\alpha_i} \cap A_{\alpha_{i+1}}$ from $x_0$ to $\gamma(t_i)$. We also let $\xi_0$ and $\xi_n$ be $c_{x_0}$ (the constant path at $x_0$). Consider the composition

$$\gamma_i := \xi_{i-1} \cdot \gamma|_{[t_{i-1}, t_i]} \cdot \xi_i^{-1}.$$

Then $\gamma_i$ a loop in $A_{\alpha_i}$ based at $x_0$ so it represent an element in $[\gamma_i] \in \pi_1(X_{\alpha_i})$. Note that we have

$$\begin{aligned}
\gamma_1 \cdots \gamma_n &\simeq (\xi_0 \gamma|_{[t_0,t_1]} \xi_1^{-1}) \cdot (\xi_1 \gamma|_{[t_1,t_2]} \xi_2^{-1}) \cdots (\xi_{n-1} \gamma|_{[t_{n-1},t_n]} \xi_n^{-1}) \\
&\simeq \xi_0 \cdot \gamma|_{[t_0,t_1]} \cdot \gamma|_{[t_1,t_2]} \cdots \gamma_{[t_{n-1},t_n]} \cdot \xi_n^{-1} \\
&\simeq c_{x_0} \cdot \gamma \cdot c_{x_0}^{-1} \\
&\simeq \gamma
\end{aligned}$$

So $([\gamma_1], \cdots, [\gamma_n])$ is a factorization of $[\gamma]$.

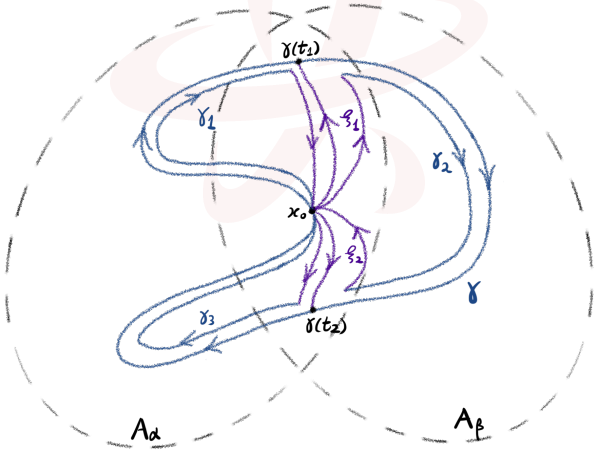


Figure 11.1

(2) Fix $g \in \pi_1(X)$. Consider the set of all factorizations of $g$ and the equivalence relation $\sim$ on this set generated by the following three collections of relations:

- Reduction: Suppose $g_i = e_{\alpha_i}$. Then $(\cdots, g_{i-1}, g_i, g_{i+1}, \cdots) \sim (\cdots, g_{i-1}, g_{i+1}, \cdots)$.
- Merge: Suppose $g_i, g_{i+1}$ belongs to the same group $\pi_1(A_{\alpha_i})$. Then $(\cdots, g_i, g_{i+1}, \cdots) \simeq (\cdots, g_i g_{i+1}, \cdots)$.
- Replacement: Suppose $g_i = f_{\alpha,\beta}(h)$ for some $h \in \pi_1(A_\alpha \cap A_\beta)$. Let $g_i' = f_{\beta,\alpha}(h)$. Then $(\cdots, g_i, \cdots) \simeq (\cdots, g_i', \cdots)$.



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We call an equivalence class under $\sim$ a factorization class. Then each factorization class corresponds to a single element in $(\underset{\alpha\in S}{*}\pi_1(A_\alpha,x_0))/N(B)$. We just need to show that any two factorizations of the same $g$ are equivalent to each other. This will be done in several steps:

**Step 1. Any decomposition of a loop $\gamma$ induces a unique factorization class of $[\gamma]$:** Let $(t_0,\cdots,t_n)$ be a decomposition of $\gamma$. We have constructed a factorization $([\gamma_1],\cdots,[\gamma_n])$ in the proof of (1). And we need to show that the equivalence class of this factorization does not depend on the auxiliary choices of $\alpha_*,\xi_*$.

Suppose we replace $\xi_i$ by another path $\xi_i'$ in $A_{\alpha_i}\cap A_{\alpha_{i+1}}$. Then $\xi_i(\xi_i')^{-1}$ is a loop in $A_{\alpha_i}\cap A_{\alpha_{i+1}}$ so represents an element $h\in\pi_1(A_{\alpha_i}\cap A_{\alpha_{i+1}})$. The new factorization is given by

$$(\cdots,[\gamma_i]\cdot f_{\alpha_i,\alpha_{i+1}}(h),f_{\alpha_{i+1},\alpha_i}(h^{-1})\cdot[\gamma_{i+1}],\cdots).$$

This is equivalent to the original factorization via a replacement and a merge. So the choice of $\xi_*$ do not affect the factorization class.

Suppose $\gamma|_{[t_i,t_{i+1}]}$ is contained in some $A_{\alpha_i'}$ and we use $A_{\alpha_i'}$ instead of $A_{\alpha_i}$ to construct the factorization. Then $\gamma(t_{i+1})$ belongs to $A_{\alpha_i}\cap A_{\alpha_i'}\cap A_{\alpha_{i+1}}$, which is path-connected by our assumption. So we may choose $\xi_{i+1}$ to be a path in $A_{\alpha_i}\cap A_{\alpha_i'}\cap A_{\alpha_{i+1}}$. Similarly, we choose $\xi_i$ to be a path in $A_{\alpha_i}\cap A_{\alpha_i'}\cap A_{\alpha_{i-1}}$. Then $\gamma_i$ is actually a loop in $A_{\alpha_i}\cap A_{\alpha_i'}$ so it represents an element $h\in\pi_1(A_{\alpha_i}\cap A_{\alpha_i'})$. By changing $A_{\alpha_i}$ to $A_{\alpha_i'}$, we are changing the factorization from $(\cdots,f_{\alpha_i,\alpha_i'}(h),\cdots)$ to $(\cdots,f_{\alpha_i',\alpha_i}(h),\cdots)$. This doesn't affect the factorization class as well.

**Step 2. Passing to a subdivision doesn't change the factorization class:** We say $(t_0',\cdots,t_m')$ is a subdivision of $(t_0,\cdots,t_n)$ if $\{t_0',\cdots,t_m'\}\subset\{t_0,\cdots,t_n\}$. With suitable choices of $\xi_*,\alpha_*$, the factorization induced by $(t_0',\cdots,t_m')$ can be converted to the factorization induced by $\{t_0,\cdots,t_n\}$ by a sequence of merges. So this doesn't change the factorization class.

**Step 3. Any loop $\gamma$ induces a unique factorization class of $[\gamma]$:** Any two factorizations of the same $\gamma$ can be made identical after passing to suitable subdivisions.

**Step 4. Factorization class is unchanged under a small homotopy.** A homotopy $H:I\times I\to X$ from $\gamma$ to $\gamma'$ is called a small homotopy if there exists $0=t_0<t_1<\cdots<t_n=1$ such that $H(I\times[t_i,t_{i+1}])\subset A_{\alpha_i}$ for some $\alpha_i$. To construct factorizations from $\gamma$, $\gamma'$, we need to pick a path $\xi_i$ from $x_0$ to $\gamma(t_i)$ and a path $\xi_i'$ from $x_0$ to $\gamma'(t_i)$. For each $i$, we set $\xi_i'$ to be the composition $\xi_i\cdot H|_{I\times\{t_i\}}$. Then $H|_{I\times[t_i,t_{i+1}]}$ provides a homotopy between the loops $\xi_i\cdot\gamma|_{[t_i,t_{i+1}]}\cdot\xi_{i+1}^{-1}$ and $\xi_i'\cdot\gamma'|_{[t_i,t_{i+1}]}\cdot(\xi_{i+1}')^{-1}$ So the factorizations of $\gamma$ and $\gamma'$ constructed in this way are actually equal to each other.

**Step 5. Factorization class is unchanged under a general homotopy.** Let $H:I\times I\to X$ be a homotopy class. By applying the Lebesgue lemma to the open cover $\{H^{-1}(A_\alpha)\}$, we can find $0=s_0<s_1<\cdots<s_m=1$ such that $H|_{[s_i,s_{i+1}]\times I}$ is a small homotopy for any $i$.

**Step 6. Any two factorizations of the same $g\in\pi_1(X)$ are equivalent to each other.** Suppose we have two factorizations $([\gamma_1],\cdots,[\gamma_n])$ and $([\gamma_1'],\cdots,[\gamma_m'])$ of the same $g$. Then they are induced by the loops $\gamma=\gamma_1\cdots\gamma_n$ and $\gamma'=\gamma_1'\cdots\gamma_m'$ respectively. Since $\gamma\simeq\gamma'$. We see that these two factorizations belong to the same factorization class. □

## 11.4 Applications of the Seifert-Van Kampen theorem

Let's see some applications of the Seifert-Van Kampen theorem.



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**Proposition 11.19.** *Suppose $X$ has an open cover $\bigcup_{\alpha\in S} A_\alpha$ such that: (i) $\bigcap_{\alpha\in S} A_\alpha \neq \varnothing$, (ii) $A_\alpha$ is simply-connected for any $\alpha$, (iii) $A_\alpha \cap A_\beta$ is path-connected for any $\alpha, \beta$. Then $X$ is simply connected.*

*Proof.* By (i) and (ii), $X$ is path-connected. We pick $x_0 \in \bigcap_{\alpha\in S} A_\alpha \neq \varnothing$. Then by theorem 11.17 (1), there is a surjective map

$$\underset{\alpha}{*}\, \pi_1(A_\alpha) \to \pi_1(X).$$

Since $*_\alpha\, \pi_1(A_\alpha)$ is trivial, $\pi_1(X)$ is trivial as well. □

**Corollary 11.20.** *For any $n \geq 2$, $S^n$ is simply-connected.*

*Proof.* $S^n = U \bigcup V$ where $U = S^n \setminus \{(1, 0, \cdots, 0)\}$ and $V = S^n \setminus \{(-1, 0, \cdots, 0)\}$. Then $U, V$ are both homeomorphic to $\mathbb{R}^n$ so they are simply-connected, and $U \cap V$ is homeomorphic to $\mathbb{R}^n \setminus \{0\}$ so is path-connected. □

**Definition 11.21.** We say a point $x \in X$ is "nice" if there is an open neighborhood $U$ of $x$ that deformation retracts to $x$.

**Example 11.22.** *(1) Any point on a manifold is nice.*

*(2) The origin in the Hawaiian earring*

$$X := \bigcup_{n\in\mathbb{N}^{>0}} \{(x, y) \in \mathbb{R}^2 \mid x^2 + (y - \frac{1}{n})^2 = \frac{1}{n^2}\}.$$

*is not nice.*

**Proposition 11.23.** *Let $\{(X_\alpha, x_\alpha)\}_{\alpha\in S}$ be a collection of pointed spaces. Suppose $x_\alpha$ is nice for any $\alpha$. Then we have*

$$\pi_1(\bigvee_{\alpha\in S} X_\alpha, [x_\alpha]) \cong \underset{\alpha\in S}{*}\, \pi_1(X_\alpha, x_\alpha).$$

*Proof.* By passing the the path component of $X_\alpha$ that contains $x_\alpha$, we may assume that $X_\alpha$ is path-connected. Let $V_\alpha$ be an open neighborhood of $x_\alpha$ in $X_\alpha$ that deformation retracts to $x_\alpha$. We consider the following subset of $\bigvee_{\alpha\in S} X_\alpha$

$$U_\alpha := X_\alpha \vee (\bigvee_{\beta\in S\setminus\{\alpha\}} V_\beta)$$

Then $\{U_\alpha\}$ is an open cover of $\bigvee_{\alpha\in S} X_\alpha$ and $U_\alpha$ deformation retracts to $X_\alpha$. Furthermore, for any distinct $\alpha, \beta$, $U_\alpha \cap U_\beta = V_\alpha \cap V_\beta$ is contractible so $\pi_1(U_\alpha \cap U_\beta) = \{e\}$. Similarly, one can show that $U_\alpha \cap U_\beta \cap U_\delta$ is path-connected. So we apply theorem 11.17 to conclude that

$$\pi_1(\bigvee_{\alpha\in S} X_\alpha, [x_\alpha]) \cong \underset{\alpha\in S}{*}\, \pi_1(U_\alpha, x_\alpha) \cong \underset{\alpha\in S}{*}\, \pi_1(X_\alpha, x_\alpha).$$

□

**Example 11.24.** *(1) $\pi_1(\bigvee_n S^1) \cong F_n$.*

*(2) Let $X = \mathbb{R}^m \setminus \{n \text{ points}\}$. Then $X$ deformation retracts to $\bigvee_n S^{m-1}$ so $\pi_1(X) \cong F_n$ when $m = 2$ and $X$ is simply-connected when $m \geq 3$.*



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**Exercise B.** *Let* $A = \{(\frac{4}{2n+1}, 0) \mid n = 1, 2, 3, \cdots\}$.

*(1) Show that* $\mathbb{R}^2 \setminus A$ *deformation retracts to the Hawaiian earring*

$$\bigcup_{n \in \mathbb{N}^{>0}} \{(x, y) \in \mathbb{R}^2 \mid y^2 + (x - \frac{1}{n})^2 = \frac{1}{n^2}\}.$$

*(2) Show that there is a surjective group homomorphism from* $\pi_1(\mathbb{R}^2 \setminus A)$ *to* $\mathbb{Z}^\infty$ *(the product of infinitely many copies* $\mathbb{Z}$*). In particular,* $\pi_1(\mathbb{R}^2 \setminus A)$ *contains uncoutable many elements.*

**Exercise C.** *Let* $f : X \to Y$ *be a continuous map. The* ***mapping cone*** *of* $f$ *is defined as the quotient space*

$$Cf := ([0, 1] \times X) \sqcup Y / \sim,$$

*where* $\sim$ *is generated by the following relations*

$$(0, x) \sim (0, x') \ \forall x, x' \in X, \ \text{and} \ (1, x) \sim f(x) \ \forall x \in X.$$

*(1) Assume* $X, Y$ *are both pah-connected. Show that* $\pi_1(Cf) \cong \pi_1(Y)/N$*. Here* $N$ *is the normal closure of the image of* $f_* : \pi_1(X) \to \pi_1(Y)$*.*

*(2) Let* $f_n : S^1 \to S^1$ *be defined by* $f_n(z) = z^n$*. Find the homeomorphism type of* $Cf_0, Cf_1$ *and* $Cf_2$*. (I.e. identify them with some topological spaces that we are familiar with.)*

*(3) Compute* $\pi_1(Cf_n)$ *for* $n \geq 0$*.*

**Exercise D.** *Let* $X = U \cup V$ *be an open cover of* $X$*. Suppose* $U, V, U \cap V$ *are all path-connected. Consider the map*

$$f_{U,V} : \pi_1(U \cap V) \to \pi_1(V)$$

*induced by the inclusion* $U \cap V \hookrightarrow V$ *and the map*

$$f_U : \pi_1(U) \to \pi_1(X).$$

*induced by the inclusion* $U \hookrightarrow X$*. Suppose* $f_{U,V}$ *is surjective. Show that* $f_U$ *is also surjective. (Actually, it is also true that injectivity of* $f_{U,V}$ *implies injectivity of* $f_U$*. But this is a lot harder to show and not required.)*



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# Chapter 12

# Basic properties of CW complexes

We have seen the definition of simplicial complexes. Intuitively, it decomposes a complicated topological space into simplices $\Delta^n$. In this section, we will study a generalization called "CW complexes".

## 12.1 CW complexes: definition and examples

We start by introducing an important construction called "attaching spaces". Consider a space $A$ and its subspace $B$. We call $(A, B)$ a space pair. Given such a pair $(A, B)$ and a continuous map $f : B \to X$ to some space $X$, we consider the equivalence relation $\sim$ on the space $X \sqcup A$ generated by

$$f(x) \sim x \text{ for any } x \in B.$$

We denote the quotient space $(X \sqcup A)/\sim$ by $X \cup_f A$ (or $X \cup_B A$ if $f$ is clear from the context). We call $X \cup_f A$ the space obtained by "attaching $A$ to $X$ along its subspace $B$ via the attaching map $f$."

*Remark* 12.1. Although the notation looks symmetric. The role played by $X$ and $A$ in $X \bigcup_f A$ are essentially different. For example, we may consider the maps

$$i_X : X \hookrightarrow X \sqcup A \xrightarrow{\text{quotient map}} X \cup_f A.$$
$$i_A : A \hookrightarrow X \sqcup A \xrightarrow{\text{quotient map}} X \cup_f A.$$

Then $i_X$ is always an embedding so we may treat $X$ as a subspace of $X \cup_f A$. But $i_A$ is usually not an embedding, so $A$ is not a subspace of $X \cup_f A$. An exception is when the attaching map $f$ is also an embedding. In this case, we let $f^{-1} : f(B) \to B$ be the inverse of $f$. Then $X \cup_f A \cong A \cup_{f^{-1}} X$.

**Example 12.2.** *The space $* \cup_B A$ is exactly the quotient space $A/B$.*

$$\begin{array}{ccc} B & \longrightarrow & * \\ \downarrow & & \downarrow \\ A & \longrightarrow & A/B \end{array}$$

**Example 12.3.** *When $(A, B) = (D^n, S^{n-1})$, we call $D^n$ a **$n$-dimensional cell** (or just an $n$-cell) and say the space $X \cup_f D^n$ is obtained from $X$ by **attaching a $n$-cell** via the attaching map $f : S^{n-1} \to X$.*

$$\begin{array}{ccc} S^{n-1} & \longrightarrow & X \\ \downarrow & & \downarrow \\ D^n & \longrightarrow & X \cup_f D^n \end{array}$$



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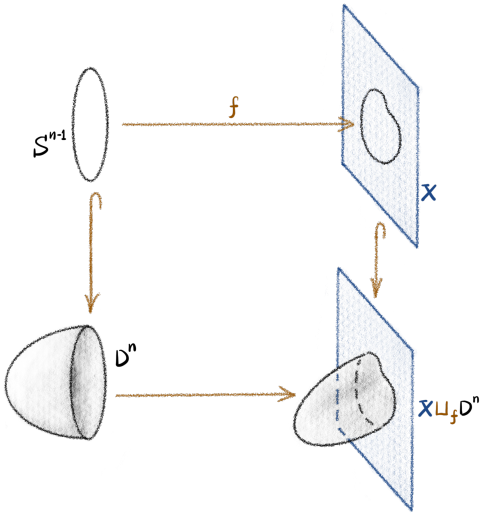


Figure 12.1

*Of course, we can generalize this a little by attaching a (finite or infinite) collection of $n$-cells $\sqcup_{\alpha\in S} D^n_\alpha$* ***simultaneously*** *to $X$, via an attaching map $\sqcup_{\alpha\in S} f_\alpha : S^{n-1}_\alpha \to X$.*

$$\begin{array}{ccc} \sqcup S^{n-1}_\alpha & \xrightarrow{\sqcup f_\alpha} & X \\ \downarrow & & \downarrow \\ \sqcup D^n_\alpha & \longrightarrow & X \cup_{\sqcup f_\alpha} (\sqcup D^n_\alpha) \end{array}$$

Now we are ready to introduce the definition of a CW complex.

**Definition 12.4.** A CW complex is a space $X$ equipped with a sequence of subspaces

$$\varnothing = \mathrm{Sk}_{-1} X \subseteq \mathrm{Sk}_0 X \subseteq \mathrm{Sk}_1 X \subseteq \cdots \subseteq X$$

such the the following condition holds.

1. $\mathrm{Sk}_n X$ is obtained from $\mathrm{Sk}_{n-1} X$ by attaching a collection of $n-$cells $\sqcup_{\alpha\in A_n} D^n_\alpha$. (In particular, $\mathrm{Sk}_0 X$ is a discrete topological space.)

$$\begin{array}{ccc} \sqcup S^{n-1}_\alpha & \longrightarrow & \mathrm{Sk}_{n-1} X \\ \downarrow & & \downarrow \\ \sqcup D^n_\alpha & \longrightarrow & \mathrm{Sk}_n X \end{array}$$

2. $X = \cup_{n=0}^{\infty} \mathrm{Sk}_n X$. And the topology on $X$ is the so-called "weak topology". That means $U$ in $X$ if and only if $U \cap \mathrm{Sk}_n X$ is open in $\mathrm{Sk}_n X$ for any $n$.

We say $X$ is finite dimensional if $X = \mathrm{Sk}_n X$ for some $n$. We say it's finite type if $A_n$ is a finite set for each $n$. (I.e., there are only finitely many cells in each dimension.) We say $X$ is a finite CW complex if it is finite dimensional and finite type. A CW structure on a topological space $X$ is a homeomorphism from $X$ to a CW complex.

*Remark* 12.5. The cell $D^n_\alpha$ of a CW complex $X$ is in general **not** a subspace of $X$. But there is indeed a well-defined map

$$\varphi_\alpha : D^n_\alpha \to \mathrm{Sk}_n X \hookrightarrow X$$



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小红书号:3189205503 qq

This map is called **the characteristic map** of the cell. In general, the map $\varphi_\alpha$ is not an embedding (because $\varphi_\alpha|_{\partial D^n_\alpha}$ just the gluing map and usually not injective). However, the restriction composition

$$\varphi_\alpha|_{\mathring{D}^n_\alpha} : \mathring{D}^n_\alpha \to X$$

is indeed always an embedding. We denote the image of this composition by $e^n_\alpha$ call it the "**open cell**". This name is a little confusing because an open cell $n$-cell $e^n_\alpha \subset X$ is an open subset of $\mathrm{Sk}_n X$ but usually not an open subset of $X$ (unless $X$ is $n$-dimensional). The open cells are useful because we have a CW decomposition

$$X = \bigcup_{n\in\mathbb{N}} \bigcup_{\alpha\in A_n} e^n_\alpha \tag{12.1}$$

of the CW complex into its open cells.

Not every space has a CW structure (e.g. the Hawaiian earring has no CW structure). And CW structure is almost never unique, the following examples illustrate this.

**Example 12.6.** *Since $S^n \cong D^n/S^{n-1} = * \cup_{S^{n-1}} D^n$, $S^n$ has a simple CW structure consisting of a single 0-cell and a single $n$-cell.*

**Example 12.7.** *$S^n$ has another important CW structure. We start with $\mathrm{Sk}_0 S^n = S^0$. And attach two $k$-cells $D^k \sqcup D^k$ to $\mathrm{Sk}_{k-1} S^n = S^{k-1}$. These two $k$-cells become the upper- and lower-hemisphere in $\mathrm{Sk}_k S^n = S^k$.*

$$\begin{array}{ccc} S^{k-1} \sqcup S^{k-1} & \xrightarrow{\mathrm{id}\sqcup\mathrm{id}} & \mathrm{Sk}_{k-1} S^n = S^{k-1} \\ \downarrow & & \downarrow \text{meridian} \\ D^n \sqcup D^n & \longrightarrow & \mathrm{Sk}_k S^n = S^k. \end{array}$$

*So we get another CW structure on $S^n$ with two $k$-cells for each $0 \le k \le n$.*

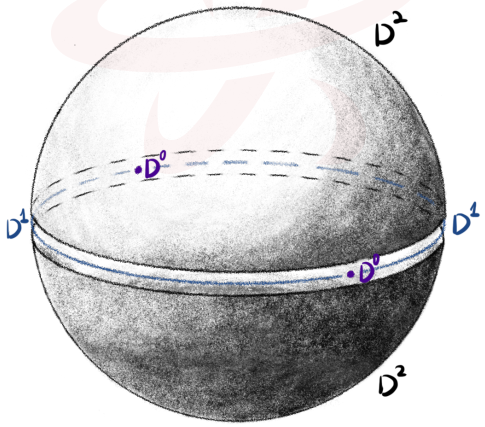


Figure 12.2

**Example 12.8.** *The CW structure in example 12.7 is compatible with the antipodal map on $S^n$. (I.e., the antipodal map sends a $k$-cell to the other $k$-cell). Therefore, by taking quotient, we obtain a CW structure on $\mathbb{RP}^n$ with exactly one cell in each dimension $0 \le k \le n$. From $\mathrm{Sk}_k S^n = S^k$, we can deduce that $\mathrm{Sk}_k \mathbb{RP}^n = \mathbb{RP}^k$. And the $k$-cell is attached to $\mathrm{Sk}_{k-1}\mathbb{RP}^n = \mathbb{RP}^{k-1}$ via the 2-fold covering map $S^{k-1} \to \mathbb{RP}^{k-1}$.*

$$\begin{array}{ccc} S^{k-1} & \xrightarrow{\text{2-fold covering map}} & \mathrm{Sk}_{k-1}\mathbb{RP}^n = \mathbb{RP}^{k-1} \\ \downarrow & & \downarrow \\ D^k & \longrightarrow & \mathrm{Sk}_k \mathbb{RP}^n = \mathbb{RP}^k. \end{array}$$



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*Note that one can actually keep this process (ie. attaching a single $k$-dimensional cell to $\mathbb{RP}^{k-1}$ via the covering map $S^{k-1} \to \mathbb{RP}^{k-1}$) without stopping. This gives a CW complex $\mathbb{RP}^\infty$, which is finite type (since there is a single cell in each dimension) but not finite dimensional.*

**Example 12.9.** *Here is a CW structure of the torus $T^2$. Recall that $T^2$ is obtained by gluing the opposite edges of the cube $[0,1]^2$. We take a CW structure of this cube with four 0-cells (vertices), four 1-cells (edges) and a single 2-cell (face). This CW structure actually induces a CW structure of the quotient space $T^2$. In the quotient space, all four vertices are identified with each other and two pairs of edges are identified. So $T^2$ has a single 0-cell, two 1-cells and a single 2-cell. The unique 2-cell is attached to $\mathrm{Sk}_1 T^2 = S^1 \vee S^1$ via the attaching map $aba^{-1}b^{-1}$. So we have the following diagram*

$$\begin{array}{ccc} S^1 & \xrightarrow{aba^{-1}b^{-1}} & \mathrm{Sk}_1 T^2 = S^1 \vee S^1 \\ \downarrow & & \downarrow \\ D^2 & \longrightarrow & \mathrm{Sk}_2 T^2 = T^2. \end{array}$$

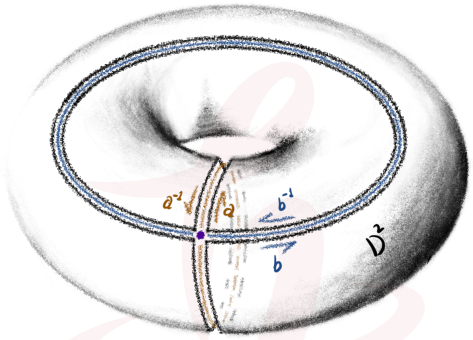


Figure 12.3

**Example 12.10.** *Similarly as Example 12.9, the Klein bottle $K$ has a CW structure with a single 0-cell, two 1-cells, and a single 2-cell with attaching map $abab^{-1}$*

Now we summarize some important properties of CW complexes. Most of them are easy to believe (although not so easy to prove). We won't give proof here. See Chapter 0 and appendix of Hatcher for more details.

**Proposition 12.11.** *Let $X$ be a CW complex. Then the following results hold.*

1. *$X$ satisfies the $T_1 \sim T_4$ separation axioms.*
2. *$X$ is locally connected and locally path connected.*
3. *Any point $x \in X$ is nice. In particular, $X$ is locally contractible. (That means for any neighborhood $U$ of $x$, there exists a smaller neighborhood $V \subset U$ that is contractible.)*
4. *$\mathrm{Sk}_1 X$ is always a graph. And the inclusion $\mathrm{Sk}_1 X \hookrightarrow X$ induces a bijection $\pi_0(\mathrm{Sk}_1 X) \leftrightarrow \pi_0(X)$.*
5. *A subset $A \subset X$ is compact if and only if $A$ is closed and $A$ intersects finitely many open cells of $X$. In particular, $X$ itself is compact if and only if it is a finite CW complex.*
6. *Suppose $\widetilde{X}$ is a $k$-fold covering space of $X$. Then the CW structure on $X$ induces a CW structure on $\widetilde{X}$. Each $n$-cell of $X$ lifts to $k$ $n$-cells of $\widetilde{X}$.*



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7. *Suppose $X, Y$ are both CW complexes with CW decomposition $X = \cup_{n\in\mathbb{N}} \cup_{\alpha\in A_n} e^n_\alpha$ and $Y = \cup_{m\in\mathbb{N}} \cup_{\beta\in B_m} e^m_\beta$. Taking the product, we get a decomposition*

$$X \times Y = \bigcup_{(\alpha,\beta)\in A_n\times B_m} \bigcup_{n,m\in\mathbb{N}} (e^n_\alpha \times e^m_\beta)$$

*Under a suitable topology on $X \times Y$ (called the CW topology), this decomposition makes $X \times Y$ into a CW complex whose $k$-cells are of the form $e^n_\alpha \times e^{k-n}_\beta$. There is a subtle point here: the CW topology on $X \times Y$ is sometimes finer than the product topology. But when $X$ and $Y$ contains only countably many cells, the two topologies on $X \times Y$ coincide (This is proved by Milnor). So we can just use the product topology.*

8. *Suppose $X$ is a finite complex. We define the **Euler number** (or Euler characteristic) of $X$ as*

$$\chi(X) := \sum_{k\geq 0} (-1)^k \cdot (\textit{number of } k\textit{-cells})$$

*A deep result is that $\chi(X)$ is a homotopy invariant. That means $\chi(X) = \chi(Y)$ if $X \simeq Y$. In particular, $\chi(X)$ is independent with the choice of a CW structure.*

9. *If $\widetilde{X}$ is a $k$-fold covering space of a finite CW complex $X$, then $\chi(\widetilde{X}) = k \cdot \chi(X)$. And we have $\chi(X \times Y) = \chi(X) \times \chi(Y)$ for any finite CW complexes $X, Y$.*

## 12.2 Fundamental group of CW complexes

Now we use the Van Kampen theorem to calculate the fundamental group of a CW complex. We let $X$ be a CW complex. Without loss of generality, we assume $X$ is connected and we pick the base point $b$ to be a 0-cell.

We first consider $\mathrm{Sk}_1 X$, which is a graph whose vertices are 0-cells and edges are 1-cells. We take a connected subgraph $T \subset \mathrm{Sk}_1 X$ that contains all vertices and no (graph-theoretic) loops. Such $T$ is called a maximal tree. We use $\{e^1_\alpha\}_{\alpha\in S}$ to denote the set of open 1-cells in $\mathrm{Sk}_1(X) \setminus T$.

**Lemma 12.12.** $\pi_1(\mathrm{Sk}_1 X, b) \cong F_S$.

*Proof.* For each $\alpha \in S$, we pick a point $b_\alpha \in e^1_\alpha$. Then we set

$$U_\alpha := \mathrm{Sk}_1 X \setminus \{b_\beta \mid \beta \in S \setminus \{\alpha\}\}$$

Then $\{U_\alpha\}_{\alpha\in S}$ is an open cover of $\mathrm{Sk}_1 X$. Each $U_\alpha$ deformation retracts to $S^1$ so we have $\pi_1(U_\alpha, b) \cong \mathbb{Z}$. Furthermore, let

$$V = \mathrm{Sk}_1 X \setminus \{b_\beta \mid \beta \in S\}.$$

Then $V$ is contractible since it deformation retracts to $T$. Note that we have

$$U_\alpha \cap U_\beta = U_\alpha \cap U_\beta \cap U_\gamma = V, \quad \alpha \neq \beta \neq \gamma \in S.$$

This allows us to apply the Van Kampen theorem to the open cover $\{U_\alpha\}_{\alpha\in S}$ and conclude that

$$\pi_1(\mathrm{Sk}_1 X, b) \cong \mathop{*}_{\alpha\in S} \pi_1(U_\alpha, b) \cong F_S.$$

(The normal closure $N$ is trivial because $U_\alpha \cap U_{\alpha'} = V$ is simply-connected.)



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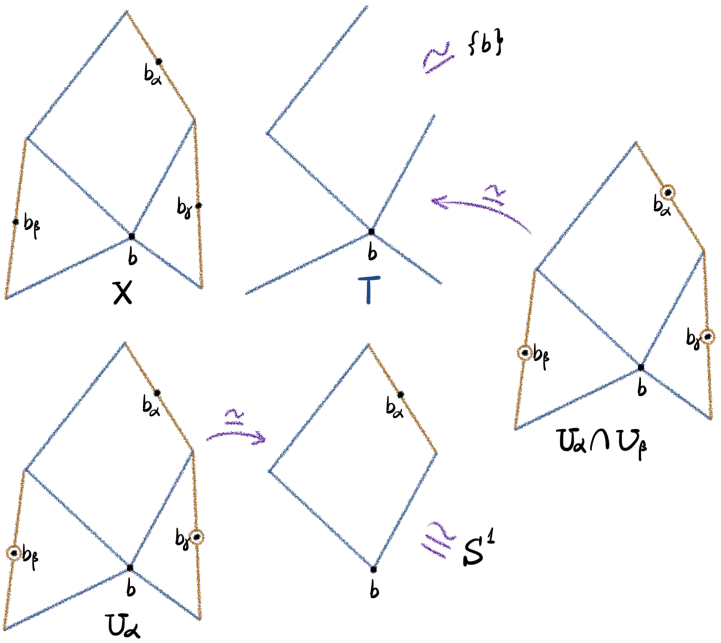


Figure 12.4

□

The next step is to compute the fundamental group of $\mathrm{Sk}_2 X$, which is obtained from $\mathrm{Sk}_1 X$ by attaching a collection of 2-cells $\{D^2_\alpha\}_{\alpha\in A_2}$. We consider the following generalized question:

**Question 12.13.** *Let $Y$ be a path-connected space and let $Z$ be obtained by attaching a collection of $n$-cells $\{D^n_\alpha\}_{\alpha\in A_n}$ to $Y$. What's the relation between $\pi_1(Y)$ and $\pi_1(Z)$?*

We start with the case $n = 2$. We fix a base point $b \in Y$. We let $f_\alpha : S^1 = \partial D^2_\alpha \to Y$ be the attaching map for $D^2_\alpha$. Let $b_\alpha = f_\alpha(1) \in Y$. With an abuse of notations, we treat $f_\alpha$ as a loop based at $b_\alpha$. To turn it into a loop based at $b$, we pick a path $\gamma_\alpha$ from $b$ to $b_\alpha$ and consider the loop $f'_\alpha := \gamma_\alpha \cdot f_\alpha \cdot \bar\gamma_\alpha$ based at $b$. This loop represents an element $[f'_\alpha]$ in $\pi_1(Z, b)$.

**Lemma 12.14.** *Let $N$ be the normal closure of $\{[f'_\alpha]\}_{\alpha\in A_2}$ in $\pi_1(Y, b)$. Then the map $\pi_1(Y, b) \to \pi_1(Z, b)$ induced by the inclusion $Y \hookrightarrow Z$ is surjective and has kernel $N$. In particular, we have $\pi_1(Z, b) \cong \pi_1(Y, b)/N$.*

*Proof.* First, we assume $A_2 = \{\alpha\}$, $b = b_\alpha$. Let $0 \in e^2_\alpha$ be the center. Then we have an open cover $Z = U \cup V$, where $U = e^2_\alpha$ and $V = Z \setminus \{0\}$. Note that

$$U \cap V \cong \mathring{D}^2 \setminus \{0\} \cong (0, 1) \times S^1 \tag{12.2}$$

is connected. We pick the point $b' \in U \cap V$ and let $f''_\alpha$ be the loop in $U \cap V$ that generates the group $\pi_1(U \cap V, b') \cong \mathbb{Z}$. By the Van-Kampen theorem, we have

$$\pi_1(Z, b') \cong \pi_1(V, b')/N([f''_\alpha]). \tag{12.3}$$

Now we observe that the deformation retract from $\mathring{D}^2\setminus\{0\}$ to $\partial D^2$ induces a deformation retract from $V$ to $Y$ which takes $b'$ to $b_\alpha = b$ and takes $f''_\alpha$ to $f'_\alpha$. By this deformation retract, we can turn (12.3) into

$$\pi_1(Z, b) \cong \pi_1(V, b)/N([f_\alpha])$$

This proves the lemma when $\gamma_\alpha$ is the constant loop $c_b$. For a general loop $\gamma_\alpha$, the elements $[\gamma_\alpha \cdot f_\alpha \cdot \bar\gamma_\alpha]$ and $[f_\alpha]$ are conjugate to each other in $\pi_1(V, b)$ so have the same normal closure.



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Next we assume $A_2 = \{\alpha\}$ and $b \neq b_\alpha$. Then we have the change of base point isomorphism

$$\gamma_\alpha^\sharp : \pi_1(Z, b_\alpha) \to \pi_1(Z, b), \quad [g] \mapsto [\gamma_\alpha \cdot g \cdot \bar{\gamma}_\alpha]$$

This allows us to deduce to the previous case and finishes the proof when $A_2 = \{\alpha\}$. By induction, we can prove the lemma when $A_2$ is finite. For the infinite case, we use the following observations to reduce to the finite case:

- For any loop $\gamma : I \to Z$, the image $\gamma(I)$ is compact so it can only intersect finitely many open cells $e^2_\alpha$.
- Given any homotopy $H : I \times I \to Z$ between two loops, the image $H(I \times I)$ is compact so it can only intersect finitely many open cells $e^2_\alpha$.

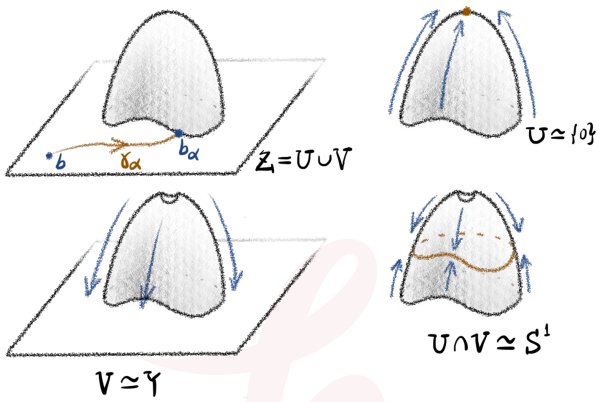


Figure 12.5

□

When $n \geq 3$, we have the following proposition.

**Lemma 12.15.** *Let $Z$ be obtained from $Y$ by attaching a collection of cells with dimension $\geq 3$. Then the inclusion $Y \hookrightarrow Z$ induces an isomorphism from $\pi_1(Y)$ to $\pi_1(Z)$.*

*Proof.* The proof is very similar to Lemma 12.14. The only essential difference is that (12.2) becomes

$$U \cap V \cong \mathring{D}^n \setminus \{0\} \cong (0,1) \times S^{n-1}.$$

Since $n \geq 3$, $U \cap V$ is simply-connected. So (12.3) now becomes

$$\pi_1(Z, b') \cong \pi_1(V, b').$$

Rest of the proof is the same. □

Combining Lemma 12.14 and Lemma 12.15, we get the following algorithm of computing $\pi_1(X)$ for a CW complex $X$:

1. Pick the base point to be a 0-cell.
2. Take a maximal tree $T \subset \mathrm{Sk}_1 X$.
3. Consider the set $\{e^1_\alpha\}_{\alpha \in S}$ of open 1-cells that are not in $T$. Fix a direction on each of them. This gives a set of generators for $\pi_1(\mathrm{Sk}_1 X) \cong F_S$.



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4. For each 2-cell $D^2_\beta$ with attaching map $f_\beta : \partial D^2_\beta \to \mathrm{Sk}_1 X$ gives an element in $\pi_1(\mathrm{Sk}_1 X) = F_S$ (i.e. a word in $S$). To explicitly describe this word, we pick a generic point $p_\alpha \in e^1_\alpha$ for each $\alpha \in S$. Then we record how the loop $f_\beta(\partial D^2_\beta)$ passes the points $\{p_\alpha\}_{\alpha\in S}$: Whenever the loop passes $p_\alpha$ from its back to its front, we add a letter $\alpha$ in the word. Whenever the loop passes $p_\alpha$ from its front to its back, we put an $\alpha^{-1}$ in the word. This gives the word $R_\beta$ we need.
5. We have $\pi_1(\mathrm{Sk}_2) = \langle S \mid R_\beta \text{ for } \beta \in A_2\rangle$.
6. Since attaching cells of dimension $\geq 3$ has no effect on $\pi_1$, we get
$$\pi_1(X) = \langle S \mid R_\beta \text{ for } \beta \in A_2\rangle.$$
7. The above process can be summarized as follows: Each 1-cell outside the maximal tree contribute a generator of $\pi_1(X)$, and each 2-cell contributes a relation in $\pi_1(X)$. Cells in dimension $\geq 3$ has no effect on $\pi_1(X)$.

**Example 12.16.** *(1) By Example 12.8, we get*
$$\pi_1(\mathbb{RP}^n) \cong \langle a \mid a^2\rangle \cong \mathbb{Z}/2.$$
*for any $n \geq 2$. We also note that $\mathbb{RP}^1 \cong S^1$ so $\pi_1(\mathbb{RP}^1) \cong \mathbb{Z}$.*

*(2) By Example 12.9, we get*
$$\pi_1(T^2) \cong \langle a, b \mid aba^{-1}b^{-1}\rangle \cong \mathbb{Z} \oplus \mathbb{Z}.$$
*This is consistent with our calculation in Example 11.2.*

*(3) By Example 12.10, the fundamental group of the Klein bottle is a nonabelian group given by*
$$\pi_1(K) \cong \langle a, b \mid abab^{-1}\rangle$$

**Definition 12.17.** A subspace $Y$ of a CW complex $X$ is called a sub-CW complex if

1. $Y$ is closed,
2. $Y = \bigcup_{n\in\mathbb{N}} \bigcup_{\alpha\in B_n} e^n_\alpha$ for some $B_n \subset A_n$.

In this case, we call $(X, Y)$ a CW pair. In particular, $Y$ is itself is a CW complex.

We have the following important property of CW pairs.

**Theorem 12.18.** *Let $(X, Y)$ be a CW pair. Then the following results hold.*

1. *Suppose $Y$ is contractible. Then $X/Y$ is homotopy equivalent to $X$.*
2. *Let $Z$ be a general topological space (not necessarily a CW complex). Let $f, g : Y \to Z$ be two continuous maps. Then*
$$f \simeq g \implies Z \cup_f X \simeq Z \cup_g X.$$

Theorem 12.18 is a consequence of the so-called "homotopy extension property" of CW pairs. We refer to [**Hatcher**] for an introduction of such property and proof of Theorem 12.18. We just illustrate the power of Theorem 12.18 by the following some examples.

**Example 12.19.** *$(D^n, S^{n-1})$ is a CW pair. Actually, we get a CW structure on $D^n$ by start with any CW structure on $S^{n-1}$ and then add an additional $n$-cell via the attaching identity map $\partial D^n = S^{n-1} \xrightarrow{\mathrm{id}} S^{n-1}$. Then $S^{n-1}$ is a sub-CW complex under this CW structure of $D^n$. By Theorem 12.18, the homotopy type of $Z \cup_f D^n$ only depends on the homotopy class of $f : S^{n-1} \to Z$. In particular, this tells us that changing the attaching map in a CW complex by a homotopy doesn't effect the homotopy type of of resulting CW complex.*



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**Example 12.20.** *Let $x_0 \neq x_1$ be two points in a connected CW complex $X$. Consider the equivalent relation $\sim$ on $X$ generated by $x_0 \sim x_1$. and the quotient space $X/\sim$. We will show that $X/\sim \simeq X \vee S^1$.*

*To prove this, we first note that one can always modify the CW structure on $X$ so that $x_0, x_1$ are 0-cells. (Exercise: think about why?) We consider the constant map $f_0 : S^0 \to X$, which sends both points to $x_0$, and the non-constant map $f_1 : S^0 \to X$, which sends two points to $x_0$ and $x_1$ respectively. Consider the spaces*

$$X_0 := X \cup_{f_0} D^1, \quad X_1 := X \cup_{f_1} D^1.$$

*Since $X$ is path-connected, $f_0$ and $f_1$ are homotopic. By Theorem 12.18, $X_0 \simeq X_1$. Also note that $(X_1, D^1)$ is a CW pair. So by Theorem 12.18, $X_1 \simeq X_1/D^1 \cong X/\sim$. Observe that $X_0 \cong X \vee S^1$. We have proved $X/\sim \simeq X \vee S^1$.*



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# Chapter 13

# Classification of closed surfaces

It is a classical result that a connected 1-dimensional manifold (without boundary) is either $\mathbb{R}$ or $S^1$. In this section, we classify all closed surfaces (i.e. compact 2-dimensional manifold without boundary). All the arguments in this section can be adapted to compact surfaces with boundary but we focus on the closed case for simplicity.

## 13.1 Constructing closed surfaces by gluing edges of a polygon

We have seen how to obtain $T^2, \mathbb{RP}^2, S^2$ by gluing edges of a square. It turns out that this construction can be generalized to give **all** connected, closed surfaces.

**Definition 13.1.** Let $P \subset \mathbb{R}^2$ be a polygon with $2n$ edges $e_1, \cdots, e_n$ (ordered counter-clockwisely). A gluing data $D$ on $P$ is a direction on each edge $e_*$ and a pairing

$$(e_{i_1}, e_{j_1}), \cdots, (e_{i_n}, e_{j_n})$$

of the edges into $n$-groups. Given a gluing data, we can construct a topological space $\Sigma_D$ by gluing together the edges in the same group in a way that respect the direction.

To conveniently describe a gluing data, we assign the same letter (e.g. the letter "a") to both edges in the same group. If an edge goes counterclockwisely, then we use the letter itself, but if it goes closewisely, then we use the inverse of the letter. Then we just record the letters for $e_1, \cdots, e_{2n}$. This gives a sequence of letters that describes the gluing data.

**Example 13.2.** *(1) If $D = aba^{-1}b^{-1}$, then $\Sigma_D \cong T^2$.*
*(2) If $D = aa^{-1}$, then $\Sigma_D \cong S^2$.*
*(3) If $D = aa$, then $\Sigma_D \cong \mathbb{RP}^2$.*
*(4) If $D = aba^{-1}b$, then $\Sigma_D$ is the following quotient space*

$$[0,1] \times S^1 / \sim, \quad (0, e^{i\theta}) \sim (1, e^{-i\theta}).$$

*i.e., the Klein bottle.*

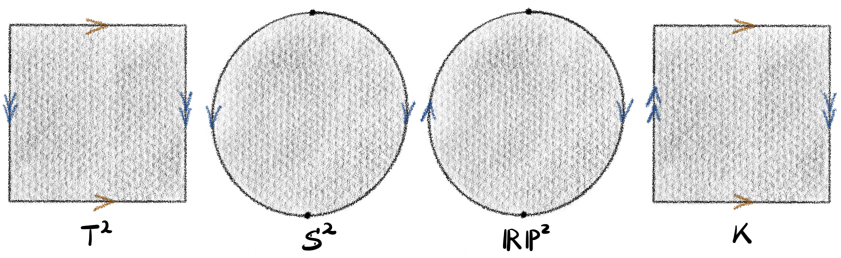


Figure 13.1



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**Lemma 13.3.** *For any gluing data $D$, the space $\Sigma_D$ is a closed surface.*

*Proof.* It's straightforward to check that $\Sigma_D$ is Hausdorff. Given $[x] \in \Sigma_D = P/\sim$, there are three possibilities of $x$: (1) $x$ belongs to the interior of $P$; (2) $x$ belongs to the interior of an edge; (3) $x$ is a vertex. In all three cases, one can find a neighborhood $U$ of $[x]$ that is homeomorphic to $\mathbb{R}^2$.

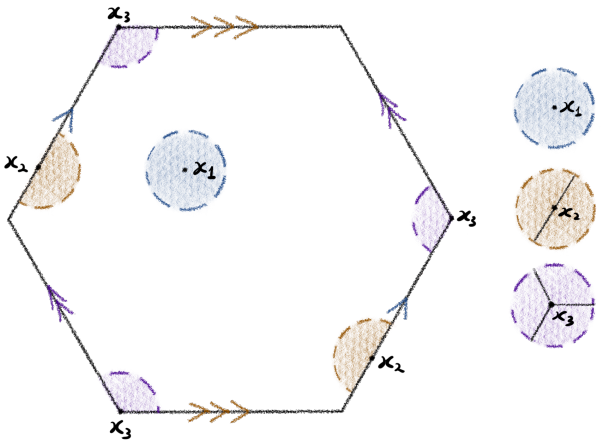


Figure 13.2

□

**Definition 13.4.** We consider the following two families of surfaces:

1. For any $g \geq 1$, let $D = a_1 b_1 a_1^{-1} b_1^{-1} a_2 b_2 a_2^{-1} b_2^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}$. Then $\Sigma_D$ is called an **orientable surface of genus $g$, denoted as $\Sigma_g$.** We also call $S^2$ to be an orientable surface of genus 0.
2. For any $g \geq 1$, we let $D' = a_1 a_1 a_2 a_2 \cdots a_g a_g$. Then $\Sigma_{D'}$ is called an **unorientable surface of genus $g$, denoted as $\Sigma_g^u$.**

Let's call these gluing data the "standard gluing data".

Now we state the main theorem of this section.

**Theorem 13.5.** *(1) Any two surfaces in definition 13.4 are not homeomorphic. Actually, they are not even homotopy equivalent.*

*(2) Any connected, closed surface is homeomorphic to one of the surfaces in definition 13.4.*

The proof of theorem 13.5 will take up most of today's lecture. Before starting the proof. We first describe these surfaces in a more intuitive way. An orientable surface of genus is just the torus $T^2$. And an unorientable surface of genus 1 is the projective plane $\mathbb{RP}^2$. An unorientable surface of genus 2 is the **Klein bottle**. (One can see this by cutting the square open along the diagonal.) To describe a surface of higher genus, we introduce an important operation on manifolds, called the "connected sum".



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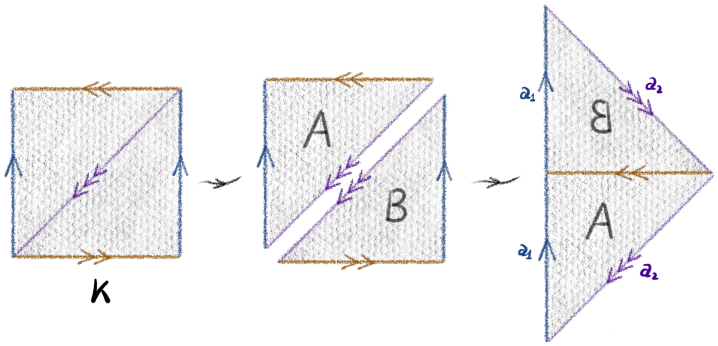


Figure 13.3

**Definition 13.6.** Let $X_1, X_2$ be closed, connected $n$-dimensional manifold. We points $x_i \in X_i$ and a homeomorphism $\varphi_i : U_i \to \mathbb{R}^n$ from some neighborhood $U_i$ of $x_i$. The connected sum of $X_1, X_2$ is defined as the following quotient space

$$X_1 \# X_2 := ((X_1 \setminus \varphi_1^{-1}(\mathring{D}^n)) \bigsqcup (X_2 \setminus \varphi_2^{-1}(\mathring{D}^n)))/\sim$$

where $\sim$ is generated by the relations

$$x \sim \varphi_2^{-1} \circ \varphi_1(x), \quad \forall x \in \varphi_1^{-1}(S^{n-1}).$$

In other words, we create two manifolds with $S^{n-1}$ as boundaries by removing an open balls $\varphi_i^{-1}(\mathring{D}^n)$ from $X_i$ (this operation is called "taking a puncture"). Then we glue the two boundaries together to form the closed manifold $X_1 \# X_2$.

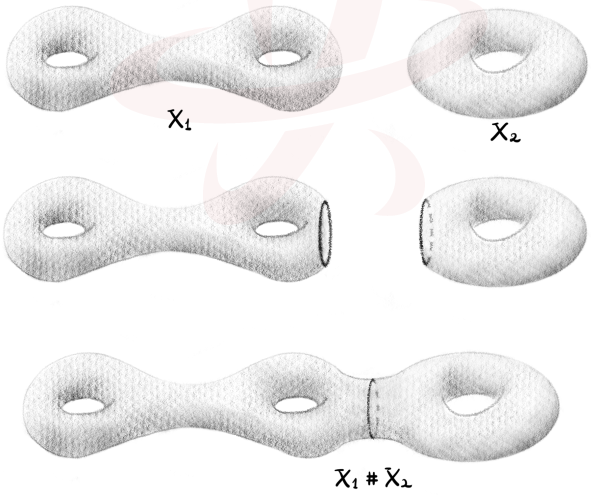


Figure 13.4

*Remark* 13.7. In the construction of $X_1 \# X_2$, we made choices of $(x_i, U_i, \varphi_i)$ for $i = 1, 2$. When $n = 2$, these choices do not affect the homeomorphism type of $X_1 \# X_2$. (This is essentially a consequence of the classification theorem of closed surface.). So there is no ambiguity in the notation $X_1 \# X_2$. When $n \geq 3$, there are two (but no more[1]) homeomorphism types in general, one is given by some $\{(x_1, U_1, \varphi_1); (x_2, U_2, \varphi_2)\}$ and the other one is obtained by $\{(x_1, U_1, \varphi_1); (x_2, U_2, r \circ \varphi_2)\}$, where $r : \mathbb{R}^n \to \mathbb{R}^n$ is the reflection $r(x_1, \cdots, x_n) := (-x_1, x_2, \cdots, x_n)$. So there is indeed a slight ambiguity. To get rid of this ambiguity, one

[1]For topological manifolds, this is a consequence of a deep theorem called the "annulus theorem". For smooth manifolds, the well-definedness of connected sum is much easier to prove.



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needs to specify the orientations on $X_i$. (We will soon learn the definition of orientations on manifolds).

**Exercise A.** *(1) Show that an orientable surface with genus $g$ is homeomorphic $\#_n T^2$ (the connected sum of $g$ copies of $T^2$). (Hint: try to construct the once-punctured torus $T^2 \setminus \mathring{D}^2$ by gluing some edges of a pentagon). (2) Show that an unorientable surface with genus $g$ is homeomorphic to $\#_g \mathbb{RP}^2$.*

Now prove theorem 13.5 (1) using the fundamental group.

**Lemma 13.8.** *(1) Let $\Sigma_D$ be an orientable surface of genus $g$. Then*

$$\pi_1(\Sigma_D) \cong \langle a_1, \cdots, a_g, b_1, \cdots, b_g \mid a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1} \rangle.$$

*(2) Let $\Sigma_D$ be an unorientable surface of genus $g$. Then*

$$\pi_1(\Sigma_D) \cong \langle a_1, \cdots, a_g \mid a_1^2 a_2^2 \cdots a_g^2 \rangle.$$

*Proof.* For any gluing data $D$, $\Sigma_D$ is a two-dimensional CW complex. Its 0-cells are equivalent classes of vertices; 1-cells are equivalent classes of edges. And the $P$ correspond to the unique 2-cell. For those $D$ in definition 13.4, all vertices are glued together so there is a unique 0-cell, which we choose as the base point. (This is not true for a general $D$.) Since each 1-cell gives a generator and each 2-cell gives a relation. The lemma is proved. □

One challenge is that the group can have different presentations. So even though the presentations in lemma 13.8 looks very different, we still need to show that they do not give the same group. To do this, we introduce the following definition.

**Definition 13.9.** Let $X$ a topological space. Suppose $X$ is path-connected. We define **the first homology group** $H_1(X;\mathbb{Z})$ to be the abelianization of its fundamental group $\pi_1(X)$. Suppose $X$ is not path-connected. We define $H_1(X;\mathbb{Z})$ to be the direct sum of the homology groups of all its path-components.

*Proof of theorem 13.5 (1).* By lemma 13.8, we have

$$H_1(\Sigma_D;\mathbb{Z}) \cong \mathbb{Z}^{2g}.$$

if $\Sigma_D$ is an orientable surface of genus $g$. And if $\Sigma_D$ is an unorientable surface of genus $g$, then we have

$$H_1(\Sigma_D) \cong \mathbb{Z}^g/(\mathbb{Z}\cdot(2,2,\cdots,2)) \cong \mathbb{Z}^g/(\mathbb{Z}\cdot(2,0,\cdots,0)) \cong \mathbb{Z}/2 \oplus \mathbb{Z}^{g-1}.$$

Since all these abelian groups are not isomorphic, the corresponding surfaces are not homotopy equivalent. □

**Exercise B.** *Given $n \in \mathbb{N}$, consider the space $X_n := ([0,1]\times S^1)/\sim$ where $\sim$ is generated by $(0,z) \sim (1,z^n)$ for all $z \in S^1$. Find a CW structure on $X_n$ and compute $H_1(X_n;\mathbb{Z})$. Show that $X_n$ is not homotopy equivalent to $X_m$ if $n \neq m$.*

**Definition 13.10.** Let $\Sigma$ be a closed surface. A (polygon) presentation of $\Sigma$ is a homeomorphism from $\Sigma$ to $\Sigma_D$ for some gluing data $D$. If the gluing data belongs to one of those in definition 13.4, we say the the presentation is standard.

The proof of theorem 13.5 (2) is a lot harder. It is implied by the following two propositions.

**Proposition 13.11.** *Any closed surface $\Sigma$ has a presentation.*



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**Proposition 13.12.** *Given any nonstandard presentation $\Sigma_D$, there is a (unique) standard $D_{st}$ such that $\Sigma_D \cong \Sigma_{D_{st}}$.*

The proof proposition 13.11 will be discussed in next subsection.

*Proof of proposition 13.12.* We have the following three operations that changes the data $D$ without changing the homeomorphism type of $\Sigma_D$:

1. Cyclic permute $D$. Interchange $a$ and $a^{-1}$. Replace $D$ by $D^{-1}$ (i.e. invert the order of the letters and invert all of them.)
2. Surgery of type A: Suppose $D = D_1 a a^{-1} D_2$. Then we change it into $D' = D_1 D_2$. (E.g. $abcc^{-1}ab \mapsto abab$.) Geometrically, this means the following Suppose two adjacent edges of $P$ are glued and they have opposite directions. Then we can just glue them and form a new polygon $P'$.
3. Surgery of type B: Suppose $D = D_1 a D_2 D_3 a^{-1} D_4$. Then we change it into $D' = D_1 D_4 a'^{-1} D_3 D_2 a'$. This corresponds to the following operations: We draw a line segment $a'$ that connects two (non-adjacent) vertices of $P$ and cut $P$ open along it. This gives two polygons $P_1, P_2$. Then we glue them together along a pair of edges $a$ and form $P'$. (Draw a picture.) Similarly, suppose $D = D_1 a D_2 D_3 a D_4$. Then we can change it into $D' = D_1 D_3^{-1} a' D_4^{-1} D_2 a'$. (This corresponds to the case that two edges for $a$ are in the same direction.)

By an element but tedious argument (see [**Youchengye**]), one can show that any $D$ can be transformed to some standard $D_{st}$ in definition 13.4. □

By proposition 13.12, we can perform a sequence of operation to turn a nonstandard $\Sigma_D$ into a standard one $\Sigma_{D_{st}}$. However, this process is usually tedious and we want a more efficient way to directly determine the homeomorphism type of $\Sigma_D$ (i.e. the orientability and the genus $g(\Sigma_D)$). To do this, we introduce the following definition.

**Definition 13.13.** Let $D$ be a gluing data. Then we define two invariants for a general $D$.

1. Suppose $D$ contains at least one pair of edges which are glued in the same direction. (I.e. $D$ contains $\cdots a \cdots a \cdots$ or $\cdots a^{-1} \cdots a^{-1} \cdots$ for some letter $a$.) Then we say $D$ is unorientable. Otherwise, we say $D$ is orientable.
2. Define the Euler number of $\Sigma_D$ by

$$\chi(\Sigma_D) := \#\{\text{equivalent classes of vertices}\} - \#\{\text{equivalent classes of edges}\} + 1$$

It's straightforward to check that both quantities are preserved under the three operations in the proof of proposition 13.12. So the orientablity and the Euler characteristic only depends on the homeomorphism type of $\Sigma_D$. Furthermore, a simple calculation (via $D_{st}$) gives us the relation between Euler characteristic and genus

$$\chi(\Sigma) = \begin{cases} 2 - 2g(\Sigma) & \text{if } \Sigma \text{ is orientable} \\ 2 - g(\Sigma) & \text{if } \Sigma \text{ is unorientable} \end{cases}. \tag{13.1}$$

This allows us to quickly determine the genus and orientably of a nonstandard $\Sigma_D$.

**Example 13.14.** *(1) $D = ab^{-1}a^{-1}bc^{-1}c$. Then $D$ contains no paired edges with same direction. So $\Sigma_D$ is orientable. And $\chi(\Sigma_D) = 2 - 3 + 1 = 0$. So $2 - 2g(\Sigma_D) = 0$. I.e. $\Sigma_D \cong T^2$.*

*(2) $D = ab^{-1}abcc$. Then $\Sigma_D$ is unorientable. And $\chi(\Sigma_D) = 1 - 3 + 1 = -1$. So $g(\Sigma_D) = 3$ and $\Sigma_D \cong \#_3 \mathbb{RP}^2$*

**Exercise C.** *Let $\Sigma$ be a connected closed surface (orientable or unorientable). Take $k$ disjoint points $x_1, x_2, \cdots, x_k \in \Sigma$. Show that*

$$\Sigma \setminus \{x_1, \cdots, x_k\} \simeq \bigvee_{1+k-\chi(\Sigma)} S^1.$$



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## 13.2 Simplicial complexes

We have learnt the definition of a CW complex. Now we introduce a more combinatorial construction, called the simplical complex.

**Definition 13.15.** A simplicial complex is a pair $K = (V, S)$, where $V$ is a set and $S$ is a collections of subsets of $V$, i.e.,

$$S \subset P(V) := \{\text{subsets of } V\}.$$

$V$ is called the set of vertices and $S$ is called the set of simplices. They need to satisfy the following conditions:

1. For any $v \in V$, we have $\{v\} \in S$.
2. For any $\sigma \in S$, we have $|\sigma| < \infty$.
3. Suppose $\sigma \in S$ and $\tau \subset \sigma$. Then $\tau \in S$.

We call $\sigma \in S$ an (combinatorial) $n$-simplex if $|\sigma| = n + 1$.

**Definition 13.16.** Let $\sigma = \{v_0, \cdots, v_n\}$ be an $n$-simplex. We use $\sigma$ as a basis and define the real vector space $\mathbb{R}^\sigma$ (so an element in $\mathbb{R}^\sigma$ is a formal linear combination of $v_i$, and two different linear combinations always give different vectors). The geometric $n$-simplex associated to $\sigma$ is defined as

$$\Delta^\sigma := \{\sum_{i=0}^{n} a_i v_i \mid a_i \in [0,1], \sum_{i=0}^{n} a_i = 1\} \subset \mathbb{R}^\sigma$$

We also denote $\Delta^{\{v_0,\cdots,v_n\}}$ by $\Delta^n$ and call it the standard geometric $n$-simplex. We here use the subspace topology of the standard topology on $\mathbb{R}^\sigma$.

**Example 13.17.** *$\Delta^{\{v_0\}}$ is a single point $v_0$. $\Delta^{\{v_0,v_1\}}$ is a line segment with vertices $v_0, v_1$. $\Delta^{\{v_0,v_1,v_2\}}$ is a triangle with vertices $v_0, v_1, v_2$. $\Delta^{\{v_0,v_1,v_2,v_3\}}$ is a tetrahedra with vertices $v_0, v_1, v_2, v_3$.*

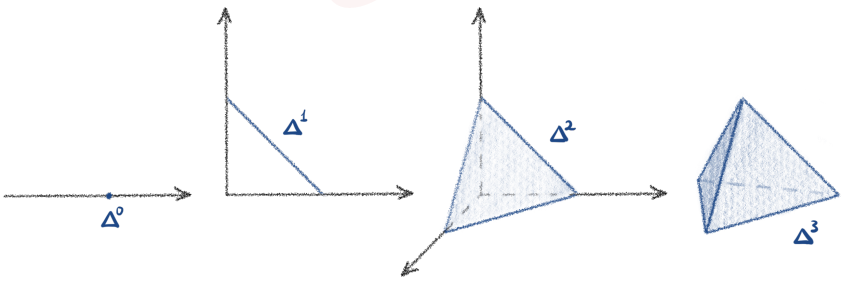


Figure 13.5

Note the given any subsets $\eta \subset \sigma$. We have the so-called "face inclusion map"

$$i_{\eta,\sigma} : \Delta^\eta \to \Delta^\sigma, \quad \sum_{i=0}^{n} a_i v_i \mapsto \sum_{i=0}^{n} a_i v_i.$$

(Geometrically, $\Delta^\eta$ can be thought as a face of $\Delta^\sigma$ of codimension $|\sigma| - |\eta|$, spanned by vertices that belongs to $\eta$. And the map $i_{\eta,\sigma}$ is just the inclusion map.)



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**Definition 13.18.** (1) Let $K = (V, S)$ be a simplical complex. The geometric realization of $(V, S)$, denoted by $|K|$ is the following space

$$|K| := (\bigsqcup_{\sigma \in S} \Delta^\sigma)/\sim .$$

Here $\sim$ is generated by $i_{\eta,\sigma}(x) \sim x$ for any $\eta \subset \sigma \in S$ and any $x \in \Delta^\eta$. In other words, $|K|$ is obtained by gluing the geometric $n$-simplices $\{\Delta^\sigma\}_{\sigma \in S}$ together along their common faces.

(2) A **trianglulation** of a topological space $X$ is a homeomorphism from $X$ to $|K|$ for some simplical complex $K$.

Geometrically, we can construct $|K|$ as follows: We start with the set of vertices $V$. Then we attach geometric simplices inductively according to their dimensions. For each combinatorial $n$-simplex $\sigma = \{v_{i_0}, v_{i_2}, \cdots, v_{i_n}\} \in S$, we attach a geometric $n$-simplex. From this description, we see that $|K|$ has a natural structure of a CW complex. The $l$-dimensional cells are the $l$-dimensional simplicies. And the $l$-skeleton is given by

$$\mathrm{Sk}_l |K| = \bigcup_{\sigma \in S,\ |\sigma| \le l+1} \Delta^\sigma.$$

**Example 13.19.** *Suppose $K = (V, S)$, where $V = \{1, 2, \cdots, n\}$ and $S = P(V) \setminus V$. Then*

$$|K| = \bigcup_{\sigma \subset V, \sigma \neq V} \Delta^\sigma = \partial \Delta^V \cong S^{n-1}.$$

We have the following deep theorem regarding triangulation of surfaces.

**Theorem 13.20** (Radó (1925))**.** *Every surface (compact or not, with or without boundary) has a triangulation.*

The proof is very delicate so we skip. (See Armstrong for a sketch of idea). (Draw a picture for a triangulation of $T^2$.)

*Remark* 13.21. It's natural to ask whether any $n$-dimensional manifold has a triangulation. When $n = 3$, the answer is yes (Moise, 1952). When $n = 4$, the answer is no (Casson, 1990). When $n \ge 5$, the answer is also no. (Manolescu, 2012). Note that if we assume $M$ has a smooth structure. Then it is always triangulable. (Whitehead, 1940).

*Proof of proposition 13.11.* Let $\Sigma$ be a connected, closed surface. By theorem 13.20. $\Sigma$ has a triangulation. Since $\Sigma$ is a 2-manifold without boundary, each 1-simplex is contained in exactly two 2-simplices.

So $\Sigma$ is obtained by gluing together a collection of two simplices along their edges. Since $\Sigma$ is compact, there are only finitely many two simplices $\Delta^2_1, \cdots, \Delta^2_n$. Now we construct the polygon $P$ inductively as follows: We pick any $\Delta^2_{i_1}$ and set it to be $P_1$. Suppose we have constructed

$$P_m = \Delta^2_{i_1} \cup \cdots \cup \Delta^2_{i_m}$$

for $m < n$. Since every edge is contained in two 2-simplices, every edge of $P_m$ must be contained in some other $\Delta^2_i$. We claim that there is at least one edge $e_m$ that is contained in some $\Delta^2_{i_{m+1}} \notin \{\Delta^2_{i_1}, \cdots \cup \Delta^2_{i_m}\}$. Otherwise, all the edges of $P_m$ are glued in pairs and we obtain a closed surface $\Sigma' \subset \Sigma$. This contradicts with the correctness of $\Sigma$. So the claim is proved.

We set $P_{m+1} = P_m \cup_{e_m} \Delta^2_{i_{m+1}}$. Keep this construction, we obtain $P = P_n$. And $\Sigma$ is obtained by gluing edges of $P$ in pairs. So $\Sigma \cong \Sigma_D$ for some gluing data $D$ on $P$. □

We end this section by the following classification theorem for compact surface with boundary, which can be proved similarly as theorem 13.5. (I.e, we first use theorem 13.20 to prove that any compact surface with boundary can be obtained by gluing some (but not all) edges of a polygon in pairs. Then we repeat the combinatorial proof of proposition 13.12.)



小红书
小红书号:3189205503 qq

**Theorem 13.22.** *Let $\Sigma_1, \Sigma_2$ be two compact surfaces with boundary. Suppose they have the same orientability, Euler number and same number of boundary components. Then $\Sigma_1 \cong \Sigma_2$.*

Some extra exercises:

**Exercise D.** *(1) Suppose $D = aab^{-1}c^{-1}bc$. Find (the genus and orientability of) $\Sigma_D$.*
*(2) Suppose $D = ab^{-1}a^{-1}c^{-1}bc$. Find (the genus and orientability of) $\Sigma_D$.*
*(3) Suppose $D = ad^{-1}bc^{-1}dab^{-1}c$. Find (the genus and orientability of) $\Sigma_D$.*

**Exercise E.** *Let $\Sigma$ be a connected, closed surface. (1) Suppose $\Sigma$ is orientable. Construct a compact 3-manifold $M$ with boundary such that $\partial M \cong \Sigma$.*
*(2) Suppose $\Sigma$ is unorientable and genus of $\Sigma$ is even. Construct a compact 3-manifold $M$ with boundary such that $\partial M \cong \Sigma$. (Actually, when the genus is odd, you can't find such a $M$. You will learn how to prove this in algebraic topology class.)*

**Exercise F.** *Let $M$ be a manifold. We define the configuration space*

$$\mathrm{Con}^n(M) := \{(x_1, \cdots, x_n) \in M \times \cdots \times M \mid x_i \neq x_j \text{ for any } i \neq j\}/\sim$$

*where $\sim$ is generated by*

$$(x_1, \cdots, x_n) \sim (x_1', \cdots, x_n') \text{ if } \{x_1, \cdots, x_n\} = \{x_1', \cdots, x_n'\}.$$

*(1) Show that $\mathrm{Con}^2(S^1) \cong$the open Mobius band.*
*(2) Show that $\mathrm{Con}^2(\mathbb{R}^2) \simeq S^1$.*

## 13.3 Orientation on manifolds

We have mentioned the orientability of a surface in definition 13.4 and definition 13.13. In this section, we give two equivalent definitions of orientations on manifolds. The first one makes use of triangulation (works for any dimension), and the second one makes use of fundamental groups (only works for surfaces).

We start with the first approach.

**Definition 13.23.** Let $K = (V, S)$ be a simplical complex and let $\sigma = \{v_0, \cdots, v_k\} \subset V$ be a $k$-simplex. We define an orientation on $\sigma$ as follows.

- Suppose $k = 0$. Then an orientation on $\sigma$ is an element in the set $\{+, -\}$.
- Suppose $k > 0$. Then an orientation on $\sigma$ is an element in the set

$$\{\text{orders of the vertices } v_0, \cdots, v_k\}/\text{even permutation}.$$

Let $\eta$ be a codimension-1 face of $\sigma$. I.e., $\eta \subset \sigma$ and $|\eta| = |\sigma| - 1$. Then an orientation on $\sigma$ induces an orientation on $\eta$ as follows: Suppose the orientation on $\sigma$ is given by $[v_{i_0}, \cdots, v_{i_k}]$ and suppose $\sigma \setminus \eta = v_{i_l}$. Then the induced orientation on $\eta$ is given by

$$(-1)^l [v_{i_0}, \cdots, \widehat{v_{i_l}}, \cdots, v_{i_k}].$$

Here $\widehat{v_{i_l}}$ means we remove $v_{i_l}$ from the vertices. And the minus sign means orientation reversal.

**Example 13.24.** *Let $\sigma = \{a, b, c\}$. Then $\sigma$ has two orientation $o = [a, b, c]$ and $-o = [a, c, b]$. $\sigma$ has three faces $\eta_0 = \{a, b\}, \eta_1 = \{b, c\}, \eta_2 = \{a, c\}$. The orientation on these faces induced by $o$ is given by $[a, b], [b, c], [c, a]$ respectively.*



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小红书号:3189205503 qq

**Definition 13.25.** Let $M$ be a $n$-dimensional manifold with a triangulation $M \cong |K|$. An orientation on $M$ is an orientation on each $n$-simplex of $K$. These orientations should be compatible to each other in the following sense:

- Given any two $n$-simplices $\sigma_1, \sigma_2$ that share a common $n-1$-simplex $\eta$ as a face, the orientations $o_{\eta_1}$ and $o_{\eta_2}$ induces different orientations on $\eta$.

Note that then every $(n-1)$-simplex is a face of one or two $n$-simplices. Suppose $M$ has no boundary. Then every $(n-1)$-simplex is a face of exactly two $n$-simplices. We say $M$ is **orientable** if it has an orientation. And we say $M$ is **unorientable** if it doesn't have an orientation. This turns out to be independent with the choice of the triangulation $K$.

Now we move on to the second approach which works for all 2-dimensional manifolds (including noncompact surfaces and surfaces with boundary).

**Definition 13.26.** Let $X$ be a (possibly noncompact) manifold without boundary. We make the following definitions.

1. For any $x \in X$, an orientation of $X$ at $x$ is a generator $o_x$ of $\pi_1(U \setminus \{x\})$, where $U$ is a neighborhood that is homeomorphic to $\mathbb{R}^2$.

2. Let $U, U'$ be two neighborhoods of $x$ that are both homeomorphic to $\mathbb{R}^2$. We can pick a third neighborhood $V \subset U \cap U'$ s.t. $V \cong \mathbb{R}^2$. Then the inclusion map induce an isomoprhism
$$\pi_1(U \setminus \{x\}) \xrightarrow{\cong} \pi_1(V \setminus \{x\}) \xrightarrow{\cong} \pi_1(U' \setminus \{x\}).$$
If a generator $o_x \in \pi_1(U \setminus \{x\})$ is mapped to $o'_x \in \pi_1(U \setminus \{x\})$ under this isomorphism, then we treat $o_x$ and $o'_x$ as the same orientation.

3. Let $U \subset X$ be a open set and let $\varphi : U \to \mathbb{R}^2$ be a homeomorphism. Given $x_1, x_2 \in U$, we consider the following homeomorphism
$$U \to U, \quad x \mapsto \varphi^{-1}(\varphi(x) - \varphi(x_1) + \varphi(x_2)),$$
which sends $x_1$ to $x_2$ and hence induces an isomorphism
$$\pi_1(U \setminus \{x_1\}) \xrightarrow{\cong} \pi_1(U \setminus \{x_2\}).$$
If a generator $o_{x_1} \in \pi_1(U \setminus \{x_1\})$ is sent to $o_{x_2} \in \pi_1(U \setminus \{x_2\})$ under this isomorphism, then we say $o_{x_1}$ and $o_{x_2}$ are compatible under $U$.

4. Let $\{o_x\}_{x \in X}$ be a collection of orientations of $X$ at $x$. We say it is locally compatible if there exists an open cover $\{U_\alpha\}$ of $X$ such that $U_\alpha \cong \mathbb{R}^2$ for any $\alpha$ and for any $x, x' \in U_\alpha$, $o_x$ and $o_{x'}$ are compatible under $U\alpha$.

5. An orientation of $X$ is a locally compatible collection $\{o_x\}_{x \in X}$.

6. We say $X$ is orientable if it has an orientation. And we say $X$ is unorientable otherwise.

7. For a surface $X'$ with boundary, we say $X'$ is orientable if $X' \setminus \partial X'$, which is a manifold without boundary, is orientable.

**Lemma 13.27.** *Let $X$ be a surface. Suppose the Mobius band $M$ can be embedded into $X$. Then $X$ is unorientable.*



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小红书号:3189205503 qq

*Proof.* Since any orientation on $X$ induces an orientation on $M$. It suffices to show that the mobius band

$$M = ([0,1] \times [0,1])/(0,s) \sim (1, 1-s)$$

is not orientable. We suppose $M$ has an orientation $\{o_x\}_{x \in M}$. We let $x$ goes around the loop $\gamma : [0,1] \to M$ defined by $\gamma(t) = (t, \frac{1}{2})$. Using the locally compatibility condition, we see that $o_{\gamma(0)}$ determines $o_{\gamma(t)}$ for any $t$ and that $o_{\gamma(1)} = -o_{\gamma(0)}$. This is a contradiction. (Draw a picture.) □

**Corollary 13.28.** *Let $D$ be a gluing data. Then $\Sigma_D$ is orientable (in the sense of definition 13.26 if and only if $D$ is orientable (in the sense of definition 13.13).*

*Proof.* Suppose $D$ contains a pair of edges glued together in the same direction. Then there exists an embedding of the Mobius band into $\Sigma_D$. Suppose $D$ does not contain such pairs. Then the standard orientation on the polygon $P$ can be glued along the edges to give an orientation on $\Sigma_D$. □



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# Chapter 14

# Classification of covering spaces

In this chapter, we give a classification theorem of covering spaces. Roughly speaking, such result build a 1-1 correspondence:

$$\{\text{path-connected covering spaces of } X\}/\text{isomorphism} \longleftrightarrow \{\text{subgroups of } \pi_1(X)\}/\text{conjugacy}$$

## 14.1 Lifting criteria

We start with the following lemma.

**Lemma 14.1.** *Let $p : (\widetilde{X}, \widetilde{x}_0) \to (X, x_0)$ be a covering map. Consider the induced map*

$$p_* : \pi_1(\widetilde{X}, \widetilde{x}_0) \to \pi_1(X, x_0).$$

*Then the following conclusion holds.*

1. *$p_*$ is injective.*
2. *$[\gamma] \in p_*(\pi_1(\widetilde{X}, \widetilde{x}_0))$ if and only if $\gamma$ can be lifted to a loop in $\widetilde{X}$ based at $\widetilde{x}_0$.*
3. *Suppose $\widetilde{X}$ and $X$ are path-connected. Then the number of sheets of $p$ (i.e. the number $|p^{-1}(x)|$, which doesn't depend on $x$ since $X$ is path-connected) equals the index $[\pi_1(X, x_0), p_*(\pi_1(\widetilde{X}, \widetilde{x}_0))]$.*

*Proof.* (1) Suppose $p_*([\widetilde{\gamma}]) = 0$. Let $\gamma = p \circ \widetilde{\gamma}$ and let $c_{x_0}$ be the constant loop at $x_0$. Then $\gamma \simeq c_{x_0}$. Note that $\widetilde{\gamma}$ and $c_{\widetilde{x}_0}$ lifts $\gamma$ and $c_{x_0}$ respectively and they both starts at $\widetilde{x}_0$. By the path-lifting property Corollary 10.8, we have $\widetilde{\gamma} \simeq c_{\widetilde{x}_0}$. So $[\widetilde{\gamma}] = 0$.

(2) Suppose $[\gamma]$ is in the image of $p_*$. Then $\gamma \simeq p \circ \widetilde{\gamma}'$ for some loop $\widetilde{\gamma}'$ based at $\widetilde{x}_0$. Let $\widetilde{\gamma}$ be the lift of $\gamma$ that starts from $\widetilde{x}_0$. Then by the path-lifting property again, $\widetilde{\gamma} \simeq \widetilde{\gamma}'$. In particular, $\widetilde{\gamma}$ is also a loop based at $\widetilde{x}_0$. This proves one direction. The other direction is straightforward.

(3) Let $H = p_*(\pi_1(\widetilde{X}, \widetilde{x}_0))$. We build a bijection $\varphi : \{H[\gamma]\} \to p^{-1}(x_0)$ as follows. For any $[\gamma] \in \pi_1(X, x_0)$, we take its lift $\widetilde{\gamma}$ that starts at $\widetilde{x}_0$. Then we define $\varphi(H[\alpha]) := \widetilde{\gamma}(1)$. There are three things to check:

- $\varphi$ is well-defined: We consider $\gamma, \gamma'$ with $H[\gamma] = H[\gamma']$. Then $\gamma \cdot \bar{\gamma}' \in H$. By (2), we can lift $\gamma \cdot \bar{\gamma}'$ to a loop $\delta : I \to \widetilde{X}$ based at $\widetilde{x}_0$. Note that $\delta|_{[0,\frac{1}{2}]}$ and $\bar{\delta}|_{[0,\frac{1}{2}]}$ lifts $\gamma$ and $\gamma'$ respectively. And they both starts at $\widetilde{x}_0$. Therefore, $\gamma$ and $\gamma'$ gives the same definition of $\varphi$, namely $\delta(\frac{1}{2})$.
- $\varphi$ is surjective: This is because $\widetilde{X}$ is path-connected.
- $\varphi$ is injective: Suppose $\varphi(H[\gamma]) = \varphi(H[\gamma'])$. Let $\widetilde{\gamma}$ and $\widetilde{\gamma}'$ be lifts of $\gamma$ and $\gamma'$ that start at $\widetilde{x}_0$. Then $[\gamma] \cdot [\bar{\gamma}'] \in H$ because it is the image of the loop $[\widetilde{\gamma} \cdot \bar{\widetilde{\gamma}}']$.



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□

**Example 14.2.** *Let $X = S^1 \vee S^1$. Then $\pi_1(X) = \langle a, b \mid \varnothing \rangle \cong F_2$. Let $\widetilde{X} =$ We have a 3-fold covering map $p : \widetilde{X} = S^1 \vee S^1 \vee S^1 \to X$. $p_*(\pi_1(\widetilde{X}) \cong F_3$ is an index 3-subgroup of $\pi_1(X)$ generated by $a, b^2, bab$. (Picture.)*

Now we introduce the following extremely useful lifting criteria:

**Theorem 14.3.** *Let $p : (\widetilde{X}, \widetilde{x}_0) \to (X, x_0)$ be a covering map and let $f : (Y, y_0) \to (X, x_0)$ be a continuous map. Suppose $Y$ is path-connected and locally path-connected. Then there exists a lift $\widetilde{f}$ of $f$ with $f(y_0) = \widetilde{x}_0$ if and only if $f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\widetilde{X}, \widetilde{x}_0))$*

$$\begin{array}{ccc} & & (\widetilde{X}, \widetilde{x}_0) \\ & \overset{\widetilde{f}}{\nearrow} & \big\downarrow p \\ (Y, y_0) & \xrightarrow{f} & (X, x_0) \end{array}.$$

*Moreover, such lift $\widetilde{f}$ with $\widetilde{f}(y_0)$ is unique in this case.*

*Proof.* One direction is clear. Suppose such $\widetilde{f}$ exists. Then $f_* = \widetilde{f}_* \circ p_*$. So the image of $f_*$ is contained in the image of $p_*$.

Now we assume $f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\widetilde{X}, \widetilde{x}_0))$ and construct the lift $\widetilde{f} : (Y, y_0) \to (\widetilde{X}, \widetilde{x}_0)$ explicitly.

We will construct $\widetilde{f}$ explicitly as follows. For any $y \in Y$, we take a path $\eta : I \to Y$ from $y_0$ to $y$. Let $\gamma = f \circ \eta$ and let $\widetilde{\gamma}$ be a lift that starts at $\widetilde{x}_0$. Then we define $\widetilde{f}(y) := \widetilde{\gamma}(0)$. There are several things to check:

- $\widetilde{f}$ is well-defined: Take any two paths $\eta, \eta'$ from $y_0$ to $y$. Then
$$[\gamma \cdot \bar{\gamma}'] = f_*([\eta \cdot \bar{\eta}']) \in f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(\widetilde{X}, \widetilde{x}_0)).$$
By Lemma 14.1, $\eta \cdot \bar{\eta}'$ can be lifted to a loop $\delta$ in $\widetilde{X}$ based at $\widetilde{x}_0$. Since $\delta|_{[0,\frac{1}{2}]}$ and $\bar{\delta}|_{[0,\frac{1}{2}]}$ lifts $\gamma$ and $\gamma'$ respectively and they both starts at $\widetilde{x}_0$, we see that $\widetilde{f}(y)$, defined either using $\eta$ or $\eta'$, equals $\delta(\frac{1}{2})$.
- $\widetilde{f}$ lifts $f$ and satisfies $\widetilde{f}(y_0) = \widetilde{x}_0$: This is straightforward to check.
- $\widetilde{f}$ is continuous: Take $y \in Y$. Let $U$ be an evenly covered neighborhood of $f(y)$ and let $\widetilde{U} \subset p^{-1}(U)$ be the sheet that contains $\widetilde{f}(y)$. We take a path-connected neighborhood $V$ of $y$ such that $f(V) \subset U$ (this is possible since $Y$ is locally path-connected.) We claim that $\widetilde{f}(y') \in \widetilde{U}$ for any $y' \in V$. To see this, we take a path $\eta : I \to Y$ from $y_0$ to $y$ and a path $\eta' : I \to V$ from $y$ to $y'$. Then we can use $\eta$ to define $\widetilde{f}(y)$ and use $\eta \cdot \eta'$ to define $\widetilde{f}(y')$. From this, we see that $\widetilde{f}(y)$ and $\widetilde{f}(y')$ belongs to the same path-component of $p^{-1}(U)$. So $\widetilde{f}(y') \in \widetilde{U}$. We see that $\widetilde{f}|_V = p_U^{-1} \circ f$. Here $p_U^{-1}$ denotes the inverse of the homeomorphism $\widetilde{U} \xrightarrow{p} U$.

It remains to show that any two lifts $\widetilde{f}$ and $\widetilde{f}'$ of $f$ which equals other at a single point are equal everywhere. To prove this, we consider the set

$$W := \{y \in Y \mid \widetilde{f}(y) = \widetilde{f}'(y)\}.$$

Then $W$ is nonempty. For any $y \in W$, we take a small neighborhood $U_1$ of $\widetilde{f}(y) = \widetilde{f}'(y)$ such that $p|_{U_1}$ is injective. Then we take a neighborhood $V$ of $y$ such that $\widetilde{f}(V), \widetilde{f}'(V) \subset U_1$. Since $p \circ \widetilde{f} = p \circ \widetilde{f}'$, we see that $\widetilde{f}|_V = \widetilde{f}'|_V$. This shows that $W$ is open. For any $y \notin W$, the



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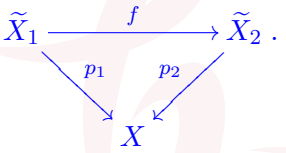
points $\widetilde{f}(y)$ and $\widetilde{f}'(y)$ are different points in $p^{-1}(f(y))$. Since $p$ is a covering map, we can find a neighborhood $U_2$ of $\widetilde{f}(y)$ and a neighborhood $U_3$ of $\widetilde{f}'(y)$ such that $U_2 \cap U_3 = \varnothing$. Let $V'$ be a small neighborhood of $y$ such that $\widetilde{f}(V') \subset U_2$ and $\widetilde{f}'(V') \subset U_3$. Then $V' \cap W = \varnothing$. So $W$ is also closed. Since $Y$ is connected, $W = Y$ so $\widetilde{f} = \widetilde{f}'$. □

**Example 14.4.** *For $n \geq 2$, let $f : S^n \to S^1$ be a continuous map. Then by Theorem 14.3, $f$ can be lifted to a map $\widetilde{f} : S^n \to \mathbb{R}$. Since $\mathbb{R}$ is contractible, $\widetilde{f}$ is null-homotopic. So $f$ is alao null-homotopic. This argument shows that any map from a high-dimensional sphere to a 1-dimensional sphere is null-homotoopic. But there can be maps from high a dimensional sphere to a low dimensional sphere which is not null-homotopic. For example, as we will learn in algebraic topology class, the Hopf map $\eta : S^3 \to S^2$ is not null-homotopic. (See Example 7.34.)*

## 14.2 Classification theorem

In this section, we will always assume $X$ is path-connected and locally path-connected. If $X$ is locally path-connected but not path-connected. Then we can apply the classification result to each path component of $X$.

**Definition 14.5.** Given covering spaces $p_1 : \widetilde{X}_1 \to X$ and $p_2 : \widetilde{X}_2 \to X$. A **morphism** from $\widetilde{x}_1$ to $\widetilde{x}_2$ is a continuous map $f$ that fits into the following commutative diagram




If such $f$ is a homeomorphism, then we call it an **isomorphism**.

**Proposition 14.6.** *Given covering spaces $p_1 : \widetilde{X}_1 \to X$ and $p_2 : \widetilde{X}_2 \to X$. Take base points $x_0 \in X$, $\widetilde{x}_1 \in p_1^{-1}(x_0)$ and $\widetilde{x}_2 \in p_1^{-1}(x_0)$. Suppose $X$, $\widetilde{X}_1$ and $\widetilde{X}_2$ are all path-connected and also suppose $X$ is locally path-connected. Then the following conclusion holds.*

1. *There exists a morphism $f : \widetilde{X}_1 \to \widetilde{X}_2$ with $f(\widetilde{x}_1) = \widetilde{x}_2$ if and only if*

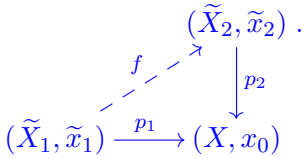
$$p_{1,*}(\pi_1(\widetilde{X}_1, \widetilde{x}_1)) \subseteq p_{2,*}(\pi_1(\widetilde{X}_2, \widetilde{x}_2)). \tag{14.1}$$

   *Furthermore, such a morphism $f$ is unique (if exists). And it is an isomoprhism if and only if $=$ in 14.1 holds.*

2. *There exists a morphism $f : \widetilde{X}_1 \to \widetilde{X}_2$ if and only if there exists some $[\gamma] \in \pi_1(X, x)$ such that*

$$[\gamma] \cdot p_{1,*}(\pi_1(\widetilde{X}_1, \widetilde{x}_1))[\gamma]^{-1} \subseteq p_{2,*}(\pi_1(\widetilde{X}_2, \widetilde{x}_2)). \tag{14.2}$$

*Proof.* (1) Finding such a morphism is equivalent to solving the lifting problem




So the first assertion and the uniqueness directly follows from Theorem 14.3. Suppose "=" holds. Then there also exists a morphism $g : \widetilde{X}_2 \to \widetilde{X}_1$ with $g(\widetilde{x}_2) = \widetilde{x}_1$. The $g \circ f$ is an automorphism



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小红书号:3189205503 qq

on $\widetilde{X}_1$ which fixes $\widetilde{x}_1$. By uniqueness, we see that $g \circ f = \mathrm{Id}_{\widetilde{X}_1}$. Similarly, $f \circ g = \mathrm{Id}_{\widetilde{X}_2}$. So $f$ is an isomorphism.

(2) Suppose there exists such $\gamma$. We lift $\gamma$ to a loop $\widetilde{\gamma}$ in $\widetilde{X}_2$ that starts at $\widetilde{x}_2$. Set $\widetilde{x}_2' = \widetilde{\gamma}(1)$. Then the following diagram commutes

$$\begin{array}{ccc} \pi_1(\widetilde{X}_2, \widetilde{x}_2') & \xrightarrow[\cong]{\widetilde{\gamma}^\#} & \pi_1(\widetilde{X}_2, \widetilde{x}_2) \\ \downarrow{\scriptstyle p_{2,*}} & & \downarrow{\scriptstyle p_{2,*}} \\ \pi_1(X, x_0') & \xrightarrow[\cong]{\gamma^\#} & \pi_1(X, x_0) \end{array}$$

Here $\widetilde{\gamma}^\#, \gamma^\#$ are the change of basepoint isomorphisms. By definition of $\gamma^\#$, we have

$$\gamma^\#(\alpha) = [\gamma] \cdot [\alpha] \cdot [\gamma]^{-1}, \quad \forall \alpha \in \pi_1(X, x_0').$$

So we see that

$$p_{2,*}(\pi_1(\widetilde{X}_2, \widetilde{x}_2)) = [\gamma] \cdot p_{2,*}(\pi_1(\widetilde{X}_2, \widetilde{x}_2')) \cdot [\gamma]^{-1}.$$

So the condition (14.2) translates to

$$p_{1,*}(\pi_1(\widetilde{X}_1, \widetilde{x}_1)) \subseteq p_{2,*}(\pi_1(\widetilde{X}_2, \widetilde{x}_2')).$$

By (1), there exists a morphism $f : \widetilde{X}_1 \to \widetilde{X}_2$ with $f(\widetilde{x}_1) = \widetilde{x}_2'$. This proves one direction. The other direction is similar (but easier). □

Given $X$, we consider the following set

$$\mathcal{C}(X) := \{\text{path-connected covering spaces of } X\}/\text{isomorphism}$$

with a "quasi-order" defined by

$$[\widetilde{X}_2] \leq [\widetilde{X}_1] \iff \text{There exists a morphism } \widetilde{X}_1 \to \widetilde{X}_2$$

(A quasi order on a set is a binary relation $\leq$ which satisfies reflecxivity axiom: $a \leq a$ for any $a$, the transitivity axiom $a \leq b, b \leq c \implies a \leq c$, but **not** the antisymmetry axiom $a \leq b, b \leq a \implies a = b$.) And the set

$$\mathcal{G}(X) := \{\text{subgroups of } \pi_1(X, x_0)\}/\text{conjugate}$$

with the quasi order defined by

$$[H] \leq [G] \iff H \subset G \textbf{ for some representatives } H, G.$$

Then we have a map

$$\psi : \mathcal{C}(X) \to \mathcal{G}(X) \tag{14.3}$$

defined by $\psi([\widetilde{X}]) := [p_*(\pi_1(\widetilde{X}, \widetilde{x}_0))$. Here $p$ is the covering map and $\widetilde{x}_0$ is any point in $p^{-1}(x_0)$. Proposition 14.6 can be rephrased as follows.

**Corollary 14.7.** *For any path-connected, locally path-connected space $X$, the map $\psi$ is injective and order **reversing**.*

Since the minimal element in $\mathcal{G}(X)$ is given by the trivial group, the corresponding element in $\mathcal{C}(X)$ is the maximum element. This motivates the following definition.

**Definition 14.8.** A covering space $p : \widetilde{X} \to X$ is called a universal covering space if $\widetilde{X}$ is simply-connected.



小红书号:3189205503 qq

By Corollary 14.7, the universal covering space $\widetilde{X}$ of $X$ (if exists) is unique up to isomorphism. And $\widetilde{X}$ has a morphism to any other path-connected covering spaces $\widetilde{X}'$ of $X$. It is not hard to prove that a morphism to a path-connected covering space is a covering map. So $\widetilde{X}$ is a covering space of all path-connected covering space $\widetilde{X}'$. This is the reason why we call $\widetilde{X}$ the universal covering space.

**Example 14.9.** *(1) The universal cover of $S^1$ is $\mathbb{R}$.*

*(2) The universal cover of $\mathbb{C}\setminus\{0\}$ is $\mathbb{C}$, with the covering map $\mathbb{C}\to\mathbb{C}\setminus\{0\}$ given by $z\mapsto e^z$.*

*(3) Let $\Sigma$ be a closed surface. Then the universal cover $\widetilde{\Sigma}$ of $\Sigma$ is divided into the following three cases*

$$\widetilde{\Sigma}=\begin{cases} S^2 & \text{if } \chi(\sigma)>0 \quad (\text{i.e. } \Sigma=S^2,\mathbb{RP}^2)\\ \mathbb{C} & \text{if } \chi(\sigma)=0 \quad (\text{i.e. } \Sigma=T^2, \textit{Klein bottle})\\ \mathring{D}^2 & \text{if } \chi(\sigma)=0 \quad (\text{i.e. } \textit{other surfaces})\end{cases}$$

*You may wonder why we need to distinguish $\mathring{D}$ and $\mathbb{C}$, which are homeomorphic to each other. Here is the reason: if we consider additional structures on $\Sigma$ (e.g. the complex structure or hyperbolic structure). Then the complex plane $\mathbb{C}$ and the unit disk $\mathring{D}$ are different.*

As a natural question, one may wonder when does a space $X$ have a universal cover. To address this question, we make the following definition.

**Definition 14.10.** A space $X$ is semilocally simply-connected if for any $x\in X$. There exists a neighborhood $U$ such that the map $\pi_1(U,x)\to\pi_1(X,x)$, which is induced by the inclusion $U\hookrightarrow X$, is trivial. In other words, any loop in $U$ based at $x$ is contractible in $X$.

**Theorem 14.11.** *(1) Suppose $X$ has a universal cover. Then $X$ is semilocally simply-connected.*

*(2) Suppose $X$ is path-connected, locally path-connected and semilocally simply-connected. Then $X$ has a universal cover.*

*Proof.* (1) Let $p:\widetilde{X}\to X$ be a universal covering space. Then for any point $x\in X$ and any point $\widetilde{x}\in p^{-1}(x)$, there exists a neighborhood $U$ of $x$ and a neighborhood $\widetilde{U}$ of $\widetilde{x}$ such that the restriction $p|_U:\widetilde{U}\to U$ is a homeomorphism. So the map $\pi_1(U,x)\to\pi_1(X,x)$ has a decomposition

$$\pi_1(U,x)\xrightarrow{(p|_{U,*})^{-1}}\pi_1(\widetilde{U},x)\to\pi_1(\widetilde{X},\widetilde{x})\xrightarrow{p_*}\pi_1(X,x).$$

Since $\pi_1(\widetilde{X},\widetilde{x})=0$, this composition is trivial.

(2) (sketched proof) We give the explicitly construction $\widetilde{X}$ as follows. Pick a base point $x_0\in X$ and consider the set

$$\widetilde{X}:=\{\gamma:I\to X\mid\gamma(0)=x_0\}/\text{homotopy}. \tag{14.4}$$

We give the set $\widetilde{X}$ a topology as follows: For any path $\gamma:I\to X$ starting at $x_0$ and any neighborhood $U$ of $\gamma(1)$. We consider the following subset

$$U_\gamma:=\{[\gamma\cdot\eta]\mid\eta:I\to U,\eta(0)=\gamma(1)\}\subset\widetilde{X}.$$

The following collection

$$\{U_{[\gamma]}\mid\gamma:I\to X, U\text{is a path-connected neighborhood of }\gamma(1)\}$$

can be shown to be a basis. We use this collection as a basis and define the topology on $\widetilde{X}$. There is a map $p:\widetilde{X}\to X$ defined by $p([\gamma]):=\gamma(1)$. We refer to [**Hatcher**] for a detailed explanation why $p$ is a covering map and why $\widetilde{X}$ is simply-connected. □



小红书号:3189205503 qq

**Example 14.12.** *(1) Locally contractible implies semilocally simply-connected. So any CW complex or any manifold is semilocally simply-connected. Hence any connected CW complex or connected manifold has a universal covering space.*

*(2) The Hawaiian earring is not semilocally simply-connected so it doesn't have a universal covering space.*

**Proposition 14.13.** *Let $X$ be a path-connected, locally path-connected, semi simply-connected space and let $x_0$ be a point. Then given any subgroup $H$ of $\pi_1(X, x_0)$. There exists a covering map $p : (\widetilde{X}_H, \widetilde{x}_0) \to (X, x_0)$ such that $p_*(\pi_1(\widetilde{X}_H, \widetilde{x}_0)) = H$.*

*Proof.* We can construct $\widetilde{X}_H$ explicitly as follows. Let $\widetilde{X}$ be the universal coveing space constructed in (14.4). We define an equivalence relation $\sim_H$ on $\widetilde{X}$ as follows: To points $[\gamma]$ and $[\gamma']$ in $\widetilde{X}$ are equivalent if and only if $\gamma(1) = \gamma'(1)$ and $[\gamma \cdot \bar{\gamma}'] \in H$. Then we take the quotient space $\widetilde{X}_H := \widetilde{X}/\sim_H$. Let $p : \widetilde{X}_H$ be defined as $p([\gamma]) := \gamma(1)$. We refer to [**Hatcher**] for a proof why $p$ is a covering map and why $p_*(\pi_1(\widetilde{X}, [c_{x_0}])) = H$. □

Putting Proposition 14.13 and Corollary 14.7 together, we finally get the classification theorem of covering space.

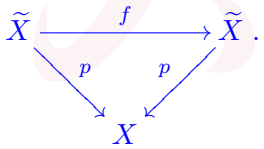
**Theorem 14.14.** *Let $X$ be a path-connected, locally path-connected, semilocally simply-connected space. Then the map $\psi : \mathcal{C}(X) \to \mathcal{G}(X)$ is bijective and order-reversing. In particular, we have a one-to-one correspondence*

$$\{\text{path-connected covering spaces of } X\}/\text{isomorphism} \longleftrightarrow \{\text{subgroups of } \pi_1(X)\}/\text{conjugation}.$$

*Furthermore, under this correspondence, $n$-fold covering spaces correspond to index-$n$ subgroups.*

One may wonder what kind of covering space does normal subgroups corresponds to. To answer this question, we make the following definition.

**Definition 14.15.** Let $p : \widetilde{X} \to X$ be a covering map. An covering transformation of $\widetilde{X}$ is an isomorphism from $\widetilde{X}$ to itself. I.e., a homeomorphism $f : \widetilde{X} \to \widetilde{X}$ that fits into the following commutative diagram




We use $G(\widetilde{X})$ to denote the group of covering transformations on $\widetilde{X}$.

**Definition 14.16.** We say $\widetilde{X}$ is a **normal covering space** (or regular covering space) if for any two points $\widetilde{x}, \widetilde{x}'$ with $p(\widetilde{x}) = p(\widetilde{x}')$, there exists a covering transformation $f : \widetilde{X} \to \widetilde{X}$ such that $f(\widetilde{x}) = \widetilde{x}'$.

**Proposition 14.17.** *Let $p : (\widetilde{X}, \widetilde{x}_0) \to (X, x_0)$ be a covering map. Assume that $\widetilde{X}$ is path-connected and that $X$ is locally path-connected. Then $\widetilde{X}$ is a normal covering space if and only if $p_*(\pi_1(\widetilde{X}, \widetilde{x}_0))$ is a normal subgroup of $\pi_1(\widetilde{X}, \widetilde{x}_0)$. In this case, we have*

$$G(X) \cong \pi_1(X, x_0)/p_*(\pi_1(\widetilde{X}, \widetilde{x}_0)). \tag{14.5}$$

*Proof.* Suppose $p$ is a normal covering map. Take any $[\gamma] \in \pi_1(X, x_0)$. We lift it to a path $\widetilde{\gamma}$ in $\widetilde{X}$ with $\widetilde{\gamma}(0) = \widetilde{x}_0$. Let $\widetilde{\gamma}(1) = \widetilde{x}_1$. Then as we have seen in the proof Proposition 14.6, we have

$$p_*(\pi_1(\widetilde{X}, \widetilde{x}_0)) = [\gamma] p_*(\pi_1(\widetilde{X}, \widetilde{x}_1))[\gamma]^{-1}.$$

Since there exists an isomorphism $\widetilde{X} \to \widetilde{X}$ sending $\widetilde{x}_0$ to $\widetilde{x}_1$, by Proposition 14.6, we must have

$$p_*(\pi_1(\widetilde{X}, \widetilde{x}_0)) = p_*(\pi_1(\widetilde{X}, \widetilde{x}_0')).$$



小红书号:3189205503 qq

This implies

$$p_*(\pi_1(\widetilde{X}, \widetilde{x}_0)) = [\gamma] \cdot p_*(\pi_1(\widetilde{X}, \widetilde{x}_0)) \cdot [\gamma]^{-1}.$$

Hence $p_*(\pi_1(\widetilde{X}, \widetilde{x}_0))$ is a normal subgroup. This proves the only if part. We refer to [**Hatcher**] for the proof of the if part, and the proof of (14.5). □

**Example 14.18.** *Let $\rho : G \times \widetilde{X} \to \widetilde{X}$ be a covering space action and let $X = \widetilde{X}/G$. (For example $G$ may be a finite, discrete group and $\rho$ is a free action.) Then as we mentioned in Example 10.5 $p : \widetilde{X} \to X$ is a covering map. Actually more is true: $p$ is a normal covering map. And the group of covering transformation is exactly the group $G$. Therefore, we see that $p_*(\pi_1(\widetilde{X}))$ is a normal subgroup of $\pi_1(X)$. And (14.5) tells us*

$$G \cong \pi_1(X)/p_*(\pi_1(\widetilde{X})).$$

*In particular, when $\widetilde{X}$ is simply-connected, we get the following result.*

**Corollary 14.19.** *Let $\widetilde{X}$ be a simply-connected, locally path-connected space with a covering space action by a group $G$. Then one has $\pi_1(\widetilde{X}/G) \cong G$.*

**Exercise A.** *Prove the following group theoretic results using covering space theory.*

*(1) Let $G$ be a free group of rank $k < \infty$ and let $H$ be a subgroup of index $m < \infty$. Show that $H$ is a free group of rank $1 + km - m$. (Similar argument can be used to prove the prove the Nielsen–Schreier theorem: Any subgroup of a free group is a free group. But this not required in this homework)*

*(2) Let $G$ be a finitely generated group and let $H$ be a subgroup. Suppose the index of $H$ is finite. Show that $H$ is also finitely generated. Given a counter example to show that $H$ may not be finitely generated if the index of $H$ is infinite.*

**Exercise B.** *Realize the group $G = \mathbb{Z}/2 * (\mathbb{Z} \oplus \mathbb{Z})$ as the fundamental group of a 2-dimensional CW-complex. Then classify all index-2 subgroups of $G$ up to isomorphism.*

**Exercise C.** *Consider the group $G = \mathbb{Z}/2 * \mathbb{Z}/3$ and its abelianization $G/[G, G] = \mathbb{Z}/2 \oplus \mathbb{Z}/3$. First realize $G$ as the fundamental group of a 2-dimensional CW-complex. Then use the covering space theory to prove that the kernel of the quotient map $G \to G/[G, G]$ is isomorphic to a free group of rank 2.*

**Exercise D.** *(1) Suppose $n \geq 2$. Show that any continuous map $\mathbb{RP}^n \to T^n$ is null-homotopic.*

*(2) Let $p : \widetilde{X} \to X$ be a covering map. Suppose $\widetilde{X}$ is connected. Show that $p$ is null-homotopic if and only if $\widetilde{X}$ is contractible.*

**Exercise E.** *Let $\widetilde{\Sigma}, \Sigma$ be two closed, connected surface that satisfy $\chi(\widetilde{\Sigma}) = k \cdot \chi(\Sigma)$ for some integer $k \geq 1$. Prove the following conclusions.*

*(1) Suppose $\Sigma$ is orientable. Show that there exists a $k$-fold covering map $p : \widetilde{\Sigma} \to \Sigma$ if and only if $\widetilde{\Sigma}$ is also orientable.*

*(2) Suppose $\Sigma$ is unorientable and $k$ is odd. Show that there exists a $k$-fold covering map $p : \widetilde{\Sigma} \to \Sigma$ if and only if $\widetilde{\Sigma}$ is also unorientable. (Hint: To show $\widetilde{\Sigma}$ must be unorientable, embed the mobius band into $\Sigma$ and think about its preimage under $p$.)*

*(3) Suppose $\Sigma$ is unorientable and $k$ is even. Show that there always exists a $k$-fold covering map $p : \widetilde{\Sigma} \to \Sigma$ (regardless whether $\widetilde{\Sigma}$ is orientable or not).*



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# Chapter 15

# Introduction to the simplical homology

## 15.1 Definition of the simplical homology group

Now we give a very quick introduction to simplicial homology theory, the main reference is [**MunkresAT**]. We will have spend a whole semester to discuss homology theory in the algebraic topology class.

Let $K = (V, S)$ be a simplicial complex and let

$$|K| = \bigcup_{\sigma \in S} \Delta^\sigma$$

be its geometric realization. (See definition 13.18.) Our goal is to define a combinatorial invariant of the topological space $|K|$ using $K$.

**Definition 15.1.** Let $\sigma \in S$ be a $n$-simplex. We define an orientation on $\sigma$ as follows:

- Suppose $n \geq 1$. An orientation on $\sigma$ is an element of the following set

$$\{\text{orders of elements in } \sigma\}/\text{even permutations.}$$

- Suppose $n = 0$. An orientation on $\sigma$ is an assignment of a $+$ or $-$ sign to $\sigma$.

In each case, a simplex $\sigma$ has exactly two orientations. An **oriented $n$-simplex** is an $n$-simplex $\sigma$ together with an orientation on it. With an abuse of notation, we continuous to use $\sigma$ to denote an oriented $n$-simplex and we use $\overline{\sigma}$ to denote its orientation reversal (i.e., the same simplex equipped with the other orientation). We also use $(v_0, \cdots, v_n)$ to denote an oriented $n$-simplex.

**Example 15.2.** *(1) Given a 0-simplex $\sigma = \{v_0\}$, it has two orientations $+(v_0)$ and $-(v_0)$.*

*(2) Given a 1-simplex $\sigma = \{v_0, v_1\}$, it has two orientations $(v_0, v_1)$ and $(v_1, v_0)$. Geometrically, an orientation determines which vertex is the starting point of $\Delta^\sigma$ and which vertex is the terminal point of $\Delta^\sigma$.*

*(3) Given a 2-simplex $\sigma = \{v_0, v_1, v_2\}$, it has two orientations $(v_0, v_1, v_2)$ and $(v_1, v_0, v_2)$. If we fix an embedding $\Delta^\sigma \hookrightarrow \mathbb{R}^2$ by sending $v_0, v_1, v_2$ to $(0, 0), (1, 0), (0, 1)$ respectively. Then $(v_0, v_1, v_2)$ means we are ordering the vertices counterclockwisely and $(v_1, v_0, v_2)$ means that we order the vertices clockwisely.*

**Definition 15.3.** Given $K = (V, S)$ and $n \geq 0$, we let $S_n \subset S$ be the set of (**unoriented**) $n$-simplices and let $\widetilde{S}_n$ be the set of **oriented** $n$-simplices. (So $|\widetilde{S}_n| = 2|S_n|$.) We consider the



小红书
小红书号:3189205503 qq

free abelian group generated by $\widetilde{S}_n$, denoted as $\mathbb{Z}\langle\widetilde{S}_n\rangle$. It has a subgroup $\mathbb{Z}\langle\{\sigma+\overline{\sigma}\}_{\sigma\in\widetilde{S}}\rangle$. We define the $n$-th simplical chain group for $K$ as the quotient group

$$C_n(K) := \mathbb{Z}\langle\widetilde{S}_n\rangle/\mathbb{Z}\langle\{\sigma+\overline{\sigma}\}_{\sigma\in\widetilde{S}_n}\rangle.$$

An element in $C_n(K)$ is called an $n$-chain. In other words, an $n$-chain is a formal linear combination of oriented $n$-simplices

$$\sum_{i=0}^{k} a_i\sigma_i, \quad a_i\in\mathbb{Z}, \sigma_i\in\widetilde{S}_n,$$

subject to the relation

$$a_1\sigma_1+\cdots+a_i\sigma_i+\cdots a_n\sigma_n = a_1\sigma_1+\cdots-a_i\overline{\sigma_i}+\cdots a_n\sigma_n.$$

**Example 15.4.** *Let $K=(V,S)$ where $V=\{v_0,v_1,v_2,v_3\}$ and $S=\{\sigma \mid \sigma\subsetneq V\}$. Then $|K|\cong S^2$ (i.e. the boundary of the tetrahedron). We have the following description of the chain groups:*

- *$C_0(K)=Z\langle(v_0),(v_1),(v_2),(v_3)\rangle\cong\mathbb{Z}^4$. This corresponds to the 4 vertices of the tetrahedron.*
- *$C_1(K)=\mathbb{Z}\langle(v_0,v_1),(v_0,v_2),(v_0,v_3),(v_1,v_2),(v_1,v_3),(v_2,v_3)\rangle\cong\mathbb{Z}^6$. This corresponds to the 6 edges of the tetrahedron.*
- *$C_2(K)=\mathbb{Z}\langle(1,2,3),(0,2,3),(0,1,3),(0,1,2)\rangle\cong\mathbb{Z}^4$. This corresponds to the 4 faces of the tetrahedron.*
- *$C_n(K)=0$ for any $n\neq 0,1,2$.*

**Definition 15.5.** The $n$-th boundary map $\partial_n: C_n(K)\to C_{n-1}(K)$ is the group homomorphism defined by

$$\partial(v_0,\cdots,v_n) := \sum_{i=0}^{n}(-1)^i(v_0,\cdots,\hat{v_i},\cdots,v_n).$$

Here $(v_0,\cdots,\hat{v_i},\cdots,v_n)$ means that we remove the vertex $\hat{v_i}$ and keep the order of the other vertices. An element in $\ker\partial_n$ is called an $n$-cycle.

**Lemma 15.6.** *The composition $\partial_n\circ\partial_{n+1}: C_{n+1}(K)\to C_{n-1}(K)$ is always zero.*

*Proof.* We have

$$\begin{aligned}\partial_n\circ\partial_{n+1}(v_0,\cdots,v_n) &= \partial_n(\sum_{i=0}^{n+1}(-1)^i(v_0,\cdots,\hat{v_i},\cdots,v_n))\\ &= \sum_{0\le i<j\le n}((-1)^{i+j-1}+(-1)^{i+j})(v_0,\cdots,\hat{v_i},\cdots,\hat{v_j},\cdots,v_n))\\ &= 0\end{aligned}$$

□

**Definition 15.7.** A **chain complex** $(C_*,\partial_*)$ is a sequence

$$\cdots\to C_{n+1}\xrightarrow{\partial_{n+1}} C_n\xrightarrow{\partial_n} C_{n-1}\xrightarrow{\partial_{n-1}} C_{n-2}\to\cdots$$



小红书
小红书号:3189205503 qq

of morphisms between abalian groups such that $\partial_{n+1} \circ \partial_n = 0$. The $n$-th homology of the chain complex is defined as

$$H_n(C_*) := \frac{\ker(\partial_n : C_n \to C_{n-1})}{\text{image}(\partial_{n+1} : C_{n+1} \to C_n)}.$$

An element in $\ker(\partial_n$ is called an $n$-cycle, an element in image is called an $n$-boundary, an element in $H_n(C_*)$ is called a $n$-th homology class. So a $n$-th homology class is represented by a $n$-cycle. Two $n$-cycles that differ by a $n$-boundary represent the same homology class.

**Definition 15.8.** The chain complex $(C_n(K), \partial_n)$ (where $\partial_n$ is defined as in definition 15.5) is called the simplical chain complex of a simplical complex $K$. It's $n$-th homology group is called the $n$-th homology of $K$, denoted as $H_n(K)$.

The following theorem is highly nontrivial and we discuss the idea of its proof later.

**Definition 15.9.** Let $X$ be a triangulable space and let $X \cong |K|$ be a triangulation. We define the **$n$-th simplical homology** of $X$ as

$$H_n(X) := H_n(K).$$

The following theorem is highly nontrivial and we will discuss the idea of its proof later.

**Theorem 15.10.** *(1) The simplical homology $H_n(X)$ does not depend on the choice of the triangulation $K$. So it is a topological invariant of $X$.*

*(2) Given any continuous map $f : X \to Y$ and any $n \in \mathbb{N}$, we have a well-defined homomorphism $f_* : H_n(X) \to H_n(Y)$ (called the induced map by $f$). It satisfies the following properties*

*(i) $f_* \circ g_* = (f \circ g)_*$;*

*(ii) The identity map on $X$ induces the identity map on $H_n(X)$;*

*(iii) Suppose $f \simeq f'$. Then $f_* = f'_*$.*

**Corollary 15.11.** *Any homotopy equivalence between two triangulable spaces induces an isomorphism on their simplical homology groups.*

**Example 15.12.** *Let $K$ be the simplical complex as in example 15.4 with $|K| \cong S^2$. One can compute that*

$$\text{image}\,\partial_1 = \{\sum_{i=0}^{3} a_i(v_i) \mid \sum_{i=0}^{3} a_i = 0\}.$$

$$\ker \partial_1 = \text{image}\,\partial_2$$

$$\ker \partial_2 = \mathbb{Z}\langle (1,2,3) - (0,2,3) + (0,1,3) - (0,1,2)\rangle.$$

*So $H_n(S^2) = H_n(K) \cong \mathbb{Z}$ when $n = 0, 2$ and $H_n(S^2) = 0$ for other $n$.*

**Exercise A.** *Draw a triangulation $K$ of $T^2$. Use this triangulation to compute $H_n(T^2)$ for any $n \geq 0$. (The most challenging part is the computation of $H_1(K)$. Here is the hint: we can let $K$ be induced by a triangulation of the square $[0,1]^2$. Any 1-cycle can be pushed to the boundary of this square by adding a suitable boundary, so we only need to consider those 1-cycles that are contained in $\partial[0,1]^2$.*

*To show two cycles do not differ by a boundary, draw a loop $\gamma : S^1 \to T^2$ that doesn't pass any vertices of $K$. Given any 1-cycle*

$$\xi \in \ker(\partial : C_1(K) \to C_0(K),$$

*we can count the number of intersection points between $\xi$ and $\gamma$ (with sign and weight). Show that this number is unchanged if we add $\xi$ by a boundary.)*



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小红书号:3189205503 qq

**Exercise B.** *Consider the simplical complex $K = (V, S)$, where $V = \{0, \cdots, n\}$ and $S = \{\sigma \mid \sigma \subsetneq V\}$. As we have seen in example 13.19, $|K| \cong S^{n-1}$.*

*(1) Use definition to show that $H_0(K) \cong \mathbb{Z}$.*

*(2) Use definition to show that $H_{n-1}(K) \cong \mathbb{Z}$. (Hint: any $(n-2)$-simplex is contained in exactly two $(n-1)$-simplices.)*

*(3) Use definition to show that $H_m(K) = 0$ for any $1 \leq m \leq n-2$. (Hint: for $0 \leq m \leq n-1$, let $C'_m(K)$ be the subgroup of $C_m(K)$ consisting of linear combinations of oriented simplices that do not have $n$ as a vertex. Construct an isomorphism of abelian groups*

$$C'_{m-1}(K) \oplus C'_m(K) \cong C_m(K).$$

*Try to understand the boundary map $\partial$ under this decomposition.)*

## 15.2 The simplical approximation theorem

We sketch the proof of theorem 15.10 in this section. We start with some algebraic definitions.

**Definition 15.13.** Let $(C_n, \partial_n)$ and $(D_n, \partial'_n)$ be two chain complexes. A chain map is a collection of homomorphisms $\tilde{f} = \{f_n : C_n \to D_n\}$ such that $\partial'_n \circ f_n = f_{n-1} \circ \partial_n$. In other words, the following diagram commutes

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & C_{n+1} & \xrightarrow{\partial_{n+1}} & C_n & \xrightarrow{\partial_n} & C_{n-1} & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle f_{n+1}} & & \downarrow{\scriptstyle f_n} & & \downarrow{\scriptstyle f_{n-1}} & & \\
\cdots & \longrightarrow & D_{n+1} & \xrightarrow{\partial'_{n+1}} & D_n & \xrightarrow{\partial'_n} & D_{n-1} & \longrightarrow & \cdots .
\end{array}$$

Any chain map $\{f_n\}$ sends a cycle to a cyle and sends a boundary to boundary. So it induces a map

$$f_* : H_n(C_*) \to H_n(D_*).$$

**Definition 15.14.** Given two chain maps $\tilde{f} = \{f_n\}$ and $\tilde{g} = \{g_n\}$ from $(C_n, \partial_n)$ to $(D_n, \partial'_n)$, a chain homotopy between them is a sequence of maps $\{h_n : C_n \to D_{n+1}\}$ such that

$$f_n - g_n = \partial'_{n+1} \circ h_n + h_{n-1} \circ \partial_n.$$

We say $\tilde{f}$ and $\tilde{g}$ are chain homotopic (denoted as $\tilde{f} \simeq \tilde{g}$) if such chain homotopy exists.

**Lemma 15.15.** *Suppose $\tilde{f} \simeq \tilde{g}$. Then the induced maps $f_*$ and $g_* : H_n(C_*) \to H_n(D_*)$ are equal.*

*Proof.* Take any homology class $[\alpha] \in H_n(C_*)$. Represented by a $n$-cycle $\alpha \in \ker \partial_n$. We have

$$f_n(\alpha) - g_n(\alpha) = \partial'_{n+1} \circ h_n(\alpha)$$

So $[f_n(\alpha)] = [g_n(\alpha)] \in H_n(D_*)$. □

**Definition 15.16.** (1) A chain map $\tilde{f} : C_* \to D_*$ is called a chain homotopy equivalence if there exists a chain map $\tilde{g} : D_* \to C_*$ such that $\tilde{f} \circ \tilde{g} \simeq \mathrm{Id}_{D_*}$ and $\tilde{g} \circ \tilde{f} \simeq \mathrm{Id}_{C_*}$. Two chains are chain homotopic if there exists a chain homotopy equivalence between them.

(2) A chain map $\tilde{f} : C_* \to D_*$ is called a quasi-isomorphism if the induced map $f_* : H_n(C_*) \to H_n(D_*)$ is an isomorphism for any $n$.

**Corollary 15.17.** *(1) Any chain homotopy equivalence is a quasi-isomorphism.*

*(2) Suppose $C_*$ and $D_*$ are are chain-homotopic to each other. Then $H_n(C_*) \cong H_n(D_*)$ for any $n$.*



小红书
小红书号:3189205503 qq

Now we return to the discussion of simplical complexes.

**Definition 15.18.** Let $K = (V, S)$ be simplical complex. Its barycentric subdivision $K^{(1)} = \{V^{(1)}, S^{(n)}\}$ is defined as follows: We set $V^{(1)} = S$ and set

$$S^{(1)} := \{\{\sigma_1, \cdots, \sigma_m\} \subset S \mid \sigma_1 \subsetneq \sigma_2 \subsetneq \cdots \subsetneq \sigma_m\}$$

Geometrically, this means that we add a vertex in the center of each simplex. (Draw a picture) We have a canonical homeomorphism $|K^{(1)}| \cong |K|$.

We define a chain map $\{\eta_n : C_n(K) \to C_n(K^{(1)})\}$ inductively as follows:

- For $n = 0$, we define $\eta_0(v) := (v)$;
- For $n = 1$, let $\sigma = (v_0, v_1)$, then we define
$$\eta_1(\sigma) = (v_0, \sigma) + (\sigma, v_1).$$
- Suppose $\eta_q$ is already defined for $q \leq n-1$ and satisfies $\partial_q^{(1)} \circ \eta_q = \eta_{q-1} \circ \partial_q$. For any $n$-simplex $\sigma$, suppose we have
$$\eta_{n-1}(\partial_n \sigma) = \sum a_{i_0,\cdots,i_{n-1}}(\sigma_{i_0}, \sigma_{i_2}, \cdots, \sigma_{i_{n-1}}).$$
Then we define $\eta_n(\sigma) := \sum a_{i_0,\cdots,i_{n-1}}(\sigma_{i_0}, \sigma_{i_2}, \cdots, \sigma_{i_{n-1}}, \sigma)$.
One can check that this is actually well-defined and satisfies $\partial_n^{(1)} \circ \eta_n = \eta_{n-1} \circ \partial_n$. Actually, it's not hard to check that
$$\eta_n(\sigma) = \sum_{\varnothing \neq \sigma_0 \subsetneq \sigma_2 \subsetneq \cdots \subsetneq \sigma_n = \sigma} \pm(\sigma_0, \sigma_2, \cdots, \sigma_n) \tag{15.1}$$

**Lemma 15.19.** *The chain map $\{\eta_n\}$ is a quasi-isomorphism. In particular, it induces an isomorphism $H_n(K) \cong H_n(K^{(1)})$ for any $n \in \mathbb{N}$.*

*Proof.* See [**Youchengye**]. □

Let $K^{(m)}$ be the $m$-th barycentric subdivision of $K$. Then we can use lemma 15.19 to fix an indentification between the homology groups

$$H_n(K) \cong H_n(K^{(1)}) \cong \cdots \cong H_n(K^{(m)})$$

for any $m > 0$.

**Definition 15.20.** (1) Let $K = (V, S)$ and $K' = (V', S')$ be two simplical complexes. A simplical map from $K$ to $K'$ is a map $\rho : V \to V'$ such that for any $\sigma \in S$, we have $\rho(\sigma) \in S'$.

(2) A simplical map $\rho$ from $K$ to $K'$ induces a chain map $r\tilde{h}o : C_*(K) \to C_*(K')$, defined by

$$\tilde{\rho}(v_0, \cdots, v_n) = \begin{cases} (\rho(v_0), \cdots, \rho(v_n)) & \text{if} \rho(v_i) \neq \rho(v_j) \forall i \neq j \\ 0 & \text{otherwise} \end{cases}$$

So it induces a map $\rho_* : H_n(K) \to H_n(K')$.

(3) A simplical map $\rho$ from $K$ to $K'$ also induces a continuous map $|\rho| : |K| \to |K'|$ defined by

$$\sum a_i v_i \in \Delta^\sigma \mapsto \sum a_i \rho(v_i) \in \Delta^{\rho(\sigma)}.$$

A continuous map from $f : |K| \to |K'|$ is called a simplical map if $f = |\rho|$ from simplical map $\rho$.

(4) Let $f : |K| \to |K'|$ be a continuous map. We say a simplical map $\rho : K \to K'$ is a simplical approximation of $f$ if the following condition is satisfied:



小红书
小红书号:3189205503 qq

- For any $x \in |K|$ we have $|\rho|(x) \in \Delta^{\sigma(f(x))}$, where

$$\sigma(f(x)) = \min\{\sigma \in S \mid f(x) \in \Delta^{\sigma}\}.$$

Note that this condition implies that $f \simeq |\rho|$.

The following theorem is the key technical result to prove the invariance of homology.

**Theorem 15.21** (Simplicial approximation theorem)**.** *Let $f : |K| \to |K'|$ be a continuous map between the geometric realization of $K$ and $K'$. Then for any $m$ large enough. The map $f : |K^{(m)}| \cong |K| \to |K'|$ has a simplical approximation $\rho : K^{(m)} \to K'$.*

*Proof.* See Armstrong page 168. □

*Idea of the proof of theorem 15.10.* Given any continuous $f : |K| \to |K'|$, we define the induced map $f_* : H_n(|K|) \to H_n(|K'|)$ as follows: We pick $m \gg 0$ such that $f : |K^{(m)}| \to |K'|$ has a simplical approximation $\rho$. Then we define

$$f_* : H_n(|K|) = H_n(K) \cong H_n(K^{(m)}) \xrightarrow{\rho_*} H_n(K') = H_n(|K'|).$$

Furthermore, suppose $\rho$ and $\rho' : K^{(m)} \to K'$ are simplical approximations of $f$ and $g$ respectively. And suppose there is a homotopy $h : I \times |K| \to |K'|$ from $f$ to $g$. Then we can use a simplical approximation of $h$ to find a chain homotopy between the chain maps $\tilde{\rho}$ and $\tilde{\rho'}$. So we have $f_* = g_*$. □

*Remark* 15.22. For any triangulable space $X$, we have a canonical isomorphism

$$H_0(X) \cong \mathbb{Z}\langle \pi_0(X) \rangle \tag{15.2}$$

where $\pi_0(X)$ denotes the set of path-components for $X$. We also have an canonical isomorphism

$$H_1(X) \cong \bigoplus_{[b] \in \pi_0(X)} \pi_1(X, b)/[\pi_1(X, b), \pi_1(X, b)]$$

where we take one base point $b$ in each path component.

## 15.3 Applications of the simplical homology

In this section, we discuss some first applications of the simplical homology.

As we have proved in exercise B, the homology of $S^m$ is given by

$$H_m(S^n) = \begin{cases} \mathbb{Z} \oplus \mathbb{Z} & \text{if } m = n = 0 \\ \mathbb{Z} & \text{if } n \neq m = 0 \text{ or } n = m \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

Therefore, we get some immediate corollaries of corollary 15.11

**Corollary 15.23.** *Suppose $n \neq m$. Then $S^n \not\simeq S^m$.*

**Corollary 15.24.** *Suppose $n \neq m$. Then $\mathbb{R}^n \ncong \mathbb{R}^m$.*

*Proof.* $\mathbb{R}^n \setminus \{\text{a point}\} \simeq S^{n-1}$, while $\mathbb{R}^m \setminus \{\text{a point}\} \simeq S^{m-1}$. □

**Corollary 15.25.** *For any $n \geq 1$, $S^{n-1}$ is not a retract of $D^n$.*



小红书 小红书号:3189205503 qq

*Proof.* The case $n = 1$ can be proved using path-connectedness of $D^1$ so we assume $n \geq 2$. Suppose there is a retraction $r : D^n \to S^{n-1}$. Let $i : S^{n-1} \to D^n$ be the inclusion map. Then $r \circ i = \mathrm{Id}_{S^{n-1}}$. This implies the composition

$$H_{n-1}(S^{n-1}) \xrightarrow{i_*} H_{n-1}(D^n) \xrightarrow{r_*} H_{n-1}(S^{n-1})$$

is the identity map. In particular, $i_*$ must be injective. This is impossible because $H_{n-1}(S^{n-1}) \cong \mathbb{Z}$ while $H_{n-1}(D^n) \cong H_{n-1}(\text{point}) = 0$. □

As we have seen in the proof of corollary 1.5. corollary 15.25 directly implies the Brower fixed point theorem.

**Theorem 15.26** (Brouwer fixed point theorem)**.** *Any continous map $f : D^n \to D^n$ has a fixed point.*

Actually, there is a vast generalization of the Brouwer fixed point theorem called the Lefschetz fixed point theorem. To state it, we first note that given any abelian group $A$, we can define the simplical chain complex with coefficient $A$ by $C_n(|K|; A) := C_n(K) \otimes A$ with boundary map $\partial_n \otimes \mathrm{Id}_A$. The resulting homology group

$$H(|K|; A) := \frac{\ker(\partial_n \otimes A : C_n(K; A) \to C_{n-1}(K; A))}{\mathrm{image}(\partial_{n+1} \otimes A : C_{n+1}(K; A) \to C_n(K; A))}$$

In particular, we can take $A = \mathbb{Q}$. Then $H_n(|K|; \mathbb{Q})$ is a vector space over $\mathbb{Q}$.

**Definition 15.27.** Let $f : X \to X$ be a continuous map from a compact, triangulable space $X$ to itself. The Lefschetz number of $f$ is defined as

$$\mathrm{Lef}(f) := \sum_{i \in \mathbb{N}} \mathrm{Trace}(f_* : H_n(X; \mathbb{Q})) \to H_n(X; \mathbb{Q})).$$

(Compactness implies that any triangulation of $X$ is finite. So $H_n(X; \mathbb{Q})$ is finite dimensional for each $n$ and is nonzero for only finitely many $n$.)

**Theorem 15.28** (Lefschetz fixed point theorem)**.** *Let $f : X \to X$ be a continuous map from a compact, triangulable space to itself. Suppose $\mathrm{Lef}(f) \neq 0$. Then $f$ has a fixed point.*

*Proof.* Any compact, triangulable space is metrizable. We pick a compatible metric on $X$. Suppose $f : X \to X$ has no fixed point. By compactness, we have $d(f(x), x) \geq \epsilon$ for some $\epsilon > 0$. We take a triangulation $|K| \cong X$. By replacing $K$ with $K^{(l)}$ for some $l \gg 0$, we can assume that the diameter of any simplex $\Delta^\sigma \subset |K|$ is less than $\frac{\epsilon}{4}$. We apply theorem 15.21 to the map $f : |K| \to |K|$ and get a simplical approximation $\rho : |K^{(m)}| \to |K|$ of $f$. Then we have

$$d(f(x), |\rho|(x)) \leq \frac{\epsilon}{4}.$$

This implies that

$$|\rho|(\Delta^\sigma) \cap \Delta^\sigma = \varnothing. \tag{15.3}$$

Note that the map $f_* : H_n(K^{(m)}; \mathbb{Q}) \to H_n(K^{(m)}; \mathbb{Q})$ is induced by the composition

$$C_*(K^{(m)}; \mathbb{Q}) \xrightarrow{\tilde{\rho}} C_*(K; \mathbb{Q}) \xrightarrow{\eta} C_*(K^{(1)}; \mathbb{Q}) \xrightarrow{\eta} C_*(K^{(2)}; \mathbb{Q}) \to \cdots \to C_*(K^{(m)}; \mathbb{Q})$$

where $\eta$ is the quasi-isomorphism defined in (eq. (15.1)). We denote this compositing by

$$f_n : C_n(K^{(m)}; \mathbb{Q}) \to C_*(K^{(m)}; \mathbb{Q}).$$

Then (eq. (15.3)) implies the diagonal entries of $f_{n,*}$ are all zero. So we have

$$\sum_{n \in \mathbb{N}} (-1)^n \mathrm{Trace}(f_n) = 0$$

By exercise C, this is exactly the Lefschetz number of $f$. □



小红书
小红书号:3189205503 qq

**Exercise C.** *This problem is the key algebraic lemma in the proof of the Lefschetz fixed point theorem. Consider a chain complex $V_\bullet$ of vector spaces over $\mathbb{Q}$:*

$$0 = V_{n+1} \xrightarrow{\partial_{n+1}} V_n \xrightarrow{\partial_n} V_{n-1} \xrightarrow{\partial_{n-1}} V_{n-2} \to \cdots \to V_1 \xrightarrow{\partial_1} V_0 \xrightarrow{\partial_0} V_{-1} = 0.$$

*In other words, $V_m$ is a finite dimensional vector space and $\partial_m : V_m \to V_{m-1}$ is a linear map that satisfies $\partial_m \circ \partial_{m-1} = 0$. Let $f = \{f_m : V_m \to V_m\}_{0 \le m \le n}$ be a chain map. (I.e., we have $f_m \circ \partial_{m+1} = \partial_{m+1} \circ f_{m+1}$.) Then $f$ induces a map*

$$f_{m,*} : H_m(V_\bullet) \to H_m(V_\bullet).$$

*on the $m$-th homology $H_m(V_\bullet) := \ker \partial_m / \operatorname{Image} \partial_{m+1}$. Prove the following relation*

$$\sum_{m=0}^{n} (-1)^m \operatorname{Trace}(f_m) = \sum_{m=0}^{n} (-1)^m \operatorname{Trace}(f_{m,*}).$$

*So the Lefschetz number of a chain map equals the Lefschetz number of the induced map on homology.*

**Corollary 15.29.** *Let $f : X \to X$ be a continuous map on a compact, triangulable space $X$. Then we have*

*(1) Suppose $X$ is contractible. Then $f$ has a fixed point.*

*(2) Suppose $\chi(X) \neq 0$ and $f \simeq \mathrm{Id}_X$. Then $f$ has a fixed point.*

*Proof.* (1) Since $X$ is contractible, $X$ is path-connected. By (a variation of) equation (eq. (15.2)), we have $H_0(X;\mathbb{Q}) \cong \mathbb{Q}$ and the map $f_* : H_0(X;\mathbb{Q}) \to H_0(X;\mathbb{Q})$ is the identity map. Since $X$ is contractible, $H_n(X;\mathbb{Q}) = 0$ for any $n > 0$. So $\operatorname{Lef}(f) = 1$.

(2) By exercise C again, we have

$$\operatorname{Lef}(f) = \operatorname{Lef}(\mathrm{Id}_X) = \sum_{i \in \mathbb{N}} (-1)^n \dim(C_n(K;\mathbb{Q})) = \sum_{i \in \mathbb{N}} (-1)^n |S_n| = \chi(X) \neq 0.$$

□

**Exercise D.** *Let $n \ge 1$. Any continuous map $f : S^n \to S^n$ induces a map*

$$f_* : \mathbb{Z} \cong H_n(S^n) \to H_n(S^n) \cong \mathbb{Z}.$$

*The mapping degree of $f$ is defined as*

$$\boldsymbol{deg}(f) := f_*(1) \in \mathbb{Z}.$$

*This is a generalization of the winding number to higher dimensions.*

*(1) Consider the antipodal map $f_0 : S^n \to S^n$ defined by $f_0(x) = -x$. Show that $\deg(f_0) = (-1)^{n-1}$. (Hint: use Lefschetz fixed point theorem.)*

*(2) Suppose $n$ is even, show that $f_0$ is not homotopy to the identity map.*

*(3) Suppose $n$ is even. Show that there does not exist a continuous function $g : S^n \to S^n$ such that for any $x \in S^n$, we have*

$$g(x) \notin \{x, -x\}.$$



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